

# **$q$ -Pochhammer Symbols and $q$ -Binomial Coefficients**

**Algebra, Combinatorics, Analysis, Arithmetic, and Summation Theory**

A self-contained, proof-oriented monograph

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# Abstract

The finite and infinite  $q$ -Pochhammer symbols and the Gaussian, or  $q$ -binomial, coefficients form a small set of notational primitives from which a remarkable amount of mathematics can be built. They simultaneously encode weighted subsets, partitions inside rectangles, lattice paths, subspaces over finite fields, Schubert cells in Grassmannians, noncommutative binomial expansions,  $q$ -difference operators, basic hypergeometric summations, theta products, partition identities, and congruences at roots of unity.

This monograph develops those themes from first principles. It proves the algebraic laws of shifted factorials; the polynomial, reciprocal, and cyclotomic structure of Gaussian coefficients; finite and infinite  $q$ -binomial theorems; weighted generalizations through elementary and complete symmetric functions; combinatorial and finite-geometric models; unimodality by  $\mathfrak{sl}_2$  representation theory; the  $q$ -Gauss,  $q$ -Chu–Vandermonde, and  $q$ -Pfaff–Saalschütz summations; Jackson calculus and the  $q$ -gamma and  $q$ -beta functions; Jacobi’s triple product, Euler’s pentagonal theorem, and the Rogers–Ramanujan identities through Bailey pairs; cyclotomic factorization,  $q$ -Lucas and  $q$ -Babbage congruences; negative upper indices and  $q$ -Newton interpolation; a recent family of central  $q$ -binomial/Lambert-series evaluations; geometric Lagrange filters with exact Gaussian residual moments; half-base Mersenne normalizations and Prouhet–Thue–Morse cancellation; and geometric uniform sums,  $q$ -cumulants, finite spline approximants, and the Fabius–Rvachev specializations. Every displayed theorem used in the main development is accompanied by a proof, and the final appendices collect formulas, limiting transitions, computational recipes, and a per-result record of what has and has not yet been formalized in Lean.

# Preface and source map

The aim is not merely to list identities, but to expose the mechanisms that repeatedly generate them. Three such mechanisms dominate:

1. finite products split into residue classes, blocks, or initial and tail factors;
2. coefficient extraction turns product factorizations into convolution identities;
3. functional equations determine analytic  $q$ -series from one initial value.

Once these mechanisms are visible, many formulas that initially appear unrelated become instances of the same principle.

The supplied archive was used throughout. In particular, the weighted subset framework of Konvalina motivates [Chapter 4](#); Kupershmidt’s treatment of the  $q$ -Newton binomial informs [Chapters 11](#) and [17](#); Straub’s seminar on  $q$ -binomial congruences is reflected in [Chapter 16](#); the formalized  $q$ -series project of Lau, Lee, and Ono motivates the algebraic-versus-analytic distinctions in [Chapter 20](#) and supplies a modern Bailey-pair route to the Rogers–Ramanujan identities; and the central  $q$ -binomial paper of Chern, Dilcher, and Jiu is developed, with a complete proof, in [Chapter 18](#). The archive’s reference chapters and encyclopedia snapshots were checked against the standard sources listed in the bibliography. The current *ProveIt* Fabius-function documentation [38], including its main exposition and non-formalized research-frontier drafts, supplies the geometric-node interpolation rows, exact residual moments, half-base identities, formal filters, and dyadic Thue–Morse connections expanded independently in [Chapters 21](#) to [23](#). Online research also led to the recent automated identity-translation framework of Zhong and Zhao [40] and the Isabelle formalization of combinatorial  $q$ -analogues by Eberl [36]. Additional material was drawn from the classical books and articles of Andrews, Askey, Bailey, Gasper, Rahman, Rota, Goldman, Reiner, Stanton, White, and others, as well as the NIST Digital Library of Mathematical Functions.

## Two regimes for $q$

Throughout the finite algebraic chapters,  $q$  may be an indeterminate and identities are understood in a polynomial or rational-function ring. In analytic chapters, unless stated otherwise,

$$q \in \mathbb{C}, \quad 0 < |q| < 1.$$

For the Jackson integral and positivity statements we usually assume  $0 < q < 1$ . A capital  $Q$  is used for the cardinality of a finite field, so  $Q$  is a prime power and normally  $Q > 1$ .

## Conventions

Empty products equal 1, empty sums equal 0, and

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = 0 \quad \text{when } k < 0 \text{ or } k > n$$

---

for nonnegative integral  $n$ . Multiple parameters are compressed as

$$(a_1, \dots, a_r; q)_n = \prod_{j=1}^r (a_j; q)_n.$$

The notation  $(a; q)_\infty$  is analytic when  $|q| < 1$ ; when formal power series are intended, that will be said explicitly.

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## **Part I**

# **The finite algebra**

# Chapter 1

## The algebra of $q$ -shifted factorials

### 1.1 Definitions and first examples

**Definition 1.1.** For  $n \in \mathbb{N}_0$  the  $q$ -shifted factorial, or  $q$ -Pochhammer symbol, is

$$(a; q)_n = \prod_{j=0}^{n-1} (1 - aq^j), \quad (a; q)_0 = 1.$$

For parameters  $a_1, \dots, a_r$  we write

$$(a_1, \dots, a_r; q)_n = \prod_{i=1}^r (a_i; q)_n.$$

When  $0 < |q| < 1$ , the infinite symbol is

$$(a; q)_\infty = \prod_{j=0}^{\infty} (1 - aq^j).$$

The word “shifted” refers to the geometric progression  $a, aq, aq^2, \dots$ . The first few values are

$$(a; q)_1 = 1 - a, \quad (a; q)_2 = (1 - a)(1 - aq), \quad (a; q)_3 = (1 - a)(1 - aq)(1 - aq^2).$$

Important specializations include

$$(q; q)_n = \prod_{j=1}^n (1 - q^j), \quad (-q; q)_n = \prod_{j=1}^n (1 + q^j), \quad (a; q^r)_n = \prod_{j=0}^{n-1} (1 - aq^{rj}).$$

### 1.2 Concatenation, dissection, and reversal

**Proposition 1.2** (Concatenation law). For  $m, n \in \mathbb{N}_0$ ,

$$(a; q)_{m+n} = (a; q)_m (aq^m; q)_n.$$

Consequently, whenever the denominator is nonzero,

$$\frac{(a; q)_{m+n}}{(a; q)_m} = (aq^m; q)_n.$$

*Proof.* Split the defining product at  $j = m$ :

$$\prod_{j=0}^{m+n-1} (1 - aq^j) = \left( \prod_{j=0}^{m-1} (1 - aq^j) \right) \left( \prod_{j=m}^{m+n-1} (1 - aq^j) \right).$$

In the second product put  $j = m + r$ , where  $0 \leq r < n$ . □

**Proposition 1.3** (Dissection into residue classes). *For integers  $r, n \geq 1$ ,*

$$(a; q)_{rn} = \prod_{s=0}^{r-1} (aq^s; q^r)_n.$$

*If  $|q| < 1$ , then also*

$$(a; q)_{\infty} = \prod_{s=0}^{r-1} (aq^s; q^r)_{\infty}.$$

*Proof.* Every integer  $j$  with  $0 \leq j < rn$  has a unique representation  $j = s + rt$  with  $0 \leq s < r$  and  $0 \leq t < n$ . Reordering the finite product gives the first identity. For the infinite identity, the factors are absolutely summably close to 1 because  $\sum_{j \geq 0} |aq^j| < \infty$ , so the product may be regrouped into the  $r$  residue classes. □

**Corollary 1.4** (Dissection with a remainder). *Let  $r \geq 1$  and  $n \geq 0$  be integers, let  $0 \leq u < r$ , and set  $N = rn + u$ . Then*

$$(a; q)_N = \prod_{s=0}^{u-1} (aq^s; q^r)_{n+1} \prod_{s=u}^{r-1} (aq^s; q^r)_n.$$

*For  $u = 0$  this is Proposition 1.3.*

*Proof.* Division with remainder by  $r$  is a bijection between the nonnegative integers  $j$  and the pairs  $(s, \ell)$  with  $0 \leq s < r$  and  $\ell \geq 0$ , given by  $j = s + r\ell$ . Fix a residue  $s$  and count the  $\ell$  for which  $j = s + r\ell$  lies in the exponent set  $\{0, \dots, N-1\}$  of the defining product. The condition  $s + r\ell < rn + u$  reads

$$r\ell < rn + (u - s).$$

If  $s < u$ , then  $0 < u - s \leq u < r$ , so

$$rn < rn + (u - s) \leq rn + r = r(n + 1).$$

As  $r\ell$  runs through the multiples of  $r$ , the inequality  $r\ell < rn + (u - s)$  therefore holds exactly for  $\ell = 0, 1, \dots, n$ , which is  $n + 1$  values.

If  $s \geq u$ , then  $u - s \leq 0$ , so the inequality implies  $r\ell < rn$ , that is  $\ell \leq n - 1$ ; conversely every such  $\ell$  satisfies  $s + r\ell \leq s + r(n - 1) = rn + (s - r) < rn \leq N$ . Hence  $\ell$  ranges over  $0, \dots, n - 1$ , which is  $n$  values.

The defining product is finite and its factors commute, so it may be reordered according to the residue of  $j$  modulo  $r$ :

$$(a; q)_N = \prod_{s=0}^{u-1} \prod_{\ell=0}^n (1 - aq^s(q^r)^\ell) \prod_{s=u}^{r-1} \prod_{\ell=0}^{n-1} (1 - aq^s(q^r)^\ell).$$

Each inner product is by definition a  $q^r$ -shifted factorial with argument  $aq^s$ , namely  $(aq^s; q^r)_{n+1}$  in the first family and  $(aq^s; q^r)_n$  in the second. This is the asserted identity. When  $u = 0$  the first family is empty and the second has all  $r$  residues, recovering Proposition 1.3. □

*Remark 1.5.* Two consistency checks are worth recording. The total number of linear factors on the right is

$$u(n+1) + (r-u)n = rn + u = N,$$

as it must be, since  $(a; q)_N$  is a product of  $N$  linear factors in  $a$ . At the other extreme  $n = 0$  the corollary reads  $(a; q)_u = \prod_{s=0}^{u-1} (aq^s; q^r)_1 = \prod_{s=0}^{u-1} (1 - aq^s)$ , again the definition. Combining the corollary with [Proposition 1.2](#) in the form  $(a; q)_{rn+u} = (a; q)_{rn} (aq^{rn}; q)_u$  and dividing by [Proposition 1.3](#) recovers the identity  $\prod_{s=0}^{u-1} (1 - aq^{s+rn}) = (aq^{rn}; q)_u$ , so the two decompositions agree.

**Proposition 1.6** (Base reversal). *For  $n \geq 0$  and  $a \neq 0$ ,*

$$(a; q)_n = (-a)^n q^{\binom{n}{2}} (a^{-1}; q^{-1})_n.$$

*Equivalently,*

$$(a; q^{-1})_n = (-a)^n q^{-\binom{n}{2}} (a^{-1}; q)_n.$$

*Proof.* Factor  $-aq^j$  from each term:

$$1 - aq^j = -aq^j (1 - a^{-1}q^{-j}).$$

Multiplying for  $0 \leq j < n$  gives the factor  $(-a)^n q^{0+1+\dots+(n-1)} = (-a)^n q^{\binom{n}{2}}$ . The remaining product is  $(a^{-1}; q^{-1})_n$ .  $\square$

### 1.3 Negative indices

The concatenation law suggests the only extension to negative integers that preserves

$$(a; q)_{m+n} = (a; q)_m (aq^m; q)_n.$$

**Definition 1.7.** For  $n \geq 1$ , define

$$(a; q)_{-n} = \frac{1}{(aq^{-n}; q)_n},$$

provided no factor in the denominator vanishes.

**Proposition 1.8** (Closed form for a negative index). *For  $n \geq 1$  and  $a \neq 0$ ,*

$$(a; q)_{-n} = \frac{(-1)^n a^{-n} q^{\binom{n+1}{2}}}{(q/a; q)_n}.$$

*Moreover, for all integers  $m, n$  for which both sides are defined,*

$$(a; q)_{m+n} = (a; q)_m (aq^m; q)_n.$$

*Proof.* By definition,

$$(aq^{-n}; q)_n = \prod_{j=0}^{n-1} (1 - aq^{-n+j}).$$

Factor  $-aq^{-n+j}$  from each factor. Their product is

$$(-a)^n q^{-n^2 + \binom{n}{2}} = (-a)^n q^{-\binom{n+1}{2}},$$

and the remaining product is

$$\prod_{j=0}^{n-1} (1 - a^{-1}q^{n-j}) = (q/a; q)_n.$$

Taking the reciprocal proves the formula. The extended concatenation identity follows by cancelling the overlapping finite factors; the cases in which one of  $m, n, m + n$  is negative reduce directly to the definition.  $\square$

**Corollary 1.9** (A terminating factor). *If  $0 \leq k \leq N$ , then*

$$(q^{-N}; q)_k = (-1)^k q^{-Nk + \binom{k}{2}} \frac{(q; q)_N}{(q; q)_{N-k}}.$$

*In particular,  $(q^{-N}; q)_k = 0$  for  $k > N$ .*

*Proof.* Apply base reversal to  $(q^{-N}; q)_k$ :

$$(q^{-N}; q)_k = (-1)^k q^{-Nk + \binom{k}{2}} (q^{N-k+1}; q)_k.$$

The last factor is  $(q; q)_N / (q; q)_{N-k}$ . For  $k > N$ , the defining product contains the factor  $1 - q^0$ .  $\square$

## 1.4 Partial fractions and logarithmic derivatives

**Theorem 1.10** (Finite partial-fraction decomposition). *Assume  $q^i \neq q^j$  for  $0 \leq i < j \leq n$ . Then*

$$\frac{1}{(z; q)_{n+1}} = \sum_{j=0}^n \frac{(-1)^j q^{\binom{j+1}{2}}}{(q; q)_j (q; q)_{n-j}} \frac{1}{1 - zq^j}.$$

*Proof.* The left side has simple poles at  $z = q^{-j}$ ,  $0 \leq j \leq n$ . The coefficient of  $(1 - zq^j)^{-1}$  is

$$A_j = \left. \frac{1 - zq^j}{(z; q)_{n+1}} \right|_{z=q^{-j}} = \frac{1}{\prod_{r \neq j} (1 - q^{r-j})}.$$

For  $r < j$ , put  $s = j - r$  and obtain

$$\prod_{r=0}^{j-1} (1 - q^{r-j}) = \prod_{s=1}^j (1 - q^{-s}) = (-1)^j q^{-\binom{j+1}{2}} (q; q)_j.$$

The factors with  $r > j$  multiply to  $(q; q)_{n-j}$ . Thus  $A_j$  is the displayed coefficient. Subtracting the proposed right side from the left gives a rational function with no poles and tending to 0 at infinity, hence it is identically zero.  $\square$

**Proposition 1.11** (Logarithmic derivatives). *Away from its zeros,*

$$\frac{\partial}{\partial a} \log(a; q)_n = - \sum_{j=0}^{n-1} \frac{q^j}{1 - aq^j}.$$

*If  $a = a(x)$  is differentiable, then*

$$\frac{d}{dx} \log(a(x); q)_n = -a'(x) \sum_{j=0}^{n-1} \frac{q^j}{1 - a(x)q^j}.$$

*Proof.* Differentiate the logarithm of the finite product term by term.  $\square$

## 1.5 Infinite products and Lambert series

**Theorem 1.12** (Convergence and zero set). *Let  $0 < |q| < 1$ . The product  $(a; q)_\infty$  converges locally uniformly as a function of  $a \in \mathbb{C}$  and therefore defines an entire function. Its zeros are precisely*

$$a = q^{-j}, \quad j = 0, 1, 2, \dots,$$

*and all are simple.*

*Proof.* On a compact set  $K \subset \mathbb{C}$  let  $M = \max_{a \in K} |a|$ . Then

$$\sum_{j=0}^{\infty} \sup_{a \in K} |aq^j| \leq \frac{M}{1 - |q|} < \infty.$$

The standard convergence criterion for infinite products shows that  $\prod_j (1 - aq^j)$  converges uniformly on  $K$ ; equivalently, one may apply uniform convergence to the tails after choosing an analytic branch of the logarithm near 1. Uniform limits of holomorphic functions are holomorphic, so the limit is entire.

If  $a = q^{-j}$ , one factor vanishes. Conversely, if no factor vanishes, then the tail product is nonzero because the sum of the absolute deviations from 1 converges, while the finite initial product is nonzero. At  $a = q^{-j}$  exactly one factor vanishes and its derivative is  $-q^j \neq 0$ , whereas all other factors are nonzero; the zero is simple.  $\square$

### 1.5.1 Complex orders

Finite and negative integral indices extend naturally to a complex order once an infinite product is available. Fix  $0 < |q| < 1$  and a branch of the logarithm at  $q$ , denoted by  $\log q$ , and write  $q^\alpha = \exp(\alpha \log q)$ .

**Definition 1.13.** For  $\alpha \in \mathbb{C}$ , define

$$(a; q)_\alpha := \frac{(a; q)_\infty}{(aq^\alpha; q)_\infty}, \quad (1.1)$$

whenever the denominator is nonzero.

**Proposition 1.14** (Complex concatenation). *Whenever the displayed quantities are defined,*

$$(a; q)_{\alpha+\beta} = (a; q)_\alpha (aq^\alpha; q)_\beta. \quad (1.2)$$

*In particular,*

$$(a; q)_{\alpha+1} = (1 - aq^\alpha)(a; q)_\alpha. \quad (1.3)$$

*For integral  $\alpha$ , (1.1) agrees with the finite and negative-index definitions already given.*

*Proof.* By definition,

$$(a; q)_\alpha (aq^\alpha; q)_\beta = \frac{(a; q)_\infty}{(aq^\alpha; q)_\infty} \frac{(aq^\alpha; q)_\infty}{(aq^{\alpha+\beta}; q)_\infty} = \frac{(a; q)_\infty}{(aq^{\alpha+\beta}; q)_\infty}.$$

This is (1.2). Taking  $\beta = 1$  gives (1.3). If  $\alpha = n \geq 0$ , the tail factorization  $(a; q)_\infty = (a; q)_n (aq^n; q)_\infty$  recovers the finite product. If  $\alpha = -n < 0$ , the same relation, applied after shifting by  $-n$ , gives  $(a; q)_{-n} = 1/(aq^{-n}; q)_n$ .  $\square$

**Proposition 1.15** (Derivative with respect to the order). *On every domain avoiding the poles of (1.1),*

$$\frac{\partial}{\partial \alpha} \log(a; q)_\alpha = (\log q) \sum_{j=0}^{\infty} \frac{aq^{\alpha+j}}{1 - aq^{\alpha+j}}. \quad (1.4)$$

*The series is locally uniformly convergent.*

*Proof.* Only the denominator in (1.1) depends on  $\alpha$ . Differentiate its locally uniformly convergent logarithm term by term:

$$-\frac{\partial}{\partial \alpha} \sum_{j \geq 0} \log(1 - aq^{\alpha+j}) = (\log q) \sum_{j \geq 0} \frac{aq^{\alpha+j}}{1 - aq^{\alpha+j}}.$$

On a compact set avoiding the poles, the denominators of the tail are bounded away from zero and the numerators form a geometric majorant, proving local uniform convergence.  $\square$

**Theorem 1.16** (Lambert-series expansion). *If  $|q| < 1$  and  $|a| < 1$ , then*

$$\log(a; q)_\infty = - \sum_{m=1}^{\infty} \frac{a^m}{m(1 - q^m)}.$$

*Consequently,*

$$\frac{\partial}{\partial a} \log(a; q)_\infty = - \sum_{m=1}^{\infty} \frac{a^{m-1}}{1 - q^m} = - \sum_{j=0}^{\infty} \frac{q^j}{1 - aq^j}.$$

*All series converge locally uniformly in  $|a| < 1$ .*

*Proof.* For  $|a| < 1$ , use  $\log(1 - u) = - \sum_{m \geq 1} u^m/m$  and absolute convergence:

$$\sum_{j \geq 0} \sum_{m \geq 1} \frac{|a|^m |q|^{jm}}{m} = \sum_{m \geq 1} \frac{|a|^m}{m(1 - |q|^m)} < \infty.$$

Thus the order of summation may be exchanged:

$$\sum_{j \geq 0} \log(1 - aq^j) = - \sum_{m \geq 1} \frac{a^m}{m} \sum_{j \geq 0} q^{jm}.$$

The geometric sum gives the first formula. Local uniform convergence permits termwise differentiation. Expanding  $(1 - aq^j)^{-1}$  geometrically gives the second form of the derivative.  $\square$

**Corollary 1.17** (Divisor sums). *As a formal power series,*

$$-q \frac{d}{dq} \log(q; q)_\infty = \sum_{n=1}^{\infty} \sigma_1(n) q^n,$$

where  $\sigma_1(n) = \sum_{d|n} d$ .

*Proof.* Logarithmic differentiation gives

$$-q \frac{d}{dq} \log(q; q)_\infty = \sum_{j \geq 1} \frac{jq^j}{1 - q^j} = \sum_{j \geq 1} \sum_{r \geq 1} jq^{jr}.$$

The coefficient of  $q^n$  is the sum of all  $j$  dividing  $n$ .  $\square$



## 1.6 The dilogarithmic limit

The singular regime  $q \rightarrow 1^-$  turns an infinite product into an exponential of order  $(1 - q)^{-1}$ . The dilogarithm is the natural leading term.

**Theorem 1.18** (Asymptotic expansion near  $q = 1$ ). *Fix  $0 < \rho < 1$  and an integer  $N \geq 1$ . Uniformly for  $|a| \leq \rho$ , as  $t \rightarrow 0^+$ ,*

$$\log(a; e^{-t})_\infty = -\frac{\text{Li}_2(a)}{t} + \frac{1}{2} \log(1 - a) - \sum_{r=1}^{N-1} \frac{B_{2r}}{(2r)!} t^{2r-1} \text{Li}_{2-2r}(a) + O_{\rho,N}(t^{2N-1}),$$

where  $B_{2r}$  are Bernoulli numbers and  $\text{Li}_s(a) = \sum_{m \geq 1} a^m / m^s$  for  $|a| < 1$ .

*Proof.* By Theorem 1.16,

$$\log(a; e^{-t})_\infty = -\sum_{m \geq 1} \frac{a^m}{m(1 - e^{-mt})}.$$

The Bernoulli expansion at the origin is

$$\frac{1}{1 - e^{-x}} = \frac{1}{x} + \frac{1}{2} + \sum_{r=1}^{N-1} \frac{B_{2r}}{(2r)!} x^{2r-1} + R_N(x),$$

with  $R_N(x) = O_N(x^{2N-1})$  for  $0 \leq x \leq 1$ . Substitution of the displayed polynomial terms gives respectively

$$-\frac{1}{t} \sum_{m \geq 1} \frac{a^m}{m^2}, \quad -\frac{1}{2} \sum_{m \geq 1} \frac{a^m}{m} = \frac{1}{2} \log(1 - a),$$

and the polylogarithmic terms in the statement.

It remains to bound the total remainder. Split the  $m$ -sum at  $m \leq t^{-1}$ . In this range,

$$\sum_{m \leq t^{-1}} \frac{|a|^m}{m} |R_N(mt)| \leq C_N t^{2N-1} \sum_{m \geq 1} \rho^m m^{2N-2} = O_{\rho,N}(t^{2N-1}).$$

For  $m > t^{-1}$ , subtracting finitely many asymptotic terms leaves a quantity bounded by a polynomial in  $mt$  plus  $(1 - e^{-mt})^{-1}$ ; the factor  $\rho^m$  makes the tail exponentially small in  $1/t$ , hence smaller than every power of  $t$ . This proves the uniform remainder estimate.  $\square$

## 1.7 Classical shifted factorials as a limit

**Proposition 1.19** (Finite classical limit). *For fixed  $x \in \mathbb{C}$  and  $n \in \mathbb{N}_0$ ,*

$$\lim_{q \rightarrow 1} \frac{(q^x; q)_n}{(1 - q)^n} = x(x + 1) \cdots (x + n - 1) = (x)_n,$$

where the limit is taken through values for which a consistent branch of  $q^x$  is chosen.

*Proof.* Write

$$\frac{(q^x; q)_n}{(1 - q)^n} = \prod_{j=0}^{n-1} \frac{1 - q^{x+j}}{1 - q}.$$

Each factor tends to  $x + j$ , since  $q^u = 1 + u(q - 1) + o(q - 1)$  for fixed  $u$ .  $\square$

## Chapter 2

# $q$ -integers and Gaussian coefficients

### 2.1 Definitions

**Definition 2.1.** For  $n \in \mathbb{N}_0$  define the  $q$ -integer and  $q$ -factorial by

$$[n]_q = \frac{1 - q^n}{1 - q} = 1 + q + \cdots + q^{n-1}, \quad [n]_q! = \prod_{j=1}^n [j]_q.$$

For  $0 \leq k \leq n$ , the *Gaussian coefficient* is

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = \frac{(q; q)_n}{(q; q)_k (q; q)_{n-k}} = \frac{[n]_q!}{[k]_q! [n-k]_q!}.$$

Outside  $0 \leq k \leq n$  it is defined to be 0.

The notation is initially a rational expression, but the recurrences below show that it is a polynomial in  $q$  with nonnegative integral coefficients.

**Proposition 2.2** (Product forms). For  $0 \leq k \leq n$ ,

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = \frac{(q^{n-k+1}; q)_k}{(q; q)_k} = \prod_{j=1}^k \frac{1 - q^{n-k+j}}{1 - q^j}.$$

*Proof.* Cancel  $(q; q)_{n-k}$  from  $(q; q)_n$  using [Proposition 1.2](#). □

### 2.2 The two Pascal recurrences

**Theorem 2.3** ( $q$ -Pascal recurrences). For  $1 \leq k \leq n-1$ ,

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = \begin{bmatrix} n-1 \\ k-1 \end{bmatrix}_q + q^k \begin{bmatrix} n-1 \\ k \end{bmatrix}_q, \tag{2.1}$$

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = q^{n-k} \begin{bmatrix} n-1 \\ k-1 \end{bmatrix}_q + \begin{bmatrix} n-1 \\ k \end{bmatrix}_q. \tag{2.2}$$

Both identities remain valid at the boundary under the zero convention.

*Proof.* Put the two terms on the right of (2.1) over the common denominator  $(q; q)_k(q; q)_{n-k}$ . Their numerator is

$$(q; q)_{n-1}((1 - q^k) + q^k(1 - q^{n-k})) = (q; q)_{n-1}(1 - q^n) = (q; q)_n.$$

The proof of (2.2) is the same, using  $q^{n-k}(1 - q^k) + (1 - q^{n-k}) = 1 - q^n$ .  $\square$

**Corollary 2.4** (Polynomiality and positivity). *For all  $0 \leq k \leq n$ , the Gaussian coefficient  $\begin{bmatrix} n \\ k \end{bmatrix}_q$  belongs to  $\mathbb{N}_0[q]$ .*

*Proof.* Induct on  $n$ . The boundary values are 1. Either recurrence in Theorem 2.3 expresses an interior coefficient as a sum of two polynomials with nonnegative integral coefficients.  $\square$

## 2.3 Symmetry, reciprocity, and degree

**Theorem 2.5** (Elementary structure). *For  $0 \leq k \leq n$ ,*

- (i)  $\begin{bmatrix} n \\ k \end{bmatrix}_q = \begin{bmatrix} n \\ n-k \end{bmatrix}_q$ ;
- (ii)  $\begin{bmatrix} n \\ k \end{bmatrix}_{q^{-1}} = q^{-k(n-k)} \begin{bmatrix} n \\ k \end{bmatrix}_q$ ;
- (iii)  $\begin{bmatrix} n \\ k \end{bmatrix}_q$  has degree  $k(n-k)$ , constant term 1, and leading coefficient 1;
- (iv) its coefficient sequence is palindromic.

*Proof.* Symmetry is immediate from the factorial definition. For reciprocity, use

$$(q^{-1}; q^{-1})_m = (-1)^m q^{-\binom{m+1}{2}} (q; q)_m$$

for  $m = n, k, n-k$ . The signs cancel and the exponent is

$$-\binom{n+1}{2} + \binom{k+1}{2} + \binom{n-k+1}{2} = -k(n-k).$$

Since  $\begin{bmatrix} n \\ k \end{bmatrix}_q$  is a polynomial with constant term 1, reciprocity says

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = q^{k(n-k)} \begin{bmatrix} n \\ k \end{bmatrix}_{q^{-1}}.$$

Hence its degree is at most  $k(n-k)$  and its leading coefficient equals its constant coefficient. The product form shows that the order at infinity is exactly  $k(n-k)$ , so the degree is exact. The same reciprocal identity equates coefficients at complementary degrees and proves palindromicity.  $\square$

**Corollary 2.6** (Classical limit). *For fixed  $n, k$ ,*

$$\lim_{q \rightarrow 1} \begin{bmatrix} n \\ k \end{bmatrix}_q = \binom{n}{k}.$$

*Proof.* Use the  $q$ -factorial form and  $[j]_q \rightarrow j$ .  $\square$

## 2.4 Multinomial coefficients and flags

**Definition 2.7.** If  $n_1, \dots, n_r \geq 0$  and  $n_1 + \dots + n_r = n$ , define

$$\left[ \begin{matrix} n \\ n_1, \dots, n_r \end{matrix} \right]_q = \frac{(q; q)_n}{(q; q)_{n_1} \cdots (q; q)_{n_r}}.$$

**Proposition 2.8** (Factorization into binomials). *Let  $N_j = n_j + n_{j+1} + \dots + n_r$ . Then*

$$\left[ \begin{matrix} n \\ n_1, \dots, n_r \end{matrix} \right]_q = \prod_{j=1}^{r-1} \left[ \begin{matrix} N_j \\ n_j \end{matrix} \right]_q.$$

*In particular, every  $q$ -multinomial coefficient is a polynomial with nonnegative integral coefficients.*

*Proof.* Write out the factorial ratios. Consecutive numerator and denominator factors telescope:

$$\prod_{j=1}^{r-1} \frac{(q; q)_{N_j}}{(q; q)_{n_j} (q; q)_{N_{j+1}}} = \frac{(q; q)_n}{\prod_{j=1}^r (q; q)_{n_j}}.$$

Positivity follows from [Corollary 2.4](#). □

## 2.5 Adjacent ratios and efficient recurrences

**Proposition 2.9.** *For  $0 \leq k < n$ ,*

$$\frac{\left[ \begin{matrix} n \\ k+1 \end{matrix} \right]_q}{\left[ \begin{matrix} n \\ k \end{matrix} \right]_q} = \frac{1 - q^{n-k}}{1 - q^{k+1}}.$$

*For  $0 \leq k \leq n-1$ ,*

$$\frac{\left[ \begin{matrix} n \\ k \end{matrix} \right]_q}{\left[ \begin{matrix} n-1 \\ k \end{matrix} \right]_q} = \frac{1 - q^n}{1 - q^{n-k}}.$$

*Proof.* Cancel common factors in the product or factorial definitions. □

These ratios are useful numerically, but near roots of unity they hide cancellations. The polynomial recurrences of [Theorem 2.3](#) are then preferable.

## 2.6 Coefficient moments

Write

$$\left[ \begin{matrix} n \\ k \end{matrix} \right]_q = \sum_{j=0}^D c_j q^j, \quad D = k(n-k),$$

and choose an exponent  $X$  with probability  $\Pr(X = j) = c_j / \left[ \begin{matrix} n \\ k \end{matrix} \right]_q$ . The combinatorial meaning of  $X$  will be established in [Chapter 5](#); its moments can already be found algebraically.

**Theorem 2.10** (Mean and variance). *For the preceding distribution,*

$$\mathbb{E}X = \frac{k(n-k)}{2}, \quad \text{Var}(X) = \frac{k(n-k)(n+1)}{12}.$$

*Equivalently,*

$$\left. \frac{d}{dq} \left[ \begin{matrix} n \\ k \end{matrix} \right]_q \right|_{q=1} = \frac{k(n-k)}{2} \left( \begin{matrix} n \\ k \end{matrix} \right).$$

*Proof.* The palindromic identity  $c_j = c_{D-j}$  immediately gives  $\mathbb{E}X = D/2$  and the derivative formula.

For the variance, use the moment-generating function

$$M(t) = \frac{[n]_k e^t}{\binom{n}{k}} = \prod_{i=1}^k \frac{e^{(n-k+i)t} - 1}{(n-k+i)t} \frac{it}{e^{it} - 1}.$$

The local expansion

$$\log \frac{e^u - 1}{u} = \frac{u}{2} + \frac{u^2}{24} + O(u^3)$$

gives

$$\log M(t) = \frac{t}{2} \sum_{i=1}^k (n-k) + \frac{t^2}{24} \sum_{i=1}^k ((n-k+i)^2 - i^2) + O(t^3).$$

The first sum is  $k(n-k)$ . The second equals

$$k(n-k)^2 + 2(n-k) \frac{k(k+1)}{2} = k(n-k)(n+1).$$

Since the first and second derivatives of  $\log M$  at 0 are the mean and variance, respectively, the result follows.  $\square$

## Chapter 3

# Finite $q$ -binomial theorems and inversion

### 3.1 Euler's finite product expansions

**Theorem 3.1** (Finite  $q$ -binomial theorem). *For  $n \geq 0$ ,*

$$(z; q)_n = \sum_{k=0}^n (-1)^k q^{\binom{k}{2}} \begin{bmatrix} n \\ k \end{bmatrix}_q z^k, \quad (3.1)$$

$$(-z; q)_n = \sum_{k=0}^n q^{\binom{k}{2}} \begin{bmatrix} n \\ k \end{bmatrix}_q z^k. \quad (3.2)$$

*Proof.* It is enough to prove the first identity; replace  $z$  by  $-z$  for the second. The case  $n = 0$  is clear. Suppose the formula holds for  $n$ . Then

$$(z; q)_{n+1} = (1 - zq^n)(z; q)_n.$$

The coefficient of  $z^k$  on the right is

$$(-1)^k q^{\binom{k}{2}} \left( \begin{bmatrix} n \\ k \end{bmatrix}_q + q^{n-k+1} \begin{bmatrix} n \\ k-1 \end{bmatrix}_q \right).$$

By the second  $q$ -Pascal recurrence, the parenthesis is  $\begin{bmatrix} n+1 \\ k \end{bmatrix}_q$ . This completes the induction.  $\square$

**Corollary 3.2** (Alternating sums). *For  $n > 0$ ,*

$$\sum_{k=0}^n (-1)^k q^{\binom{k}{2}} \begin{bmatrix} n \\ k \end{bmatrix}_q = 0.$$

*More generally,*

$$\sum_{k=0}^n (-1)^k q^{\binom{k}{2} + mk} \begin{bmatrix} n \\ k \end{bmatrix}_q = (q^m; q)_n.$$

*Proof.* Set  $z = 1$  and  $z = q^m$ , respectively, in (3.1).  $\square$

**Remark 3.3** (Formal version). Over every commutative ring, including  $q = 0$ , roots of unity, zero divisors, and positive characteristic, `Fabius.sum_gaussianBinomial_alternating_mul_pow` proves the second display for all  $m, n \in \mathbb{N}$ , and `Fabius.sum_gaussianBinomial_alternating` proves the first with its exact row-zero boundary 1. Both declarations are in `QBinomialInversion.lean`; neither uses a quotient or cancellation hypothesis.

**Corollary 3.4** (Weighted subset sum). For  $0 \leq k \leq n$ ,

$$\sum_{0 \leq i_1 < \dots < i_k \leq n-1} q^{i_1 + \dots + i_k} = q^{\binom{k}{2}} \begin{bmatrix} n \\ k \end{bmatrix}_q.$$

Equivalently,

$$\sum_{1 \leq j_1 < \dots < j_k \leq n} q^{j_1 + \dots + j_k} = q^{\binom{k+1}{2}} \begin{bmatrix} n \\ k \end{bmatrix}_q.$$

*Proof.* Expand  $\prod_{i=0}^{n-1} (1 + zq^i)$  by choosing either 1 or  $zq^i$  from each factor, and compare the coefficient of  $z^k$  with (3.2).  $\square$

**Remark 3.5** (Formal version). The zero-based and one-based formulas are respectively `Fabius.sum_pow_sum_powersetCard_eq_gaussianBinomial` and `Fabius.sum_pow_sum_powersetCard_Icc_eq_gaussianBinomial` in `BitPositionQBinomial.lean`. They are total in  $n, k \in \mathbb{N}$  over every commutative ring: the first uses `range n`, and the second literally uses the interval `Icc 1 n`. The latter is obtained by translating each chosen index upward by one, which contributes the extra factor  $q^k$ .

## 3.2 Reciprocal finite products

**Theorem 3.6** (Reciprocal finite  $q$ -binomial theorem). For  $n \geq 0$  and  $|z| < 1$ ,

$$\frac{1}{(z; q)_{n+1}} = \sum_{k=0}^{\infty} \begin{bmatrix} n+k \\ k \end{bmatrix}_q z^k.$$

The identity also holds formally in  $\mathbb{Z}[q][[z]]$ .

*Proof.* Let  $H_n(z) = \sum_{k \geq 0} \begin{bmatrix} n+k \\ k \end{bmatrix}_q z^k$ . The second  $q$ -Pascal recurrence gives

$$\begin{bmatrix} n+k \\ k \end{bmatrix}_q - q^n \begin{bmatrix} n+k-1 \\ k-1 \end{bmatrix}_q = \begin{bmatrix} n+k-1 \\ k \end{bmatrix}_q.$$

Therefore

$$(1 - zq^n)H_n(z) = H_{n-1}(z),$$

where  $H_0(z) = \sum_{k \geq 0} z^k = (1 - z)^{-1}$ . Iterating yields

$$H_n(z) = \frac{1}{(1 - z)(1 - zq) \cdots (1 - zq^n)}.$$

All operations are valid formally; analytically they are justified for  $|z| < 1$ .  $\square$

**Remark 3.7** (Formal version and analytic boundary). For every commutative ring  $R$ , every  $q \in R$ , and every  $n \in \mathbb{N}$ , `Fabius.prod_one_sub_qPow_X_mul_gaussianBinomialGeneratingSeries` in `SymmetricFunctionGenerating.lean` proves

$$\left( \prod_{j=0}^n (1 - q^j X) \right) \text{PowerSeries.mk} \left( k \mapsto \begin{bmatrix} n+k \\ k \end{bmatrix}_q \right) = 1 \quad \text{in } R[[X]].$$

This is denominator-free and total at  $q = 0$ , roots of unity, zero divisors, positive characteristic, and the zero ring. It proves the formal power-series half of [Theorem 3.6](#); no current declaration here evaluates that series at a complex  $z$  or establishes convergence for  $|z| < 1$ . The theorem therefore remains only partially formalized as stated.

### 3.3 $q$ -Vandermonde convolution

**Theorem 3.8** ( $q$ -Vandermonde convolution). For  $m, n, r \in \mathbb{N}_0$ ,

$$\begin{bmatrix} m+n \\ r \end{bmatrix}_q = \sum_{j \in \mathbb{Z}} q^{(m-j)(r-j)} \begin{bmatrix} m \\ j \end{bmatrix}_q \begin{bmatrix} n \\ r-j \end{bmatrix}_q, \quad (3.3)$$

$$= \sum_{j \in \mathbb{Z}} q^{j(n-r+j)} \begin{bmatrix} m \\ j \end{bmatrix}_q \begin{bmatrix} n \\ r-j \end{bmatrix}_q. \quad (3.4)$$

Only finitely many terms are nonzero.

*Proof.* Factor

$$(-z; q)_{m+n} = (-z; q)_m (-zq^m; q)_n$$

and use (3.2). The coefficient of  $z^r$  on the left is  $q^{\binom{r}{2}} \begin{bmatrix} m+n \\ r \end{bmatrix}_q$ . The term with  $j$  chosen from the first factor and  $r-j$  from the second has exponent

$$\binom{j}{2} + m(r-j) + \binom{r-j}{2}.$$

Subtracting  $\binom{r}{2}$  gives  $(m-j)(r-j)$ , proving (3.3). Splitting the product in the reverse order gives (3.4).  $\square$

**Corollary 3.9** (Central form). For  $m, n \geq 0$ ,

$$\begin{bmatrix} m+n \\ m \end{bmatrix}_q = \sum_j q^{j^2} \begin{bmatrix} m \\ j \end{bmatrix}_q \begin{bmatrix} n \\ j \end{bmatrix}_q.$$

*Proof.* Set  $r = m$  in (3.3) and put  $\ell = m - j$ . Then the exponent becomes  $\ell^2$ , while symmetry gives  $\begin{bmatrix} m \\ m-\ell \end{bmatrix}_q = \begin{bmatrix} m \\ \ell \end{bmatrix}_q$  and the second factor is  $\begin{bmatrix} n \\ \ell \end{bmatrix}_q$ .  $\square$

**Corollary 3.10** (Shifted central form). For  $N \in \mathbb{N}_0$  and  $k \in \mathbb{Z}$ ,

$$\begin{bmatrix} 2N \\ N+k \end{bmatrix}_q = \sum_{\ell \in \mathbb{Z}} q^{\ell^2 + \ell k} \begin{bmatrix} N \\ \ell \end{bmatrix}_q \begin{bmatrix} N \\ \ell+k \end{bmatrix}_q, \quad (3.5)$$

where a Gaussian coefficient whose lower index falls outside  $\{0, 1, \dots, \text{upper index}\}$  is read as 0, so that only finitely many terms are nonzero. The case  $k = 0$  is [Corollary 3.9](#) with  $m = n = N$ .

*Proof.* If  $|k| > N$  both sides vanish: the left side because  $N+k \notin \{0, \dots, 2N\}$ , and the right side because a nonzero term requires both  $0 \leq \ell \leq N$  and  $0 \leq \ell+k \leq N$ , which forces  $|k| \leq N$ .

Assume therefore  $-N \leq k \leq N$  and set  $r = N+k \in \{0, \dots, 2N\}$ . [Theorem 3.8](#), in the form (3.3) with  $m = n = N$ , gives

$$\begin{bmatrix} 2N \\ N+k \end{bmatrix}_q = \sum_{j \in \mathbb{Z}} q^{(N-j)(N+k-j)} \begin{bmatrix} N \\ j \end{bmatrix}_q \begin{bmatrix} N \\ N+k-j \end{bmatrix}_q.$$

Substitute  $\ell = N-j$ . Then  $N+k-j = \ell+k$ , so the exponent becomes  $\ell(\ell+k) = \ell^2 + \ell k$ ; the symmetry  $\begin{bmatrix} N \\ N-\ell \end{bmatrix}_q = \begin{bmatrix} N \\ \ell \end{bmatrix}_q$  of [Theorem 2.5](#) (i) turns the first Gaussian coefficient into  $\begin{bmatrix} N \\ \ell \end{bmatrix}_q$ , and the second is  $\begin{bmatrix} N \\ \ell+k \end{bmatrix}_q$ . This is (3.5).  $\square$



*Remark 3.11* (Formal version). `FiniteQBinomialCore.lean` defines the zero-extended integer coefficient `Fabius.gaussianBinomialInt`; its cast, negative-index, above-row, and total-reflection laws are `Fabius.gaussianBinomialInt_ofNat`, `Fabius.gaussianBinomialInt_eq_zero_of_neg`, `Fabius.gaussianBinomialInt_eq_zero_of_lt`, and `Fabius.gaussianBinomialInt_symm`. `QBinomialVandermonde.lean` proves the signed formula first on the finite range  $0 \leq \ell \leq N$  as `Fabius.gaussianBinomial_two_mul_int_shifted_central`, and then literally over all  $\ell \in \mathbb{Z}$  as `Fabius.gaussianBinomial_two_mul_int_shifted_central_finsum`. The Lean power is written  $q^{\ell^{\natural}(\ell+k)^{\natural}}$ , with  $t^{\natural} = \max(t, 0)$ ; every possibly nonzero term has  $\ell, \ell + k \geq 0$ , so its exponent is exactly  $\ell(\ell + k)$ . Thus the formal theorem is total in  $k$ , including  $|k| > N$ , over every commutative semiring and at  $q = 0$ .

### 3.4 $q$ -binomial inversion

**Theorem 3.12** ( $q$ -binomial inversion). *For sequences  $(a_n)$  and  $(b_n)$  in a commutative ring containing  $q$ , the relations*

$$b_n = \sum_{k=0}^n \begin{bmatrix} n \\ k \end{bmatrix}_q a_k$$

*are equivalent to*

$$a_n = \sum_{k=0}^n (-1)^{n-k} q^{\binom{n-k}{2}} \begin{bmatrix} n \\ k \end{bmatrix}_q b_k.$$

*Proof.* Substitute the first relation into the proposed inverse. The coefficient of  $a_j$  becomes

$$\sum_{k=j}^n (-1)^{n-k} q^{\binom{n-k}{2}} \begin{bmatrix} n \\ k \end{bmatrix}_q \begin{bmatrix} k \\ j \end{bmatrix}_q.$$

The factorial definition gives

$$\begin{bmatrix} n \\ k \end{bmatrix}_q \begin{bmatrix} k \\ j \end{bmatrix}_q = \begin{bmatrix} n \\ j \end{bmatrix}_q \begin{bmatrix} n-j \\ k-j \end{bmatrix}_q.$$

Put  $r = n - k$ . The coefficient is

$$\begin{bmatrix} n \\ j \end{bmatrix}_q \sum_{r=0}^{n-j} (-1)^r q^{\binom{r}{2}} \begin{bmatrix} n-j \\ r \end{bmatrix}_q.$$

By [Corollary 3.2](#), this equals 0 when  $n > j$  and 1 when  $n = j$ . Thus the composition is the identity. The same argument in the opposite order proves equivalence.  $\square$

**Corollary 3.13** (Matrix orthogonality). *For  $0 \leq j \leq n$ ,*

$$\sum_{k=j}^n (-1)^{n-k} q^{\binom{n-k}{2}} \begin{bmatrix} n \\ k \end{bmatrix}_q \begin{bmatrix} k \\ j \end{bmatrix}_q = \delta_{n,j}.$$

*Proof.* This is the kernel computation in the preceding proof.  $\square$

*Remark 3.14* (Formal version and scaled strengthening). The scalar algebra is formalized in `QBinomialInversion.lean`. Its public surface consists exhaustively of the four definitions `Fabius.gaussianBinomialKernel`, `Fabius.gaussianBinomialInverseKernel`, `Fabius.scaledGaussianBinomialKernel`, and `Fabius.scaledGaussianBinomialInverseKernel`, and the seven theorems

`Fabius.gaussianBinomial_mul`, `Fabius.sum_gaussianBinomial_alternating_mul_pow`, `Fabius.sum_gaussianBinomial_alternating`, `Fabius.gaussianBinomialKernel_left_orthogonality`, `Fabius.gaussianBinomialKernel_right_orthogonality`, `Fabius.scaledGaussianBinomialKernel_left_orthogonality`, and `Fabius.scaledGaussianBinomialKernel_right_orthogonality`. The chain identity already holds over every commutative semiring. The alternating and orthogonality results hold over every commutative ring. Both multiplication orders are stated on the total interval  $\text{Icc } j \ n$ , including the empty case  $n < j$ . The scaled forward and inverse entries are

$$s^{n-k} \begin{bmatrix} n \\ k \end{bmatrix}_q, \quad (-s)^{n-k} q^{\binom{n-k}{2}} \begin{bmatrix} n \\ k \end{bmatrix}_q,$$

for an arbitrary independent  $s$ ; neither  $s$  nor  $q$  is assumed nonzero or invertible.

The common finite-sum argument is isolated in `FiniteTriangularTransform.lean`, whose exhaustive public surface is the definition `Fabius.lowerTriangularTransform` and the theorem `Fabius.lowerTriangularTransform_comp`. For `[Semiring R] [AddCommMonoid M] [Module R M]`, the theorem converts a total ordered kernel convolution on  $\text{Icc } j \ n$  into equality of the whole composite sequence function with its input. It assumes neither commutativity of  $R$  nor additive inverses in  $M$ . The Gaussian kernels are zero-extended above the row, so the empty case  $n < j$  fits this generic total interface without a separate range hypothesis.

The module-valued sequence theorem is formalized in `QBinomialTransform.lean`, whose exhaustive public surface is the four definitions `Fabius.scaledGaussianBinomialTransform`, `Fabius.scaledGaussianBinomialInverseTransform`, `Fabius.gaussianBinomialTransform`, and `Fabius.gaussianBinomialInverseTransform`, together with the four theorems `Fabius.scaledGaussianBinomialInverseTransform_transform`, `Fabius.scaledGaussianBinomialTransform_inverseTransform`, `Fabius.scaledGaussianBinomial_inversion`, and `Fabius.gaussianBinomial_inversion`. The two forward-transform definitions require only `[Semiring R] [AddCommMonoid M] [Module R M]`; the two signed inverse definitions require `[Ring R]` with the same target assumptions. All four theorems require exactly `[CommRing R] [AddCommMonoid M] [Module R M]`. The first two give both compositions as equalities of whole sequence functions and the last two give the scaled and classical iff statements. Taking  $M = R$  yields [Theorem 3.12](#); the formal theorem is stronger in allowing arbitrary module-valued sequences and requires no topology or convergence.

The scaled characteristic polynomial and the  $q$ -difference functional are formalized in `QDifferenceAnnihilation.lean`, whose public surface is exhaustively the four theorems `Fabius.sum_scaledGaussianBinomialInverseKernel_mul_pow`, `Fabius.sum_gaussianBinomialInverseKernel_mul_geometric_pow`, `Fabius.qDifference_sum_eval2_eq_map_coeff_mul`, and `Fabius.qDifference_sum_eval2_eq_zero_of_degree_lt`. Writing

$$K_{q,s}^{-1}(n, k) = (-s)^{n-k} q^{\binom{n-k}{2}} \begin{bmatrix} n \\ k \end{bmatrix}_q,$$

the first theorem is the homogeneous identity

$$\sum_{k=0}^n K_{q,s}^{-1}(n, k) z^k = \prod_{j=0}^{n-1} (z - sq^j). \quad (3.6)$$

It holds over every commutative ring for arbitrary  $q, s, z$ . With  $s = 1$  and  $z = q^d$ , the second theorem gives the complete moment table

$$\sum_{k=0}^n K_{q,1}^{-1}(n, k) (q^k)^d = \prod_{j=0}^{n-1} (q^d - q^j), \quad (3.7)$$

so the row annihilates every monomial of degree  $d < n$ .

The last two theorems make the polynomial statement strictly stronger than the scalar lemma usually quoted in Bailey inversion. If  $A$  is a semiring,  $R$  a commutative ring,  $\varphi: A \rightarrow R$  a semiring homomorphism, and  $P \in A[X]$  has degree at most  $n$ , then

$$\sum_{k=0}^n K_{q,1}^{-1}(n, k) P^\varphi(q^k) = \varphi([X^n]P) \prod_{j=0}^{n-1} (q^n - q^j), \quad (3.8)$$

where  $P^\varphi$  denotes coefficientwise scalar extension. If  $\deg P < n$ , the left side is zero. These degree statements include  $n = 0$  and the zero polynomial. Nodes may collide and the surviving factor in (3.8) may vanish: no division, distinctness, nonzero or invertible base, domain, characteristic, topology, or convergence hypothesis is present.

The global geometric-numerator bridge `Fabius.geometricQBinomialWeightNumerator_eq_scaledGaussianBinomialInverseKernel` is owned by `GeometricQBinomialLagrange.lean`. It identifies the denominator-free numerator with  $K_{q,q}^{-1}(n, k)$  for every pair of natural indices, including indices above the diagonal; equivalently, its generating polynomial is the  $s = q$  specialization of (3.6). This global identity does not remove the field, finite-node injectivity, or in-range assumptions from separately normalized Lagrange quotients.

Finally, `QBinomialInversionSpecializations.lean` has exactly two public definitions and four public theorems: `Fabius.qGaussianResidualCoeff`, `Fabius.qGaussianReconstructionCoeff`, `Fabius.qGaussianResidualCoeff_eq`, `Fabius.qGaussianReconstructionCoeff_eq`, `Fabius.qGaussianReconstructionCoeff_residualCoeff_delta`, and `Fabius.qGaussianResidualCoeff_reconstructionCoeff_delta`. At base  $q^2$  and scale  $-q$ , the coefficients are

$$(-q)^{n-k} \begin{bmatrix} n \\ k \end{bmatrix}_{q^2}, \quad q^{(n-k)^2} \begin{bmatrix} n \\ k \end{bmatrix}_{q^2},$$

The two definitions and these two pointwise closed forms require only an arbitrary, possibly noncommutative ring. Exactly the two convolution-delta theorems require a commutative ring, and prove that both total finite orders are the Kronecker delta.

### 3.5 A $q$ -Cauchy product identity

**Theorem 3.15** ( $q$ -Cauchy identity). *For  $u, v$  in a commutative ring,*

$$(uv; q)_n = \sum_{k=0}^n \begin{bmatrix} n \\ k \end{bmatrix}_q (u; q)_k v^k (v; q)_{n-k}.$$

*Proof.* Induct on  $n$ . The assertion is clear at  $n = 0$ . Let the right side be  $C_n(u, v)$ . Use

$$\begin{bmatrix} n+1 \\ k \end{bmatrix}_q = q^{n+1-k} \begin{bmatrix} n \\ k-1 \end{bmatrix}_q + \begin{bmatrix} n \\ k \end{bmatrix}_q$$

and split  $C_{n+1}$  accordingly. In the first part shift  $k \mapsto k+1$ , use  $(u; q)_{k+1} = (u; q)_k (1 - uq^k)$ , and combine with the second part after using  $(v; q)_{n+1-k} = (v; q)_{n-k} (1 - vq^{n-k})$ . The bracket that remains is

$$(1 - vq^{n-k}) + vq^{n-k} (1 - uq^k) = 1 - uvq^n,$$

independent of  $k$ . Therefore

$$C_{n+1}(u, v) = (1 - uvq^n) C_n(u, v) = (uv; q)_{n+1}.$$

□

*Remark 3.16.* Taking  $u = 0$  recovers the tautology  $(0; q)_k = 1$  inside a weighted expansion; taking special limits of  $u$  or  $v$  produces several finite convolution formulas used later in the proof of the  $q$ -Pfaff–Saalschütz summation.

*Remark 3.17* (Formal version and strengthenings). The theorem is `Fabius.finite_qCauchy_identity` in `QBinomialCauchy.lean`. It is stated for arbitrary  $q, u, v$  in every commutative ring and every  $n \in \mathbb{N}$ , so it includes  $n = 0$ ,  $q = 0$ , roots of unity, positive characteristic, and zero divisors. The proof is entirely polynomial and assumes neither cancellation nor nonvanishing. The reflected strengthening `Fabius.finite_qCauchy_identity_reflected` states

$$(uv; q)_n = \sum_{k=0}^n \begin{bmatrix} n \\ k \end{bmatrix}_q (v; q)_k v^{n-k} (u; q)_{n-k}.$$

The same module defines the denominator-free  $q$ -Bernstein term

$$B_{n,k}^{(q)}(x) = \begin{bmatrix} n \\ k \end{bmatrix}_q x^k (x; q)_{n-k}$$

as `Fabius.qBernsteinBasis` and proves the total partition of unity

$$\sum_{k=0}^n B_{n,k}^{(q)}(x) = 1$$

as `Fabius.sum_qBernsteinBasis`. Together with `Fabius.finite_qCauchy_second_identity`, described at [Lemma 9.7](#), these are exhaustively the one public definition and four public theorems of `QBinomialCauchy.lean`.

## Chapter 4

# Weighted binomial coefficients and symmetric functions

The ordinary and Gaussian binomial coefficients are instances of a broader construction in which each available object carries a weight. This point of view, developed systematically by Konvalina, places subset selection, multiset selection, Stirling numbers, and Gaussian coefficients in one framework.

### 4.1 Elementary and complete weighted coefficients

**Definition 4.1.** Let  $\mathbf{w} = (w_1, \dots, w_n)$  be elements of a commutative ring. Define

$$C_k^n(\mathbf{w}) = \sum_{1 \leq i_1 < \dots < i_k \leq n} w_{i_1} \cdots w_{i_k},$$

$$S_k^n(\mathbf{w}) = \sum_{1 \leq i_1 \leq \dots \leq i_k \leq n} w_{i_1} \cdots w_{i_k}.$$

We set  $C_0^n = S_0^n = 1$ . Thus  $C_k^n$  is the elementary symmetric function  $e_k(w_1, \dots, w_n)$  and  $S_k^n$  is the complete homogeneous symmetric function  $h_k(w_1, \dots, w_n)$ .

**Theorem 4.2** (Weighted Pascal recurrences). For  $n, k \geq 1$ ,

$$C_k^n(\mathbf{w}) = C_k^{n-1}(\mathbf{w}) + w_n C_{k-1}^{n-1}(\mathbf{w}), \quad (4.1)$$

$$S_k^n(\mathbf{w}) = S_k^{n-1}(\mathbf{w}) + w_n S_{k-1}^{n-1}(\mathbf{w}). \quad (4.2)$$

*Proof.* A  $k$ -element subset of  $\{1, \dots, n\}$  either omits  $n$ , contributing  $C_k^{n-1}$ , or contains  $n$ , in which case removal of  $n$  leaves a  $(k-1)$ -element subset and the weight acquires a factor  $w_n$ . This proves (4.1).

For multisets, separate those with no occurrence of  $n$  from those with at least one occurrence. Removing one occurrence of  $n$  from the latter leaves an arbitrary multiset of size  $k-1$  drawn from all  $n$  types. This proves (4.2).  $\square$

**Theorem 4.3** (Generating products). One has

$$\prod_{i=1}^n (1 + w_i z) = \sum_{k=0}^n C_k^n(\mathbf{w}) z^k, \quad (4.3)$$

$$\prod_{i=1}^n \frac{1}{1 - w_i z} = \sum_{k=0}^{\infty} S_k^n(\mathbf{w}) z^k. \quad (4.4)$$

The second identity is formal in  $z$ , or analytic when  $|w_i z| < 1$  for all  $i$ .

*Proof.* In the first product, choosing  $w_i z$  from exactly the factors indexed by a subset  $I$  produces  $z^{|I|} \prod_{i \in I} w_i$ . Summing over subsets gives (4.3). For the second, expand each factor geometrically:

$$\frac{1}{1 - w_i z} = \sum_{r_i \geq 0} w_i^{r_i} z^{r_i}.$$

The coefficient of  $z^k$  sums over weak compositions  $r_1 + \cdots + r_n = k$ , which are equivalent to multisets of size  $k$ .  $\square$

**Corollary 4.4** (Elementary–complete orthogonality). *For  $m \geq 0$ ,*

$$\sum_{j=0}^m (-1)^j C_j^m(\mathbf{w}) S_{m-j}^n(\mathbf{w}) = \delta_{m,0}.$$

*Proof.* Multiply  $\prod_i (1 - w_i z)$  by its reciprocal and compare the coefficient of  $z^m$ , using Theorem 4.3 with  $z$  replaced by  $-z$  in the elementary generating function.  $\square$

**Remark 4.5** (Formal version and exact boundary). For a finite family  $a : \iota \rightarrow R$ , the evaluator and its bridge to Mathlib’s universal `MvPolynomial.esymm` are

```
elementarySymmetricEval
elementarySymmetricEval_eq_eval_esymm.
```

The adjoining-variable recurrences are

```
elementarySymmetricEval_option_succ
completeHomogeneousEval_option_succ.
```

They prove, respectively,

$$e_{r+1}(x, a) = x e_r(a) + e_{r+1}(a), \quad h_{r+1}(x, a) = x h_r(x, a) + h_{r+1}(a).$$

Thus both weighted Pascal recurrences of Theorem 4.2 are exact, including their zero-degree boundaries, over every commutative semiring.

The exhaustive public surface of `SymmetricFunctionOrthogonality.lean` is its one definition above and the following six theorems, all in namespace `Fabius`:

```
elementarySymmetricEval_eq_eval_esymm
elementarySymmetricEval_zero
elementarySymmetricEval_comp_equiv
elementarySymmetricEval_option_succ
elementarySymmetricEval_fin_succ
sum_elementarySymmetricEval_mul_completeHomogeneousEval.
```

The first five structural theorems hold over a commutative semiring; the last proves exactly

$$\sum_{j=0}^m (-1)^j e_j(a) h_{m-j}(a) = \begin{cases} 1, & m = 0, \\ 0, & m > 0, \end{cases}$$

over every commutative ring, for every finite index type and every  $m \in \mathbb{N}$ . It therefore includes the empty family and degree zero and uses no division, domain, characteristic, or nonvanishing hypothesis.

The exhaustive public surface of `SymmetricFunctionGenerating.lean` consists of the two definitions

```

elementarySymmetricGeneratingSeries
completeHomogeneousGeneratingSeries

```

and the six theorems

```

coeff_elementarySymmetricGeneratingSeries
coeff_completeHomogeneousGeneratingSeries
elementarySymmetricGeneratingSeries_eq_prod
elementarySymmetricGeneratingSeries_neg_mul_completeHomogeneousGeneratingSeries
completeHomogeneousGeneratingSeries_eq_invOfUnit_elementarySymmetricGeneratingSeries_neg
prod_one_sub_qPow_X_mul_gaussianBinomialGeneratingSeries.

```

The definitions, coefficient formulas, and finite product  $E_a(X) = \prod_i (1 + a_i X)$  hold over every commutative semiring. Over every commutative ring the remaining theorems prove

$$E_{-a}(X)H_a(X) = 1, \quad H_a(X) = \text{invOfUnit}(E_{-a}(X), 1),$$

as well as the geometric-alphabet reciprocal identity recorded after [Theorem 3.6](#). Hence the polynomial elementary identity and the formal-power-series version of the complete identity in [Theorem 4.3](#) are exact. The theorem as labelled remains **partial** because its analytic evaluation and convergence assertion under  $|w_i z| < 1$  has not been formalized.

The complementary finite-index interface in `CompleteHomogeneousGenerating.lean` consists of the definition `completeHomogeneousGeneratingSeriesOn` and the six theorems

```

coeff_completeHomogeneousGeneratingSeriesOn
completeHomogeneousGeneratingSeriesOn_empty
completeHomogeneousGeneratingSeriesOn_insert
one_sub_mul_completeHomogeneousGeneratingSeriesOn_insert
prod_one_sub_mul_completeHomogeneousGeneratingSeriesOn
completeHomogeneousGeneratingSeriesOn_eq_invOfUnit_prod.

```

The coefficient, empty-family, and adjoining-variable recurrence results hold over every commutative semiring. Over every commutative ring, the remaining results clear one or all finite linear denominators and identify the series with Mathlib’s canonical `PowerSeries.invOfUnit`; zero divisors are allowed. Thus this module and `SymmetricFunctionGenerating.lean` give complementary formal proofs of the complete-product half of [Theorem 4.3](#), while the latter also supplies the elementary product and elementary–complete reciprocity.

There is also a finite-family analytic companion. The sole public theorem `completeHomogeneousEvalOn_isBigO_pow` in `CompleteHomogeneousAsymptotics.lean` transfers coordinatewise  $\mathcal{O}(g)$  bounds through every fixed degree  $r$ : the corresponding complete homogeneous evaluation is  $\mathcal{O}(g^r)$ . It includes degree zero and requires no nonvanishing assumption on  $g$ . This asymptotic bound does not evaluate either formal series at  $z$ , establish its convergence, or close the analytic clause of the labelled theorem.

## 4.2 Classical, Stirling, and Gaussian specializations

**Theorem 4.6** (A specialization table). *The following identities hold.*

(i) If  $w_i = 1$ , then

$$C_k^n = \binom{n}{k}, \quad S_k^n = \binom{n+k-1}{k}.$$

(ii) If  $w_i = q^{i-1}$ , then

$$C_k^n = q^{\binom{k}{2}} \begin{bmatrix} n \\ k \end{bmatrix}_q, \quad S_k^n = \begin{bmatrix} n+k-1 \\ k \end{bmatrix}_q.$$

(iii) For the unsigned Stirling numbers of the first kind  $c(n, k)$  and Stirling numbers of the second kind  $\left\{ \begin{smallmatrix} n \\ k \end{smallmatrix} \right\}$ ,

$$c(n+1, k+1) = C_{n-k}^n(1, 2, \dots, n),$$

$$\left\{ \begin{smallmatrix} n+k \\ n \end{smallmatrix} \right\} = S_k^n(1, 2, \dots, n).$$

*Proof.* For  $w_i = 1$ ,  $C_k^n$  counts  $k$ -subsets. The multiset count is the stars-and-bars number  $\binom{n+k-1}{k}$ .

For  $w_i = q^{i-1}$ , the elementary identity is [Corollary 3.4](#). For the complete identity, [Theorem 4.3](#) gives

$$\prod_{i=0}^{n-1} \frac{1}{1 - zq^i} = \frac{1}{(z; q)_n}.$$

The coefficient of  $z^k$  is  $\begin{bmatrix} n+k-1 \\ k \end{bmatrix}_q$  by [Theorem 3.6](#).

Finally,

$$x(x+1) \cdots (x+n) = \sum_{j=0}^{n+1} c(n+1, j) x^j.$$

The coefficient of  $x^{k+1}$  is  $e_{n-k}(1, 2, \dots, n)$ . The standard ordinary generating function

$$\prod_{i=1}^n \frac{1}{1 - iz} = \sum_{k \geq 0} \left\{ \begin{smallmatrix} n+k \\ n \end{smallmatrix} \right\} z^k$$

follows from the recurrence for Stirling numbers of the second kind, or by induction on  $n$ ; comparison with (4.4) proves the last formula.  $\square$

### 4.3 Principal specialization of Schur polynomials

The two Gaussian entries of [Theorem 4.6\(ii\)](#) are the one-column and the one-row case of a single product formula. Stating it requires no symmetric-function theory beyond determinants, because a Schur polynomial can be *defined* by its bialternant.

**Definition 4.7.** Let  $m \geq 1$  and let  $\lambda = (\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_m \geq 0)$  be a partition with at most  $m$  parts, padded by zeros. Put

$$\mu_j = \lambda_j + m - j \quad (1 \leq j \leq m), \quad \delta = (m-1, m-2, \dots, 0),$$

so that  $\mu_1 > \mu_2 > \cdots > \mu_m \geq 0$ , and  $\delta$  is the value of  $\mu$  at  $\lambda = 0$ . For indeterminates  $x_1, \dots, x_m$  set

$$a_\mu(x_1, \dots, x_m) = \det(x_i^{\mu_j})_{i,j=1}^m,$$

and define the *Schur polynomial*

$$s_\lambda(x_1, \dots, x_m) = \frac{a_\mu(x_1, \dots, x_m)}{a_\delta(x_1, \dots, x_m)}.$$

Finally put  $n(\lambda) = \sum_{j=1}^m (j-1)\lambda_j$ .



**Lemma 4.8** (Vandermonde alternant). *For indeterminates  $y_1, \dots, y_m$ ,*

$$\det(y_i^{j-1})_{i,j=1}^m = \prod_{1 \leq i < j \leq m} (y_j - y_i).$$

*Consequently  $a_\delta(x_1, \dots, x_m) = \prod_{1 \leq i < j \leq m} (x_i - x_j)$ , and  $a_\delta$  divides  $a_\mu$  in  $\mathbb{Z}[x_1, \dots, x_m]$ , so  $s_\lambda$  is a polynomial with integer coefficients.*

*Proof.* Induct on  $m$ ; for  $m = 1$  both sides equal 1. Regard  $D = \det(y_i^{j-1})_{i,j=1}^m$  as a polynomial in  $y_m$  of degree at most  $m - 1$ . If  $y_m = y_i$  for some  $i < m$ , two rows coincide and  $D = 0$ ; hence  $\prod_{i < m} (y_m - y_i)$  divides  $D$ , and by the degree bound  $D = c \prod_{i < m} (y_m - y_i)$  with  $c$  free of  $y_m$ . Expanding along the last row, the coefficient of  $y_m^{m-1}$  in  $D$  is the cofactor at position  $(m, m)$ , namely  $\det(y_i^{j-1})_{i,j=1}^{m-1}$ , which by induction equals  $\prod_{1 \leq i < j \leq m-1} (y_j - y_i)$ . Multiplying proves the formula.

Reversing the order of the columns of  $(x_i^{m-j})_{i,j}$  turns the exponents  $m - j$  into  $j - 1$  and multiplies the determinant by the sign  $(-1)^{\binom{m}{2}}$  of the reversal permutation, so

$$a_\delta = (-1)^{\binom{m}{2}} \prod_{1 \leq i < j \leq m} (x_j - x_i) = \prod_{1 \leq i < j \leq m} (x_i - x_j).$$

For the divisibility, interchanging  $x_i$  and  $x_j$  interchanges two rows of  $a_\mu$ , so  $a_\mu$  changes sign; setting  $x_i = x_j$  therefore gives  $a_\mu = 0$ , and  $x_i - x_j$  divides  $a_\mu$  in  $\mathbb{Z}[x_1, \dots, x_m]$ . The polynomials  $x_i - x_j$  with  $i < j$  are pairwise non-associate primes in that unique factorization domain, so their product, which is  $a_\delta$ , divides  $a_\mu$ .  $\square$

**Theorem 4.9** (Principal specialization of a Schur polynomial). *Let  $\lambda$  have at most  $m$  parts. Then, as an identity in  $\mathbb{Q}(q)$ ,*

$$s_\lambda(1, q, q^2, \dots, q^{m-1}) = q^{n(\lambda)} \prod_{1 \leq i < j \leq m} \frac{1 - q^{\lambda_i - \lambda_j + j - i}}{1 - q^{j - i}}.$$

*The left side lies in  $\mathbb{Z}[q]$ , so by Corollary 20.11 the identity persists at every complex  $q$  once the right-hand quotient has been reduced; as displayed it is to be read at those  $q$  with  $q^d \neq 1$  for  $1 \leq d \leq m - 1$ .*

*Proof.* Specialize  $x_i = q^{i-1}$ , so that  $x_i^{\mu_j} = q^{(i-1)\mu_j}$ . Writing  $y_j = q^{\mu_j}$ , the matrix  $(x_i^{\mu_j})_{i,j}$  has entries  $y_j^{i-1}$ , hence is the transpose of  $(y_i^{j-1})_{i,j}$ . A determinant is invariant under transposition, so Lemma 4.8 gives

$$a_\mu(1, q, \dots, q^{m-1}) = \prod_{1 \leq i < j \leq m} (q^{\mu_j} - q^{\mu_i}).$$

Because  $\lambda$  is weakly decreasing we have  $\mu_1 > \dots > \mu_m \geq 0$ , so  $\mu_i - \mu_j > 0$  whenever  $i < j$ , and we extract from each difference the *smaller* power:

$$q^{\mu_j} - q^{\mu_i} = q^{\mu_j} (1 - q^{\mu_i - \mu_j}).$$

Hence

$$a_\mu(1, q, \dots, q^{m-1}) = q^{A(\mu)} \prod_{1 \leq i < j \leq m} (1 - q^{\mu_i - \mu_j}), \quad A(\mu) = \sum_{1 \leq i < j \leq m} \mu_j = \sum_{j=1}^m (j-1)\mu_j,$$

the last equality because  $\mu_j$  is contributed once for each  $i < j$ . Applying this to  $\mu = \delta$ , where  $\delta_i - \delta_j = j - i$ , gives

$$a_\delta(1, q, \dots, q^{m-1}) = q^{A(\delta)} \prod_{1 \leq i < j \leq m} (1 - q^{j-i}),$$

a nonzero element of  $\mathbb{Q}(q)$ . Dividing,

$$s_\lambda(1, q, \dots, q^{m-1}) = q^{A(\mu) - A(\delta)} \prod_{1 \leq i < j \leq m} \frac{1 - q^{\mu_i - \mu_j}}{1 - q^{j-i}}.$$

Finally  $\mu_i - \mu_j = \lambda_i - \lambda_j + j - i$  and

$$A(\mu) - A(\delta) = \sum_{j=1}^m (j-1)(\mu_j - \delta_j) = \sum_{j=1}^m (j-1)\lambda_j = n(\lambda).$$

The left side is a polynomial with integer coefficients by [Lemma 4.8](#); the transfer to arbitrary complex  $q$  is [Lemma 20.10](#) together with [Corollary 20.11](#).  $\square$

**Corollary 4.10** (The one-column and one-row shadows). *Let  $m = n \geq 1$ .*

(i) *For  $0 \leq k \leq n$  and  $\lambda = (1^k) = (1, \dots, 1, 0, \dots, 0)$ ,*

$$s_{(1^k)}(1, q, \dots, q^{n-1}) = q^{\binom{k}{2}} \begin{bmatrix} n \\ k \end{bmatrix}_q.$$

(ii) *For  $k \geq 0$  and  $\lambda = (k)$ ,*

$$s_{(k)}(1, q, \dots, q^{n-1}) = \begin{bmatrix} n + k - 1 \\ k \end{bmatrix}_q.$$

Since  $s_{(1^k)} = e_k$  and  $s_{(k)} = h_k$  [21], these are exactly the two Gaussian entries of [Theorem 4.6\(ii\)](#), that is,  $C_k^m(\mathbf{w})$  and  $S_k^n(\mathbf{w})$  at  $w_i = q^{i-1}$ .

*Proof.* (i) Here  $\lambda_i = 1$  for  $i \leq k$  and  $\lambda_i = 0$  for  $i > k$ , so  $n(\lambda) = \sum_{i \leq k} (i-1) = \binom{k}{2}$ , and  $\lambda_i - \lambda_j$  vanishes except when  $i \leq k < j$ , where it equals 1. The product of [Theorem 4.9](#) therefore collapses to

$$\prod_{i=1}^k \prod_{j=k+1}^n \frac{1 - q^{j-i+1}}{1 - q^{j-i}} = \prod_{i=1}^k \frac{1 - q^{n-i+1}}{1 - q^{k-i+1}},$$

the inner product telescoping. The numerators are  $(1 - q^n)(1 - q^{n-1}) \cdots (1 - q^{n-k+1}) = (q; q)_n / (q; q)_{n-k}$  and the denominators are  $(1 - q^k) \cdots (1 - q) = (q; q)_k$ , so the product is  $\begin{bmatrix} n \\ k \end{bmatrix}_q$ .

(ii) Here  $\lambda_1 = k$  and  $\lambda_j = 0$  for  $j \geq 2$ , so  $n(\lambda) = 0$  and only the pairs  $(1, j)$  contribute:

$$\prod_{j=2}^n \frac{1 - q^{k+j-1}}{1 - q^{j-1}} = \frac{(q; q)_{n+k-1} / (q; q)_k}{(q; q)_{n-1}} = \begin{bmatrix} n + k - 1 \\ k \end{bmatrix}_q. \quad \square$$

**Remark 4.11.** Both hypotheses of [Theorem 4.9](#) are needed as stated. If  $\lambda$  had more than  $m$  parts,  $\mu$  would not be strictly decreasing and  $a_\mu$  would vanish identically. And the displayed quotient is meaningful only where  $q^{j-i} \neq 1$  for all  $1 \leq i < j \leq m$ ; at the excluded roots of unity, and at  $q = 1$ , one must cancel first, exactly as prescribed by [Corollary 20.11](#). After cancellation the limit  $q \rightarrow 1$  is

$$s_\lambda(1, 1, \dots, 1) = \prod_{1 \leq i < j \leq m} \frac{\lambda_i - \lambda_j + j - i}{j - i},$$

by  $\lim_{q \rightarrow 1} (1 - q^a) / (1 - q^b) = a/b$ ; this is Weyl's dimension formula for  $\mathrm{GL}_m$ , and it stands to [Theorem 4.9](#) exactly as [Corollary 2.6](#) stands to the Gaussian coefficient.

## 4.4 Weighted inversion

**Theorem 4.12** (Symmetric-function inversion). *Let  $(a_m)$  and  $(b_m)$  be sequences. The transformations*

$$b_m = \sum_{j=0}^m S_{m-j}^n(\mathbf{w}) a_j$$

and

$$a_m = \sum_{j=0}^m (-1)^{m-j} C_{m-j}^n(\mathbf{w}) b_j$$

are inverse to each other.

*Proof.* The generating functions  $A(z) = \sum a_m z^m$  and  $B(z) = \sum b_m z^m$  satisfy

$$B(z) = A(z) \prod_{i=1}^n (1 - w_i z)^{-1}.$$

Multiplication by  $\prod_i (1 - w_i z)$  gives the proposed inverse coefficient formula. Equivalently, use the orthogonality in [Corollary 4.4](#).  $\square$

*Remark 4.13* (Formal version and module-valued strengthening). The exhaustive public surface of `SymmetricFunctionTransform.lean` is the four definitions

```
Fabius.completeHomogeneousKernel
Fabius.signedElementaryKernel
Fabius.completeHomogeneousTransform
Fabius.signedElementaryTransform
```

and the five theorems

```
Fabius.completeHomogeneousKernel_left_orthogonality
Fabius.completeHomogeneousKernel_right_orthogonality
Fabius.signedElementaryTransform_completeHomogeneousTransform
Fabius.completeHomogeneousTransform_signedElementaryTransform
Fabius.weightedSymmetricFunction_inversion.
```

For a finite family  $w : \iota \rightarrow R$ , the complete-homogeneous kernel and transform need only `[CommSemiring R]`; the signed-elementary kernel and transform require `[CommRing R]`. Both kernel orthogonalities, both inverse compositions, and the inversion iff use exactly `[CommRing R] [AddCommMonoid M] [Module R M]` with `[Fintype ι]`. The compositions are equalities of whole sequence functions, so `Fabius.weightedSymmetricFunction_inversion` proves [Theorem 4.12](#) exactly and strengthens it to arbitrary module-valued sequences.

Both kernels are extended by zero above the diagonal, including the empty outer interval. Their composition proofs reuse `Fabius.lowerTriangularTransform_comp`; together with the definition `Fabius.lowerTriangularTransform`, these are exhaustively the one theorem and one definition of `FiniteTriangularTransform.lean`. The generic theorem itself only assumes a semiring acting on an additive commutative monoid and a total ordered finite-convolution identity. No division, topology, domain, characteristic, or nonvanishing condition occurs.

## 4.5 The subset–subspace analogy

The choice  $w_i = q^{i-1}$  is more than a convenient specialization. It replaces the Boolean lattice of subsets by the lattice of subspaces over a finite field. Ordinary binomial coefficients count subsets by cardinality; Gaussian coefficients count subspaces by dimension. The two-term weighted recurrence mirrors the decomposition of a subspace according to its relation with a fixed hyperplane. This analogy is made exact in [Chapter 6](#).

## Chapter 5

# Partitions, words, paths, and inversion statistics

### 5.1 Binary words

Let  $\mathcal{W}_{n,k}$  be the set of binary words of length  $n$  containing  $k$  ones. Define

$$\text{inv}(w) = \# \{(i, j) : i < j, w_i = 1, w_j = 0\}.$$

**Theorem 5.1** (Inversion generating function). *For  $0 \leq k \leq n$ ,*

$$\left[ \begin{matrix} n \\ k \end{matrix} \right]_q = \sum_{w \in \mathcal{W}_{n,k}} q^{\text{inv}(w)}.$$

*Proof.* Let  $F_{n,k}(q)$  denote the right side. If the last letter is 1, deletion of it gives a word in  $\mathcal{W}_{n-1,k-1}$  and creates no new inversion. If the last letter is 0, deletion gives a word in  $\mathcal{W}_{n-1,k}$  and the final zero forms an inversion with each of the  $k$  ones, contributing a factor  $q^k$ . Hence

$$F_{n,k} = F_{n-1,k-1} + q^k F_{n-1,k}.$$

The boundary values are 1, so  $F_{n,k}$  and  $\left[ \begin{matrix} n \\ k \end{matrix} \right]_q$  obey the same recurrence and initial conditions.  $\square$

**Corollary 5.2.** *The random variable in [Theorem 2.10](#) is the inversion number of a uniformly random word in  $\mathcal{W}_{n,k}$ .*

*Proof.* The coefficient of  $q^j$  in [Theorem 5.1](#) counts words with  $j$  inversions, while  $|\mathcal{W}_{n,k}| = \binom{n}{k}$ .  $\square$

### 5.2 Partitions inside a rectangle

A partition  $\lambda = (\lambda_1, \dots, \lambda_k)$  is said to fit in a  $k \times m$  rectangle if

$$m \geq \lambda_1 \geq \dots \geq \lambda_k \geq 0.$$

Its size is  $|\lambda| = \lambda_1 + \dots + \lambda_k$ .

**Theorem 5.3** (Rectangular partition generating function). *For  $m, k \geq 0$ ,*

$$\left[ \begin{matrix} m+k \\ k \end{matrix} \right]_q = \sum_{\lambda \subseteq k \times m} q^{|\lambda|}.$$

*Proof.* Take a word  $w \in \mathcal{W}_{m+k,k}$  and let the positions of its ones be  $1 \leq i_1 < \cdots < i_k \leq m+k$ . Define

$$\lambda_j = m + j - i_j \quad (1 \leq j \leq k).$$

Then  $0 \leq \lambda_j \leq m$  and

$$\lambda_j - \lambda_{j+1} = i_{j+1} - i_j - 1 \geq 0.$$

Thus  $\lambda$  fits in the rectangle. Conversely,  $i_j = m + j - \lambda_j$  recovers a strictly increasing sequence of positions, so this is a bijection. The number  $\lambda_j$  is exactly the number of zeros to the right of the  $j$ th one, hence  $|\lambda| = \text{inv}(w)$ . Apply [Theorem 5.1](#).  $\square$

**Corollary 5.4** (Conjugation and complement). *Conjugating Ferrers diagrams proves  $\left[ \begin{smallmatrix} m+k \\ k \end{smallmatrix} \right]_q = \left[ \begin{smallmatrix} m+k \\ m \end{smallmatrix} \right]_q$ . Taking the complement in the rectangle proves*

$$\left[ \begin{smallmatrix} m+k \\ k \end{smallmatrix} \right]_{q^{-1}} = q^{-mk} \left[ \begin{smallmatrix} m+k \\ k \end{smallmatrix} \right]_q.$$

*Proof.* Conjugation exchanges a  $k \times m$  rectangle with an  $m \times k$  rectangle without changing size. Complementation sends size  $r$  to  $mk - r$ , so the generating polynomial is reciprocal of degree  $mk$ .  $\square$

### 5.3 Lattice paths and area

Encode a binary word by a path from  $(0, 0)$  to  $(m, k)$ , reading 0 as an east step and 1 as a north step. With a suitable choice of side, the number of unit squares between the path and the boundary is the inversion number.

**Corollary 5.5** (Area generating function). *The polynomial  $\left[ \begin{smallmatrix} m+k \\ k \end{smallmatrix} \right]_q$  is the generating function of monotone lattice paths from  $(0, 0)$  to  $(m, k)$  by the area under the path:*

$$\left[ \begin{smallmatrix} m+k \\ k \end{smallmatrix} \right]_q = \sum_P q^{\text{area}(P)}.$$

*Proof.* Under the word–path encoding, every north step contributes the number of later east steps, which is precisely the number of squares in its row between the path and the chosen boundary. Summing over north steps gives  $\text{inv}(w)$ .  $\square$

### 5.4 Durfee squares and the central convolution

The largest square in the northwest corner of a partition diagram is its Durfee square.

**Theorem 5.6** (Durfee-square decomposition). *For  $m, n \geq 0$ ,*

$$\left[ \begin{smallmatrix} m+n \\ m \end{smallmatrix} \right]_q = \sum_{j=0}^{\min(m,n)} q^{j^2} \left[ \begin{smallmatrix} m \\ j \end{smallmatrix} \right]_q \left[ \begin{smallmatrix} n \\ j \end{smallmatrix} \right]_q.$$

*Proof.* This is [Corollary 3.9](#); we give the partition proof. A partition in an  $m \times n$  rectangle with Durfee square of side  $j$  consists of the  $j \times j$  square, a partition to its right fitting in a  $j \times (n-j)$  rectangle, and a partition below it fitting in an  $(m-j) \times j$  rectangle. By [Theorem 5.3](#), their generating functions are

$$q^{j^2} \left[ \begin{smallmatrix} n \\ j \end{smallmatrix} \right]_q \left[ \begin{smallmatrix} m \\ j \end{smallmatrix} \right]_q.$$

The Durfee size is unique, so summing over  $j$  counts every partition exactly once.  $\square$

## 5.5 Words with several letters

Let  $\mathcal{W}(n_1, \dots, n_r)$  be the words containing  $n_i$  copies of the letter  $i$ , ordered by  $1 < \dots < r$ . Let  $\text{inv}(w)$  count pairs  $i < j$  of positions with  $w_i > w_j$ .

**Theorem 5.7** (Multiset inversion enumerator). *If  $n = n_1 + \dots + n_r$ , then*

$$\left[ \begin{matrix} n \\ n_1, \dots, n_r \end{matrix} \right]_q = \sum_{w \in \mathcal{W}(n_1, \dots, n_r)} q^{\text{inv}(w)}.$$

*Proof.* Choose the positions of the largest letter  $r$ . If their binary indicator has inversion number  $s$ , then  $s$  counts precisely the inversions in which the larger letter is  $r$ . The remaining positions contain a word on  $1, \dots, r-1$ , and its inversions are independent. Therefore the generating function factors as

$$\left[ \begin{matrix} n \\ n_r \end{matrix} \right]_q \left[ \begin{matrix} n - n_r \\ n_1, \dots, n_{r-1} \end{matrix} \right]_q.$$

Iterating and using [Proposition 2.8](#) proves the formula.  $\square$

## 5.6 Rogers–Szegő polynomials

**Definition 5.8.** The Rogers–Szegő polynomial is

$$H_n(z; q) = \sum_{k=0}^n \left[ \begin{matrix} n \\ k \end{matrix} \right]_q z^k.$$

**Proposition 5.9** (Basic recurrence). *One has*

$$H_{n+1}(z; q) = zH_n(z; q) + H_n(qz; q), \quad H_0(z; q) = 1.$$

Moreover, if  $|q| < 1$ , then

$$\sum_{n=0}^{\infty} \frac{H_n(z; q)}{(q; q)_n} t^n = \frac{1}{(t; q)_{\infty} (zt; q)_{\infty}}$$

for  $|t| < 1$  and  $|zt| < 1$ .

*Proof.* Insert the first  $q$ -Pascal recurrence into  $H_{n+1}$ :

$$\sum_k \left( \left[ \begin{matrix} n \\ k-1 \end{matrix} \right]_q + q^k \left[ \begin{matrix} n \\ k \end{matrix} \right]_q \right) z^k = zH_n(z; q) + H_n(qz; q).$$

For the generating function, use

$$\frac{\left[ \begin{matrix} n \\ k \end{matrix} \right]_q}{(q; q)_n} = \frac{1}{(q; q)_k (q; q)_{n-k}}.$$

Writing  $n = k + j$  and separating the double sum gives

$$\left( \sum_{j \geq 0} \frac{t^j}{(q; q)_j} \right) \left( \sum_{k \geq 0} \frac{(zt)^k}{(q; q)_k} \right).$$

Each factor equals the reciprocal infinite product by Euler's identity, proved later in [Corollary 8.2](#); alternatively, the equality follows already by taking  $n \rightarrow \infty$  in [Theorem 3.6](#). This gives the stated product.  $\square$

**Proposition 5.10** (Three-term recurrence). *For  $n \geq 1$ ,*

$$H_{n+1}(z; q) = (1 + z)H_n(z; q) - z(1 - q^n)H_{n-1}(z; q).$$

*Proof.* Both sides are polynomials in  $z$ , so it suffices to compare the coefficient of  $z^k$  for every  $k \in \mathbb{Z}$ , using the convention  $\begin{bmatrix} n \\ k \end{bmatrix}_q = 0$  for  $k < 0$  or  $k > n$ . On the left that coefficient is  $\begin{bmatrix} n+1 \\ k \end{bmatrix}_q$ ; on the right it is

$$\begin{bmatrix} n \\ k \end{bmatrix}_q + \begin{bmatrix} n \\ k-1 \end{bmatrix}_q - (1 - q^n) \begin{bmatrix} n-1 \\ k-1 \end{bmatrix}_q.$$

We claim that for every  $k \in \mathbb{Z}$  and every  $n \geq 1$ ,

$$(1 - q^n) \begin{bmatrix} n-1 \\ k-1 \end{bmatrix}_q = (1 - q^{n+1-k}) \begin{bmatrix} n \\ k-1 \end{bmatrix}_q.$$

Granting the claim, the right-hand coefficient equals

$$\begin{bmatrix} n \\ k \end{bmatrix}_q + \left(1 - (1 - q^{n+1-k})\right) \begin{bmatrix} n \\ k-1 \end{bmatrix}_q = \begin{bmatrix} n \\ k \end{bmatrix}_q + q^{n+1-k} \begin{bmatrix} n \\ k-1 \end{bmatrix}_q,$$

which is the second  $q$ -Pascal recurrence (2.2) with  $n$  replaced by  $n+1$ ; by Theorem 2.3 that recurrence is valid for all  $k$  under the zero convention, and the right-hand coefficient is therefore  $\begin{bmatrix} n+1 \\ k \end{bmatrix}_q$ , as required.

To prove the claim, let first  $1 \leq k \leq n$ , so that  $k-1 \geq 0$  and  $n-k \geq 0$  and both Gaussian coefficients are given by factorial quotients. From  $(q; q)_n = (1 - q^n)(q; q)_{n-1}$  and  $(q; q)_{n-k+1} = (1 - q^{n-k+1})(q; q)_{n-k}$ ,

$$\frac{\begin{bmatrix} n \\ k-1 \end{bmatrix}_q}{\begin{bmatrix} n-1 \\ k-1 \end{bmatrix}_q} = \frac{(q; q)_n (q; q)_{n-k}}{(q; q)_{n-1} (q; q)_{n-k+1}} = \frac{1 - q^n}{1 - q^{n-k+1}},$$

and cross-multiplication gives the claim. For  $k \leq 0$  both sides vanish, since then  $k-1 < 0$  and both Gaussian coefficients are zero. For  $k = n+1$  the left side vanishes because  $\begin{bmatrix} n-1 \\ n \end{bmatrix}_q = 0$  and the right side vanishes because  $1 - q^{n+1-k} = 1 - q^0 = 0$ . For  $k > n+1$  every term is zero. This exhausts all  $k$  and completes the proof.  $\square$

**Corollary 5.11** (Dilation formula). *For  $n \geq 1$ ,*

$$H_n(qz; q) = H_n(z; q) - z(1 - q^n)H_{n-1}(z; q).$$

*Proof.* By Proposition 5.9,  $H_n(qz; q) = H_{n+1}(z; q) - zH_n(z; q)$ . Substituting Proposition 5.10 for  $H_{n+1}(z; q)$  and cancelling  $zH_n(z; q)$  gives the formula.  $\square$

*Remark 5.12.* The recurrence of Proposition 5.9 moves the argument, from  $z$  to  $qz$ ; the one above does not. It is therefore a three-term recurrence in the index  $n$  alone, of the shape familiar from orthogonal polynomial sequences, and it evaluates  $H_0, H_1, \dots, H_N$  at a fixed numerical  $z$  in  $\mathcal{O}(N)$  ring operations, without ever forming the Gaussian coefficients. At  $q = 0$  every  $\begin{bmatrix} n \\ k \end{bmatrix}_q$  equals 1 by Proposition 19.11, the recurrence becomes  $H_{n+1} = (1 + z)H_n - zH_{n-1}$ , and its solution is the geometric sum  $1 + z + \dots + z^n$ , as it must be.



## Chapter 6

# Finite fields, Grassmannians, and rank enumeration

### 6.1 Counting subspaces

**Theorem 6.1** (Gaussian coefficients count subspaces). *Let  $Q$  be a prime power. The number of  $k$ -dimensional linear subspaces of  $\mathbb{F}_Q^n$  is*

$$\begin{bmatrix} n \\ k \end{bmatrix}_Q.$$

*Proof.* Count ordered linearly independent  $k$ -tuples in  $\mathbb{F}_Q^n$ . The first vector may be any nonzero vector, the second any vector outside the span of the first, and so on, giving

$$(Q^n - 1)(Q^n - Q) \cdots (Q^n - Q^{k-1}).$$

Each  $k$ -dimensional subspace has

$$(Q^k - 1)(Q^k - Q) \cdots (Q^k - Q^{k-1})$$

ordered bases. Dividing and cancelling  $Q^i$  in corresponding factors yields

$$\prod_{i=0}^{k-1} \frac{Q^{n-i} - 1}{Q^{k-i} - 1} = \prod_{j=1}^k \frac{1 - Q^{n-k+j}}{1 - Q^j} = \begin{bmatrix} n \\ k \end{bmatrix}_Q.$$

□

**Theorem 6.2** (Geometric Pascal recurrence). *Let  $H$  be a fixed hyperplane in  $\mathbb{F}_Q^n$ . Then*

$$\begin{bmatrix} n \\ k \end{bmatrix}_Q = \begin{bmatrix} n-1 \\ k \end{bmatrix}_Q + Q^{n-k} \begin{bmatrix} n-1 \\ k-1 \end{bmatrix}_Q.$$

*Proof.* A  $k$ -subspace  $U$  is either contained in  $H$  or not. The contained subspaces contribute  $\begin{bmatrix} n-1 \\ k \end{bmatrix}_Q$ . If  $U \not\subseteq H$ , then  $W = U \cap H$  has dimension  $k-1$ . Fixing  $W$ , the quotient  $\mathbb{F}_Q^n/W$  has dimension  $n-k+1$ , and  $H/W$  is a hyperplane. The subspace  $U/W$  is a line not contained in  $H/W$ . The total number of lines in  $\mathbb{F}_Q^{n-k+1}$  minus those in its hyperplane is

$$\frac{Q^{n-k+1} - 1}{Q - 1} - \frac{Q^{n-k} - 1}{Q - 1} = Q^{n-k}.$$

There are  $\begin{bmatrix} n-1 \\ k-1 \end{bmatrix}_Q$  choices for  $W$ , proving the formula. □

## 6.2 Flags and $q$ -multinomial coefficients

**Theorem 6.3** (Partial flags). *Let  $d_i = n_1 + \cdots + n_i$ . The number of flags*

$$0 = V_0 \subset V_1 \subset \cdots \subset V_r = \mathbb{F}_Q^n, \quad \dim(V_i/V_{i-1}) = n_i,$$

is

$$\begin{bmatrix} n \\ n_1, \dots, n_r \end{bmatrix}_Q.$$

*Proof.* Choose  $V_1$  in  $\begin{bmatrix} n \\ n_1 \end{bmatrix}_Q$  ways. In the quotient  $\mathbb{F}_Q^n/V_1$ , choose  $V_2/V_1$  in  $\begin{bmatrix} n-n_1 \\ n_2 \end{bmatrix}_Q$  ways, and continue. The product is the factorization in [Proposition 2.8](#).  $\square$

## 6.3 Matrices of fixed rank

**Theorem 6.4** (Rank enumeration). *The number of  $m \times n$  matrices of rank  $r$  over  $\mathbb{F}_Q$  is*

$$N_{m,n,r}(Q) = \begin{bmatrix} m \\ r \end{bmatrix}_Q \begin{bmatrix} n \\ r \end{bmatrix}_Q |\mathrm{GL}_r(\mathbb{F}_Q)|,$$

where

$$|\mathrm{GL}_r(\mathbb{F}_Q)| = \prod_{i=0}^{r-1} (Q^r - Q^i).$$

Equivalently,

$$N_{m,n,r}(Q) = \frac{\prod_{i=0}^{r-1} (Q^m - Q^i)(Q^n - Q^i)}{\prod_{i=0}^{r-1} (Q^r - Q^i)}.$$

*Proof.* Every rank- $r$  matrix  $A$  can be factored as  $A = UV$ , where  $U$  is an  $m \times r$  matrix of full column rank and  $V$  is an  $r \times n$  matrix of full row rank. There are  $\prod_{i=0}^{r-1} (Q^m - Q^i)$  choices for  $U$  and  $\prod_{i=0}^{r-1} (Q^n - Q^i)$  choices for  $V$ . Two pairs  $(U, V)$  and  $(U', V')$  give the same  $A$  exactly when

$$U' = UG, \quad V' = G^{-1}V$$

for a unique  $G \in \mathrm{GL}_r(\mathbb{F}_Q)$ . Thus every matrix is counted  $|\mathrm{GL}_r|$  times. Substituting the subspace-count formulas gives the first expression.  $\square$

## 6.4 Schubert cells and a topological meaning

Let  $\mathrm{Gr}(k, n)$  be the complex Grassmannian of  $k$ -planes in  $\mathbb{C}^n$ . Its Schubert cell decomposition is indexed by partitions inside a  $k \times (n - k)$  rectangle.

**Theorem 6.5** (Poincaré polynomial of the Grassmannian). *The integral cohomology of  $\mathrm{Gr}(k, n)$  is concentrated in even degrees, and its Poincaré polynomial is*

$$P_{\mathrm{Gr}(k,n)}(t) = \sum_{j \geq 0} \mathrm{rank} H^{2j}(\mathrm{Gr}(k, n); \mathbb{Z}) t^{2j} = \begin{bmatrix} n \\ k \end{bmatrix}_{t^2}.$$

*Proof.* Fix the standard flag in  $\mathbb{C}^n$ . Every  $k$ -plane has a unique row-reduced echelon matrix. Its pivot positions determine a binary word with  $k$  ones, hence a partition  $\lambda$  inside the  $k \times (n - k)$  rectangle. Once the pivots are fixed, the entries allowed to vary freely occupy exactly the boxes of  $\lambda$ ; the corresponding stratum is therefore an affine complex space  $\mathbb{C}^{|\lambda|}$ . Ordering the strata by inclusion of diagrams yields a CW decomposition with one real cell of dimension  $2|\lambda|$  for each  $\lambda$ . Since all cells have even dimension, every cellular boundary map is zero. Thus the rank of  $H^{2j}$  is the number of such partitions of size  $j$ . Apply Theorem 5.3 with  $q = t^2$ .  $\square$

*Remark 6.6* (The “same polynomial” principle). For a prime power  $Q$ , evaluating  $\left[ \begin{smallmatrix} n \\ k \end{smallmatrix} \right]_q$  at  $q = Q$  counts rational points of a Grassmannian over  $\mathbb{F}_Q$ . Evaluating it at  $q = t^2$  records the even Betti numbers of the complex Grassmannian. This is an elementary instance of the broad principle that polynomial point counts over finite fields often shadow topological invariants over  $\mathbb{C}$ .

## 6.5 Growth over finite fields

**Proposition 6.7** (Dimension-dominant bounds). *For  $Q > 1$  and  $0 \leq k \leq n$ ,*

$$Q^{k(n-k)} \leq \left[ \begin{smallmatrix} n \\ k \end{smallmatrix} \right]_Q \leq \frac{Q^{k(n-k)}}{(Q^{-1}; Q^{-1})_\infty}.$$

*Proof.* Rewrite the product as

$$\left[ \begin{smallmatrix} n \\ k \end{smallmatrix} \right]_Q = Q^{k(n-k)} \prod_{j=1}^k \frac{1 - Q^{-(n-k+j)}}{1 - Q^{-j}}.$$

Each ratio is at least 1 because  $n - k + j \geq j$ , proving the lower bound. For the upper bound, discard the numerator factors and note

$$\prod_{j=1}^k (1 - Q^{-j}) \geq \prod_{j=1}^{\infty} (1 - Q^{-j}) = (Q^{-1}; Q^{-1})_\infty.$$

$\square$

## Chapter 7

# Symmetry and unimodality

Palindromicity follows immediately from reciprocity. Unimodality is subtler: the coefficients rise toward the middle and then fall. A concise conceptual proof comes from the representation theory of  $\mathfrak{sl}_2$ .

### 7.1 A representation-theoretic lemma

Let  $\mathfrak{sl}_2(\mathbb{C})$  have generators  $E, F, H$  with

$$[H, E] = 2E, \quad [H, F] = -2F, \quad [E, F] = H.$$

**Lemma 7.1** (Structure of finite-dimensional  $\mathfrak{sl}_2$ -modules). *Every finite-dimensional complex  $\mathfrak{sl}_2$ -module is a direct sum of irreducibles  $L_d$  indexed by  $d \in \mathbb{N}_0$ . The module  $L_d$  has dimension  $d + 1$  and  $H$ -weights*

$$d, d - 2, \dots, -d,$$

*each with multiplicity one.*

*Proof.* We include the standard argument. Restrict a representation to the compact real form  $\mathfrak{su}_2$ . A finite-dimensional Lie-algebra representation integrates to the simply connected compact group  $SU(2)$  by exponentiating the representing matrices; the Lie relations ensure consistency of the local exponentials, and simple connectedness removes monodromy. Starting from any Hermitian inner product, average it over  $SU(2)$  with normalized Haar measure. The result is invariant. Hence the orthogonal complement of every invariant subspace is invariant, proving complete reducibility.

It remains to classify an irreducible module  $V$ . In the invariant Hermitian structure, a suitable scalar multiple of  $H$  is self-adjoint, so  $H$  is diagonalizable. Choose an eigenvector  $v$  whose eigenvalue  $d$  has maximal real part. Since  $E$  raises an  $H$ -weight by 2, maximality gives  $Ev = 0$ . The commutation relations imply by induction

$$HF^j v = (d - 2j)F^j v, \quad EF^j v = j(d - j + 1)F^{j-1} v.$$

Because  $V$  is finite-dimensional, let  $m$  be maximal with  $F^m v \neq 0$ . Applying  $E$  to  $F^{m+1} v = 0$  gives

$$(m + 1)(d - m)F^m v = 0,$$

so  $d = m \in \mathbb{N}_0$ . The vectors  $v, Fv, \dots, F^d v$  have distinct weights and are linearly independent. Their span is invariant under  $E, F, H$  and is nonzero; irreducibility makes it all of  $V$ . The displayed weights and multiplicities follow.  $\square$

## 7.2 Exterior powers and Gaussian coefficients

Let  $L_{n-1}$  have a weight basis  $v_0, \dots, v_{n-1}$  with  $Hv_i = (n-1-2i)v_i$ .

**Proposition 7.2** (Exterior-power character). *For  $0 \leq k \leq n$ ,*

$$\text{ch} \left( \bigwedge^k L_{n-1} \right) (x) = x^{k(n-k)} \begin{bmatrix} n \\ k \end{bmatrix}_{x^{-2}}.$$

*Proof.* A basis of the exterior power consists of  $v_{i_1} \wedge \dots \wedge v_{i_k}$  with  $0 \leq i_1 < \dots < i_k \leq n-1$ . Its weight is

$$k(n-1) - 2(i_1 + \dots + i_k).$$

Therefore

$$\text{ch} = x^{k(n-1)} \sum_{i_1 < \dots < i_k} (x^{-2})^{i_1 + \dots + i_k}.$$

Use [Corollary 3.4](#) with  $q = x^{-2}$ . The prefactor becomes

$$x^{k(n-1)} x^{-2 \binom{k}{2}} = x^{k(n-k)}.$$

□

**Theorem 7.3** (Unimodality of Gaussian coefficients). *If*

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = c_0 + c_1 q + \dots + c_D q^D, \quad D = k(n-k),$$

*then*

$$c_0 \leq c_1 \leq \dots \leq c_{\lfloor D/2 \rfloor}$$

*and  $c_j = c_{D-j}$ . Thus the coefficient sequence is symmetric and unimodal.*

*Proof.* Symmetry was proved in [Theorem 2.5](#). By [Lemma 7.1](#), decompose

$$\bigwedge^k L_{n-1} \cong \bigoplus_{d \geq 0} m_d L_d.$$

All weights in this exterior power have the same parity as  $D$ . The character of one summand  $L_d$ , placed inside the ambient sequence of weights

$$D, D-2, \dots, -D,$$

contributes a consecutive block of ones centered at weight 0: it is zero until weight  $d$  is reached, then one at every step down to  $-d$ , then zero again. As one moves from weight  $D$  toward 0, such a block can turn on but cannot turn off. A nonnegative sum of these centered blocks therefore has weakly increasing multiplicities toward the middle.

By [Proposition 7.2](#), the multiplicity of weight  $D-2j$  is exactly  $c_j$ . Hence  $c_0 \leq c_1 \leq \dots \leq c_{\lfloor D/2 \rfloor}$ . □

*Remark 7.4.* The proof explains more than the inequality: the successive differences  $c_j - c_{j-1}$  record multiplicities of irreducible summands with prescribed highest weight. Stronger strict-unimodality results require additional arguments and have a small family of exceptional cases; ordinary unimodality is exactly what the  $\mathfrak{sl}_2$  decomposition supplies for free.

## **Part II**

# **Infinite series and basic hypergeometric summation**

## Chapter 8

# The infinite $q$ -binomial theorem

### 8.1 A functional-equation proof

**Theorem 8.1** (Infinite  $q$ -binomial theorem). *Let  $|q| < 1$  and  $|z| < 1$ . Then*

$$\sum_{n=0}^{\infty} \frac{(a; q)_n}{(q; q)_n} z^n = \frac{(az; q)_{\infty}}{(z; q)_{\infty}}.$$

*The convergence is locally uniform in  $(a, z)$  on compact subsets of  $\mathbb{C} \times \{|z| < 1\}$ .*

*Proof.* Set

$$F(z) = \sum_{n \geq 0} c_n z^n, \quad c_n = \frac{(a; q)_n}{(q; q)_n}.$$

Since  $c_n/c_{n-1} = (1 - aq^{n-1})/(1 - q^n) \rightarrow 1$ , the radius of convergence is 1 unless the series terminates; in every case it converges for  $|z| < 1$ . The coefficient recurrence is equivalent to

$$(1 - q^n)c_n = (1 - aq^{n-1})c_{n-1}.$$

Comparing coefficients shows that

$$(1 - z)F(z) = (1 - az)F(qz), \quad F(0) = 1.$$

Now put  $G(z) = (az; q)_{\infty}/(z; q)_{\infty}$ . Using  $(z; q)_{\infty} = (1 - z)(qz; q)_{\infty}$  gives the same functional equation and  $G(0) = 1$ .

To prove uniqueness, iterate the functional equation:

$$F(z) = \prod_{j=0}^{N-1} \frac{1 - azq^j}{1 - zq^j} F(q^N z).$$

As  $N \rightarrow \infty$ ,  $q^N z \rightarrow 0$ , so  $F(q^N z) \rightarrow 1$ . The finite product tends to  $G(z)$ . Hence  $F = G$ . □

**Corollary 8.2** (Euler's two infinite identities). *For  $|q| < 1$ ,*

$$(z; q)_{\infty} = \sum_{n=0}^{\infty} \frac{(-1)^n q^{\binom{n}{2}}}{(q; q)_n} z^n, \quad z \in \mathbb{C}, \tag{8.1}$$

$$\frac{1}{(z; q)_{\infty}} = \sum_{n=0}^{\infty} \frac{z^n}{(q; q)_n}, \quad |z| < 1. \tag{8.2}$$

*Proof.* Set  $a = 0$  in [Theorem 8.1](#) to obtain (8.2). For (8.1), start with the finite identity

$$(z; q)_N = \sum_{n=0}^N (-1)^n q^{\binom{n}{2}} \begin{bmatrix} N \\ n \end{bmatrix}_q z^n.$$

For fixed  $n$ ,  $\begin{bmatrix} N \\ n \end{bmatrix}_q \rightarrow 1/(q; q)_n$  as  $N \rightarrow \infty$ . The limiting series converges for every  $z$  because its successive-term ratio has magnitude asymptotic to  $|z||q|^n$ . To justify passage to the limit uniformly on  $|z| \leq R$ , note that

$$\left| \begin{bmatrix} N \\ n \end{bmatrix}_q \right| \leq \frac{\prod_{j \geq 1} (1 + |q|^j)}{(\inf_{r \geq 0} |(q; q)_r|)^2},$$

where the denominator is positive because  $(q; q)_r \rightarrow (q; q)_\infty \neq 0$ . Thus the summands, extended by zero for  $n > N$ , are dominated by a constant multiple of  $|q|^{\binom{n}{2}} R^n$ . Dominated convergence, together with local uniform convergence of the products, now permits passage to the limit.  $\square$

**Corollary 8.3** (A ratio expansion). *For  $|z| < 1$ ,*

$$\frac{(a_1 z; q)_\infty \cdots (a_r z; q)_\infty}{(b_1 z; q)_\infty \cdots (b_s z; q)_\infty}$$

*can be expanded by multiplying  $r$  copies of (8.1) and  $s$  copies of (8.2). In particular, every coefficient is a finite convolution of  $q$ -Pochhammer ratios.*

*Proof.* Each numerator factor has an entire series expansion and each reciprocal denominator factor converges for  $|b_j z| < 1$ . Cauchy multiplication is valid in their common disk of absolute convergence.  $\square$

## 8.2 Formal and analytic interpretations

**Proposition 8.4** (Formal version). *If  $q$  is an indeterminate, then*

$$\frac{(az; q)_\infty}{(z; q)_\infty} = \sum_{n \geq 0} \frac{(a; q)_n}{(q; q)_n} z^n$$

*holds in the  $q$ -adically complete ring  $\mathbb{Q}(a)[[q]][[z]]$ . More generally, it remains valid after any specialization into a complete topological ring for which the displayed products and series converge.*

*Proof.* The products are well defined  $q$ -adically: apart from their initial factors, the  $j$ th factors differ from 1 by a multiple of  $q^j$ . For a formal power series  $F(z) = \sum c_n z^n$ , the equation

$$(1 - z)F(z) = (1 - az)F(qz), \quad c_0 = 1,$$

determines  $c_n$  recursively by  $(1 - q^n)c_n = (1 - aq^{n-1})c_{n-1}$ , because  $1 - q^n$  is a unit in  $\mathbb{Q}(a)[[q]]$ . Both sides satisfy this equation, so their coefficients agree. The same uniqueness argument works in every stated complete specialization.  $\square$



### 8.3 Limits in the number of factors

**Proposition 8.5.** For fixed  $k$  and  $|q| < 1$ ,

$$\lim_{n \rightarrow \infty} \begin{bmatrix} n \\ k \end{bmatrix}_q = \frac{1}{(q; q)_k}.$$

More generally,

$$\lim_{n \rightarrow \infty} \begin{bmatrix} n+k \\ k \end{bmatrix}_q = \frac{1}{(q; q)_k}.$$

*Proof.* Use

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = \frac{(q^{n-k+1}; q)_k}{(q; q)_k}.$$

Every numerator factor tends to 1. □

*Remark 8.6.* The two Euler identities are therefore the stable limits of the finite product and reciprocal-product formulas. This stability is one reason finite  $q$ -binomial identities so often serve as polynomial approximations to infinite  $q$ -series identities.

## Chapter 9

# Basic hypergeometric series

### 9.1 Definition and convergence

**Definition 9.1.** The basic hypergeometric series is

$${}_r\phi_s \left[ \begin{matrix} a_1, \dots, a_r \\ b_1, \dots, b_s \end{matrix}; q, z \right] = \sum_{n=0}^{\infty} \frac{(a_1, \dots, a_r; q)_n}{(q, b_1, \dots, b_s; q)_n} \left( (-1)^n q^{\binom{n}{2}} \right)^{1+s-r} z^n,$$

provided no denominator factor vanishes. If one numerator parameter is  $q^{-N}$ , the series terminates at  $n = N$ .

**Proposition 9.2** (Elementary convergence classification). *Assume the series does not terminate and the denominator parameters avoid poles.*

- (i) If  $r \leq s$ , the series converges for every  $z$ .
- (ii) If  $r = s + 1$ , its radius of convergence is 1.
- (iii) If  $r > s + 1$ , the terms generally fail to tend to zero for  $z \neq 0$ ; such series are normally interpreted only when terminating or formally.

*Proof.* Let  $T_n$  be the  $n$ th summand. Since  $(1 - aq^n) \rightarrow 1$  for each fixed parameter,

$$\frac{T_{n+1}}{T_n} = z \frac{\prod_{i=1}^r (1 - a_i q^n)}{(1 - q^{n+1}) \prod_{j=1}^s (1 - b_j q^n)} (-1)^{1+s-r} q^{n(1+s-r)}.$$

The ratio tends to 0 when  $r \leq s$ , to  $z$  when  $r = s + 1$ , and has unbounded magnitude when  $r > s + 1$  and  $z \neq 0$ . Apply the ratio test.  $\square$

The infinite  $q$ -binomial theorem is the evaluation

$${}_1\phi_0 \left[ \begin{matrix} a \\ - \end{matrix}; q, z \right] = \frac{(az; q)_{\infty}}{(z; q)_{\infty}}.$$

When  $s = r - 1$  the factor  $((-1)^n q^{\binom{n}{2}})^{1+s-r}$  in the definition is identically 1, so that

$${}_r\phi_{r-1} \left[ \begin{matrix} a_1, \dots, a_r \\ b_1, \dots, b_{r-1} \end{matrix}; q, z \right] = \sum_{n=0}^{\infty} \frac{(a_1; q)_n \cdots (a_r; q)_n}{(q; q)_n (b_1; q)_n \cdots (b_{r-1}; q)_n} z^n.$$

Under the scaling  $a_i = q^{\alpha_i}$ ,  $b_j = q^{\beta_j}$  this family degenerates as  $q \rightarrow 1^-$  to the ordinary hypergeometric series

$${}_rF_{r-1}\left(\begin{matrix} \alpha_1, \dots, \alpha_r \\ \beta_1, \dots, \beta_{r-1} \end{matrix}; z\right) := \sum_{n=0}^{\infty} \frac{(\alpha_1)_n \cdots (\alpha_r)_n}{n! (\beta_1)_n \cdots (\beta_{r-1})_n} z^n, \quad (\alpha)_n = \alpha(\alpha+1) \cdots (\alpha+n-1),$$

and – this is the point of the next theorem – the degeneration is valid for the full nonterminating series, not only for the terminating specializations tabulated in [Appendix B](#).

**Theorem 9.3** (Classical limit of a nonterminating  ${}_r\phi_{r-1}$ ). *Let  $r \geq 1$  and let  $\alpha_1, \dots, \alpha_r, \beta_1, \dots, \beta_{r-1} \in \mathbb{C}$  satisfy*

$$\beta_j \notin \{0, -1, -2, \dots\} \quad (1 \leq j \leq r-1).$$

*For  $0 < q < 1$  let  $q^c = e^{c \log q}$  with the real logarithm. Then there is  $q_0 \in (0, 1)$  such that for  $q_0 < q < 1$  no denominator factor of*

$$\mathcal{F}_q(z) := {}_r\phi_{r-1}\left[\begin{matrix} q^{\alpha_1}, \dots, q^{\alpha_r} \\ q^{\beta_1}, \dots, q^{\beta_{r-1}} \end{matrix}; q, z\right]$$

*vanishes, and for every  $\rho \in (0, 1)$*

$$\lim_{q \rightarrow 1^-} \mathcal{F}_q(z) = {}_rF_{r-1}\left(\begin{matrix} \alpha_1, \dots, \alpha_r \\ \beta_1, \dots, \beta_{r-1} \end{matrix}; z\right)$$

*uniformly for  $|z| \leq \rho$ . Both sides are analytic in  $|z| < 1$ . If no  $\alpha_i$  lies in  $\{0, -1, -2, \dots\}$ , both series have radius of convergence exactly 1: for  $\mathcal{F}_q$  this is [Proposition 9.2\(ii\)](#), and for  ${}_rF_{r-1}$  the ratio of consecutive terms is*

$$\frac{z \prod_{i=1}^r (\alpha_i + n)}{(n+1) \prod_{j=1}^{r-1} (\beta_j + n)} \rightarrow z \quad (n \rightarrow \infty).$$

*If some  $\alpha_i$  is a nonpositive integer, both sides reduce to the same polynomial in  $z$ ; see [Remark 9.4](#).*

*Proof.* Throughout put  $\beta_r := 1$ , so that  $(q; q)_n = (q^{\beta_r}; q)_n$  and the  $n$ th summand

$$T_n(q) := \frac{\prod_{i=1}^r (q^{\alpha_i}; q)_n}{\prod_{i=1}^r (q^{\beta_i}; q)_n} z^n$$

has equally many Pochhammer factors above and below.

*Step 1: no denominator vanishes near  $q = 1$ .* A factor  $1 - q^{\beta+m}$  with  $m \geq 0$  vanishes exactly when  $(\beta + m) \log q \in 2\pi i \mathbb{Z}$ . Writing  $\beta + m = \sigma + i\tau$  and using that  $\log q$  is real and nonzero, this forces  $\sigma = 0$  together with  $\tau \log q \in 2\pi \mathbb{Z}$ . If  $\tau = \Im \beta = 0$ , the two conditions say  $\beta = -m$ , which the hypothesis excludes for  $\beta = \beta_1, \dots, \beta_{r-1}$  and which is false for  $\beta_r = 1$ . If  $\Im \beta \neq 0$ , then  $\tau \log q$  is a nonzero multiple of  $2\pi$ , whence  $|\log q| \geq 2\pi/|\Im \beta|$ . Taking

$$q_0 = \max \{ \exp(-2\pi/|\Im \beta_j|) : \Im \beta_j \neq 0 \}$$

(and  $q_0$  arbitrary in  $(0, 1)$  if all  $\beta_j$  are real) rules out every vanishing for  $q_0 < q < 1$ .

*Step 2: termwise convergence.* Fix  $n$  and divide each of the  $2r$  Pochhammer symbols by  $(1 - q)^n$ ; the powers of  $(1 - q)$  cancel, so

$$T_n(q) = \frac{\prod_{i=1}^r (q^{\alpha_i}; q)_n (1 - q)^{-n}}{\prod_{i=1}^r (q^{\beta_i}; q)_n (1 - q)^{-n}} z^n.$$

By [Proposition 1.19](#) the  $i$ th numerator factor tends to  $(\alpha_i)_n$  and the  $i$ th denominator factor to  $(\beta_i)_n$ , with  $(\beta_r)_n = (1)_n = n!$ . The hypothesis on the  $\beta_j$  makes every  $(\beta_i)_n$  nonzero, so

$$\lim_{q \rightarrow 1^-} T_n(q) = t_n := \frac{(\alpha_1)_n \cdots (\alpha_r)_n}{n! (\beta_1)_n \cdots (\beta_{r-1})_n} z^n.$$

*Step 3: two elementary estimates.* Put  $t = -\log q > 0$ , so that  $q^w = e^{-wt}$ , and set  $h(u) = 1 - e^{-ut}$  for  $u \geq 0$ .

(a) If  $w = \sigma + i\tau$  with  $\sigma > 0$ , then

$$h(\sigma) \leq |1 - e^{-wt}| \leq h(\sigma) + |\tau|t.$$

Indeed, with  $\varrho = e^{-\sigma t} \in (0, 1)$  and  $\phi = -\tau t$ ,  $|1 - \varrho e^{i\phi}|^2 = (1 - \varrho)^2 + 2\varrho(1 - \cos \phi)$ . Discarding the nonnegative second term gives the lower bound; using  $1 - \cos \phi \leq \phi^2/2$  and  $\varrho \leq 1$  gives  $|1 - \varrho e^{i\phi}|^2 \leq (1 - \varrho)^2 + (\tau t)^2 \leq ((1 - \varrho) + |\tau|t)^2$ .

(b)  $h$  is increasing and concave on  $[0, \infty)$  with  $h(0) = 0$ , so  $u \mapsto h(u)/u$  is nonincreasing and therefore

$$\frac{h(v)}{h(u)} \leq \max\left(1, \frac{v}{u}\right) \quad (u, v > 0).$$

Moreover  $e^{ut} \geq 1 + ut$  gives  $h(u) \geq ut/(1 + ut)$ , that is,  $h(u)^{-1} \leq (ut)^{-1} + 1$ .

*Step 4: a majorant uniform in  $q$  and in  $|z| \leq \rho$ .* Set  $K = 1 + \max_i |\alpha_i| + \max_i |\beta_i|$  and fix an integer  $m_0 \geq 2K$ . For  $m \geq m_0$  the real numbers  $\sigma_i = \Re \alpha_i + m$  and  $\gamma_i = \Re \beta_i + m$  are at least  $m/2 > 0$ . The ratio of consecutive summands is

$$\frac{T_{m+1}(q)}{T_m(q)} = z \prod_{i=1}^r \frac{1 - q^{\alpha_i + m}}{1 - q^{\beta_i + m}},$$

and for  $m \geq m_0$  and every  $t > 0$ , Step 3 gives

$$\left| \frac{1 - q^{\alpha_i + m}}{1 - q^{\beta_i + m}} \right| \leq \frac{h(\sigma_i) + |\Im \alpha_i|t}{h(\gamma_i)} \leq \max\left(1, \frac{\sigma_i}{\gamma_i}\right) + |\Im \alpha_i|t \left(\frac{1}{\gamma_i t} + 1\right) \leq 1 + \frac{6K}{m} + Kt,$$

because  $\max(1, \sigma_i/\gamma_i) \leq 1 + |\Re \alpha_i - \Re \beta_i|/\gamma_i \leq 1 + 4K/m$  and  $|\Im \alpha_i|/\gamma_i \leq 2K/m$ . Put  $C = 6rK$ , fix  $\rho \in (0, 1)$ , choose  $t_1 > 0$  with  $e^{rKt_1} \leq \rho^{-1/2}$ , and set  $\theta = \rho^{1/2} < 1$ . For  $|z| \leq \rho$ ,  $0 < t \leq t_1$  and  $m \geq m_0$ ,

$$\left| \frac{T_{m+1}(q)}{T_m(q)} \right| \leq \rho \left(1 + \frac{6K}{m} + Kt_1\right)^r \leq \rho e^{rKt_1} e^{C/m} \leq \theta e^{C/m}.$$

Multiplying these bounds and using  $\sum_{m=m_0}^{n-1} m^{-1} \leq 1 + \log(n/m_0)$ ,

$$|T_n(q)| \leq |T_{m_0}(q)| \theta^{n-m_0} e^C \left(\frac{n}{m_0}\right)^C \quad (n > m_0).$$

By Step 2 the finitely many functions  $T_0, \dots, T_{m_0}$  converge as  $q \rightarrow 1^-$ , hence are bounded, say by  $B$ , on some interval  $(q_1, 1)$ ; the bound is uniform in  $|z| \leq \rho$  because  $T_n(q)$  is  $z^n$  times a function of  $q$  alone. Therefore, with

$$M_n := B \quad (n \leq m_0), \quad M_n := B e^C m_0^{-C} n^C \theta^{n-m_0} \quad (n > m_0),$$

one has  $|T_n(q)| \leq M_n$  for all  $n \geq 0$ , all  $|z| \leq \rho$  and all  $q$  with  $\max(q_0, q_1, e^{-t_1}) < q < 1$ , while  $\sum_{n \geq 0} M_n < \infty$  because  $\theta < 1$  dominates the polynomial factor  $n^C$ .

*Step 5: interchange.* The series  $\sum_n T_n(q)$  is dominated by the summable sequence  $(M_n)$  uniformly in  $q$ , so it converges uniformly in  $q$  on that interval; each term has the limit  $t_n$  by Step 2. Dominated convergence with respect to counting measure (Tannery's theorem) therefore gives

$$\lim_{q \rightarrow 1^-} \sum_{n \geq 0} T_n(q) = \sum_{n \geq 0} t_n,$$

which is the asserted limit. For the uniformity, note first that letting  $q \rightarrow 1^-$  in  $|T_n(q)| \leq M_n$  gives  $|t_n| \leq M_n$  as well. Let  $\varepsilon > 0$  and choose  $N > m_0$  with  $\sum_{n > N} M_n < \varepsilon/3$ . Writing  $T_n(q) = z^n g_n(q)$  and  $t_n = z^n g_n(1^-)$ , where  $g_n$  does not involve  $z$ , one has for every  $|z| \leq \rho$

$$\left| \sum_{n \geq 0} T_n(q) - \sum_{n \geq 0} t_n \right| \leq \sum_{n \leq N} |g_n(q) - g_n(1^-)| + 2 \sum_{n > N} M_n,$$

because  $|z|^n \leq 1$ . The first sum is a finite sum of quantities independent of  $z$ , each tending to 0 by Step 2, hence is  $< \varepsilon/3$  once  $q$  is close enough to 1. The convergence is therefore uniform on  $|z| \leq \rho$ .  $\square$

*Remark 9.4* (The terminating case). If some  $\alpha_i$  is a nonpositive integer  $-N$ , the numerator factor  $(q^{-N}; q)_n$  vanishes for  $n > N$ , the series is a finite sum, and neither the restriction  $|z| < 1$  nor Steps 3–5 are needed: applying [Proposition 1.19](#) to each of the  $N + 1$  terms suffices. That elementary mechanism is what underlies the itemized list in [Appendix B](#); [Theorem 9.3](#) is its nonterminating counterpart, and the analytic work in Steps 3–5 is exactly the price of dropping the terminating hypothesis.

## 9.2 Heine's transformation

**Theorem 9.5** (Heine transformation). *Initially for  $|z| < 1$  and  $|b| < 1$ ,*

$${}_2\phi_1 \left[ \begin{matrix} a, b \\ c \end{matrix}; q, z \right] = \frac{(b, az; q)_\infty}{(c, z; q)_\infty} {}_2\phi_1 \left[ \begin{matrix} c/b, z \\ az \end{matrix}; q, b \right].$$

*The identity extends wherever both sides are defined by analytic continuation.*

*Proof.* Use the tail-factor identity to write

$$\frac{(b; q)_n}{(c; q)_n} = \frac{(b; q)_\infty (cq^n; q)_\infty}{(c; q)_\infty (bq^n; q)_\infty}.$$

By the infinite  $q$ -binomial theorem,

$$\frac{(cq^n; q)_\infty}{(bq^n; q)_\infty} = \sum_{k \geq 0} \frac{(c/b; q)_k}{(q; q)_k} b^k q^{nk}.$$

Insert this into the  ${}_2\phi_1$  series and interchange the absolutely convergent sums:

$$\frac{(b; q)_\infty}{(c; q)_\infty} \sum_{k \geq 0} \frac{(c/b; q)_k}{(q; q)_k} b^k \sum_{n \geq 0} \frac{(a; q)_n}{(q; q)_n} (zq^k)^n.$$

The inner sum is

$$\frac{(azq^k; q)_\infty}{(zq^k; q)_\infty} = \frac{(az; q)_\infty}{(z; q)_\infty} \frac{(z; q)_k}{(az; q)_k}.$$

Collecting factors yields the claimed transformation. Both sides are meromorphic functions of the parameters, so equality extends from the initial open domain by the identity theorem.  $\square$

### 9.3 The $q$ -Gauss sum

**Theorem 9.6** ( $q$ -Gauss summation). *If  $|c/(ab)| < 1$ , then*

$${}_2\phi_1 \left[ \begin{matrix} a, b \\ c \end{matrix}; q, \frac{c}{ab} \right] = \frac{(c/a, c/b; q)_\infty}{(c, c/(ab); q)_\infty}.$$

*Proof.* Set  $z = c/(ab)$  in [Theorem 9.5](#). Then  $az = c/b$ , so one numerator parameter of the transformed series cancels its denominator parameter:

$${}_2\phi_1 \left[ \begin{matrix} c/b, c/(ab) \\ c/b \end{matrix}; q, b \right] = {}_1\phi_0 \left[ \begin{matrix} c/(ab) \\ - \end{matrix}; q, b \right] = \frac{(c/a; q)_\infty}{(b; q)_\infty}.$$

Multiplication by the prefactor in Heine's transformation gives the result. The initially stronger restrictions are removed by analytic continuation within the convergence domain.  $\square$

### 9.4 A second finite Cauchy identity

**Lemma 9.7** (Cauchy convolution II). *For  $a, b, c$  in a commutative ring,*

$$(ac; q)_n (bc; q)_n = \sum_{k=0}^n \begin{bmatrix} n \\ k \end{bmatrix}_q (a; q)_k (b; q)_k c^k (c; q)_{n-k} (abcq^k; q)_{n-k}.$$

*Proof.* Apply the  $q$ -Cauchy identity [Theorem 3.15](#) with  $u = bq^k$ ,  $v = ac$ , and length  $n - k$ :

$$(abcq^k; q)_{n-k} = \sum_{r=0}^{n-k} \begin{bmatrix} n-k \\ r \end{bmatrix}_q (bq^k; q)_r (ac)^r (ac; q)_{n-k-r}.$$

Insert this in the right side of the proposed identity. Use

$$\begin{bmatrix} n \\ k \end{bmatrix}_q \begin{bmatrix} n-k \\ r \end{bmatrix}_q = \begin{bmatrix} n \\ k+r \end{bmatrix}_q \begin{bmatrix} k+r \\ k \end{bmatrix}_q, \quad (b; q)_k (bq^k; q)_r = (b; q)_{k+r}.$$

Put  $m = k + r$  and use

$$(c; q)_{n-k} = (c; q)_{n-m} (cq^{n-m}; q)_{m-k}.$$

The double sum becomes

$$\sum_{m=0}^n \begin{bmatrix} n \\ m \end{bmatrix}_q (b; q)_m c^m (c; q)_{n-m} (ac; q)_{n-m} \sum_{k=0}^m \begin{bmatrix} m \\ k \end{bmatrix}_q (a; q)_k a^{m-k} (cq^{n-m}; q)_{m-k}.$$

In the inner sum replace  $j = m - k$  and apply [Theorem 3.15](#) with  $u = cq^{n-m}$  and  $v = a$ ; it equals  $(acq^{n-m}; q)_m$ . Hence

$$(ac; q)_{n-m} (acq^{n-m}; q)_m = (ac; q)_n$$

can be factored out. The remaining sum is

$$(ac; q)_n \sum_{m=0}^n \begin{bmatrix} n \\ m \end{bmatrix}_q (b; q)_m c^m (c; q)_{n-m} = (ac; q)_n (bc; q)_n$$

by one final application of the  $q$ -Cauchy identity.  $\square$

*Remark 9.8* (Formal version). The displayed identity is `Fabius.finite_qCauchy_second_identity` in `QBinomialCauchy.lean`. It holds for arbitrary  $q, a, b, c$  in every commutative ring and every  $n \in \mathbb{N}$ , including  $n = 0, q = 0$ , roots of unity, positive characteristic, and rings with zero divisors. The formal proof expands and reindexes a finite triangle and applies the primary and reflected finite  $q$ -Cauchy identities; it uses no quotient, cancellation, or nonvanishing assumption. This exact finite convolution does not by itself formalize the quotient-bearing  $q$ -Pfaff–Saalschütz theorem that follows.

## 9.5 The $q$ -Pfaff–Saalschütz sum

**Theorem 9.9** ( $q$ -Pfaff–Saalschütz summation). *For  $n \in \mathbb{N}_0$ ,*

$${}_3\phi_2 \left[ \begin{matrix} a, b, q^{-n} \\ c, abq^{1-n}/c \end{matrix}; q, q \right] = \frac{(c/a, c/b; q)_n}{(c, c/(ab); q)_n}.$$

*Proof.* In [Lemma 9.7](#), replace its parameters by

$$a \mapsto a, \quad b \mapsto b, \quad c \mapsto \frac{c}{ab}.$$

This gives

$$\frac{(c/a, c/b; q)_n}{(c, c/(ab); q)_n} = \sum_{k=0}^n \begin{bmatrix} n \\ k \end{bmatrix}_q (a, b; q)_k \left( \frac{c}{ab} \right)^k \frac{(c/(ab); q)_{n-k} (cq^k; q)_{n-k}}{(c, c/(ab); q)_n}.$$

We show that the  $k$ th summand is the  $k$ th term of the stated  ${}_3\phi_2$ . The tail identities give

$$\frac{(cq^k; q)_{n-k}}{(c; q)_n} = \frac{1}{(c; q)_k}.$$

Also, reversal of the two finite products gives

$$(q^{-n}; q)_k = (-1)^k q^{-nk + \binom{k}{2}} (q^{n-k+1}; q)_k,$$

$$(abq^{1-n}/c; q)_k = \left( -\frac{ab}{c} \right)^k q^{k(1-n) + \binom{k}{2}} (cq^{n-k}/(ab); q)_k.$$

Therefore

$$\begin{aligned} & \frac{(q^{-n}; q)_k q^k}{(q; q)_k (abq^{1-n}/c; q)_k} \frac{(c/(ab); q)_n}{(c/(ab); q)_{n-k}} \\ &= \left( \frac{c}{ab} \right)^k \frac{(q^{n-k+1}; q)_k}{(q; q)_k} \frac{(c/(ab); q)_n}{(c/(ab); q)_{n-k} (cq^{n-k}/(ab); q)_k} \\ &= \begin{bmatrix} n \\ k \end{bmatrix}_q \left( \frac{c}{ab} \right)^k. \end{aligned}$$

Substitution yields exactly

$$\sum_{k=0}^n \frac{(a, b, q^{-n}; q)_k}{(q, c, abq^{1-n}/c; q)_k} q^k.$$

□

## 9.6 The two $q$ -Chu–Vandermonde sums

**Corollary 9.10** ( $q$ -Chu–Vandermonde). For  $n \in \mathbb{N}_0$ ,

$${}_2\phi_1 \left[ \begin{matrix} q^{-n}, a \\ c \end{matrix}; q, \frac{cq^n}{a} \right] = \frac{(c/a; q)_n}{(c; q)_n}, \quad (9.1)$$

$${}_2\phi_1 \left[ \begin{matrix} q^{-n}, a \\ c \end{matrix}; q, q \right] = a^n \frac{(c/a; q)_n}{(c; q)_n}. \quad (9.2)$$

*Proof.* For (9.1), put  $b = q^{-n}$  in the  $q$ -Gauss sum. The infinite products telescope:

$$\frac{(cq^n, c/a; q)_\infty}{(c, cq^n/a; q)_\infty} = \frac{(c/a; q)_n}{(c; q)_n}.$$

The substitution is legitimate because  $\{|b| > |c/a|\}$  is connected and meets Heine's initial region  $|b| < 1$ , so Theorem 9.6 continues analytically to  $b = q^{-n}$ . Both sides of (9.1) are now rational in  $(a, c)$ , and they agree on the nonempty open set  $|cq^n/a| < 1$  where the  $q$ -Gauss hypothesis holds; a rational function vanishing on a nonempty open subset of  $\mathbb{C}^2$  vanishes identically, so the identity holds wherever both sides are defined. Termination alone would not give this: it makes the two sides defined, not equal.

For (9.2), let  $b \rightarrow 0$  in Theorem 9.9, with its other numerator parameter equal to  $a$ . Then  $(b; q)_k \rightarrow 1$  and  $(abq^{1-n}/c; q)_k \rightarrow 1$ , so the left side becomes the displayed  ${}_2\phi_1$ . On the right,

$$\frac{(c/b; q)_n}{(c/(ab); q)_n} \rightarrow a^n$$

by comparing the leading factors of the two degree- $n$  polynomials in  $1/b$ .  $\square$

**Lemma 9.11** (Reversal of a terminating  ${}_2\phi_1$ ). Let  $n \in \mathbb{N}_0$  and let  $a, c, z$  be nonzero with  $(c; q)_n \neq 0$  and  $(q^{1-n}/a; q)_n \neq 0$ . Then

$${}_2\phi_1 \left[ \begin{matrix} q^{-n}, a \\ c \end{matrix}; q, z \right] = (-1)^n q^{-\binom{n+1}{2}} z^n \frac{(a; q)_n}{(c; q)_n} {}_2\phi_1 \left[ \begin{matrix} q^{-n}, q^{1-n}/c \\ q^{1-n}/a \end{matrix}; q, \frac{cq^{n+1}}{az} \right]. \quad (9.3)$$

Both series terminate at  $n$ , so (9.3) is an identity between rational functions of  $a, c, z$  over  $\mathbb{Q}(q)$ ; no convergence hypothesis enters. The substitution  $(a, c) \mapsto (q^{1-n}/c, q^{1-n}/a)$  is an involution, and applying (9.3) twice returns the original series.

*Proof.* Since  $r = 2$  and  $s = 1$ , the factor  $((-1)^j q^{\binom{j}{2}})^{1+s-r}$  in the definition of  ${}_2\phi_1$  equals 1, so the left side is  $\sum_{j=0}^n T_j$  with

$$T_j = \frac{(q^{-n}; q)_j (a; q)_j}{(q; q)_j (c; q)_j} z^j.$$

We reverse the order of summation, writing  $j = n - i$  with  $0 \leq i \leq n$ .

First record the reversal rule for a single finite shifted factorial. Let  $x \neq 0$ . The concatenation law Proposition 1.2 gives  $(x; q)_n = (x; q)_{n-i} (xq^{n-i}; q)_i$ , and base reversal Proposition 1.6, applied to the second factor, gives

$$(xq^{n-i}; q)_i = (-x)^i q^{(n-i)i + \binom{i}{2}} (q^{1-n}/x; q)_i.$$

Because  $(n-i)i + \binom{i}{2} = ni - \binom{i+1}{2}$ , these combine into

$$(x; q)_{n-i} = (x; q)_n \frac{(-1)^i x^{-i} q^{\binom{i+1}{2} - ni}}{(q^{1-n}/x; q)_i}. \quad (9.4)$$



Now apply (9.4) to each of the four shifted factorials occurring in  $T_{n-i}$ . The four bases  $q^{-n}, a, q, c$  reflect to  $q, q^{1-n}/a, q^{-n}, q^{1-n}/c$  respectively, the first and third of which are again members of the list (the reflection of  $q^{-n}$  is  $q$  and that of  $q$  is  $q^{-n}$ , while  $q^{1-n}/a$  and  $q^{1-n}/c$  are new). Explicitly,

$$\begin{aligned} (q^{-n}; q)_{n-i} &= (q^{-n}; q)_n \frac{(-1)^i q^{\binom{i+1}{2}}}{(q; q)_i}, & (a; q)_{n-i} &= (a; q)_n \frac{(-1)^i a^{-i} q^{\binom{i+1}{2} - ni}}{(q^{1-n}/a; q)_i}, \\ (q; q)_{n-i} &= (q; q)_n \frac{(-1)^i q^{\binom{i+1}{2} - ni - i}}{(q^{-n}; q)_i}, & (c; q)_{n-i} &= (c; q)_n \frac{(-1)^i c^{-i} q^{\binom{i+1}{2} - ni}}{(q^{1-n}/c; q)_i}. \end{aligned}$$

Form the quotient  $[(q^{-n}; q)_{n-i}(a; q)_{n-i}] / [(q; q)_{n-i}(c; q)_{n-i}]$ . The four factors  $(-1)^i$  cancel in pairs. The four powers of  $q$  contribute

$$\binom{i+1}{2} + \left( \binom{i+1}{2} - ni \right) - \left( \binom{i+1}{2} - ni - i \right) - \left( \binom{i+1}{2} - ni \right) = ni + i,$$

the scalars contribute  $a^{-i}c^i$ , and the reflected shifted factorials contribute

$$\frac{(q^{-n}; q)_i (q^{1-n}/c; q)_i}{(q; q)_i (q^{1-n}/a; q)_i}.$$

Multiplying by  $z^{n-i} = z^n z^{-i}$  and collecting the  $i$ -dependent scalars as  $(cq^{n+1}/(az))^i$ , we obtain

$$T_{n-i} = \frac{(q^{-n}; q)_n (a; q)_n}{(q; q)_n (c; q)_n} z^n \frac{(q^{-n}; q)_i (q^{1-n}/c; q)_i}{(q; q)_i (q^{1-n}/a; q)_i} \left( \frac{cq^{n+1}}{az} \right)^i.$$

By Corollary 1.9 with  $N = k = n$ ,

$$(q^{-n}; q)_n = (-1)^n q^{-n^2 + \binom{n}{2}} (q; q)_n = (-1)^n q^{-\binom{n+1}{2}} (q; q)_n,$$

so the constant in front is  $(-1)^n q^{-\binom{n+1}{2}} z^n (a; q)_n / (c; q)_n$ . Summing over  $0 \leq i \leq n$  gives (9.3). The final assertion follows because the reflected parameters are  $a' = q^{1-n}/c$  and  $c' = q^{1-n}/a$ , so that  $c'q^{n+1}/(a'z') = [(q^{1-n}/a)q^{n+1}] / [(q^{1-n}/c) \cdot cq^{n+1}/(az)] = z$ , and the reflection is therefore an involution on the parameter pair.  $\square$

**Proposition 9.12** (The second  $q$ -Chu–Vandermonde sum by reversal). *Identity (9.2) is a formal consequence of (9.1): it follows from Lemma 9.11 alone, without Theorem 9.9, without Lemma 9.7, and without any limiting process.*

*Proof.* Fix  $n \in \mathbb{N}_0$  and nonzero parameters  $A, C$  with  $(A; q)_n \neq 0$  and  $(C; q)_n \neq 0$ . Put

$$a = \frac{q^{1-n}}{C}, \quad c = \frac{q^{1-n}}{A}, \quad z = \frac{cq^n}{a} = \frac{Cq^n}{A}.$$

These are nonzero, and by Proposition 1.6 the hypotheses of Lemma 9.11 hold:  $(c; q)_n = (q^{1-n}/A; q)_n = (-A)^{-n} q^{-\binom{n}{2}} (A; q)_n \neq 0$  and  $(q^{1-n}/a; q)_n = (C; q)_n \neq 0$ . Since  $az = cq^n$ , the reflected argument in (9.3) is  $cq^{n+1}/(az) = q$ , and the reflected parameters are  $q^{1-n}/c = A$  and  $q^{1-n}/a = C$ . Hence (9.3) reads

$${}_2\phi_1 \left[ \begin{matrix} q^{-n}, a \\ c \end{matrix}; q, \frac{cq^n}{a} \right] = (-1)^n q^{-\binom{n+1}{2}} \left( \frac{cq^n}{a} \right)^n \frac{(a; q)_n}{(c; q)_n} {}_2\phi_1 \left[ \begin{matrix} q^{-n}, A \\ C \end{matrix}; q, q \right].$$

The left side is  $(c/a; q)_n / (c; q)_n$  by (9.1), and  $c/a = C/A$ . Solving for the remaining series and using  $a/(cq^n) = A/(Cq^n)$ ,

$${}_2\phi_1 \left[ \begin{matrix} q^{-n}, A \\ C \end{matrix}; q, q \right] = (-1)^n q^{\binom{n+1}{2}} \left( \frac{A}{Cq^n} \right)^n \frac{(C/A; q)_n}{(a; q)_n}.$$

By Proposition 1.6 once more,  $(a; q)_n = (q^{1-n}/C; q)_n = (-C)^{-n} q^{-\binom{n}{2}} (C; q)_n$ . Substituting and cancelling  $C^n$  and the two signs,

$${}_2\phi_1 \left[ \begin{matrix} q^{-n}, A \\ C \end{matrix}; q, q \right] = q^{\binom{n+1}{2} + \binom{n}{2} - n^2} A^n \frac{(C/A; q)_n}{(C; q)_n} = A^n \frac{(C/A; q)_n}{(C; q)_n},$$

because  $\binom{n+1}{2} + \binom{n}{2} = \frac{n^2+n}{2} + \frac{n^2-n}{2} = n^2$ . This is (9.2) at the pair  $(A, C)$ , which was an arbitrary admissible pair. The auxiliary condition  $(A; q)_n \neq 0$  excludes only a proper algebraic subset of the parameter space, and both sides of (9.2) are rational functions of  $(A, C)$  there; agreeing off a proper algebraic subset, they agree wherever both are defined. This is the same identity-theorem step used for (9.1) in the proof of Corollary 9.10. Every step used only finite products, so, after multiplication by  $A^{-n}$  is undone and denominators are cleared, (9.2) holds over any commutative ring in which  $q$  and  $A$  are invertible.  $\square$

## 9.7 A hierarchy of mechanisms

The proofs in this chapter form a hierarchy:

$$\text{finite } q\text{-Pascal} \implies q\text{-Cauchy} \implies q\text{-Pfaff-Saalschütz},$$

while

$$\begin{aligned} &\text{infinite } q\text{-binomial} \implies \text{Heine} \implies q\text{-Gauss} \\ &\implies \text{first } q\text{-Chu-Vandermonde} \implies \text{second } q\text{-Chu-Vandermonde}, \end{aligned}$$

the last arrow being Proposition 9.12. The finite chain is valid in a commutative ring after denominators are cleared. The infinite chain requires convergence or a suitable complete formal topology, but only as far as its third arrow: the first  $q$ -Chu-Vandermonde sum is already a rational, hence purely algebraic, identity, as the proof of Corollary 9.10 records, and by Lemma 9.11 the fourth arrow is finite-product algebra. Thus (9.2) does not depend on Theorem 9.9, and the two derivations of it given here are genuinely independent. This distinction becomes important in computer formalization; see Chapter 20.

## 9.8 A hierarchy of mechanisms

The proofs in this chapter form a hierarchy:

$$\text{finite } q\text{-Pascal} \implies q\text{-Cauchy} \implies q\text{-Pfaff-Saalschütz},$$

while

$$\text{infinite } q\text{-binomial} \implies \text{Heine} \implies q\text{-Gauss} \implies \text{first } q\text{-Chu-Vandermonde}.$$

The finite chain is valid in a commutative ring after denominators are cleared. The infinite chain requires convergence or a suitable complete formal topology. This distinction becomes important in computer formalization; see Chapter 20.

## Chapter 10

# Bilateral series and Ramanujan's ${}_1\psi_1$ sum

### 10.1 Definition

For integral  $n$ , negative  $q$ -shifted factorials are understood as in [Proposition 1.8](#). Define

$${}_1\psi_1 \left[ \begin{matrix} a \\ b \end{matrix}; q, z \right] = \sum_{n \in \mathbb{Z}} \frac{(a; q)_n}{(b; q)_n} z^n.$$

The positive tail behaves like a geometric series in  $z$ , while the negative tail behaves like a geometric series in  $b/(az)$ .

**Proposition 10.1** (Convergence annulus). *If denominator factors do not vanish, the  ${}_1\psi_1$  series converges absolutely in*

$$|b/a| < |z| < 1.$$

*Proof.* For  $n \rightarrow +\infty$ , the ratio of consecutive terms tends to  $z$ . For the negative tail put  $n = -m$  and use the negative-index formula. The ratio of successive terms in  $m$  tends to  $b/(az)$ . The ratio test gives the annulus.  $\square$

**Theorem 10.2** (Ramanujan's  ${}_1\psi_1$  summation). *For  $|b/a| < |z| < 1$ ,*

$${}_1\psi_1 \left[ \begin{matrix} a \\ b \end{matrix}; q, z \right] = \frac{(q, b/a, az, q/(az); q)_\infty}{(b, q/a, z, b/(az); q)_\infty}.$$

*Proof.* We first prove the formula for  $b = q^{m+1}$ , where  $m \in \mathbb{N}_0$ . In that case  $1/(b; q)_n = 0$  for  $n < -m$  in the limiting sense, so shift  $n = k - m$ :

$$\begin{aligned} \sum_{n \in \mathbb{Z}} \frac{(a; q)_n}{(q^{m+1}; q)_n} z^n &= z^{-m} \frac{(a; q)_{-m}}{(q^{m+1}; q)_{-m}} \sum_{k=0}^{\infty} \frac{(aq^{-m}; q)_k}{(q; q)_k} z^k \\ &= z^{-m} \frac{(q; q)_m}{(aq^{-m}; q)_m} \frac{(aq^{-m}z; q)_\infty}{(z; q)_\infty}. \end{aligned}$$

The second equality is the infinite  $q$ -binomial theorem. Reverse the finite products:

$$\begin{aligned} (aq^{-m}; q)_m &= (-a)^m q^{-m(m+1)/2} (q/a; q)_m, \\ (aq^{-m}z; q)_m &= (-az)^m q^{-m(m+1)/2} (q/(az); q)_m. \end{aligned}$$

Since  $(aq^{-m}z; q)_\infty = (aq^{-m}z; q)_m (az; q)_\infty$ , we obtain

$$\frac{(q; q)_m}{(q/a; q)_m} \frac{(q/(az); q)_m (az; q)_\infty}{(z; q)_\infty}.$$

On the other hand, substituting  $b = q^{m+1}$  in the proposed product and cancelling tails gives exactly the same expression:

$$\frac{(q; q)_\infty}{(q^{m+1}; q)_\infty} \frac{(q^{m+1}/a; q)_\infty}{(q/a; q)_\infty} \frac{(q/(az); q)_\infty}{(q^{m+1}/(az); q)_\infty} \frac{(az; q)_\infty}{(z; q)_\infty}.$$

For fixed  $a, z$  with  $|z| < 1$ , both sides are analytic functions of  $b$  in a neighborhood of 0 contained in  $|b| < |az|$ . Absolute convergence is locally uniform there by [Proposition 10.1](#). For all sufficiently large  $m$ , the points  $b = q^{m+1}$  lie in that neighborhood, and they accumulate at 0. The identity theorem therefore proves equality near 0. Both sides are meromorphic functions of  $b$  throughout the disk  $|b| < |az|$ : the series gives the meromorphic continuation termwise on compacta avoiding denominator zeros, and the product quotient is visibly meromorphic. Uniqueness of meromorphic continuation proves the identity everywhere in that disk where both sides are defined, which is precisely the asserted annulus for fixed  $a$  and  $z$ .  $\square$

*Remark 10.3.* Ramanujan's sum contains the infinite  $q$ -binomial theorem and several theta-product identities as limiting or specialized cases. It is a model example of a bilateral identity: the second inequality in the convergence annulus is controlled by the positive tail, while the first is controlled by the negative tail.

# Chapter 11

## $q$ -difference calculus and the quantum binomial

### 11.1 The $q$ -derivative

Assume  $q \neq 1$ . For  $x \neq 0$  define

$$D_q f(x) = \frac{f(x) - f(qx)}{(1 - q)x}.$$

At  $x = 0$  one uses the continuous extension when it exists.

**Proposition 11.1** (Elementary rules). *For  $n \geq 1$ ,*

$$D_q x^n = [n]_q x^{n-1}.$$

*The product rules are*

$$\begin{aligned} D_q(fg)(x) &= g(x)D_q f(x) + f(qx)D_q g(x), \\ &= f(x)D_q g(x) + g(qx)D_q f(x). \end{aligned}$$

*Moreover,*

$$D_q(f(q^r x)) = q^r (D_q f)(q^r x)$$

*for integral  $r$ .*

*Proof.* The monomial identity follows from  $(x^n - q^n x^n)/((1 - q)x) = [n]_q x^{n-1}$ . For the first product rule, insert and subtract  $f(qx)g(x)$  in  $f(x)g(x) - f(qx)g(qx)$ . For the second, insert and subtract  $f(x)g(qx)$ . The scaling rule follows directly from the definition.  $\square$

**Theorem 11.2** ( $q$ -Leibniz rule). *For  $n \in \mathbb{N}_0$ ,*

$$D_q^n(fg)(x) = \sum_{k=0}^n \begin{bmatrix} n \\ k \end{bmatrix}_q (D_q^k f)(q^{n-k} x) (D_q^{n-k} g)(x).$$

*Proof.* Induct on  $n$ . The assertion is the ordinary product at  $n = 0$ . Apply  $D_q$  to the  $k$ th term at level  $n$ . By the product and scaling rules, it contributes

$$q^{n-k} \begin{bmatrix} n \\ k \end{bmatrix}_q (D_q^{k+1} f)(q^{n-k} x) (D_q^{n-k} g)(x)$$

to index  $k + 1$ , and

$$\begin{bmatrix} n \\ k \end{bmatrix}_q (D_q^k f)(q^{n-k+1}x)(D_q^{n-k+1}g)(x)$$

to index  $k$ . The total coefficient of index  $k$  is

$$\begin{bmatrix} n \\ k \end{bmatrix}_q + q^{n-k+1} \begin{bmatrix} n \\ k-1 \end{bmatrix}_q = \begin{bmatrix} n+1 \\ k \end{bmatrix}_q$$

by the second  $q$ -Pascal recurrence. □

## 11.2 $q$ -Taylor and Newton interpolation

Define the  $q$ -falling power based at  $a$  by

$$(x-a)_q^m = \prod_{j=0}^{m-1} (x - q^j a), \quad (x-a)_q^0 = 1.$$

**Lemma 11.3.** For  $m \geq 1$ ,

$$D_q(x-a)_q^m = [m]_q (x-a)_q^{m-1}.$$

Consequently,

$$D_q^j (x-a)_q^m = \frac{[m]_q!}{[m-j]_q!} (x-a)_q^{m-j}$$

for  $0 \leq j \leq m$ .

*Proof.* Let  $P_m(x) = (x-a)_q^m$ . Then

$$P_m(qx) = q^m (x - q^{-1}a) \prod_{j=0}^{m-2} (x - q^j a).$$

Subtracting from  $P_m(x) = (x - q^{m-1}a) \prod_{j=0}^{m-2} (x - q^j a)$  gives

$$P_m(x) - P_m(qx) = (1 - q^m)xP_{m-1}(x).$$

Divide by  $(1-q)x$ . Iteration proves the second formula. □

**Theorem 11.4** ( $q$ -Taylor formula). Assume  $[j]_q \neq 0$  for  $1 \leq j \leq N$ . If  $f$  is a polynomial of degree at most  $N$ , then

$$f(x) = \sum_{k=0}^N \frac{(D_q^k f)(a)}{[k]_q!} (x-a)_q^k.$$

*Proof.* The polynomials  $(x-a)_q^k$ ,  $0 \leq k \leq N$ , are monic of distinct degrees and hence form a basis. Write  $f = \sum c_k (x-a)_q^k$ . Apply  $D_q^j$  and set  $x = a$ . By Lemma 11.3, every term with  $k > j$  still contains the factor  $x-a$ , terms with  $k < j$  vanish after too many derivatives, and the  $k = j$  term equals  $c_j [j]_q!$ . Thus  $c_j = (D_q^j f)(a) / [j]_q!$ . □

**Remark 11.5.** When the nodes  $a, aq, aq^2, \dots$  are distinct, this is a Newton interpolation formula on that geometric grid. In every case covered by the theorem, its basis zeros follow a multiplicative rather than an additive progression.

### 11.3 The quantum plane

**Theorem 11.6** (Noncommutative  $q$ -binomial theorem). *Let  $X, Y$  belong to an associative algebra and satisfy*

$$YX = qXY.$$

*Then*

$$(X + Y)^n = \sum_{k=0}^n \begin{bmatrix} n \\ k \end{bmatrix}_q X^{n-k} Y^k.$$

*Proof.* Induct on  $n$  and multiply on the right by  $X + Y$ . Since  $Y^k X = q^k X Y^k$ , the coefficient of  $X^{n+1-k} Y^k$  becomes

$$q^k \begin{bmatrix} n \\ k \end{bmatrix}_q + \begin{bmatrix} n \\ k-1 \end{bmatrix}_q = \begin{bmatrix} n+1 \\ k \end{bmatrix}_q$$

by the first  $q$ -Pascal recurrence. □

**Corollary 11.7** (Noncommutative  $q$ -multinomial theorem). *Let  $r \geq 1$  and let  $x_1, \dots, x_r$  be elements of an associative algebra over a commutative ring in which  $q$  is a central element, and suppose that*

$$x_j x_i = q x_i x_j \quad (1 \leq i < j \leq r).$$

*Then, for every  $n \geq 0$ ,*

$$(x_1 + \dots + x_r)^n = \sum_{\substack{n_1, \dots, n_r \geq 0 \\ n_1 + \dots + n_r = n}} \begin{bmatrix} n \\ n_1, \dots, n_r \end{bmatrix}_q x_1^{n_1} x_2^{n_2} \dots x_r^{n_r},$$

*where on the right the factors are written in increasing order of index. Equivalently, the  $q$ -multinomial coefficients are exactly the structure constants that expand powers of  $x_1 + \dots + x_r$  in the ordered monomial basis of the quantum affine  $r$ -space.*

*Proof.* Induct on  $r$ . For  $r = 1$  there is nothing to prove, and  $r = 2$  is [Theorem 11.6](#).

Let  $r \geq 2$  and set  $X = x_1$  and  $Y = x_2 + \dots + x_r$ . Because  $x_j x_1 = q x_1 x_j$  for every  $j \geq 2$  and  $q$  is central,

$$YX = \sum_{j=2}^r x_j x_1 = q \sum_{j=2}^r x_1 x_j = qXY,$$

so [Theorem 11.6](#) applies to the pair  $(X, Y)$  and gives

$$(x_1 + \dots + x_r)^n = \sum_{k=0}^n \begin{bmatrix} n \\ k \end{bmatrix}_q x_1^{n-k} (x_2 + \dots + x_r)^k.$$

Substituting  $n_1 = n - k$  and using the symmetry  $\begin{bmatrix} n \\ k \end{bmatrix}_q = \begin{bmatrix} n \\ n-k \end{bmatrix}_q$  of [Theorem 2.5](#), this reads

$$(x_1 + \dots + x_r)^n = \sum_{n_1=0}^n \begin{bmatrix} n \\ n_1 \end{bmatrix}_q x_1^{n_1} (x_2 + \dots + x_r)^{n-n_1}.$$

The elements  $x_2, \dots, x_r$  satisfy the same commutation relations among themselves, so the inductive hypothesis applies to them and gives, with  $m = n - n_1$ ,

$$(x_2 + \dots + x_r)^m = \sum_{\substack{n_2, \dots, n_r \geq 0 \\ n_2 + \dots + n_r = m}} \begin{bmatrix} m \\ n_2, \dots, n_r \end{bmatrix}_q x_2^{n_2} \dots x_r^{n_r}.$$

Since the coefficients are central, substitution produces precisely the ordered monomials  $x_1^{n_1} x_2^{n_2} \cdots x_r^{n_r}$ , the coefficient of  $x_1^{n_1} \cdots x_r^{n_r}$  being

$$\begin{bmatrix} n \\ n_1 \end{bmatrix}_q \begin{bmatrix} n - n_1 \\ n_2, \dots, n_r \end{bmatrix}_q = \frac{(q; q)_n}{(q; q)_{n_1} (q; q)_{n-n_1}} \cdot \frac{(q; q)_{n-n_1}}{(q; q)_{n_2} \cdots (q; q)_{n_r}} = \begin{bmatrix} n \\ n_1, \dots, n_r \end{bmatrix}_q$$

by the definition of the  $q$ -multinomial coefficient preceding [Proposition 2.8](#); the factor  $(q; q)_{n-n_1}$  telescopes away, exactly as in the proof of that proposition. No factor was transposed at any stage after the expansion, so the monomials are produced in increasing order of index, as asserted.  $\square$

*Remark 11.8* (The same coefficient read combinatorially). Expanding  $(x_1 + \cdots + x_r)^n$  without using any relation produces the  $r^n$  words  $x_{w_1} x_{w_2} \cdots x_{w_n}$  with  $w \in \{1, \dots, r\}^n$ . Sorting such a word into increasing order uses one adjacent transposition for each inversion of  $w$ , and each transposition contributes exactly one factor  $q$ ; hence

$$x_{w_1} x_{w_2} \cdots x_{w_n} = q^{\text{inv}(w)} x_1^{n_1} \cdots x_r^{n_r},$$

where  $n_i$  is the multiplicity of the letter  $i$  in  $w$ . Comparing coefficients with [Corollary 11.7](#) recovers the multiset inversion enumerator of [Theorem 5.7](#), and conversely that theorem proves [Corollary 11.7](#) again. The quantum affine space is therefore a purely algebraic encoding of the inversion statistic, just as [Theorem 11.6](#) encodes [Theorem 5.1](#).

**Corollary 11.9** (Factorization of  $q$ -exponentials). *Assume  $|q| < 1$ . Define*

$$e_q(x) = \sum_{n=0}^{\infty} \frac{x^n}{[n]_q!} = \frac{1}{((1-q)x; q)_{\infty}}, \quad |(1-q)x| < 1,$$

and

$$E_q(x) = \sum_{n=0}^{\infty} \frac{q^{\binom{n}{2}} x^n}{[n]_q!} = (-(1-q)x; q)_{\infty}, \quad x \in \mathbb{C}.$$

Then  $e_q(x)E_q(-x) = 1$  wherever the first series is defined. If  $YX = qXY$ , then, as an identity of formal series in total degree,

$$e_q(X + Y) = e_q(X)e_q(Y).$$

*Proof.* The product formulas and the stated convergence domains follow from Euler's identities after replacing  $z$  by  $(1-q)x$ . Their product is 1. For the noncommutative factorization, apply [Theorem 11.6](#) degree by degree and use

$$\frac{1}{[n]_q!} \begin{bmatrix} n \\ k \end{bmatrix}_q = \frac{1}{[n-k]_q! [k]_q!}.$$

Reindex with  $r = n - k$  to obtain the Cauchy product of  $e_q(X)$  and  $e_q(Y)$ , with all  $X$  factors preceding all  $Y$  factors. Since every total degree receives only finitely many contributions, the formal rearrangement is valid.  $\square$

**Proposition 11.10** (Eigenfunction properties). *Under the hypotheses and within the convergence domains of [Corollary 11.9](#), for scalar  $\lambda$ ,*

$$D_q e_q(\lambda x) = \lambda e_q(\lambda x), \quad D_q E_q(\lambda x) = \lambda E_q(q\lambda x).$$

*Proof.* Differentiate the series term by term and use  $D_q x^n = [n]_q x^{n-1}$ .  $\square$



## 11.4 Jackson's $q$ -integral

Assume  $0 < q < 1$ . Define

$$\int_0^a f(t) d_q t = (1-q)a \sum_{n=0}^{\infty} q^n f(aq^n),$$

whenever the series converges.

**Theorem 11.11** (Fundamental theorem of  $q$ -calculus). *Let*

$$F(x) = \int_0^x f(t) d_q t.$$

*Whenever the defining series converges locally uniformly,*

$$D_q F(x) = f(x).$$

*Conversely, if  $G(aq^n) \rightarrow G(0)$ , then*

$$\int_0^a D_q G(t) d_q t = G(a) - G(0).$$

*Proof.* From the definition,

$$F(x) = (1-q)x \sum_{n \geq 0} q^n f(xq^n),$$

while

$$F(qx) = (1-q)x \sum_{n \geq 1} q^n f(xq^n).$$

Their difference is  $(1-q)xf(x)$ , proving the first identity. For the second,

$$\begin{aligned} \int_0^a D_q G(t) d_q t &= (1-q)a \sum_{n \geq 0} q^n \frac{G(aq^n) - G(aq^{n+1})}{(1-q)aq^n} \\ &= \sum_{n \geq 0} (G(aq^n) - G(aq^{n+1})), \end{aligned}$$

which telescopes to  $G(a) - G(0)$ . □

**Corollary 11.12** (Jackson integration by parts). *Under convergence assumptions,*

$$\int_0^a f(t) D_q g(t) d_q t = f(a)g(a) - f(0)g(0) - \int_0^a g(qt) D_q f(t) d_q t.$$

*Proof.* Use the product rule  $D_q(fg) = fD_q g + g(q \cdot) D_q f$  and integrate with [Theorem 11.11](#). □

**Example 11.13** (Monomials). For  $\Re(s) > -1$ ,

$$\int_0^a t^s d_q t = (1-q)a^{s+1} \sum_{n \geq 0} q^{n(s+1)} = \frac{a^{s+1}}{[s+1]_q},$$

where  $[u]_q = (1 - q^u)/(1 - q)$  for complex  $u$ .

## Chapter 12

# The $q$ -gamma and $q$ -beta functions

Throughout this chapter  $0 < q < 1$  unless stated otherwise.

### 12.1 Definition and functional equation

**Definition 12.1.** For  $x > 0$ , define

$$\Gamma_q(x) = (1 - q)^{1-x} \frac{(q; q)_\infty}{(q^x; q)_\infty}.$$

With  $q^z = \exp(z \log q)$ , the same product quotient defines a meromorphic function of  $z \in \mathbb{C}$ ; its possible poles occur where  $q^{z+j} = 1$  for some  $j \in \mathbb{N}_0$ . Whenever complex arguments of  $\Gamma_q$  occur below, this meromorphic continuation is intended.

**Theorem 12.2** (Basic properties). For  $x > 0$ ,

$$\Gamma_q(1) = 1, \quad \Gamma_q(x+1) = [x]_q \Gamma_q(x).$$

Consequently, for  $n \in \mathbb{N}$ ,

$$\Gamma_q(n) = [n-1]_q!.$$

The recurrence extends as an identity of meromorphic functions of a complex argument.

*Proof.* The value at 1 is immediate. The tail identity  $(q^x; q)_\infty = (1 - q^x)(q^{x+1}; q)_\infty$  gives

$$\frac{\Gamma_q(x+1)}{\Gamma_q(x)} = (1 - q)^{-1}(1 - q^x) = [x]_q.$$

Iteration gives the integral values. The same tail identity is valid for complex  $x$ , and hence proves the meromorphic recurrence wherever both sides are finite; equality elsewhere follows in the sense of meromorphic continuation.  $\square$

**Proposition 12.3** (Logarithmic derivatives and convexity). For  $x > 0$ ,

$$\begin{aligned} \frac{d}{dx} \log \Gamma_q(x) &= -\log(1 - q) + (\log q) \sum_{n=0}^{\infty} \frac{q^{n+x}}{1 - q^{n+x}}, \\ \frac{d^2}{dx^2} \log \Gamma_q(x) &= (\log q)^2 \sum_{n=0}^{\infty} \frac{q^{n+x}}{(1 - q^{n+x})^2} > 0. \end{aligned}$$

Thus  $\Gamma_q$  is strictly log-convex.

*Proof.* Differentiate the locally uniformly convergent logarithm of the defining infinite product. A second differentiation gives the positive series.  $\square$

## 12.2 A $q$ -Bohr–Mollerup theorem

**Theorem 12.4** ( $q$ -Bohr–Mollerup characterization). *The function  $\Gamma_q$  is the unique function  $f : (0, \infty) \rightarrow (0, \infty)$  satisfying*

$$f(1) = 1, \quad f(x+1) = [x]_q f(x),$$

*and such that  $\log f$  is convex.*

*Proof.* Existence follows from [Theorem 12.2](#) and [proposition 12.3](#). For uniqueness let  $f$  have the stated properties and fix  $0 < x \leq 1$ . Convexity on the interval from  $n$  to  $n+1$  gives

$$f(n+x) \leq f(n)^{1-x} f(n+1)^x.$$

A comparison of the adjacent secant slopes on  $[n+x, n+1]$  and  $[n+1, n+2]$  gives

$$f(n+x) \geq \frac{f(n+1)}{[n+1]_q^{1-x}}.$$

Using the recurrence,

$$f(n+x) = f(x) \prod_{j=0}^{n-1} [x+j]_q, \quad f(n) = [n-1]_q!, \quad f(n+1) = [n]_q!,$$

we obtain

$$\frac{[n]_q!}{[n+1]_q^{1-x} \prod_{j=0}^{n-1} [x+j]_q} \leq f(x) \leq \frac{[n-1]_q! [n]_q^x}{\prod_{j=0}^{n-1} [x+j]_q}.$$

The ratio of the upper bound to the lower bound is

$$\left( \frac{[n+1]_q}{[n]_q} \right)^{1-x} \rightarrow 1.$$

Hence the bounds determine  $f(x)$  uniquely for  $0 < x \leq 1$ , and the recurrence determines it on all of  $(0, \infty)$ . Since  $\Gamma_q$  satisfies the same conditions,  $f = \Gamma_q$ .  $\square$

## 12.3 Classical limit

**Theorem 12.5** (Limit to Euler's gamma function). *For every  $x > 0$ ,*

$$\lim_{q \rightarrow 1^-} \Gamma_q(x) = \Gamma(x).$$

*The convergence is locally uniform on  $(0, \infty)$ .*

*Proof.* It suffices by the functional equations to consider  $0 < x \leq 1$ . The bounds in the proof of [Theorem 12.4](#) hold for  $\Gamma_q(x)$ . Fix  $n$  and let  $q \rightarrow 1^-$ . Since  $[u]_q \rightarrow u$ , they become

$$\begin{aligned} \frac{n!}{(n+1)^{1-x} x(x+1) \cdots (x+n-1)} &\leq \liminf_{q \rightarrow 1^-} \Gamma_q(x) \\ &\leq \limsup_{q \rightarrow 1^-} \Gamma_q(x) \leq \frac{(n-1)! n^x}{x(x+1) \cdots (x+n-1)}. \end{aligned}$$

Both outer expressions tend to  $\Gamma(x)$  as  $n \rightarrow \infty$ . To see this directly, use Euler's beta integral:

$$B(x, n) = \int_0^1 t^{x-1} (1-t)^{n-1} dt = \frac{(n-1)!}{x(x+1) \cdots (x+n-1)}.$$

After  $t = u/n$ , dominated convergence gives  $n^x B(x, n) \rightarrow \int_0^\infty u^{x-1} e^{-u} du = \Gamma(x)$ . The lower expression differs from the upper by a factor tending to 1. This proves pointwise convergence. We record the uniformity argument. On a compact interval  $x \in [\delta, 1]$ , the upper outer bound can be written as

$$n^x B(x, n) = \int_0^n u^{x-1} \left(1 - \frac{u}{n}\right)_+^{n-1} du.$$

For  $n \geq 2$ , its integrand is dominated uniformly in  $x$  by  $u^{\delta-1}$  on  $(0, 1]$  and by  $e^{-u/2}$  on  $[1, \infty)$ . More explicitly, after extending the kernel by 0 for  $u > n$ ,

$$\sup_{\delta \leq x \leq 1} \left| u^{x-1} \left(1 - \frac{u}{n}\right)_+^{n-1} - u^{x-1} e^{-u} \right| \rightarrow 0$$

for every  $u > 0$ , while this supremum is bounded by an integrable multiple of  $u^{\delta-1} \mathbf{1}_{(0,1]}(u) + e^{-u/2} \mathbf{1}_{[1,\infty)}(u)$ . The dominated convergence theorem therefore gives

$$\sup_{\delta \leq x \leq 1} |n^x B(x, n) - \Gamma(x)| \rightarrow 0.$$

The lower outer bound differs by the factor  $(n/(n+1))^{1-x}$ , which also tends uniformly to 1. For fixed  $n$ , the original  $q$ -bounds converge uniformly as  $q \rightarrow 1^-$  because they contain only finitely many continuous factors. A uniform squeeze proves local uniform convergence on  $(0, 1]$ , and the recurrence extends it to arbitrary compact subsets of  $(0, \infty)$ .  $\square$

## 12.4 The Jackson $q$ -beta integral

**Definition 12.6.** For  $x, y > 0$ , define

$$B_q(x, y) = \int_0^1 t^{x-1} \frac{(qt; q)_\infty}{(q^y t; q)_\infty} d_q t.$$

**Theorem 12.7** ( $q$ -beta evaluation). For  $x, y > 0$ ,

$$B_q(x, y) = \frac{\Gamma_q(x) \Gamma_q(y)}{\Gamma_q(x+y)}.$$

*Proof.* Expand the Jackson integral:

$$\begin{aligned} B_q(x, y) &= (1-q) \sum_{n \geq 0} q^{nx} \frac{(q^{n+1}; q)_\infty}{(q^{n+y}; q)_\infty} \\ &= (1-q) \frac{(q; q)_\infty}{(q^y; q)_\infty} \sum_{n \geq 0} \frac{(q^y; q)_n}{(q; q)_n} q^{nx}. \end{aligned}$$

The infinite  $q$ -binomial theorem evaluates the sum as  $(q^{x+y}; q)_\infty / (q^x; q)_\infty$ . Thus

$$B_q(x, y) = (1-q) \frac{(q; q)_\infty (q^{x+y}; q)_\infty}{(q^x; q)_\infty (q^y; q)_\infty},$$

which is exactly the gamma quotient after substituting the definition of  $\Gamma_q$ .  $\square$

**Corollary 12.8** (Recurrences). *For  $x, y > 0$ ,*

$$B_q(x+1, y) = \frac{[x]_q}{[x+y]_q} B_q(x, y), \quad B_q(x, y+1) = \frac{[y]_q}{[x+y]_q} B_q(x, y).$$

*Proof.* Apply the functional equation for  $\Gamma_q$  to the quotient in [Theorem 12.7](#). □

## 12.5 Multiplication and reflection-type formulas

**Theorem 12.9** ( $q$ -multiplication formula). *For an integer  $m \geq 1$  and  $x > 0$ ,*

$$\Gamma_q(mx) \prod_{j=1}^{m-1} \Gamma_{q^m} \left( \frac{j}{m} \right) = [m]_q^{mx-1} \prod_{j=0}^{m-1} \Gamma_{q^m} \left( x + \frac{j}{m} \right).$$

*Proof.* Multiply the definitions on the right. By dissection,

$$\prod_{j=0}^{m-1} (q^{mx+j}; q^m)_\infty = (q^{mx}; q)_\infty.$$

Hence

$$\prod_{j=0}^{m-1} \Gamma_{q^m} \left( x + \frac{j}{m} \right) = (1 - q^m)^{(m+1)/2 - mx} \frac{(q^m; q^m)_\infty^m}{(q^{mx}; q)_\infty}.$$

Similarly,

$$\prod_{j=1}^{m-1} \Gamma_{q^m} \left( \frac{j}{m} \right) = (1 - q^m)^{(m-1)/2} \frac{(q^m; q^m)_\infty^m}{(q; q)_\infty}.$$

Taking the ratio and using  $[m]_q = (1 - q^m)/(1 - q)$  produces the formula. □

**Proposition 12.10** (Reflection-type product). *For  $0 < x < 1$ ,*

$$\Gamma_q(x) \Gamma_q(1-x) = (1-q) \frac{(q; q)_\infty^2}{(q^x, q^{1-x}; q)_\infty}.$$

*Proof.* Multiply the two defining products; the powers of  $1 - q$  combine to one. □

*Remark 12.11.* Unlike Euler's reflection formula, the right side is naturally a theta quotient rather than an elementary trigonometric function. Jacobi's triple product in the next part makes that theta structure explicit.

## **Part III**

# **Theta products and partition identities**

## Chapter 13

# Jacobi's triple product

### 13.1 A bilateral theta series

For  $z \neq 0$  and  $|q| < 1$ , define

$$\Theta(z; q) = \sum_{n \in \mathbb{Z}} q^{\binom{n}{2}} z^n.$$

The series converges locally uniformly on  $\mathbb{C}^*$  because its terms decay quadratically in both directions.

**Theorem 13.1** (Jacobi triple product). *For  $z \neq 0$  and  $|q| < 1$ ,*

$$\sum_{n \in \mathbb{Z}} q^{\binom{n}{2}} z^n = (-z, -q/z, q; q)_\infty, \quad (13.1)$$

$$\sum_{n \in \mathbb{Z}} (-1)^n q^{\binom{n}{2}} z^n = (z, q/z, q; q)_\infty. \quad (13.2)$$

*Proof.* It suffices to prove (13.1); replace  $z$  by  $-z$  for (13.2). Put

$$F(z) = (-z; q)_\infty (-q/z; q)_\infty (q; q)_\infty.$$

The products converge locally uniformly on  $\mathbb{C}^*$ , so  $F$  has a Laurent expansion  $F(z) = \sum_{n \in \mathbb{Z}} c_n z^n$ . The shift identities give

$$F(qz) = \frac{1 + z^{-1}}{1 + z} F(z) = z^{-1} F(z).$$

Comparing Laurent coefficients yields

$$c_{n+1} = q^n c_n, \quad c_n = q^{\binom{n}{2}} c_0$$

for every integer  $n$ .

It remains to find  $c_0$ . Euler's product expansions give

$$(-z; q)_\infty = \sum_{r \geq 0} \frac{q^{\binom{r}{2}} z^r}{(q; q)_r}, \quad (-q/z; q)_\infty = \sum_{s \geq 0} \frac{q^{\binom{s+1}{2}} z^{-s}}{(q; q)_s}.$$

Hence

$$c_0 = (q; q)_\infty \sum_{r \geq 0} \frac{q^{r^2}}{(q; q)_r^2}.$$

The sum is the generating function obtained by decomposing an arbitrary partition according to the side  $r$  of its Durfee square: the square contributes  $q^{r^2}$ , while the diagrams to its right and below are independent partitions with at most  $r$  parts, each having generating function  $1/(q; q)_r$ . Therefore

$$\sum_{r \geq 0} \frac{q^{r^2}}{(q; q)_r^2} = \frac{1}{(q; q)_\infty},$$

so  $c_0 = 1$ . Thus  $F(z) = \sum_{n \in \mathbb{Z}} q^{\binom{n}{2}} z^n$ . □

**Corollary 13.2** (Quasi-periodicity and zeros). *The theta function satisfies*

$$\Theta(qz; q) = z^{-1} \Theta(z; q).$$

*Its zeros are the simple points  $z = -q^m$ ,  $m \in \mathbb{Z}$ .*

*Proof.* The functional equation follows either from reindexing the bilateral series or from the proof above. The product in (13.1) vanishes exactly when  $1 + zq^j = 0$  or  $1 + q^{j+1}/z = 0$ , which together give  $z = -q^m$  for  $m \in \mathbb{Z}$ . Exactly one factor vanishes at each such point. □

## 13.2 A finite triple product

Jacobi's identity has an exact polynomial refinement. The truncated product  $(z; q)_N (q/z; q)_N$  is again a bilateral theta-like sum, with the factor  $1/(q; q)_\infty$  replaced by a central Gaussian coefficient; letting  $N \rightarrow \infty$  then reproves Theorem 13.1 from finite algebra plus a single application of dominated convergence.

**Theorem 13.3** (Finite triple product). *For  $N \in \mathbb{N}_0$  and  $z \neq 0$ ,*

$$(z; q)_N (q/z; q)_N = \sum_{k=-N}^N (-1)^k q^{\binom{k}{2}} \begin{bmatrix} 2N \\ N+k \end{bmatrix}_q z^k. \quad (13.3)$$

Here  $\binom{k}{2}$  denotes  $k(k-1)/2$  for every integer  $k$ , negative  $k$  included; equivalently  $\binom{-m}{2} = \binom{m+1}{2}$  for  $m \geq 0$ . Both sides lie in  $\mathbb{Z}[q][z, z^{-1}]$  and (13.3) is an identity there; in particular it carries no convergence hypothesis and holds with  $q$  an indeterminate.

*Proof.* By Theorem 3.1, (3.1),

$$(z; q)_N = \sum_{i=0}^N (-1)^i q^{\binom{i}{2}} \begin{bmatrix} N \\ i \end{bmatrix}_q z^i,$$

and, replacing  $z$  by  $q/z$  in the same identity and using  $\binom{j}{2} + j = \binom{j+1}{2}$ ,

$$(q/z; q)_N = \sum_{j=0}^N (-1)^j q^{\binom{j+1}{2}} \begin{bmatrix} N \\ j \end{bmatrix}_q z^{-j}.$$

Multiply the two expansions and collect the pairs contributing to  $z^k$ , namely those with  $i = j + k$ . Since  $(-1)^{j+k}(-1)^j = (-1)^k$ , and since  $\begin{bmatrix} N \\ j+k \end{bmatrix}_q = 0$  unless  $0 \leq j+k \leq N$ , the inner sum may be extended over all  $j \in \mathbb{Z}$ :

$$(z; q)_N (q/z; q)_N = \sum_{k \in \mathbb{Z}} (-1)^k z^k \sum_{j \in \mathbb{Z}} q^{\binom{j+k}{2} + \binom{j+1}{2}} \begin{bmatrix} N \\ j+k \end{bmatrix}_q \begin{bmatrix} N \\ j \end{bmatrix}_q.$$



The two binomial exponents combine cleanly. As an identity between polynomials in  $j$  and  $k$ ,

$$\binom{j+k}{2} + \binom{j+1}{2} - \binom{k}{2} = \frac{1}{2}[(j+k)(j+k-1) + j(j+1) - k(k-1)] = j^2 + jk,$$

and being a polynomial identity it is valid for negative  $k$  as well, with the convention above. On a term with  $\begin{bmatrix} N \\ j \end{bmatrix}_q \begin{bmatrix} N \\ j+k \end{bmatrix}_q \neq 0$  one has  $0 \leq j \leq N$  and  $0 \leq j+k \leq N$ , so  $j^2 + jk = j(j+k) \geq 0$  and no negative power of  $q$  is created; terms with  $j(j+k) < 0$  carry a vanishing Gaussian coefficient. Hence the inner sum equals

$$q^{\binom{k}{2}} \sum_{j \in \mathbb{Z}} q^{j^2 + jk} \begin{bmatrix} N \\ j \end{bmatrix}_q \begin{bmatrix} N \\ j+k \end{bmatrix}_q = q^{\binom{k}{2}} \begin{bmatrix} 2N \\ N+k \end{bmatrix}_q$$

by [Corollary 3.10](#). Therefore

$$(z; q)_N (q/z; q)_N = \sum_{k \in \mathbb{Z}} (-1)^k q^{\binom{k}{2}} \begin{bmatrix} 2N \\ N+k \end{bmatrix}_q z^k,$$

and the summand vanishes for  $|k| > N$ , which is (13.3). Both sides are Laurent polynomials in  $z$  with coefficients in  $\mathbb{Z}[q]$  because each Gaussian coefficient is such a polynomial by [Theorem 2.5](#).  $\square$

**Example 13.4.** At  $N = 1$  the coefficients are  $\begin{bmatrix} 2 \\ 0 \end{bmatrix}_q = \begin{bmatrix} 2 \\ 2 \end{bmatrix}_q = 1$  and  $\begin{bmatrix} 2 \\ 1 \end{bmatrix}_q = 1 + q$ , so (13.3) reads

$$(1-z)(1-q/z) = -qz^{-1} + (1+q) - z,$$

which is correct. The coefficient  $q^{\binom{-1}{2}} = q$  of  $z^{-1}$  shows the convention for negative  $k$  at work. At  $N = 2$  one gets

$$(z; q)_2 (q/z; q)_2 = q^3 z^{-2} - q(1+q+q^2+q^3)z^{-1} + (1+q+2q^2+q^3+q^4) - (1+q+q^2+q^3)z + qz^2,$$

the coefficients being  $\begin{bmatrix} 4 \\ 0 \end{bmatrix}_q = \begin{bmatrix} 4 \\ 4 \end{bmatrix}_q = 1$ ,  $\begin{bmatrix} 4 \\ 1 \end{bmatrix}_q = \begin{bmatrix} 4 \\ 3 \end{bmatrix}_q = 1+q+q^2+q^3$  and  $\begin{bmatrix} 4 \\ 2 \end{bmatrix}_q = 1+q+2q^2+q^3+q^4$ , with the outer  $q$ -powers  $q^{\binom{-2}{2}} = q^3$ ,  $q^{\binom{-1}{2}} = q$  and  $q^{\binom{2}{2}} = q$ .

**Corollary 13.5** (Second proof of Jacobi's triple product). *Let  $|q| < 1$  and  $z \neq 0$ . Letting  $N \rightarrow \infty$  in (13.3) and multiplying by  $(q; q)_\infty$  yields (13.2). The passage to the limit is uniform on every closed annulus  $r \leq |z| \leq R$  with  $0 < r \leq R$ .*

*Proof.* Extend the summand of (13.3) by zero, that is, read  $\begin{bmatrix} 2N \\ N+k \end{bmatrix}_q = 0$  for  $|k| > N$ , so that the right side is a sum over all  $k \in \mathbb{Z}$ .

A bound uniform in both  $N$  and  $k$ . By [Proposition 2.2](#),

$$\begin{bmatrix} 2N \\ N+k \end{bmatrix}_q = \frac{(q; q)_{2N}}{(q; q)_{N+k} (q; q)_{N-k}} \quad (|k| \leq N).$$

Put

$$\mu = \inf_{m \geq 0} |(q; q)_m|, \quad M = \prod_{j \geq 1} (1 + |q|^j) < \infty.$$

No factor of  $(q; q)_m$  vanishes, because  $|q| < 1$  forces  $q^j \neq 1$  for  $j \geq 1$ , and  $(q; q)_m \rightarrow (q; q)_\infty$ , which is nonzero by [Theorem 1.12](#). A convergent sequence of nonzero numbers with nonzero limit is bounded away from 0, so  $\mu > 0$ ; and  $|(q; q)_m| \leq M$  for every  $m$ . Hence

$$\left| \begin{bmatrix} 2N \\ N+k \end{bmatrix}_q \right| \leq \frac{M}{\mu^2} \quad \text{for all } N \geq 0 \text{ and all } k \in \mathbb{Z}. \quad (13.4)$$

*A summable dominating function.* For every integer  $k$  one has  $\binom{k}{2} = \frac{1}{2}(k^2 - k) \geq \frac{1}{2}(k^2 - |k|)$ , with equality for  $k \geq 0$ , so  $|q|^{\binom{k}{2}} \leq |q|^{(k^2 - |k|)/2}$ . On the annulus  $r \leq |z| \leq R$  we have  $|z^k| \leq R^{|k|} + r^{-|k|}$ . Therefore, for all  $N$  and all  $z$  in the annulus,

$$\left| (-1)^k q^{\binom{k}{2}} \begin{bmatrix} 2N \\ N+k \end{bmatrix}_q z^k \right| \leq g(k) := \frac{M}{\mu^2} |q|^{(k^2 - |k|)/2} (R^{|k|} + r^{-|k|}),$$

and  $\sum_{k \in \mathbb{Z}} g(k) < \infty$ , since  $|q|^{k^2/2}$  decays faster than any geometric sequence grows.

*The termwise limit.* For each fixed  $k$ , using  $(q; q)_m \rightarrow (q; q)_\infty \neq 0$  again,

$$\begin{bmatrix} 2N \\ N+k \end{bmatrix}_q = \frac{(q; q)_{2N}}{(q; q)_{N+k} (q; q)_{N-k}} \xrightarrow{N \rightarrow \infty} \frac{(q; q)_\infty}{(q; q)_\infty^2} = \frac{1}{(q; q)_\infty}.$$

This is [Theorem 19.7](#) when  $k = 0$ ; the argument is unchanged for fixed  $k \neq 0$ , since  $N \pm k \rightarrow \infty$  with  $N$ . Dominated convergence for series, applied on the counting measure of  $\mathbb{Z}$  with dominating function  $g$ , therefore gives

$$\lim_{N \rightarrow \infty} \sum_{k \in \mathbb{Z}} (-1)^k q^{\binom{k}{2}} \begin{bmatrix} 2N \\ N+k \end{bmatrix}_q z^k = \frac{1}{(q; q)_\infty} \sum_{k \in \mathbb{Z}} (-1)^k q^{\binom{k}{2}} z^k,$$

The convergence is moreover uniform on the annulus: given  $\varepsilon > 0$ , choose  $K$  with  $\sum_{|k| > K} g(k) < \varepsilon/3$ , which bounds the two tails simultaneously for every  $N$  and every  $z$  in the annulus because  $g$  depends on neither; the remaining  $2K + 1$  terms converge because  $\begin{bmatrix} 2N \\ N+k \end{bmatrix}_q \rightarrow 1/(q; q)_\infty$  is a limit of numbers, and  $|z^k| \leq R^K + r^{-K}$  there. On the left of (13.3),  $(z; q)_N (q/z; q)_N \rightarrow (z; q)_\infty (q/z; q)_\infty$  locally uniformly on  $\mathbb{C}^*$  by [Theorem 1.12](#). Multiplying by  $(q; q)_\infty$  we obtain

$$(q, z, q/z; q)_\infty = \sum_{k \in \mathbb{Z}} (-1)^k q^{\binom{k}{2}} z^k,$$

which is (13.2). □

*Remark 13.6* (Two mechanisms for one identity). [Theorem 13.1](#) was proved above from a functional equation together with the Durfee-square evaluation of the constant Laurent coefficient. [Corollary 13.5](#) proves it instead from the polynomial identity (13.3), whose only inputs are [Theorem 3.1](#) and [Theorem 3.8](#). The second route separates algebra from analysis completely: (13.3) holds in  $\mathbb{Z}[q][z, z^{-1}]$ , and the entire analytic content is the single limit  $\begin{bmatrix} 2N \\ N+k \end{bmatrix}_q \rightarrow 1/(q; q)_\infty$ , which is the same finite-to-infinite stabilization recorded coefficientwise in [Proposition 14.5](#) and columnwise in [Theorem 19.6](#). That separation is exactly the architecture recommended in [Chapter 20](#). Note also that (13.3) is genuinely finer than (13.2): its coefficients are the central Gaussian coefficients  $\begin{bmatrix} 2N \\ N+k \end{bmatrix}_q$ , which by [Theorem 2.5](#) are palindromic polynomials of degree  $(N+k)(N-k)$ , whereas in the limit every one of them collapses to the single  $k$ -independent factor  $1/(q; q)_\infty$ ; what survives of the polynomial is not its constant term but its stabilised coefficient sequence, namely the partition numbers of [Proposition 14.5](#).

### 13.3 Euler's pentagonal number theorem

**Theorem 13.7** (Pentagonal number theorem). *For  $|q| < 1$ ,*

$$(q; q)_\infty = \sum_{n \in \mathbb{Z}} (-1)^n q^{n(3n-1)/2} = 1 + \sum_{n=1}^{\infty} (-1)^n \left( q^{n(3n-1)/2} + q^{n(3n+1)/2} \right).$$

*Proof.* In (13.2), replace the base  $q$  by  $q^3$  and set  $z = q$ . The product side is

$$(q; q^3)_\infty (q^2; q^3)_\infty (q^3; q^3)_\infty = (q; q)_\infty$$

by dissection. The series exponent is

$$3 \binom{n}{2} + n = \frac{n(3n-1)}{2}.$$

Pair the terms  $n$  and  $-n$  to obtain the second form. □

**Corollary 13.8** (Euler's partition recurrence). *Let  $p(0) = 1$  and  $p(n) = 0$  for  $n < 0$ . Then for  $n \geq 1$ ,*

$$p(n) = \sum_{j=1}^{\infty} (-1)^{j-1} \left( p\left(n - \frac{j(3j-1)}{2}\right) + p\left(n - \frac{j(3j+1)}{2}\right) \right).$$

*Proof.* The generating function for partitions is  $\sum_{n \geq 0} p(n)q^n = 1/(q; q)_\infty$ . Multiply this by the pentagonal expansion and compare the coefficient of  $q^n$ . □

### 13.4 Jacobi's cubic identity

**Theorem 13.9** (Jacobi identity). *For  $|q| < 1$ ,*

$$(q; q)_\infty^3 = \sum_{n=0}^{\infty} (-1)^n (2n+1) q^{n(n+1)/2}.$$

*Proof.* Differentiate (13.2) with respect to  $z$  and set  $z = 1$ . Since

$$(z; q)_\infty = (1-z)(qz; q)_\infty,$$

the derivative of the product side at  $z = 1$  is  $-(q; q)_\infty^3$ . On the series side it is

$$\sum_{m \in \mathbb{Z}} m (-1)^m q^{\binom{m}{2}}.$$

Pair  $m = -n$  with  $m = n+1$  for  $n \geq 0$ . Their sum is

$$-(2n+1)(-1)^n q^{n(n+1)/2}.$$

Cancelling the common minus sign proves the identity. □

### 13.5 A theta form of the $q$ -gamma reflection product

**Corollary 13.10.** *For  $0 < x < 1$ ,*

$$\Gamma_q(x) \Gamma_q(1-x) = (1-q) \frac{(q; q)_\infty^3}{\sum_{n \in \mathbb{Z}} (-1)^n q^{\binom{n}{2} + xn}}.$$

*Proof.* Set  $z = q^x$  in (13.2):

$$\sum_{n \in \mathbb{Z}} (-1)^n q^{\binom{n}{2} + xn} = (q^x, q^{1-x}, q; q)_\infty.$$

Substitute this in Proposition 12.10. □

# Chapter 14

## Partitions and product–sum identities

### 14.1 Unrestricted, distinct, and odd parts

**Theorem 14.1** (Basic partition products). *Let  $p(n)$  count unrestricted partitions,  $d(n)$  partitions into distinct parts, and  $o(n)$  partitions into odd parts. Then*

$$\begin{aligned}\sum_{n \geq 0} p(n)q^n &= \frac{1}{(q; q)_\infty}, \\ \sum_{n \geq 0} d(n)q^n &= (-q; q)_\infty, \\ \sum_{n \geq 0} o(n)q^n &= \frac{1}{(q; q^2)_\infty}.\end{aligned}$$

Moreover,  $d(n) = o(n)$  for every  $n$ .

*Proof.* For unrestricted partitions, the part  $j$  may occur  $0, 1, 2, \dots$  times and contributes the geometric factor  $(1 - q^j)^{-1}$ . For distinct partitions, the part  $j$  occurs zero or one time and contributes  $1 + q^j$ . For odd parts, retain only the odd geometric factors. Finally,

$$(-q; q)_\infty = \prod_{j \geq 1} \frac{1 - q^{2j}}{1 - q^j} = \frac{(q^2; q^2)_\infty}{(q; q)_\infty} = \frac{1}{(q; q^2)_\infty},$$

so the two generating functions agree coefficientwise. □

**Remark 14.2** (A bijection). The equality  $d(n) = o(n)$  also has a direct proof. In a partition into distinct parts, write each part uniquely as  $2^r u$  with  $u$  odd. Replace that part by  $2^r$  copies of  $u$ . Distinct powers of two ensure that the process is reversible by binary expansion of the multiplicity of every odd part.

### 14.2 Bounded partitions

**Proposition 14.3.** *The following generating functions are equal:*

$$\frac{1}{(q; q)_r} = \sum_{n \geq 0} p_{\leq r}(n)q^n,$$

where  $p_{\leq r}(n)$  counts partitions with largest part at most  $r$ , or equivalently partitions with at most  $r$  parts.

*Proof.* The product allows parts  $1, \dots, r$  with arbitrary multiplicity. Conjugation of Ferrers diagrams exchanges the condition “largest part at most  $r$ ” with “at most  $r$  parts.”  $\square$

**Corollary 14.4** (Durfee decomposition of all partitions).

$$\frac{1}{(q; q)_{\infty}} = \sum_{r=0}^{\infty} \frac{q^{r^2}}{(q; q)_r^2}.$$

*Proof.* Decompose every partition by the side  $r$  of its Durfee square. The diagrams to the right and below are independent partitions with at most  $r$  parts. This is the identity used in the proof of Jacobi’s triple product.  $\square$

### 14.3 Finite approximations and stabilization

**Proposition 14.5.** *For fixed  $N$ , the coefficient of  $q^N$  in  $\begin{bmatrix} m+k \\ k \end{bmatrix}_q$  equals  $p(N)$  whenever  $m, k \geq N$ .*

*Proof.* The Gaussian coefficient counts partitions inside a  $k \times m$  rectangle. Any partition of  $N$  has at most  $N$  parts and largest part at most  $N$ , so for  $m, k \geq N$  the rectangle restriction is vacuous.  $\square$

*Remark 14.6.* Thus  $\begin{bmatrix} m+k \\ k \end{bmatrix}_q$  converges coefficientwise to  $1/(q; q)_{\infty}$  as both sides of the rectangle tend to infinity. This finite-to-infinite passage is a recurring bridge between Gaussian polynomials and partition identities.

## Chapter 15

# Bailey pairs and the Rogers–Ramanujan identities

### 15.1 Bailey pairs

For brevity, write  $(a)_n = (a; q)_n$  in this chapter.

**Definition 15.1.** Sequences  $(\alpha_n)_{n \geq 0}$  and  $(\beta_n)_{n \geq 0}$  form a *Bailey pair relative to  $a$*  if

$$\beta_n = \sum_{r=0}^n \frac{\alpha_r}{(q)_{n-r}(aq)_{n+r}} \quad (n \geq 0).$$

### 15.2 Bailey inversion

**Lemma 15.2** ( $q$ -difference annihilation). *If  $P(x)$  is a polynomial of degree less than  $N$ , then*

$$\sum_{s=0}^N (-1)^{N-s} q^{\binom{N-s}{2}} \begin{bmatrix} N \\ s \end{bmatrix}_q P(q^s) = 0.$$

*Proof.* By linearity it suffices to take  $P(x) = x^m$  with  $0 \leq m < N$ . Put  $t = N - s$  and use the finite  $q$ -binomial theorem:

$$\begin{aligned} \sum_{s=0}^N (-1)^{N-s} q^{\binom{N-s}{2}} \begin{bmatrix} N \\ s \end{bmatrix}_q q^{ms} &= q^{mN} \sum_{t=0}^N (-1)^t q^{\binom{t}{2}} \begin{bmatrix} N \\ t \end{bmatrix}_q q^{-mt} \\ &= q^{mN} (q^{-m}; q)_N = 0, \end{aligned}$$

for the product contains the factor  $1 - q^0$ . □

*Remark 15.3* (Exact Lean counterpart and boundary). This lemma is **exact**, by `Fabius.qDifference_sum_eval2_eq_zero_of_degree_lt` in `QDifferenceAnnihilation.lean`. The formal result is stronger than the displayed field-style proof: coefficients may lie in any semiring and be mapped into an arbitrary commutative ring before evaluation,  $q$  need not be nonzero or invertible, the geometric nodes may collide, and no use of  $q^{-m}$  or division is made. Its degree comparison includes  $N = 0$  and the zero polynomial. The same module proves the scaled characteristic identity (3.6), every monomial moment (3.7), and the exact scalar-extension top-coefficient formula (3.8).

**Theorem 15.4** (Bailey inversion). *Assume  $a \neq 1$  initially. The Bailey-pair relation is equivalent to*

$$\alpha_n = \frac{1 - aq^{2n}}{1 - a} \sum_{j=0}^n \frac{(a)_{n+j}}{(q)_{n-j}} (-1)^{n-j} q^{\binom{n-j}{2}} \beta_j.$$

*At  $a = 1$ , the right side is interpreted by its removable limit.*

*Proof.* Substitute the definition of  $\beta_j$  into the proposed inverse. The coefficient of  $\alpha_r$  is

$$\frac{1 - aq^{2n}}{1 - a} \sum_{j=r}^n \frac{(a)_{n+j}}{(q)_{n-j}(q)_{j-r}(aq)_{j+r}} (-1)^{n-j} q^{\binom{n-j}{2}}.$$

If  $n = r$ , only  $j = n$  occurs, and

$$\frac{1 - aq^{2n}}{1 - a} \frac{(a)_{2n}}{(aq)_{2n}} = 1.$$

Suppose  $N = n - r > 0$  and put  $j = r + s$ . Then

$$\frac{(a)_{n+r+s}}{(aq)_{2r+s}} = (1 - a)(aq^{2r+s+1}; q)_{N-1}.$$

The coefficient becomes

$$\frac{1 - aq^{2n}}{(q)_N} \sum_{s=0}^N (-1)^{N-s} q^{\binom{N-s}{2}} \begin{bmatrix} N \\ s \end{bmatrix}_q P(q^s),$$

where  $P(x) = (aq^{2r+1}x; q)_{N-1}$  has degree  $N - 1$ . It vanishes by [Lemma 15.2](#). Thus the two triangular matrices are inverse. Since every expression is rational in  $a$  and the singularity at  $a = 1$  is removable, the limiting statement follows.  $\square$

**Corollary 15.5** (Two unit Bailey pairs). *Let  $\beta_n = \delta_{n,0}$ .*

(i) *Relative to  $a = 1$ ,*

$$\alpha_0 = 1, \quad \alpha_n = (-1)^n q^{\binom{n}{2}} (1 + q^n) \quad (n > 0).$$

(ii) *Relative to  $a = q$ ,*

$$\alpha_n = (-1)^n q^{\binom{n}{2}} [2n + 1]_q \quad (n \geq 0).$$

*Proof.* With  $\beta_j = \delta_{j,0}$ , Bailey inversion gives

$$\alpha_n = \frac{1 - aq^{2n}}{1 - a} \frac{(a)_n}{(q)_n} (-1)^n q^{\binom{n}{2}}.$$

For  $a \rightarrow 1$  and  $n > 0$ ,  $(a)_n/(1 - a) \rightarrow (q)_{n-1}$ , so the prefactor reduces to  $(1 - q^{2n})/(1 - q^n) = 1 + q^n$ . For  $a = q$ , the ratio  $(a)_n/(q)_n$  is 1 and the remaining factor is  $[2n + 1]_q$ .  $\square$

### 15.3 The limiting Bailey lemma

**Lemma 15.6** (Auxiliary finite identity). *For  $N \geq 0$ ,*

$$\sum_{k=0}^N z^k q^{k(k-1)} \begin{bmatrix} N \\ k \end{bmatrix}_q (zq^k; q)_{N-k} = 1.$$

*Proof.* Call the left side  $S_N(z)$ . Using the second  $q$ -Pascal recurrence and splitting the last factor of  $(zq^k; q)_{N+1-k}$  gives

$$S_{N+1}(z) = (1 - zq^N)S_N(z) + zq^N S_N(qz).$$

The identity is immediate for  $N = 0$ . If  $S_N$  is identically 1, the recurrence gives  $S_{N+1} = 1 - zq^N + zq^N = 1$ . Induction completes the proof.  $\square$

**Theorem 15.7** (Limiting Bailey lemma, finite form). *If  $(\alpha_n, \beta_n)$  is a Bailey pair relative to  $a$ , define*

$$\alpha'_n = a^n q^{n^2} \alpha_n, \quad \beta'_n = \sum_{j=0}^n \frac{a^j q^{j^2}}{(q)_{n-j}} \beta_j.$$

*Then  $(\alpha'_n, \beta'_n)$  is also a Bailey pair relative to  $a$ .*

*Proof.* Substitute the Bailey relation for  $\beta_j$  and interchange the finite sums. The coefficient of  $\alpha_r$  in  $\beta'_n$  is

$$a^r q^{r^2} \sum_{s=0}^N \frac{a^s q^{s^2+2rs}}{(q)_{N-s} (q)_s (aq)_{2r+s}}, \quad N = n - r.$$

Let  $A = aq^{2r+1}$ . Since

$$a^s q^{s^2+2rs} = A^s q^{s(s-1)}, \quad (aq)_{2r+s} = (aq)_{2r} (A)_s,$$

the desired Bailey coefficient  $a^r q^{r^2} / ((q)_N (aq)_{n+r})$  is equivalent, after clearing factors, to

$$\sum_{s=0}^N \begin{bmatrix} N \\ s \end{bmatrix}_q A^s q^{s(s-1)} (Aq^s)_{N-s} = 1.$$

This is [Lemma 15.6](#).  $\square$

**Corollary 15.8** (Limiting Bailey lemma, infinite form). *If the series converge absolutely, then every Bailey pair relative to  $a$  satisfies*

$$\sum_{n=0}^{\infty} a^n q^{n^2} \beta_n = \frac{1}{(aq; q)_{\infty}} \sum_{n=0}^{\infty} a^n q^{n^2} \alpha_n.$$

*Proof.* The transformed pair in [Theorem 15.7](#) satisfies

$$\beta'_N = \sum_{r=0}^N \frac{\alpha'_r}{(q)_{N-r} (aq)_{N+r}}.$$

Let  $N \rightarrow \infty$ . On the left,

$$\beta'_N \longrightarrow \frac{1}{(q)_{\infty}} \sum_{j \geq 0} a^j q^{j^2} \beta_j.$$

On the right,

$$\beta'_N \longrightarrow \frac{1}{(q)_{\infty} (aq)_{\infty}} \sum_{r \geq 0} a^r q^{r^2} \alpha_r.$$

Absolute convergence permits dominated convergence. Cancel  $(q)_{\infty}^{-1}$ .  $\square$



## 15.4 The Rogers–Ramanujan identities

**Theorem 15.9** (Rogers–Ramanujan). For  $|q| < 1$ ,

$$\sum_{n=0}^{\infty} \frac{q^{n^2}}{(q; q)_n} = \frac{1}{(q; q^5)_{\infty} (q^4; q^5)_{\infty}}, \quad (15.1)$$

$$\sum_{n=0}^{\infty} \frac{q^{n(n+1)}}{(q; q)_n} = \frac{1}{(q^2; q^5)_{\infty} (q^3; q^5)_{\infty}}. \quad (15.2)$$

*Proof.* For the first identity, begin with the unit pair relative to  $a = 1$  in [Corollary 15.5](#). Apply the finite limiting lemma once. The new pair is

$$\alpha'_n = q^{n^2} \alpha_n, \quad \beta'_n = \frac{1}{(q)_n}.$$

Applying the infinite limiting lemma to this new pair gives

$$\begin{aligned} \sum_{n \geq 0} \frac{q^{n^2}}{(q)_n} &= \frac{1}{(q)_{\infty}} \left( 1 + \sum_{r \geq 1} (-1)^r q^{2r^2 + \binom{r}{2}} (1 + q^r) \right) \\ &= \frac{1}{(q)_{\infty}} \sum_{r \in \mathbb{Z}} (-1)^r q^{(5r^2 - r)/2}. \end{aligned}$$

By Jacobi's triple product with base  $q^5$  and  $z = q^2$ ,

$$\sum_{r \in \mathbb{Z}} (-1)^r q^{(5r^2 - r)/2} = (q^2, q^3, q^5; q^5)_{\infty}.$$

Dissection gives

$$(q; q)_{\infty} = (q, q^2, q^3, q^4, q^5; q^5)_{\infty},$$

which proves (15.1).

For the second identity use the unit pair relative to  $a = q$ . After one finite limiting step,

$$\alpha'_n = (-1)^n q^{n^2 + n + \binom{n}{2}} [2n + 1]_q, \quad \beta'_n = \frac{1}{(q)_n}.$$

The infinite limiting lemma gives

$$\begin{aligned} \sum_{n \geq 0} \frac{q^{n^2 + n}}{(q)_n} &= \frac{1}{(q^2; q)_{\infty}} \sum_{r \geq 0} q^{r^2 + r} \alpha'_r \\ &= \frac{1}{(q; q)_{\infty}} \sum_{r \geq 0} (-1)^r q^{(5r^2 + 3r)/2} (1 - q^{2r+1}) \\ &= \frac{1}{(q; q)_{\infty}} \sum_{r \in \mathbb{Z}} (-1)^r q^{(5r^2 + 3r)/2}. \end{aligned}$$

In the last step, the subtracted term indexed by  $r \geq 0$  is the term with bilateral index  $-r - 1$ . Jacobi's triple product with base  $q^5$  and  $z = q^4$  gives

$$\sum_{r \in \mathbb{Z}} (-1)^r q^{(5r^2 + 3r)/2} = (q, q^4, q^5; q^5)_{\infty}.$$

Divide by the fivefold dissection of  $(q; q)_{\infty}$  to obtain (15.2). □

*Remark 15.10* (Partition interpretation). The product side of (15.1) counts partitions into parts congruent to 1 or 4 modulo 5; the product side of (15.2) counts parts congruent to 2 or 3 modulo 5. The sum sides count partitions with successive parts differing by at least 2, with an additional smallest-part condition in the second identity. Establishing the latter combinatorial interpretations requires a separate bijective or finite-polynomial argument; the proof above instead reveals the hypergeometric engine.

## **Part IV**

# **Cyclotomic arithmetic and extended indices**

## Chapter 16

# Cyclotomic factorization and congruences

The arithmetic of Gaussian coefficients is governed by cyclotomic polynomials. This is the precise sense in which a primitive root of unity plays the role of a prime. The point of this chapter is that the essential congruences do not require mysterious cancellation: they follow from product factorization, coefficient extraction, and a careful accounting of blocks.

### 16.1 Cyclotomic valuations and carries

Let  $\Phi_d(q)$  denote the  $d$ th cyclotomic polynomial, normalized by

$$q^n - 1 = \prod_{d|n} \Phi_d(q).$$

For a nonzero polynomial  $F \in \mathbb{Z}[q]$ , write  $\nu_d(F)$  for the exponent of  $\Phi_d(q)$  in its factorization.

**Theorem 16.1** (Cyclotomic factorization of a shifted factorial). *For every  $n \geq 0$ ,*

$$(q; q)_n = (-1)^n \prod_{d=1}^n \Phi_d(q)^{\lfloor n/d \rfloor}.$$

Consequently,

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = \prod_{d=1}^n \Phi_d(q)^{e_d(n,k)}, \quad e_d(n,k) = \lfloor n/d \rfloor - \lfloor k/d \rfloor - \lfloor (n-k)/d \rfloor. \quad (16.1)$$

Moreover  $e_d(n,k)$  is always either 0 or 1.

*Proof.* Since  $1 - q^j = -(q^j - 1)$  and  $q^j - 1 = \prod_{d|j} \Phi_d(q)$ ,

$$(q; q)_n = \prod_{j=1}^n (1 - q^j) = (-1)^n \prod_{j=1}^n \prod_{d|j} \Phi_d(q).$$

For fixed  $d$ , the multiples of  $d$  among  $1, \dots, n$  are  $d, 2d, \dots, \lfloor n/d \rfloor d$ , so the exponent of  $\Phi_d$  is  $\lfloor n/d \rfloor$ . Dividing the factorizations of  $(q; q)_n$ ,  $(q; q)_k$ , and  $(q; q)_{n-k}$  gives (16.1); the signs cancel.

Write  $k = ad + r$  and  $n - k = bd + s$  with  $0 \leq r, s < d$ . Then

$$e_d(n,k) = \lfloor (r+s)/d \rfloor,$$

which is 0 or 1. □

**Corollary 16.2** (Square-free cyclotomic structure). *Every Gaussian polynomial is a product of distinct cyclotomic polynomials. More precisely,*

$$\Phi_d(q) \mid \begin{bmatrix} n \\ k \end{bmatrix}_q \iff (k \bmod d) > (n \bmod d). \quad (16.2)$$

Equivalently,  $\Phi_d$  occurs exactly when adding  $k$  and  $n - k$  produces a carry in the units place in base  $d$ .

*Proof.* The first assertion follows because every exponent in (16.1) belongs to  $\{0, 1\}$ . If  $n = cd + t$  and  $k = ad + r$  with  $0 \leq r, t < d$ , then the residue of  $n - k$  is  $t - r$  when  $r \leq t$  and  $t - r + d$  when  $r > t$ . Thus the two residues add to  $t$  in the first case and to  $t + d$  in the second. The latter is precisely the carry case, and by the proof of Theorem 16.1 it is equivalent to  $e_d(n, k) = 1$ .  $\square$

**Example 16.3.** For  $n = 10$  and  $k = 4$ , the values of  $d$  for which  $4 \bmod d > 10 \bmod d$  are  $d = 5, 7, 8, 9$ . Hence

$$\begin{bmatrix} 10 \\ 4 \end{bmatrix}_q = \Phi_5(q)\Phi_7(q)\Phi_8(q)\Phi_9(q).$$

The degrees add to  $4 + 6 + 4 + 6 = 20 = 4(10 - 4)$ , as required by reciprocity.

**Corollary 16.4** (MacMahon's  $q$ -Catalan polynomial). *For  $n \geq 0$  put*

$$C_n(q) := \frac{1}{[n+1]_q} \begin{bmatrix} 2n \\ n \end{bmatrix}_q.$$

*Then  $C_n(q) \in \mathbb{Z}[q]$ ; it has degree  $n(n-1)$ , and  $C_n(1) = \frac{1}{n+1} \binom{2n}{n}$ .*

*Proof.* Since  $q^N - 1 = \prod_{d|N} \Phi_d(q)$  and  $\Phi_1(q) = q - 1$ ,

$$[n+1]_q = \frac{q^{n+1} - 1}{q - 1} = \prod_{\substack{d|n+1 \\ d>1}} \Phi_d(q).$$

Hence  $\nu_d([n+1]_q) = 1$  exactly for those  $d > 1$  dividing  $n+1$ , and  $\nu_d([n+1]_q) = 0$  for every other  $d$ , the case  $d = 1$  included: indeed  $[n+1]_q$  takes the value  $n+1 \neq 0$  at  $q = 1$ , so  $q - 1$  does not divide it.

Let  $d > 1$  divide  $n+1$ . Then  $n \equiv -1 \pmod{d}$ , so  $n \bmod d = d-1$ , while  $2n \equiv -2 \pmod{d}$  together with  $0 \leq d-2 < d$  gives  $2n \bmod d = d-2$ . Therefore

$$n \bmod d = d-1 > d-2 = 2n \bmod d,$$

and the carry criterion (16.2) of Corollary 16.2, applied with upper index  $2n$  and lower index  $n$ , gives  $\Phi_d(q) \mid \begin{bmatrix} 2n \\ n \end{bmatrix}_q$ , that is,  $\nu_d(\begin{bmatrix} 2n \\ n \end{bmatrix}_q) = 1$ . Combining the two paragraphs,

$$\nu_d\left(\begin{bmatrix} 2n \\ n \end{bmatrix}_q\right) \geq \nu_d([n+1]_q) \quad \text{for every } d \geq 1,$$

so  $[n+1]_q$  divides  $\begin{bmatrix} 2n \\ n \end{bmatrix}_q$  in  $\mathbb{Q}[q]$  and  $C_n \in \mathbb{Q}[q]$ . Moreover  $[n+1]_q = 1 + q + \cdots + q^n$  is monic and  $\begin{bmatrix} 2n \\ n \end{bmatrix}_q \in \mathbb{Z}[q]$  by Corollary 2.4, so division with remainder in  $\mathbb{Z}[q]$  produces a quotient in  $\mathbb{Z}[q]$ ; by uniqueness of the quotient in  $\mathbb{Q}[q]$  that quotient is  $C_n$ , whence  $C_n \in \mathbb{Z}[q]$ .

Finally,  $\deg \begin{bmatrix} 2n \\ n \end{bmatrix}_q = n \cdot n = n^2$  by Theorem 2.5(iii) and  $\deg [n+1]_q = n$ , so  $\deg C_n = n^2 - n$ . Evaluating the polynomial identity  $[n+1]_q C_n(q) = \begin{bmatrix} 2n \\ n \end{bmatrix}_q$  at  $q = 1$  and using Corollary 2.6 gives  $(n+1)C_n(1) = \binom{2n}{n}$ .  $\square$

*Remark 16.5.* The argument above establishes integrality and says nothing about signs. The coefficients of  $C_n(q)$  are in fact nonnegative —  $C_n$  enumerates the Dyck paths of semilength  $n$  by a suitable major-index statistic, for which see [30] — but that is a combinatorial theorem and not a consequence of cyclotomic divisibility. Compare [Corollary 5.5](#), where the corresponding statistic for unrestricted monotone lattice paths is the area and the enumeration is the unconstrained  $\begin{bmatrix} 2n \\ n \end{bmatrix}_q$ .

## 16.2 The $q$ -Lucas congruence

The following theorem is often called the  $q$ -Lucas theorem. It is naturally proved by grouping a finite product into complete root-of-unity blocks.

**Lemma 16.6** (A complete root-of-unity block). *If  $\zeta$  is a primitive  $d$ th root of unity, then*

$$\prod_{j=0}^{d-1} (1 + x\zeta^j) = 1 - (-x)^d. \quad (16.3)$$

*Proof.* The roots in  $x$  of the left side are  $-\zeta^{-j}$ , which are exactly the roots of  $1 - (-x)^d$ . Both polynomials have constant term 1, so they are equal.  $\square$

**Theorem 16.7** ( $q$ -Lucas). *Let  $a, r \geq 0$ , let  $d \geq 1$ , and let  $0 \leq b, s < d$ . Then*

$$\begin{bmatrix} ad + b \\ rd + s \end{bmatrix}_q \equiv \binom{a}{r} \begin{bmatrix} b \\ s \end{bmatrix}_q \pmod{\Phi_d(q)}. \quad (16.4)$$

Here the right side is understood as 0 when  $r > a$  or  $s > b$ .

*Proof.* It suffices to prove equality after substituting an arbitrary primitive  $d$ th root  $\zeta$  for  $q$ , because  $\Phi_d$  is the minimal polynomial of  $\zeta$  over  $\mathbb{Q}$ .

The finite  $q$ -binomial theorem gives

$$\prod_{j=0}^{ad+b-1} (1 + x\zeta^j) = \sum_{h=0}^{ad+b} \zeta^{\binom{h}{2}} \begin{bmatrix} ad + b \\ h \end{bmatrix}_\zeta x^h. \quad (16.5)$$

Group the product into  $a$  complete blocks and one incomplete block. By [Lemma 16.6](#),

$$\prod_{j=0}^{ad+b-1} (1 + x\zeta^j) = (1 - (-x)^d)^a \prod_{j=0}^{b-1} (1 + x\zeta^j).$$

The coefficient of  $x^{rd+s}$  on the right is

$$\binom{a}{r} (-1)^{r(d+1)} \zeta^{\binom{s}{2}} \begin{bmatrix} b \\ s \end{bmatrix}_\zeta. \quad (16.6)$$

Comparing (16.5) and (16.6) leaves only the phase. We claim that

$$(-1)^{r(d+1)} \zeta^{\binom{s}{2} - \binom{rd+s}{2}} = 1. \quad (16.7)$$

Indeed,

$$2 \left( \binom{rd+s}{2} - \binom{s}{2} \right) = rd(rd+2s-1).$$

If  $d$  is odd,  $r(rd + 2s - 1)$  is even, so the exponent difference is a multiple of  $d$ , while  $r(d + 1)$  is even. If  $d$  is even, then  $\zeta^{d/2} = -1$  and  $rd + 2s - 1$  is odd; hence the second factor in (16.7) equals  $(-1)^r$ , as does the first. Their product is 1. The coefficient comparison therefore gives (16.4) at  $q = \zeta$ , completing the proof.  $\square$

**Corollary 16.8** (Evaluation at  $q = -1$ ). For  $0 \leq k \leq n$ ,

$$\left[ \begin{matrix} n \\ k \end{matrix} \right]_q \Big|_{q=-1} = \begin{cases} \binom{n/2}{k/2}, & n \text{ even and } k \text{ even,} \\ 0, & n \text{ even and } k \text{ odd,} \\ \binom{(n-1)/2}{\lfloor k/2 \rfloor}, & n \text{ odd.} \end{cases}$$

*Proof.* Apply Theorem 16.7 with  $d = 2$ . Write  $n = 2a + b$  and  $k = 2r + s$ , where  $b, s \in \{0, 1\}$ . Since  $\begin{bmatrix} b \\ s \end{bmatrix}_{-1}$  is 1 when  $s \leq b$  and 0 otherwise, the three cases follow.  $\square$

### 16.3 Cyclic sieving for subsets

A triple  $(X, C, F(q))$ , where a finite cyclic group  $C = \langle c \rangle$  acts on a finite set  $X$  and  $F(q) \in \mathbb{Z}[q]$ , exhibits the *cyclic sieving phenomenon* when

$$\# \text{Fix}(c^j) = F(\omega^j)$$

for every  $j$ , where  $\omega$  is a primitive  $|C|$ th root of unity.

**Theorem 16.9** (Cyclic sieving for  $k$ -subsets). Let  $X$  be the set of  $k$ -element subsets of  $\mathbb{Z}/n\mathbb{Z}$ , and let  $C_n$  act by translation. Then

$$\left( X, C_n, \left[ \begin{matrix} n \\ k \end{matrix} \right]_q \right)$$

exhibits the cyclic sieving phenomenon.

*Proof.* Fix  $j$ , and let  $d$  be the order of translation by  $j$ . Its orbits on  $\mathbb{Z}/n\mathbb{Z}$  all have size  $d$ , and there are  $n/d$  such orbits. A subset is fixed precisely when it is a union of these orbits. Thus

$$\# \text{Fix}(c^j) = \begin{cases} \binom{n/d}{k/d}, & d \mid k, \\ 0, & d \nmid k. \end{cases} \quad (16.8)$$

The number  $\omega^j$  is a primitive  $d$ th root. Write  $n = (n/d)d$  and  $k = rd + s$  with  $0 \leq s < d$ . By  $q$ -Lucas,

$$\left[ \begin{matrix} n \\ k \end{matrix} \right]_q \Big|_{q=\omega^j} = \binom{n/d}{r} \left[ \begin{matrix} 0 \\ s \end{matrix} \right]_{\omega^j}.$$

This is 0 if  $s > 0$ , and it is  $\binom{n/d}{k/d}$  if  $s = 0$ , exactly as in (16.8).  $\square$

## 16.4 A $q$ -Babbage congruence modulo a cyclotomic square

The modulus  $\Phi_n(q)^2$  records not only the value at a primitive  $n$ th root, but also first-order contact there. The next congruence is a particularly transparent example. Its block proof is also a useful template for stronger  $q$ -congruences; see Andrews [3] and Straub [31].

**Theorem 16.10** ( $q$ -Babbage congruence). *For integers  $a \geq b \geq 0$  and  $n \geq 1$ ,*

$$\begin{bmatrix} an \\ bn \end{bmatrix}_q \equiv \begin{bmatrix} a \\ b \end{bmatrix}_{q^{n^2}} \pmod{\Phi_n(q)^2}. \quad (16.9)$$

*Proof.* Split the finite product into  $a$  consecutive blocks of length  $n$ :

$$\prod_{j=0}^{an-1} (1 + xq^j) = \prod_{i=0}^{a-1} \prod_{j=0}^{n-1} (1 + xq^{in+j}).$$

Apply the finite  $q$ -binomial theorem to each block. Comparing the coefficient of  $x^{bn}$  gives

$$q^{\binom{bn}{2}} \begin{bmatrix} an \\ bn \end{bmatrix}_q = \sum_{\substack{0 \leq k_0, \dots, k_{a-1} \leq n \\ k_0 + \dots + k_{a-1} = bn}} q^{\sum_i \binom{k_i}{2} + n \sum_i i k_i} \prod_{i=0}^{a-1} \begin{bmatrix} n \\ k_i \end{bmatrix}_q. \quad (16.10)$$

If  $0 < k_i < n$ , then  $\Phi_n(q)$  divides  $\begin{bmatrix} n \\ k_i \end{bmatrix}_q$  by Corollary 16.2. A tuple contributing to (16.10) cannot have exactly one such intermediate entry: all other entries would be 0 or  $n$ , making the sum of the  $k_i$  nonzero modulo  $n$ . Therefore every tuple containing an intermediate entry contributes a multiple of  $\Phi_n(q)^2$ .

Modulo  $\Phi_n(q)^2$ , only tuples with each  $k_i \in \{0, n\}$  survive. Such a tuple is specified by a  $b$ -element subset  $S \subseteq \{0, 1, \dots, a-1\}$ . After division by  $q^{\binom{bn}{2}}$ , its exponent is

$$n^2 \sum_{i \in S} i + b \binom{n}{2} - \binom{bn}{2} = n^2 \left( \sum_{i \in S} i - \binom{b}{2} \right).$$

Hence

$$\begin{bmatrix} an \\ bn \end{bmatrix}_q \equiv \sum_{\substack{S \subseteq \{0, \dots, a-1\} \\ |S|=b}} (q^{n^2})^{\sum_{i \in S} i - \binom{b}{2}} \pmod{\Phi_n(q)^2}.$$

The subset form of the finite  $q$ -binomial theorem identifies the sum as  $\begin{bmatrix} a \\ b \end{bmatrix}_{q^{n^2}}$ . □

*Remark 16.11* (Why the square appears). The decisive observation in the proof is combinatorial: because the target degree is a multiple of the block length, nontrivial block choices occur in pairs. Each nontrivial block supplies one factor  $\Phi_n(q)$ , so two of them supply the square. Congruences modulo  $\Phi_n^3$  require genuinely finer first- and second-order information; the  $q$ -Ljunggren congruences of Straub and subsequent refinements belong to that deeper layer.

**Corollary 16.12** (Primitive-root value and derivative). *Let  $\zeta$  be a primitive  $n$ th root of unity. Then*

$$\begin{bmatrix} an \\ bn \end{bmatrix}_{\zeta} = \begin{pmatrix} a \\ b \end{pmatrix},$$

*and the derivatives at  $q = \zeta$  of the two sides of (16.9) are equal.*

*Proof.* The first assertion follows either from  $q$ -Lucas or from (16.9), because  $\zeta^{n^2} = 1$ . If a polynomial is divisible by  $\Phi_n^2$ , it has a zero of order at least two at every primitive  $n$ th root, so both its value and first derivative vanish there. □



## Chapter 17

# Negative upper indices and geometric Newton interpolation

Ordinary binomial coefficients acquire a useful extension to negative upper arguments. Gaussian coefficients do as well, but the powers of  $q$  are essential: omitting them destroys the Pascal recurrences. The same algebra leads naturally to interpolation on a geometric grid.

### 17.1 Generalized Gaussian coefficients

**Definition 17.1.** For an integer  $N$  and  $k \geq 0$ , define

$$\begin{bmatrix} N \\ k \end{bmatrix}_q := \frac{(q^{N-k+1}; q)_k}{(q; q)_k}. \quad (17.1)$$

For  $N \geq 0$  this agrees with the usual Gaussian coefficient, including the value 0 when  $k > N$ .

**Proposition 17.2** (Negative-index reflection). For  $m \geq 1$  and  $k \geq 0$ ,

$$\begin{bmatrix} -m \\ k \end{bmatrix}_q = (-1)^k q^{-mk - \binom{k}{2}} \begin{bmatrix} m+k-1 \\ k \end{bmatrix}_q. \quad (17.2)$$

*Proof.* From the definition,

$$\begin{aligned} (q^{-m-k+1}; q)_k &= \prod_{j=0}^{k-1} (1 - q^{-(m+k-1-j)}) \\ &= (-1)^k q^{-\sum_{j=0}^{k-1} (m+k-1-j)} \prod_{j=0}^{k-1} (1 - q^{m+k-1-j}) \\ &= (-1)^k q^{-mk - \binom{k}{2}} (q^m; q)_k. \end{aligned}$$

After division by  $(q; q)_k$ , the final ratio is  $\begin{bmatrix} m+k-1 \\ k \end{bmatrix}_q$ . □

**Theorem 17.3** (Pascal recurrences for all integral upper indices). For every integer  $N$  and every  $k \geq 1$ ,

$$\begin{bmatrix} N \\ k \end{bmatrix}_q = \begin{bmatrix} N-1 \\ k-1 \end{bmatrix}_q + q^k \begin{bmatrix} N-1 \\ k \end{bmatrix}_q, \quad (17.3)$$

$$= q^{N-k} \begin{bmatrix} N-1 \\ k-1 \end{bmatrix}_q + \begin{bmatrix} N-1 \\ k \end{bmatrix}_q. \quad (17.4)$$

Both formulas are identities in  $\mathbb{Q}(q)$ .

*Proof.* Put the two terms on the right of (17.3) over the common denominator  $(q; q)_k$ . Since

$$\left[ \begin{matrix} N-1 \\ k-1 \end{matrix} \right]_q = \frac{(q^{N-k+1}; q)_{k-1}(1-q^k)}{(q; q)_k}$$

the numerator of the right side is

$$\begin{aligned} & (q^{N-k+1}; q)_{k-1} \left[ (1-q^k) + q^k(1-q^{N-k}) \right] \\ &= (q^{N-k+1}; q)_{k-1}(1-q^N) = (q^{N-k+1}; q)_k, \end{aligned}$$

which proves (17.3). For (17.4), the corresponding bracket is

$$q^{N-k}(1-q^k) + (1-q^{N-k}) = 1 - q^N.$$

The same common-denominator calculation applies.  $\square$

**Corollary 17.4** (Reciprocal product series). *For  $m \geq 1$  and  $|z| < 1$ ,*

$$\frac{1}{(z; q)_m} = \sum_{k=0}^{\infty} \left[ \begin{matrix} m+k-1 \\ k \end{matrix} \right]_q z^k = \sum_{k=0}^{\infty} (-1)^k q^{mk + \binom{k}{2}} \left[ \begin{matrix} -m \\ k \end{matrix} \right]_q z^k. \quad (17.5)$$

*The first equality also holds in the formal power-series ring  $\mathbb{Z}[q][[z]]$ .*

*Proof.* The reciprocal finite  $q$ -binomial theorem, already proved in Theorem 3.6, gives the first expansion. Substitute (17.2) to obtain the second. Formal validity follows because each coefficient of  $z^k$  is a polynomial and multiplication by the finite product  $(z; q)_m$  can be checked coefficientwise.  $\square$

## 17.2 An arbitrary upper parameter

When a branch of  $q^\alpha$  has been fixed, it is often convenient to define

$$\left[ \begin{matrix} \alpha \\ k \end{matrix} \right]_q = \frac{(q^{\alpha-k+1}; q)_k}{(q; q)_k}. \quad (17.6)$$

The proofs of the Pascal recurrences are purely algebraic in  $q^\alpha$  and therefore remain valid. A related, and usually more analytic, expansion is the following.

**Theorem 17.5** (Generalized upper-parameter series). *For  $|q| < 1$ ,  $|z| < 1$ , and any parameter  $\alpha$  for which the factors are defined,*

$$\frac{(q^{\alpha+1}z; q)_\infty}{(z; q)_\infty} = \sum_{k=0}^{\infty} \left[ \begin{matrix} \alpha+k \\ k \end{matrix} \right]_q z^k. \quad (17.7)$$

*Proof.* Apply the infinite  $q$ -binomial theorem with  $a = q^{\alpha+1}$ :

$$\frac{(q^{\alpha+1}z; q)_\infty}{(z; q)_\infty} = \sum_{k \geq 0} \frac{(q^{\alpha+1}; q)_k}{(q; q)_k} z^k.$$

The coefficient is exactly  $\left[ \begin{matrix} \alpha+k \\ k \end{matrix} \right]_q$  by (17.6).  $\square$

### 17.3 Two complex arguments and generalized binomial series

The extension in [Definition 1.13](#) also permits both arguments of a Gaussian coefficient to be complex. Throughout this section take  $0 < q < 1$  and set  $q^z = \exp(z \log q)$ , using the real logarithm of  $q$ .

**Definition 17.6.** For  $\alpha, \beta \in \mathbb{C}$ , define

$$\begin{bmatrix} \alpha \\ \beta \end{bmatrix}_q := \frac{(q^{\beta+1}, q^{\alpha-\beta+1}; q)_\infty}{(q, q^{\alpha+1}; q)_\infty}, \quad (17.8)$$

whenever the denominator is nonzero.

**Theorem 17.7** (Gamma representation and Pascal laws). *Wherever the expressions are defined,*

$$\begin{bmatrix} \alpha \\ \beta \end{bmatrix}_q = \frac{\Gamma_q(\alpha + 1)}{\Gamma_q(\beta + 1)\Gamma_q(\alpha - \beta + 1)}, \quad (17.9)$$

$$\begin{bmatrix} \alpha \\ \beta \end{bmatrix}_q = \begin{bmatrix} \alpha \\ \alpha - \beta \end{bmatrix}_q, \quad (17.10)$$

$$\begin{bmatrix} \alpha + 1 \\ \beta \end{bmatrix}_q = q^\beta \begin{bmatrix} \alpha \\ \beta \end{bmatrix}_q + \begin{bmatrix} \alpha \\ \beta - 1 \end{bmatrix}_q, \quad (17.11)$$

$$= \begin{bmatrix} \alpha \\ \beta \end{bmatrix}_q + q^{\alpha+1-\beta} \begin{bmatrix} \alpha \\ \beta - 1 \end{bmatrix}_q. \quad (17.12)$$

If  $\beta = k \in \mathbb{N}_0$ , this definition agrees with (17.6).

*Proof.* Substitution of the product definition of  $\Gamma_q$  proves (17.9); all powers of  $1 - q$  cancel. Symmetry is immediate either from that quotient or directly from (17.8). For  $\beta = k \in \mathbb{N}_0$ , tail cancellation gives

$$\frac{(q^{k+1}; q)_\infty}{(q; q)_\infty} = \frac{1}{(q; q)_k}, \quad \frac{(q^{\alpha-k+1}; q)_\infty}{(q^{\alpha+1}; q)_\infty} = (q^{\alpha-k+1}; q)_k,$$

which is exactly (17.6).

For the first Pascal law, divide every term by  $\begin{bmatrix} \alpha \\ \beta \end{bmatrix}_q$ . The gamma functional equation gives

$$\frac{\begin{bmatrix} \alpha+1 \\ \beta \end{bmatrix}_q}{\begin{bmatrix} \alpha \\ \beta \end{bmatrix}_q} = \frac{[\alpha + 1]_q}{[\alpha - \beta + 1]_q}, \quad \frac{\begin{bmatrix} \alpha \\ \beta-1 \end{bmatrix}_q}{\begin{bmatrix} \alpha \\ \beta \end{bmatrix}_q} = \frac{[\beta]_q}{[\alpha - \beta + 1]_q}.$$

The identity  $[\alpha + 1]_q = q^\beta [\alpha - \beta + 1]_q + [\beta]_q$  proves (17.11). The alternative identity  $[\alpha + 1]_q = [\alpha - \beta + 1]_q + q^{\alpha+1-\beta} [\beta]_q$  proves (17.12).  $\square$

**Corollary 17.8** (Classical generalized binomial). *If  $\alpha > \beta > -1$  are real, then*

$$\lim_{q \rightarrow 1^-} \begin{bmatrix} \alpha \\ \beta \end{bmatrix}_q = \frac{\Gamma(\alpha + 1)}{\Gamma(\beta + 1)\Gamma(\alpha - \beta + 1)}.$$

*Proof.* Apply the locally uniform  $q$ -gamma limit in [Theorem 12.5](#) to (17.9).  $\square$

**Theorem 17.9** (Generalized finite and reciprocal  $q$ -binomial theorems). *Let  $0 < q < 1$ . If  $|zq^\alpha| < 1$ , then*

$$(z; q)_\alpha = \sum_{k=0}^{\infty} (-1)^k q^{\binom{k}{2}} \begin{bmatrix} \alpha \\ k \end{bmatrix}_q z^k. \quad (17.13)$$

If  $|z| < 1$ , then

$$\frac{1}{(z; q)_\alpha} = \sum_{k=0}^{\infty} \begin{bmatrix} \alpha + k - 1 \\ k \end{bmatrix}_q z^k. \quad (17.14)$$

For a nonnegative integral  $\alpha$ , the first series terminates and becomes Euler's finite theorem; in that case the restriction  $|zq^\alpha| < 1$  can be removed because both sides are polynomials. For a positive integral  $\alpha$ , the second becomes the reciprocal finite theorem.

*Proof.* Apply the infinite  $q$ -binomial theorem with  $a = q^{-\alpha}$  and argument  $zq^\alpha$ :

$$\frac{(z; q)_\infty}{(zq^\alpha; q)_\infty} = \sum_{k \geq 0} \frac{(q^{-\alpha}; q)_k}{(q; q)_k} (zq^\alpha)^k.$$

The left side is  $(z; q)_\alpha$ . Reversing the finite numerator product gives

$$\frac{(q^{-\alpha}; q)_k}{(q; q)_k} q^{\alpha k} = (-1)^k q^{\binom{k}{2}} \begin{bmatrix} \alpha \\ k \end{bmatrix}_q,$$

which proves (17.13).

For the reciprocal formula, apply the infinite theorem with  $a = q^\alpha$  and argument  $z$ :

$$\frac{(q^\alpha z; q)_\infty}{(z; q)_\infty} = \sum_{k \geq 0} \frac{(q^\alpha; q)_k}{(q; q)_k} z^k.$$

The left side is  $(z; q)_\alpha^{-1}$  and the coefficient is  $\begin{bmatrix} \alpha + k - 1 \\ k \end{bmatrix}_q$ . The integral specializations follow from the consistency results already proved.  $\square$

## 17.4 Newton interpolation on the grid $1, q, q^2, \dots$

The usual Newton basis is adapted to an arbitrary sequence of distinct nodes. At geometric nodes its basis polynomials are  $q$ -Pochhammer symbols.

**Theorem 17.10** (Geometric Newton interpolation). *Assume  $q^i \neq q^j$  whenever  $0 \leq i < j \leq n$ . For arbitrary values  $y_0, \dots, y_n$  there is a unique polynomial  $P$  of degree at most  $n$  satisfying*

$$P(q^j) = y_j \quad (0 \leq j \leq n).$$

*It has the Newton expansion*

$$P(x) = \sum_{k=0}^n c_k \prod_{r=0}^{k-1} (x - q^r) = \sum_{k=0}^n c_k (-1)^k q^{\binom{k}{2}} (x; q^{-1})_k, \quad (17.15)$$

where

$$c_k = \sum_{j=0}^k \frac{y_j}{\prod_{\substack{0 \leq r \leq k \\ r \neq j}} (q^j - q^r)} \quad (17.16)$$

$$= \sum_{j=0}^k \frac{(-1)^j q^{-jk + \binom{j+1}{2}}}{(q; q)_j (q; q)_{k-j}} y_j. \quad (17.17)$$

*Proof.* For distinct nodes  $x_0, \dots, x_n$ , Lagrange interpolation gives existence and uniqueness. The coefficient of the Newton basis polynomial  $\prod_{r=0}^{k-1} (x - x_r)$  is the divided difference

$$y[x_0, \dots, x_k] = \sum_{j=0}^k \frac{y_j}{\prod_{r \neq j} (x_j - x_r)};$$

this follows by comparing the coefficient of  $x^k$  in the Lagrange interpolant through the first  $k+1$  nodes. Taking  $x_j = q^j$  proves (17.16).

It remains to simplify the denominator. Splitting the factors below and above  $j$  gives

$$\begin{aligned} \prod_{\substack{0 \leq r \leq k \\ r \neq j}} (q^j - q^r) &= \left[ \prod_{r=0}^{j-1} -q^r (1 - q^{j-r}) \right] \left[ \prod_{r=j+1}^k q^j (1 - q^{r-j}) \right] \\ &= (-1)^j q^{\binom{j}{2} + j(k-j)} (q; q)_j (q; q)_{k-j} \\ &= (-1)^j q^{jk - \binom{j+1}{2}} (q; q)_j (q; q)_{k-j}. \end{aligned}$$

Taking reciprocals yields (17.17). Finally,

$$\prod_{r=0}^{k-1} (x - q^r) = (-1)^k q^{\binom{k}{2}} \prod_{r=0}^{k-1} (1 - xq^{-r}) = (-1)^k q^{\binom{k}{2}} (x; q^{-1})_k,$$

which proves the second form of (17.15).  $\square$

**Corollary 17.11** (Triangular reconstruction). *The coefficients in (17.15) can be recovered successively by*

$$c_0 = y_0, \quad c_k = \frac{y_k - \sum_{r=0}^{k-1} c_r \prod_{j=0}^{r-1} (q^k - q^j)}{\prod_{j=0}^{k-1} (q^k - q^j)}. \quad (17.18)$$

*Proof.* Evaluate (17.15) at  $x = q^k$ . Every term with index greater than  $k$  vanishes, while the coefficient multiplying  $c_k$  is nonzero by the distinct-node hypothesis. Solving for  $c_k$  gives the formula.  $\square$

*Remark 17.12.* Kupershmidt's  $q$ -Newton viewpoint emphasizes that the classical binomial theorem, Newton interpolation, and  $q$ -difference calculus are not separate constructions: each is controlled by a basis whose successive factors form a geometric progression. The finite formula above is algebraic and needs no convergence hypothesis; infinite Newton series require additional growth conditions on the data.

## **Part V**

# **Central sums, limits, and effective methods**

## Chapter 18

# Central $q$ -binomial sums and Lambert series

Central Gaussian coefficients frequently become tractable only after a seemingly extraneous factor is inserted. The elementary identity in the next lemma shows why the factor  $(-q; q)_{2k}$  is natural: it converts a central coefficient at base  $q^2$  into a single quotient of shifted factorials. This chapter develops both a regular  $q$ -beta evaluation and a deleted-singularity identity proved recently by Chern, Dilcher, and Jiu [7].

### 18.1 The central-factor reduction

**Lemma 18.1.** *For every  $k \geq 0$ ,*

$$\begin{bmatrix} 2k \\ k \end{bmatrix}_{q^2} \frac{1}{(-q; q)_{2k}} = \frac{(q; q^2)_k}{(q^2; q^2)_k}. \quad (18.1)$$

*Proof.* The elementary product identity

$$(q; q)_{2k}(-q; q)_{2k} = (q^2; q^2)_{2k}$$

gives

$$\begin{aligned} \begin{bmatrix} 2k \\ k \end{bmatrix}_{q^2} \frac{1}{(-q; q)_{2k}} &= \frac{(q^2; q^2)_{2k}}{(q^2; q^2)_k^2} \frac{(q; q)_{2k}}{(q^2; q^2)_{2k}} \\ &= \frac{(q; q^2)_k (q^2; q^2)_k}{(q^2; q^2)_k^2}, \end{aligned}$$

which is the claimed quotient. □

For a complex parameter  $z$  and  $0 < q < 1$ , we use the analytic extension

$$[z]_q = \frac{1 - q^z}{1 - q},$$

with the real logarithm in  $q^z = \exp(z \log q)$ .

## 18.2 A $q$ -beta evaluation

**Theorem 18.2** (A regular central  $q$ -binomial sum). *Let  $0 < q < 1$ , and let  $\alpha \in \mathbb{C}$  be such that none of the denominators below vanishes. Then*

$$\begin{aligned} \sum_{k=0}^{\infty} \begin{bmatrix} 2k \\ k \end{bmatrix}_{q^2} \frac{q^k}{(-q; q)_{2k} [2k+1+\alpha]_q} \\ = \frac{\Gamma_{q^2}(3/2) \Gamma_{q^2}((\alpha+1)/2)}{\Gamma_{q^2}((\alpha+2)/2)}. \end{aligned} \quad (18.2)$$

*In particular, for real  $\alpha$  it is enough to exclude  $\alpha = -1, -3, -5, \dots$*

*Proof.* By [Lemma 18.1](#) and the telescoping quotient

$$\frac{1}{[2k+1+\alpha]_q} = \frac{1}{[\alpha+1]_q} \frac{(q^{\alpha+1}; q^2)_k}{(q^{\alpha+3}; q^2)_k},$$

the left side is

$$\frac{1}{[\alpha+1]_q} {}_2\phi_1 \left[ \begin{matrix} q, q^{\alpha+1} \\ q^{\alpha+3} \end{matrix}; q^2, q \right].$$

The  $q$ -Gauss sum from [Theorem 9.6](#), with base  $q^2$ , gives

$$\frac{1}{[\alpha+1]_q} \frac{(q^{\alpha+2}, q^2; q^2)_{\infty}}{(q^{\alpha+3}, q; q^2)_{\infty}}. \quad (18.3)$$

Now

$$(q^{\alpha+1}; q^2)_{\infty} = (1 - q^{\alpha+1})(q^{\alpha+3}; q^2)_{\infty}, \quad (q; q^2)_{\infty} = (1 - q)(q^3; q^2)_{\infty}.$$

Using these two relations to simplify (18.3), we obtain

$$\frac{(q^2; q^2)_{\infty} (q^{\alpha+2}; q^2)_{\infty}}{(q^3; q^2)_{\infty} (q^{\alpha+1}; q^2)_{\infty}}. \quad (18.4)$$

Finally insert

$$\Gamma_{q^2}(z) = (1 - q^2)^{1-z} \frac{(q^2; q^2)_{\infty}}{(q^{2z}; q^2)_{\infty}}.$$

In the quotient on the right of (18.2), all powers of  $1 - q^2$  cancel, and the remaining products are exactly (18.4).  $\square$

**Corollary 18.3** (Classical limit of the regular sum). *For  $\alpha \notin \{-1, -3, -5, \dots\}$ ,*

$$\sum_{k=0}^{\infty} \binom{2k}{k} \frac{1}{4^k (2k+1+\alpha)} = \frac{\Gamma(3/2) \Gamma((\alpha+1)/2)}{\Gamma((\alpha+2)/2)}. \quad (18.5)$$

*Proof.* The series is

$$\frac{1}{\alpha+1} {}_2F_1 \left( \begin{matrix} 1/2, (\alpha+1)/2 \\ (\alpha+3)/2 \end{matrix}; 1 \right).$$

Gauss's classical summation gives

$$\frac{1}{\alpha+1} \frac{\Gamma((\alpha+3)/2) \Gamma(1/2)}{\Gamma((\alpha+2)/2)}.$$

Since  $\Gamma((\alpha+3)/2) = ((\alpha+1)/2) \Gamma((\alpha+1)/2)$  and  $\Gamma(1/2)/2 = \Gamma(3/2)$ , this is (18.5). The same identity is also the  $q \rightarrow 1^-$  limit of [Theorem 18.2](#) by the  $q$ -gamma limit in [Theorem 12.5](#).  $\square$



### 18.3 Deleting a singular term

The regular formula becomes singular when  $\alpha = -2m - 1$ . Removing the offending  $k = m$  term does not merely produce a finite part of the same gamma quotient; it yields an alternating Lambert-series tail.

**Theorem 18.4** (Deleted-singularity central identity). *Let  $m \geq 0$  and  $0 < q < 1$ . Then*

$$\begin{aligned} \sum_{\substack{k \geq 0 \\ k \neq m}} \begin{bmatrix} 2k \\ k \end{bmatrix}_{q^2} \frac{q^k}{(-q; q)_{2k} [k - m]_{q^2}} \\ = \begin{bmatrix} 2m \\ m \end{bmatrix}_{q^2} \frac{(1 + q)q^m}{(-q; q)_{2m}} \sum_{r=2m+1}^{\infty} \frac{(-1)^{r-1} q^r}{[r]_q}. \end{aligned} \quad (18.6)$$

*Proof.* By Lemma 18.1, the left side is

$$(1 - q^2) \sum_{\substack{k \geq 0 \\ k \neq m}} \frac{(q; q^2)_k q^k}{(q^2; q^2)_k (1 - q^{2k-2m})}. \quad (18.7)$$

Introduce an auxiliary parameter  $a$  and define

$$\begin{aligned} F(a, q) := \frac{1}{1 - aq^{-2m}} \left( {}_2\phi_1 \left[ \begin{matrix} q, aq^{-2m} \\ aq^{-2m+2}; q^2, q \end{matrix} \right] \right. \\ \left. - \frac{(q, aq^{-2m}; q^2)_m q^m}{(q^2, aq^{-2m+2}; q^2)_m} \right). \end{aligned} \quad (18.8)$$

The elementary cancellation

$$\frac{(aq^{-2m}; q^2)_k}{(aq^{-2m+2}; q^2)_k} = \frac{1 - aq^{-2m}}{1 - aq^{2k-2m}}$$

shows that

$$F(a, q) = \sum_{\substack{k \geq 0 \\ k \neq m}} \frac{(q; q^2)_k q^k}{(q^2; q^2)_k (1 - aq^{2k-2m})}. \quad (18.9)$$

Thus (18.7) equals  $(1 - q^2) \lim_{a \rightarrow 1} F(a, q)$ .

Apply  $q$ -Gauss with base  $q^2$  to the  ${}_2\phi_1$  in (18.8). We obtain

$$\begin{aligned} F(a, q) &= \frac{1}{1 - aq^{-2m}} \left( \frac{(q^2, aq^{-2m+1}; q^2)_{\infty}}{(q, aq^{-2m+2}; q^2)_{\infty}} - \frac{(q, aq^{-2m}; q^2)_m q^m}{(q^2, aq^{-2m+2}; q^2)_m} \right) \\ &= \frac{H(a) - C_m}{1 - a}, \end{aligned} \quad (18.10)$$

where

$$\begin{aligned} H(a) &= \frac{(q^2, aq; q^2)_{\infty} (aq^{-2m+1}; q^2)_m}{(q, aq^2; q^2)_{\infty} (aq^{-2m}; q^2)_m}, \\ C_m &= \frac{(q; q^2)_m q^m}{(q^2; q^2)_m}. \end{aligned}$$

The reversal identity for finite shifted factorials gives

$$\frac{(q^{-2m+1}; q^2)_m}{(q^{-2m}; q^2)_m} = q^m \frac{(q; q^2)_m}{(q^2; q^2)_m},$$

so  $H(1) = C_m$ . Hence L'Hospital's rule yields

$$\lim_{a \rightarrow 1} F(a, q) = -H'(1) = -H(1) \left. \frac{\partial}{\partial a} \log H(a) \right|_{a=1}. \quad (18.11)$$

Differentiating the logarithms of the product factors gives

$$\lim_{a \rightarrow 1} F(a, q) = C_m \left( \sum_{r=1}^{\infty} \frac{(-1)^{r-1} q^r}{1 - q^r} + \sum_{r=1}^{2m} \frac{(-1)^{r-1} q^{-r}}{1 - q^{-r}} \right). \quad (18.12)$$

For  $r \geq 1$ ,

$$\frac{q^{-r}}{1 - q^{-r}} = -\frac{1}{1 - q^r}.$$

Consequently, the two summands in (18.12) combine, for  $1 \leq r \leq 2m$ , to  $(-1)^r$ . Their sum is 0 because  $2m$  is even. Therefore

$$\lim_{a \rightarrow 1} F(a, q) = C_m \sum_{r=2m+1}^{\infty} \frac{(-1)^{r-1} q^r}{1 - q^r}. \quad (18.13)$$

Multiply by  $1 - q^2 = (1 + q)(1 - q)$ , use  $(1 - q)/(1 - q^r) = 1/[r]_q$ , and finally use (18.1) at  $k = m$  to rewrite

$$C_m = q^m \frac{(q; q^2)_m}{(q^2; q^2)_m} = q^m \begin{bmatrix} 2m \\ m \end{bmatrix}_{q^2} \frac{1}{(-q; q)_{2m}}.$$

The result is exactly (18.6).  $\square$

*Remark 18.5* (The mechanism). The deleted term is encoded by the auxiliary parameter  $a$ . The  $q$ -Gauss product has a removable zero-over-zero singularity at  $a = 1$ , and its logarithmic derivative turns products into Lambert series. This parameter-differentiation mechanism recurs throughout the theory of harmonic-number and Lambert-series refinements of hypergeometric summations.

## 18.4 Separate finite and infinite components

The deleted sum in Theorem 18.4 naturally splits at the missing index. Each component has a simpler one-sided Lambert-series evaluation.

**Theorem 18.6** (Infinite and finite components). *Let  $m \geq 0$  and  $0 < q < 1$ , and put*

$$P_m(q) = \begin{bmatrix} 2m \\ m \end{bmatrix}_{q^2} \frac{(1 + q)q^m}{(-q; q)_{2m}}.$$

*Then*

$$\begin{aligned} & \sum_{k=m+1}^{\infty} \begin{bmatrix} 2k \\ k \end{bmatrix}_{q^2} \frac{q^k}{(-q; q)_{2k} [k - m]_{q^2}} \\ &= P_m(q) \left( \sum_{r=1}^{\infty} \frac{q^{2r-1}}{[2r-1]_q} - \sum_{r=1}^{\infty} \frac{q^{2r+2m}}{[2r+2m]_q} \right), \end{aligned} \quad (18.14)$$

and

$$\sum_{k=0}^{m-1} \begin{bmatrix} 2k \\ k \end{bmatrix}_{q^2} \frac{q^k}{(-q; q)_{2k} [k-m]_{q^2}} = -P_m(q) \sum_{r=1}^m \frac{q^{2r-1}}{[2r-1]_q}. \quad (18.15)$$

For  $m = 0$ , the finite sum and the sum on its right are both empty.

*Proof.* By Lemma 18.1, shifting  $k \mapsto k + m$  in the infinite component gives

$$\begin{aligned} & \sum_{k=m+1}^{\infty} \begin{bmatrix} 2k \\ k \end{bmatrix}_{q^2} \frac{q^k}{(-q; q)_{2k} [k-m]_{q^2}} \\ &= (1 - q^2) q^m \frac{(q; q^2)_m}{(q^2; q^2)_m} \sum_{k=1}^{\infty} \frac{(q^{2m+1}; q^2)_k q^k}{(q^{2m+2}; q^2)_k (1 - q^{2k})}. \end{aligned}$$

Introduce

$$U(a, q) = \frac{1}{1-a} \left( {}_2\phi_1 \left[ \begin{matrix} a, q^{2m+1} \\ q^{2m+2} \end{matrix}; q^2, a^{-1}q \right] - 1 \right).$$

For  $a$  near 1, absolute convergence permits termwise passage to the limit. Since

$$\frac{(a; q^2)_k}{1-a} = (aq^2; q^2)_{k-1} \quad (k \geq 1),$$

we obtain

$$\lim_{a \rightarrow 1} U(a, q) = \sum_{k=1}^{\infty} \frac{(q^{2m+1}; q^2)_k q^k}{(q^{2m+2}; q^2)_k (1 - q^{2k})}.$$

The  $q$ -Gauss summation, at base  $q^2$ , evaluates the auxiliary series as

$$U(a, q) = \frac{1}{1-a} \left( \frac{(q, a^{-1}q^{2m+2}; q^2)_{\infty}}{(a^{-1}q, q^{2m+2}; q^2)_{\infty}} - 1 \right).$$

The product quotient tends to 1 as  $a \rightarrow 1$ . L'Hospital's rule and termwise logarithmic differentiation, justified locally uniformly, give

$$\lim_{a \rightarrow 1} U(a, q) = \sum_{r=1}^{\infty} \frac{q^{2r-1}}{1 - q^{2r-1}} - \sum_{r=1}^{\infty} \frac{q^{2r+2m}}{1 - q^{2r+2m}}.$$

Multiplication by the prefactor above, followed by  $(1-q)/(1-q^j) = 1/[j]_q$  and Lemma 18.1 at  $k = m$ , proves (18.14).

To obtain the finite component, subtract (18.14) from the full deleted sum (18.6). Indeed,

$$\begin{aligned} \sum_{j=2m+1}^{\infty} \frac{(-1)^{j-1} q^j}{[j]_q} &= \sum_{r=m+1}^{\infty} \frac{q^{2r-1}}{[2r-1]_q} - \sum_{r=m+1}^{\infty} \frac{q^{2r}}{[2r]_q} \\ &= \sum_{r=1}^{\infty} \frac{q^{2r-1}}{[2r-1]_q} - \sum_{r=1}^{\infty} \frac{q^{2r+2m}}{[2r+2m]_q} - \sum_{r=1}^m \frac{q^{2r-1}}{[2r-1]_q}. \end{aligned}$$

The first two sums constitute the infinite component, leaving exactly (18.15).  $\square$

## 18.5 The classical finite part

Define the alternating harmonic number

$$\overline{H}_N = \sum_{r=1}^N \frac{(-1)^{r-1}}{r}.$$

**Theorem 18.7** (Classical deleted-singularity identity). *For every  $m \geq 0$ ,*

$$\sum_{\substack{k \geq 0 \\ k \neq m}} \binom{2k}{k} \frac{1}{4^k(k-m)} = \binom{2m}{m} \frac{\log 2 - \overline{H}_{2m}}{2^{2m-1}}. \quad (18.16)$$

*Proof.* Put  $c_k = (1/2)_k/k! = \binom{2k}{k}/4^k$ . For  $a$  away from the nonpositive integers, the series

$$T(a) = \sum_{k=0}^{\infty} \frac{c_k}{k+a}$$

converges locally uniformly and defines a meromorphic function. For  $\Re a > 0$ , termwise integration of  $(1-x)^{-1/2} = \sum_{k \geq 0} c_k x^k$  gives

$$T(a) = \int_0^1 x^{a-1} (1-x)^{-1/2} dx = B(a, 1/2) = \frac{\Gamma(a)\Gamma(1/2)}{\Gamma(a+1/2)}. \quad (18.17)$$

Analytic continuation makes (18.17) valid wherever both sides are defined. The desired deleted sum is the finite part at  $a = -m$ :

$$\lim_{a \rightarrow -m} \left( T(a) - \frac{c_m}{a+m} \right). \quad (18.18)$$

Write  $a = -m + \varepsilon$ . The standard expansions of the gamma function at a negative integer and at a regular point are

$$\begin{aligned} \Gamma(-m + \varepsilon) &= \frac{(-1)^m}{m! \varepsilon} (1 + (H_m - \gamma)\varepsilon + O(\varepsilon^2)), \\ \Gamma(1/2 - m + \varepsilon) &= \Gamma(1/2 - m) (1 + \psi(1/2 - m)\varepsilon + O(\varepsilon^2)). \end{aligned}$$

Moreover,

$$\frac{(-1)^m \Gamma(1/2)}{m! \Gamma(1/2 - m)} = c_m.$$

Thus the finite part in (18.18) is

$$c_m (H_m - \gamma - \psi(1/2 - m)). \quad (18.19)$$

Using  $\psi(1/2) = -\gamma - 2 \log 2$  and the recurrence  $\psi(z+1) = \psi(z) + 1/z$ , we find

$$\psi(1/2 - m) = -\gamma - 2 \log 2 + 2 \sum_{r=1}^m \frac{1}{2r-1}.$$

Since

$$\overline{H}_{2m} = \sum_{r=1}^m \frac{1}{2r-1} - \frac{1}{2} H_m,$$

the bracket in (18.19) is  $2(\log 2 - \overline{H}_{2m})$ . Hence the finite part equals

$$2 \frac{\binom{2m}{m}}{4^m} (\log 2 - \overline{H}_{2m}),$$

which is (18.16). □

**Proposition 18.8** (Limit of the Lambert tail). *For fixed  $m \geq 0$ ,*

$$\lim_{q \rightarrow 1^-} \sum_{r=2m+1}^{\infty} \frac{(-1)^{r-1} q^r}{[r]_q} = \log 2 - \overline{H}_{2m}. \quad (18.20)$$

*Proof.* For  $0 < q < 1$ , set  $a_r(q) = q^r / [r]_q$ . These numbers are positive and decrease with  $r$ , since

$$\frac{a_{r+1}(q)}{a_r(q)} = q \frac{[r]_q}{[r+1]_q} < 1.$$

Moreover,

$$a_r(q) = \frac{q^r}{1 + q + \dots + q^{r-1}} \leq \frac{1}{r},$$

because every summand in the denominator is at least  $q^{r-1}$ . The alternating-series remainder after  $N$  terms therefore has absolute value at most  $a_{N+1}(q) \leq 1/(N+1)$ , uniformly in  $q$ . For each fixed  $r$ ,  $a_r(q) \rightarrow 1/r$  as  $q \rightarrow 1^-$ . First pass to the limit in a finite partial sum and then let  $N \rightarrow \infty$ . The result is the tail of the alternating harmonic series,

$$\sum_{r=2m+1}^{\infty} \frac{(-1)^{r-1}}{r} = \log 2 - \overline{H}_{2m}.$$

□

## Chapter 19

# Asymptotic regimes and Bernoulli expansions

A  $q$ -analogue should explain both the deformation near  $q = 1$  and behavior far from it. Gaussian coefficients have especially clean asymptotic descriptions because their logarithms are finite sums of a single elementary function.

### 19.1 A complete expansion as $q \rightarrow 1$

Let the Bernoulli numbers be defined by

$$\frac{x}{e^x - 1} = \sum_{r=0}^{\infty} B_r \frac{x^r}{r!}.$$

**Lemma 19.1** (The basic logarithmic expansion). As  $x \rightarrow 0$ ,

$$\log \frac{1 - e^{-x}}{x} = -\frac{x}{2} + \sum_{r=1}^{\infty} \frac{B_{2r}}{2r(2r)!} x^{2r}. \quad (19.1)$$

The series converges for  $|x| < 2\pi$ .

*Proof.* Differentiate the left side:

$$\frac{d}{dx} \log \frac{1 - e^{-x}}{x} = \frac{1}{e^x - 1} - \frac{1}{x}.$$

By the Bernoulli generating function,

$$\frac{1}{e^x - 1} - \frac{1}{x} = -\frac{1}{2} + \sum_{r=1}^{\infty} B_{2r} \frac{x^{2r-1}}{(2r)!}.$$

Integrate term by term and use the value 0 at  $x = 0$ . The nearest nonzero singularities are at  $x = \pm 2\pi i$ , giving radius  $2\pi$ .  $\square$

**Theorem 19.2** (Bernoulli expansion of a Gaussian coefficient). Let  $0 \leq k \leq n$ , put  $D = k(n - k)$ , and set  $q = e^{-t}$ . As  $t \rightarrow 0$ ,

$$\log \frac{[n]_k e^{-t}}{\binom{n}{k}} = -\frac{D}{2}t + \sum_{r=1}^{\infty} \frac{B_{2r} t^{2r}}{2r(2r)!} \sum_{j=1}^k ((n - k + j)^{2r} - j^{2r}). \quad (19.2)$$

For fixed  $n, k$ , the displayed series converges whenever  $|t| < 2\pi/n$ .

*Proof.* From the product formula,

$$\begin{aligned} \left[ \begin{matrix} n \\ k \end{matrix} \right]_{e^{-t}} &= \prod_{j=1}^k \frac{1 - e^{-(n-k+j)t}}{1 - e^{-jt}} \\ &= \binom{n}{k} \prod_{j=1}^k \frac{(1 - e^{-(n-k+j)t})/((n-k+j)t)}{(1 - e^{-jt})/(jt)}. \end{aligned}$$

Take logarithms and apply [Lemma 19.1](#) to every factor. The linear term is

$$-\frac{t}{2} \sum_{j=1}^k (n-k) = -\frac{1}{2} k(n-k)t = -\frac{D}{2}t,$$

and the even terms give the sum in (19.2). Every argument of the basic series has modulus below  $2\pi$  when  $|t| < 2\pi/n$ , so the finite sum converges there.  $\square$

*Remark 19.3* (Power-sum form of the coefficients). Let  $S_m(N) = \sum_{j=1}^N j^m$  denote the  $m$ th power sum. For every integer  $m \geq 1$ , reindexing the first half of an inner sum by  $i = n - k + j$  gives

$$\sum_{j=1}^k ((n-k+j)^m - j^m) = \sum_{i=n-k+1}^n i^m - \sum_{j=1}^k j^m = S_m(n) - S_m(k) - S_m(n-k).$$

Applied with  $m = 2r$  to the inner sum of (19.2), and with  $m = 1$  to the sum  $\sum_{j=1}^k ((n-k+j) - j) = \sum_{j=1}^k (n-k)$  that produced the linear term in the proof of [Theorem 19.2](#), this turns the expansion into

$$\log \frac{\left[ \begin{matrix} n \\ k \end{matrix} \right]_{e^{-t}}}{\binom{n}{k}} = -\frac{t}{2} (S_1(n) - S_1(k) - S_1(n-k)) + \sum_{r=1}^{\infty} \frac{B_{2r} t^{2r}}{2r(2r)!} (S_{2r}(n) - S_{2r}(k) - S_{2r}(n-k)),$$

still for  $|t| < 2\pi/n$ , the factor  $-\frac{1}{2}$  being the coefficient of  $x$  in (19.1). The case  $m = 1$  of the display evaluates the linear bracket outright:  $S_1(n) - S_1(k) - S_1(n-k) = \sum_{j=1}^k (n-k) = k(n-k) = D$ .

Three features become visible in this form. First, each coefficient is a difference of Faulhaber polynomials, hence itself a polynomial in  $n$  and  $k$  with rational coefficients, of total degree exactly  $2r+1$ : since  $S_{2r}(N) = N^{2r+1}/(2r+1) + O(N^{2r})$ , the homogeneous part of top degree is  $(n^{2r+1} - k^{2r+1} - (n-k)^{2r+1})/(2r+1)$ , whose coefficient of  $n^{2r}k$  is  $-\frac{1}{2r+1} \binom{2r+1}{1}(-1) = 1$ , so no cancellation lowers the degree. Second, the expression is manifestly invariant under  $k \mapsto n-k$ , as it must be by the symmetry recorded in [Theorem 2.5](#). Third, at  $r = 1$  the identity  $S_2(N) = N(N+1)(2N+1)/6$  gives

$$\begin{aligned} S_2(n) - S_2(k) - S_2(n-k) &= \frac{n(n+1)(2n+1) - k(k+1)(2k+1)}{6} \\ &\quad - \frac{(n-k)(n-k+1)(2n-2k+1)}{6} = k(n-k)(n+1) = D(n+1), \end{aligned}$$

the value used in [Corollary 19.4](#) below.

[The second "

**Corollary 19.4** (First terms). *With  $D = k(n-k)$ ,*

$$\left[ \begin{matrix} n \\ k \end{matrix} \right]_{e^{-t}} = \binom{n}{k} \exp \left( -\frac{D}{2}t + \frac{D(n+1)}{24}t^2 + O(t^4) \right), \quad (19.3)$$

$$= \binom{n}{k} \left( 1 - \frac{D}{2}t + \left( \frac{D^2}{8} + \frac{D(n+1)}{24} \right) t^2 + O(t^3) \right). \quad (19.4)$$

*Proof.* For  $r = 1$  in (19.2),

$$\begin{aligned} \sum_{j=1}^k ((n-k+j)^2 - j^2) &= k(n-k)^2 + (n-k)k(k+1) \\ &= k(n-k)(n+1) = D(n+1). \end{aligned}$$

Since  $B_2/(2 \cdot 2!) = 1/24$ , the logarithmic formula follows. Expanding the exponential gives the second formula.  $\square$

*Remark 19.5* (Reciprocity and centering). The factor  $e^{-Dt/2}$  is forced by reciprocity:

$$\begin{bmatrix} n \\ k \end{bmatrix}_{e^{-t}} = e^{-Dt} \begin{bmatrix} n \\ k \end{bmatrix}_{e^t}.$$

Therefore the centered quantity  $e^{Dt/2} \begin{bmatrix} n \\ k \end{bmatrix}_{e^{-t}}$  is an even analytic function of  $t$ . This explains, before any calculation, why all odd powers beyond the linear term vanish from the logarithm.

## 19.2 Fixed $q$ and growing parameters

**Theorem 19.6** (Fixed-column limit). *If  $0 < |q| < 1$  and  $k$  is fixed, then*

$$\lim_{n \rightarrow \infty} \begin{bmatrix} n \\ k \end{bmatrix}_q = \frac{1}{(q; q)_k}. \quad (19.5)$$

*More precisely,*

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = \frac{1}{(q; q)_k} \left( 1 + O(|q|^{n-k+1}) \right). \quad (19.6)$$

*Proof.* The product formula gives

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = \frac{(q^{n-k+1}; q)_k}{(q; q)_k}.$$

The numerator tends to 1. Since the number of factors is fixed,

$$\prod_{j=0}^{k-1} (1 - q^{n-k+1+j}) = 1 + O(|q|^{n-k+1}),$$

which proves both assertions.  $\square$

**Theorem 19.7** (Fixed- $q$  limit with both indices growing). *Let  $0 < |q| < 1$ . For all integers  $0 \leq k \leq n$ ,*

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = \frac{(q^{k+1}; q)_\infty (q^{n-k+1}; q)_\infty}{(q; q)_\infty (q^{n+1}; q)_\infty},$$

*and consequently*

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = \frac{1}{(q; q)_\infty} \left( 1 + O(|q|^{\min(k, n-k)+1}) \right),$$

*the implied constant depending only on  $q$ . In particular,*

$$\lim_{\min(k, n-k) \rightarrow \infty} \begin{bmatrix} n \\ k \end{bmatrix}_q = \frac{1}{(q; q)_\infty}, \quad (19.7)$$



and for every fixed  $r \in \mathbb{Z}$ ,

$$\lim_{m \rightarrow \infty} \left[ \begin{matrix} 2m \\ m+r \end{matrix} \right]_q = \frac{1}{(q; q)_\infty},$$

the coefficient being read for  $m \geq |r|$ , so that  $m+r$  and  $m-r$  are admissible indices.

*Proof.* By [Theorem 1.12](#) applied at  $a = q$ , the product  $(q; q)_\infty$  converges, and it is nonzero: the zeros of  $a \mapsto (a; q)_\infty$  are the points  $q^{-j}$  with  $j \geq 0$ , and  $q = q^{-j}$  would give  $q^{j+1} = 1$ , hence  $|q| = 1$ . The same theorem at  $a = q^{N+1}$  gives  $(q^{N+1}; q)_\infty \neq 0$  for every  $N \geq 0$ . The tail factorization of [Proposition 1.2](#),

$$(q; q)_\infty = (q; q)_N (q^{N+1}; q)_\infty,$$

therefore yields  $(q; q)_N = (q; q)_\infty / (q^{N+1}; q)_\infty$ . Substituting this at  $N = n$ ,  $N = k$ , and  $N = n - k$  in the factorial form  $\left[ \begin{matrix} n \\ k \end{matrix} \right]_q = (q; q)_n / ((q; q)_k (q; q)_{n-k})$  gives the first display.

For the error term, let  $M \geq 0$ . By the elementary product estimate,

$$|(q^{M+1}; q)_\infty - 1| \leq \prod_{j \geq M+1} (1 + |q|^j) - 1 \leq \exp\left(\frac{|q|^{M+1}}{1 - |q|}\right) - 1 \leq C_q |q|^{M+1}, \quad C_q = \frac{e^{1/(1-|q|)}}{1 - |q|},$$

since  $|q|^{M+1}/(1 - |q|) \leq 1/(1 - |q|)$  and  $e^x - 1 \leq xe^{x_0}$  for  $0 \leq x \leq x_0$ . Moreover

$$|(q^{M+1}; q)_\infty| \geq \prod_{j \geq 1} (1 - |q|^j) = (|q|; |q|)_\infty > 0$$

uniformly in  $M$ . As  $\min(k, n - k) \leq n$ , applying both bounds at  $M = k$ ,  $M = n - k$ , and  $M = n$  to the first display gives the second.

The two limits follow at once: in the first,  $\min(k, n - k) \rightarrow \infty$  by hypothesis; in the second,  $n = 2m$  and  $k = m + r$  give  $\min(k, n - k) = m - |r| \rightarrow \infty$ , which is also what makes both indices admissible once  $m \geq |r|$ .  $\square$

*Remark 19.8.* The off-centre column in the regime  $q > 1$  needs no new argument. By [\(19.9\)](#) at upper index  $2m$  and lower index  $m + r$ , whose exponent is  $k(n - k) = (m + r)(m - r) = m^2 - r^2$ ,

$$\left[ \begin{matrix} 2m \\ m+r \end{matrix} \right]_q = q^{m^2-r^2} \left[ \begin{matrix} 2m \\ m+r \end{matrix} \right]_{q^{-1}},$$

and [Theorem 19.7](#) applied with  $q^{-1}$  in place of  $q$ , which is legitimate because  $0 < q^{-1} < 1$ , gives  $\left[ \begin{matrix} 2m \\ m+r \end{matrix} \right]_{q^{-1}} \rightarrow (q^{-1}; q^{-1})_\infty^{-1}$ . Hence

$$\left[ \begin{matrix} 2m \\ m+r \end{matrix} \right]_q \sim \frac{q^{m^2-r^2}}{(q^{-1}; q^{-1})_\infty},$$

which is [\(19.11\)](#) at  $r = 0$ .

**Proposition 19.9** (Uniform elementary bound). *For  $0 < q < 1$  and  $0 \leq k \leq n$ ,*

$$1 \leq \left[ \begin{matrix} n \\ k \end{matrix} \right]_q \leq \frac{1}{(q; q)_\infty}. \quad (19.8)$$

*Proof.* The lower bound follows from the partition interpretation: the constant term is 1 and every coefficient is nonnegative. The same interpretation says that  $\left[ \begin{matrix} n \\ k \end{matrix} \right]_q$  is the generating function of partitions fitting in a  $k$  by  $(n - k)$  rectangle. Dropping the rectangle restriction embeds this set into all partitions, whose generating function is  $(q; q)_\infty^{-1}$ . Evaluating at a positive  $q < 1$  gives the upper bound.  $\square$

### 19.3 The regime $q > 1$

**Corollary 19.10** (Transfer through reciprocity). *Let  $q > 1$  and  $D = k(n - k)$ . Then*

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = q^D \begin{bmatrix} n \\ k \end{bmatrix}_{q^{-1}}. \quad (19.9)$$

Consequently, for fixed  $k$ ,

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = \frac{q^{k(n-k)}}{(q^{-1}; q^{-1})_k} \left( 1 + O(q^{-n+k-1}) \right), \quad (19.10)$$

and

$$\begin{bmatrix} 2m \\ m \end{bmatrix}_q \sim \frac{q^{m^2}}{(q^{-1}; q^{-1})_\infty}. \quad (19.11)$$

*Proof.* The reciprocity identity was proved in [Theorem 2.5](#). Apply [Theorems 19.6](#) and [19.7](#) with  $q^{-1}$  in place of  $q$ .  $\square$

### 19.4 Extreme specializations

**Proposition 19.11.** *For  $0 \leq k \leq n$ ,*

$$\begin{bmatrix} n \\ k \end{bmatrix}_q \Big|_{q=0} = 1, \quad \lim_{q \rightarrow 1} \begin{bmatrix} n \\ k \end{bmatrix}_q = \binom{n}{k},$$

*and the leading term as  $q \rightarrow \infty$  is  $q^{k(n-k)}$  with coefficient 1.*

*Proof.* The partition generating polynomial has constant term 1, proving the first assertion. The second follows by replacing every ratio  $(1 - q^a)/(1 - q^b)$  in the product formula by its limit  $a/b$ . Palindromicity and degree  $k(n - k)$  show that the leading coefficient equals the constant coefficient, namely 1.  $\square$

## Chapter 20

# Computation, certification, and formal proof architecture

The identities in this book are well suited to exact computation because their natural objects are polynomials, rational functions, and formal power series. They are also well suited to proof assistants, provided algebraic identities are separated cleanly from analytic convergence. Lau, Lee, and Ono’s 2026 formalization project [20] demonstrates this architecture for  $q$ -Pochhammer symbols, Bailey pairs, Jacobi’s triple product, and the Rogers–Ramanujan identities.

### 20.1 Dynamic programming for Gaussian polynomials

A polynomial is represented here by its coefficient vector. Multiplication by  $q^h$  is therefore a shift by  $h$  places.

**Algorithm 20.1** (Pascal dynamic program). Given  $n \geq 0$  and  $0 \leq k \leq n$ :

1. Replace  $k$  by  $\min(k, n - k)$ , using symmetry.
2. Initialize  $G_0(q) = 1$  and  $G_s(q) = 0$  for  $1 \leq s \leq k$ .
3. For  $r = 1, 2, \dots, n$ , update in descending order

$$G_s(q) \leftarrow G_s(q) + q^{r-s} G_{s-1}(q) \quad (s = \min(r, k), \dots, 1).$$

4. Return  $G_k(q)$ .

**Theorem 20.2** (Correctness and complexity). *Algorithm 20.1 returns  $\begin{bmatrix} n \\ k \end{bmatrix}_q$ . If  $D = k(n - k)$  after the symmetry reduction, its coefficient-level arithmetic cost is  $O(nkD)$  and its storage cost is  $O(kD)$ .*

*Proof.* Let  $G_s^{(r)}$  denote the value after row  $r$ . We prove by induction on  $r$  that

$$G_s^{(r)} = \begin{bmatrix} r \\ s \end{bmatrix}_q \quad (0 \leq s \leq \min(r, k)).$$

The assertion is true at  $r = 0$ . Descending updates ensure that both  $G_s$  and  $G_{s-1}$  on the right are still values from row  $r - 1$ . Hence the update is

$$G_s^{(r)} = \begin{bmatrix} r-1 \\ s \end{bmatrix}_q + q^{r-s} \begin{bmatrix} r-1 \\ s-1 \end{bmatrix}_q = \begin{bmatrix} r \\ s \end{bmatrix}_q$$

by the second  $q$ -Pascal recurrence. At  $r = n$  this gives the desired output.

After replacing  $k$  by  $\min(k, n - k)$ , we have  $k \leq n/2$ . Every intermediate polynomial  $\begin{bmatrix} r \\ s \end{bmatrix}_q$  with  $s \leq k$  has degree  $s(r - s) \leq k(n - k) = D$ . There are at most  $nk$  shift-and-add updates, each touching at most  $D + 1$  coefficients. Storing  $k + 1$  such vectors uses  $O(kD)$  coefficients.  $\square$

*Remark 20.3* (Numerical specialization). When only a numerical value is required, one should not first expand the polynomial. For positive  $q$  near 1, the stable formula is

$$\log \begin{bmatrix} n \\ k \end{bmatrix}_q = \sum_{j=1}^k [\log(-\expm1((n - k + j) \log q)) - \log(-\expm1(j \log q))],$$

where  $\expm1$  avoids cancellation in  $e^x - 1$ . For exact symbolic work, the Pascal algorithm or the cyclotomic factorization is preferable.

## 20.2 A factorization algorithm

**Algorithm 20.4** (Cyclotomic factor list). For each  $1 \leq d \leq n$ , compute

$$e_d = \lfloor n/d \rfloor - \lfloor k/d \rfloor - \lfloor (n - k)/d \rfloor.$$

Return the list of  $d$  for which  $e_d = 1$ .

**Theorem 20.5** (Correctness of the factor list). *The output of Algorithm 20.4 is exactly the set of cyclotomic factors of  $\begin{bmatrix} n \\ k \end{bmatrix}_q$ , each with multiplicity one.*

*Proof.* This is Theorem 16.1: the exponent of  $\Phi_d(q)$  is  $e_d$ , and the floor calculation in that theorem proves  $e_d \in \{0, 1\}$ .  $\square$

*Remark 20.6.* The factor-list algorithm takes only  $O(n)$  integer floor operations. It is therefore far cheaper than expanding  $\begin{bmatrix} n \\ k \end{bmatrix}_q$  when the purpose is root-of-unity evaluation, divisibility testing, or congruence work. Polynomial expansion may be postponed until the selected cyclotomic factors are actually multiplied.

## 20.3 Telescoping certificates for finite sums

A term  $T(n, k)$  is called  $q$ -hypergeometric if the ratios

$$\frac{T(n, k + 1)}{T(n, k)}, \quad \frac{T(n + 1, k)}{T(n, k)}$$

are rational functions of  $q^n$  and  $q^k$  (and of any fixed parameters). Shifted-factorial terms have this property because adjacent factors cancel.

**Theorem 20.7** (Finite telescoping certificate). *Let*

$$S_n = \sum_{k=L(n)}^{U(n)} T(n, k).$$

*Suppose there are functions  $A_0(n), \dots, A_r(n)$  and a certificate  $G(n, k)$  such that*

$$\sum_{j=0}^r A_j(n) T(n + j, k) = G(n, k + 1) - G(n, k) \tag{20.1}$$

for all relevant  $k$ , after extending  $T$  by zero outside its natural finite range. If  $G(n, k)$  vanishes at both ends of the resulting finite summation interval, then

$$\sum_{j=0}^r A_j(n) S_{n+j} = 0. \quad (20.2)$$

*Proof.* Sum (20.1) over an interval containing the support of every  $T(n + j, k)$ . The left side becomes  $\sum_j A_j(n) S_{n+j}$ . The right side telescopes to the terminal value of  $G$  minus its initial value, which is 0 by hypothesis.  $\square$

**Corollary 20.8** (Identity certification). *If two finite sums satisfy the same recurrence of order  $r$  and agree for  $r$  consecutive initial values at every nonsingular parameter choice, then they are equal wherever the recurrence determines subsequent values.*

*Proof.* Subtract the two sequences. The difference satisfies the homogeneous recurrence and has  $r$  consecutive zero initial values. Induction through the recurrence gives zero thereafter.  $\square$

**Remark 20.9** (Creative telescoping). The  $q$ -Zeilberger method [23] searches algorithmically for a rational multiple  $G(n, k) = R(q^n, q^k)T(n, k)$  satisfying (20.1). Once found, the certificate is independently checkable by rational-function arithmetic. The search may be sophisticated, but verification is elementary: divide by  $T(n, k)$  and clear denominators.

## 20.4 Denominator clearing and specialization

**Lemma 20.10** (Polynomial identity principle). *Let  $R(q), S(q) \in \mathbb{Q}(q)$ . If  $R(q) = S(q)$  at infinitely many complex values of  $q$  where both sides are defined, then  $R = S$  in  $\mathbb{Q}(q)$ .*

*Proof.* Write  $R - S = A/B$  with  $A, B \in \mathbb{Q}[q]$  and  $B \neq 0$ . The hypothesis says that  $A$  has infinitely many zeros. A nonzero polynomial has only finitely many zeros, so  $A = 0$ .  $\square$

**Corollary 20.11** (Safe generic specialization). *A rational  $q$ -identity proved under generic nonvanishing assumptions remains valid after any specialization at which the denominator-cleared identity is defined. Apparent excluded values may therefore be recovered by cancellation or by taking limits.*

*Proof.* Clear all denominators to obtain a polynomial identity. Polynomial evaluation is valid at every specialization. If the original rational expressions have removable singularities, divide out common factors or take their uniquely determined limits.  $\square$

This principle is particularly important at roots of unity. One should first rewrite a Gaussian coefficient as its polynomial, not substitute a root into the factorial quotient and accept an indeterminate  $0/0$ .

## 20.5 Integer-exponent translation and automated identity discovery

A large class of Gaussian identities can be generated by starting from a terminating basic hypergeometric summation and replacing every shifted factorial whose parameter is an integral power of  $q$  by factorial blocks. This old method is now amenable to systematic symbolic search; the identity finder of Zhong and Zhao [40] organizes the necessary side conditions and backtracking choices. The proof certificate, however, remains elementary substitution.

**Lemma 20.12** (Two translation rules). *For nonnegative integers  $a, n, m, k$  with  $0 \leq k \leq m$ ,*

$$\frac{(q^{a+1}; q)_n}{(q; q)_n} = \begin{bmatrix} a+n \\ n \end{bmatrix}_q, \quad (20.1)$$

$$\frac{(q^{-m}; q)_k}{(q; q)_k} = (-1)^k q^{-mk + \binom{k}{2}} \begin{bmatrix} m \\ k \end{bmatrix}_q. \quad (20.2)$$

More generally,

$$(q^{a+1}; q)_n = \frac{(q; q)_{a+n}}{(q; q)_a}. \quad (20.3)$$

*Proof.* The factors in (20.3) are precisely  $1 - q^{a+1}, \dots, 1 - q^{a+n}$ . Division by  $(q; q)_n$  gives (20.1). For the negative rule, factor  $-q^{-m+j}$  from the  $j$ th factor:

$$\begin{aligned} (q^{-m}; q)_k &= \prod_{j=0}^{k-1} (1 - q^{-m+j}) \\ &= (-1)^k q^{-mk + \binom{k}{2}} \prod_{j=0}^{k-1} (1 - q^{m-j}) \\ &= (-1)^k q^{-mk + \binom{k}{2}} \frac{(q; q)_m}{(q; q)_{m-k}}. \end{aligned}$$

Divide by  $(q; q)_k$  to obtain (20.2). □

**Theorem 20.13** (Certified integral-parameter translation). *Consider a terminating identity in which every summand and the closed form are products and quotients of monomials in  $q$  and shifted factorials of the forms*

$$(q^{a+1}; q)_n, \quad (q^{-m}; q)_k, \quad (q; q)_r,$$

*with integer-valued parameters satisfying side conditions that keep all displayed finite indices nonnegative and all original denominators nonzero. Replacing the factors by (20.1)–(20.3), cancelling common factorial blocks, and collecting monomial powers of  $q$  produces an algebraically equivalent identity in Gaussian coefficients and residual  $q$ -factorials. If all residual factorial multiplicities cancel, the output is a pure  $q$ -binomial identity.*

*Proof.* Each replacement is an equality by Lemma 20.12. Products of equal rational functions are equal, cancellation of a common nonzero factor preserves equality under the stated generic assumptions, and collecting monomials is ordinary exponent arithmetic. Hence the transformed summand agrees term by term with the original summand, and the transformed closed form agrees with the original closed form. Summing the finitely many terms preserves equality. Finally, clearing denominators turns the result into a polynomial identity, so removable exceptional specializations are recovered by Corollary 20.11. □

**Remark 20.14** (Constraint generation). The nontrivial search problem is not verification but representation: the same factorial block can participate in several Gaussian coefficients, and the integer parameters must satisfy inequalities such as  $0 \leq k \leq m$ . An automated finder records the multiplicity of each  $(q; q)_r$ , searches for compatible triples

$$\frac{(q; q)_u}{(q; q)_v (q; q)_{u-v}} = \begin{bmatrix} u \\ v \end{bmatrix}_q,$$

and carries the corresponding linear inequalities as side conditions. Once a candidate decomposition is printed, Theorem 20.13 reduces its proof to exact algebra.

**Corollary 20.15** (Mechanically checkable certificates). *A computer-generated integral-parameter translation can be certified without trusting the search program: expand every Gaussian coefficient into factorial quotients, apply base reversal to every negative parameter, clear denominators, and compare the resulting Laurent monomials and finite products.*

*Proof.* The prescribed steps reverse the equalities in [Lemma 20.12](#). After denominator clearing, equality is decidable by polynomial arithmetic. Therefore the search code may be treated as a conjecture generator while the emitted certificate is checked independently.  $\square$

*Historical note 20.16.* Recent formal libraries reinforce the same separation of concerns. Eberl’s Isabelle development [36] formalizes Gaussian polynomials, finite and infinite  $q$ -Pochhammer symbols, Euler expansions, the  $q$ -binomial theorem, and  $q$ -Vandermonde. The Lean developments cited earlier and the Fabius-function repository [38] similarly distinguish denominator-free polynomial recurrences from analytic infinite products. This architecture is exactly what makes automated discovery compatible with proof-level reliability.

## 20.6 Formal series versus analytic series

**Theorem 20.17** (Formal-to-analytic transfer). *Suppose*

$$\sum_{n \geq 0} a_n z^n = \sum_{n \geq 0} b_n z^n$$

*coefficientwise in  $\mathbb{C}[[z]]$ . If both series converge for  $|z| < R$ , then they define the same analytic function on that disk. Conversely, if two power series converge in a disk and define the same function on a subset with an accumulation point in the disk, then their coefficients are equal.*

*Proof.* The forward implication is immediate because the two partial sums agree term by term. For the converse, the identity theorem makes the analytic functions equal throughout the disk; differentiating  $n$  times at 0 gives  $n!a_n = n!b_n$ .  $\square$

**Principle 20.18** (A scalable proof architecture). *For a mixed algebraic-analytic  $q$ -identity, the following order minimizes hidden assumptions:*

1. *define finite shifted factorials and prove their algebraic laws;*
2. *prove terminating identities after denominator clearing;*
3. *define formal infinite products by coefficient stabilization when possible;*
4. *separately establish analytic convergence for  $|q| < 1$ ;*
5. *transfer the formal coefficient identity to analytic functions;*
6. *only then take limits, differentiate parameters, or specialize at roots of unity.*

*Proof.* Each step is justified by results already proved: finite product algebra in [Chapter 1](#); polynomial identity and specialization by [Lemma 20.10](#) and [corollary 20.11](#); formal-to-analytic transfer by [Theorem 20.17](#); and the convergence theorems in [Chapters 8 to 10](#). The ordering prevents an analytic operation from being used before its convergence hypothesis has been supplied.  $\square$

## 20.7 Necessary conditions for a proposed identity

The results of this book supply, besides proofs, a set of cheap falsification tests. They are worth isolating, because each is a *necessary* condition that can be checked without expanding anything.

**Principle 20.19** (Falsification tests). *Let  $L = R$  be a proposed identity whose two sides are built from Gaussian coefficients, shifted factorials, and explicit powers of  $q$ . Each of the following is a necessary condition for  $L = R$ .*

1. Degree and order. *If  $L$  and  $R$  are polynomials in  $q$ , then  $\deg L = \deg R$  and  $\text{ord}_{q=0} L = \text{ord}_{q=0} R$ . For a monomial  $F = q^\alpha \prod_i \begin{bmatrix} n_i \\ k_i \end{bmatrix}_q$  these are  $\alpha + \sum_i k_i(n_i - k_i)$  and  $\alpha$ , by [Theorem 2.5\(iii\)](#).*
2. Reversal. *Substituting  $q^{-1}$  for  $q$  in a true identity yields a true identity, and by [Theorem 2.5\(ii\),\(iv\)](#) that substitution acts on a monomial  $F$  as above by an explicit power:  $F(q^{-1}) = q^{-c_F} F(q)$  with  $c_F = \deg F + \text{ord } F$ . If  $L$  and  $R$  are such monomials with  $c_L \neq c_R$ , then  $L = R$  forces  $L = R = 0$ .*
3. The value  $q = 0$ . *After denominators have been cleared, the substitution  $q = 0$  must give a true numerical identity; here  $\begin{bmatrix} n \\ k \end{bmatrix}_q \big|_{q=0} = 1$  by [Proposition 19.11](#).*
4. The classical shadow. *After division by the appropriate power of  $1 - q$ , the limit  $q \rightarrow 1$  must be a true identity between ordinary binomial coefficients and factorials, since  $\begin{bmatrix} n \\ k \end{bmatrix}_q \rightarrow \binom{n}{k}$  by [Corollary 2.6](#); the standard scalings are collected in [Appendix B](#).*
5. Small indices. *Once the degree bound of item (i) is known, each side is determined by finitely many coefficients, so agreement at sufficiently many small indices is a genuine test; and if both sides satisfy one and the same certified recurrence, [Corollary 20.8](#) promotes finitely many agreements to a proof.*
6. Summand ratio. *If a side is presented as a basic hypergeometric series, the quotient of consecutive summands must be the rational function of  $q^k$  that the claimed  ${}_r\phi_s$  normalization predicts; this is the defining property of a  $q$ -hypergeometric term used in [Theorem 20.7](#). A mismatch localizes the error in the parameters rather than in the evaluation.*

*Proof.* Equal polynomials have equal degrees and equal orders at  $q = 0$ , and for a monomial  $F = q^\alpha \prod_i \begin{bmatrix} n_i \\ k_i \end{bmatrix}_q$  each factor has constant term 1 and degree  $k_i(n_i - k_i)$  by [Theorem 2.5\(iii\)](#); this proves item (i). For item (ii), [Theorem 2.5\(ii\)](#) gives  $\begin{bmatrix} n_i \\ k_i \end{bmatrix}_{q^{-1}} = q^{-k_i(n_i - k_i)} \begin{bmatrix} n_i \\ k_i \end{bmatrix}_q$ , so

$$F(q^{-1}) = q^{-\alpha} \prod_i q^{-k_i(n_i - k_i)} \begin{bmatrix} n_i \\ k_i \end{bmatrix}_q = q^{-c_F} F(q), \quad c_F = 2\alpha + \sum_i k_i(n_i - k_i) = \deg F + \text{ord } F,$$

using  $F(q) = q^\alpha \prod_i \begin{bmatrix} n_i \\ k_i \end{bmatrix}_q$ . If  $L = R$ , then applying the substitution gives  $q^{-c_L} L = q^{-c_R} R = q^{-c_R} L$ , so  $(q^{-c_L} - q^{-c_R})L = 0$  in  $\mathbb{Q}(q)$ ; when  $c_L \neq c_R$  the first factor is nonzero and hence  $L = 0$ . Items (iii) and (iv) are the specializations recorded in [Proposition 19.11](#) and [Corollary 2.6](#), legitimate after denominator clearing by [Lemma 20.10](#) and [Corollary 20.11](#). Item (v) is [Corollary 20.8](#). Item (vi) is the definition of a  $q$ -hypergeometric term given before [Theorem 20.7](#).  $\square$

**Remark 20.20.** The six conditions are necessary and not sufficient: passing all of them does not prove a candidate, and only the certified recurrence of item (v), or a proof organized as in [Principle 20.18](#), does. Their value is that every failure is cheap and localizing. A misplaced  $q^{\binom{k}{2}}$  or  $q^{k(n-k)}$  is caught by (i) and (ii) before anything is expanded, a wrong sign or normalization by (iii), and a wrong classical analogue by (iv). The algorithmic search for the certificate of item (v) is the subject of [23].



*Historical note 20.21.* Formal verification forces distinctions that handwritten mathematics often suppresses: a finite product indexed by a natural number is not the same object as a bilateral product; a polynomial Gaussian coefficient is not definitionally the same as a rational factorial quotient; and a formal power-series identity does not automatically contain a proof of analytic convergence. The Lean development of Lau, Lee, and Ono [20] turns these distinctions into reusable interfaces rather than obstacles.

## **Part VI**

# **Geometric filters, dyadic structure, and the Fabius bridge**

## Chapter 21

# Geometric interpolation and exact $q$ -filters

The geometric grid

$$1, q, q^2, \dots, q^n$$

is the natural interpolation grid for  $q$ -calculus. The Newton basis on this grid was developed in [Chapter 17](#). The complementary Lagrange viewpoint is equally useful: its weights are signed Gaussian coefficients, and every moment beyond the interpolation range is again a Gaussian coefficient. This gives exact coefficient filters, closed error formulas for geometric Richardson extrapolation, and the algebraic mechanism behind several constructions in the Fabius-function literature.

### 21.1 A reversed Gaussian row

For  $0 \leq k \leq n$  define

$$N_{n,k}(q) := (-1)^{n-k} q^{\binom{n-k+1}{2}} \begin{bmatrix} n \\ k \end{bmatrix}_q, \quad (21.1)$$

and put  $N_{n,k}(q) = 0$  outside this range. Although the definition is written with a quotient notation,  $N_{n,k}(q)$  is a polynomial in  $q$  with integral coefficients up to the displayed monomial and sign.

**Theorem 21.1** (Reversed finite  $q$ -binomial theorem). *For every  $n \geq 0$ ,*

$$\sum_{k=0}^n N_{n,k}(q) z^k = \prod_{j=1}^n (z - q^j). \quad (21.2)$$

Consequently, for every integer  $d$  with  $0 \leq d \leq n$ ,

$$\sum_{k=0}^n N_{n,k}(q) q^{kd} = \begin{cases} (q; q)_n, & d = 0, \\ 0, & 1 \leq d \leq n. \end{cases} \quad (21.3)$$

*Proof.* Write the product on the right as

$$\prod_{j=1}^n (z - q^j) = z^n \prod_{j=1}^n \left(1 - \frac{q^j}{z}\right) = z^n (q/z; q)_n.$$

The finite  $q$ -binomial theorem, [Theorem 3.1](#), gives

$$z^n (q/z; q)_n = \sum_{r=0}^n (-1)^r q^{\binom{r}{2} + r} \begin{bmatrix} n \\ r \end{bmatrix}_q z^{n-r}.$$

Put  $k = n - r$ . Since  $\begin{bmatrix} n \\ r \end{bmatrix}_q = \begin{bmatrix} n \\ k \end{bmatrix}_q$  by symmetry and  $\binom{r}{2} + r = \binom{r+1}{2} = \binom{n-k+1}{2}$ , this is exactly (21.2).

Now substitute  $z = q^d$ . At  $d = 0$  the product is  $\prod_{j=1}^n (1 - q^j) = (q; q)_n$ . If  $1 \leq d \leq n$ , the factor with  $j = d$  vanishes. This proves (21.3).  $\square$

*Remark 21.2* (Denominator-free meaning). The identity (21.2) belongs to  $\mathbb{Z}[q, z]$  and therefore remains valid after mapping  $q$  into any commutative ring. No division by  $1 - q^j$  is needed. This is the robust way to use Gaussian coefficients when  $q$  is a root of unity or when the coefficient ring has zero divisors.

## 21.2 Lagrange weights at zero

Assume henceforth that  $q$  lies in a field  $K$  and that

$$q^i \neq q^j \quad (0 \leq i < j \leq n).$$

Equivalently, the nodes  $1, q, \dots, q^n$  are distinct. Define

$$\lambda_{n,k}^{(q)} := \frac{N_{n,k}(q)}{(q; q)_n} = \frac{(-1)^{n-k} q^{\binom{n-k+1}{2}}}{(q; q)_n} \begin{bmatrix} n \\ k \end{bmatrix}_q. \quad (21.4)$$

**Theorem 21.3** (Evaluation at zero on a geometric grid). *For every polynomial  $p \in K[X]$  of degree at most  $n$ ,*

$$p(0) = \sum_{k=0}^n \lambda_{n,k}^{(q)} p(q^k). \quad (21.5)$$

Thus  $\lambda_{n,k}^{(q)}$  are precisely the Lagrange interpolation weights for evaluation at 0 on the nodes  $1, q, \dots, q^n$ .

*Proof.* It is enough to verify (21.5) on the basis  $1, X, \dots, X^n$ . For  $p(X) = 1$ , (21.3) divided by  $(q; q)_n$  gives 1. For  $p(X) = X^d$  with  $1 \leq d \leq n$ , the same identity gives  $0 = p(0)$ . Linearity proves the formula for every polynomial of degree at most  $n$ .

The Lagrange interpolation functional at 0 is the unique linear combination of the point evaluations  $p \mapsto p(q^k)$  that agrees with  $p \mapsto p(0)$  on all polynomials of degree at most  $n$ : uniqueness follows from invertibility of the Vandermonde matrix of the distinct nodes. Hence the displayed coefficients are the Lagrange weights.  $\square$

**Corollary 21.4** (Explicit product form). *For  $0 \leq k \leq n$ ,*

$$\lambda_{n,k}^{(q)} = \prod_{\substack{0 \leq j \leq n \\ j \neq k}} \frac{-q^j}{q^k - q^j}. \quad (21.6)$$

*Proof.* The right side is the usual Lagrange basis polynomial associated with the node  $q^k$ , evaluated at 0. By uniqueness in Theorem 21.3, it equals (21.4). One can also simplify the product directly by splitting the factors with  $j < k$  and  $j > k$ .  $\square$

**Example 21.5.** For  $n = 2$  the weights are

$$\lambda_{2,0}^{(q)} = \frac{q^3}{(1-q)(1-q^2)}, \quad \lambda_{2,1}^{(q)} = -\frac{q(1+q)}{(1-q)(1-q^2)}, \quad \lambda_{2,2}^{(q)} = \frac{1}{(1-q)(1-q^2)}.$$

They satisfy

$$\sum_{k=0}^2 \lambda_{2,k}^{(q)} = 1, \quad \sum_{k=0}^2 \lambda_{2,k}^{(q)} q^k = 0, \quad \sum_{k=0}^2 \lambda_{2,k}^{(q)} q^{2k} = 0.$$

### 21.3 All residual moments

The vanishing moments in (21.3) are only the beginning. Every later moment has a closed form. The key fact is a universal remainder formula involving complete homogeneous symmetric functions.

**Theorem 21.6** (Universal Lagrange residual formula). *Let  $v_1, \dots, v_N$  be distinct elements of a field, let*

$$\ell_i(x) = \prod_{\substack{1 \leq j \leq N \\ j \neq i}} \frac{x - v_j}{v_i - v_j},$$

*and let  $h_r$  denote the complete homogeneous symmetric polynomial, with  $h_0 = 1$ . For every  $r \geq 0$ ,*

$$\sum_{i=1}^N \ell_i(x) v_i^{N+r} = x^{N+r} - \prod_{i=1}^N (x - v_i) h_r(x, v_1, \dots, v_N). \quad (21.7)$$

*For exponents  $0 \leq d < N$ , the left side with  $N + r$  replaced by  $d$  is simply  $x^d$ .*

*Proof.* The assertion for  $d < N$  is ordinary Lagrange interpolation. For  $r \geq 0$ , put

$$P(T) = \prod_{i=1}^N (T - v_i), \quad H_r(T) = T^{N+r} - P(T) h_r(T, v_1, \dots, v_N).$$

We first show that  $\deg H_r < N$ . Write  $e_j = e_j(v_1, \dots, v_N)$  and  $h_j = h_j(v_1, \dots, v_N)$ . Then

$$P(T) = \sum_{j=0}^N (-1)^j e_j T^{N-j}, \quad h_r(T, v_1, \dots, v_N) = \sum_{s=0}^r T^s h_{r-s}.$$

For  $0 \leq \ell \leq r$ , the coefficient of  $T^{N+r-\ell}$  in their product is

$$\sum_{j=0}^{\ell} (-1)^j e_j h_{\ell-j}.$$

By elementary-complete orthogonality, [Corollary 4.4](#), this is 1 when  $\ell = 0$  and 0 for  $1 \leq \ell \leq r$ . Hence the terms of degrees  $N, N+1, \dots, N+r$  in  $P(T)h_r(T, v_1, \dots, v_N)$  agree with those of  $T^{N+r}$ , and therefore  $\deg H_r < N$ .

At every node  $v_i$  we have  $P(v_i) = 0$ , so  $H_r(v_i) = v_i^{N+r}$ . Since  $H_r$  has degree less than  $N$ , Lagrange interpolation gives

$$H_r(x) = \sum_{i=1}^N \ell_i(x) v_i^{N+r}.$$

Substituting the definition of  $H_r$  proves (21.7). □

**Theorem 21.7** (Complete geometric moment formula). *Under the distinct-node hypothesis, for every integer  $d \geq 0$ ,*

$$\sum_{k=0}^n \lambda_{n,k}^{(q)} q^{kd} = \begin{cases} 1, & d = 0, \\ 0, & 1 \leq d \leq n, \\ (-1)^n q^{\binom{n+1}{2}} \begin{bmatrix} d-1 \\ n \end{bmatrix}_q, & d \geq n+1. \end{cases} \quad (21.8)$$

*Equivalently, with the convention that a Gaussian coefficient is zero outside its natural range,*

$$\sum_{k=0}^n \lambda_{n,k}^{(q)} q^{kd} = (-1)^n q^{\binom{n+1}{2}} \begin{bmatrix} d-1 \\ n \end{bmatrix}_q \quad (d > 0). \quad (21.9)$$

*Proof.* The cases  $0 \leq d \leq n$  were proved in [Theorem 21.3](#). Let  $d = n + 1 + r$  with  $r \geq 0$ . Apply [Theorem 21.6](#) with target  $x = 0$  and nodes  $v_k = q^k$  for  $0 \leq k \leq n$ . There are  $N = n + 1$  nodes, and the weights  $\ell_k(0)$  are  $\lambda_{n,k}^{(q)}$ . Since

$$\prod_{k=0}^n (0 - q^k) = (-1)^{n+1} q^{\binom{n+1}{2}},$$

we obtain

$$\sum_{k=0}^n \lambda_{n,k}^{(q)} q^{k(n+1+r)} = (-1)^n q^{\binom{n+1}{2}} h_r(1, q, \dots, q^n).$$

The geometric specialization in [Theorem 4.6](#) gives

$$h_r(1, q, \dots, q^n) = \begin{bmatrix} n+r \\ n \end{bmatrix}_q.$$

Because  $n + r = d - 1$ , this is the desired formula.  $\square$

**Corollary 21.8** (Scaled geometric nodes). *Let  $c \neq 0$  and use the same weights  $\lambda_{n,k}^{(q)}$  on the nodes  $cq^k$ . Then for  $d > 0$ ,*

$$\sum_{k=0}^n \lambda_{n,k}^{(q)} (cq^k)^d = c^d (-1)^n q^{\binom{n+1}{2}} \begin{bmatrix} d-1 \\ n \end{bmatrix}_q.$$

*In particular, these weights evaluate every polynomial of degree at most  $n$  at 0 independently of the nonzero scale  $c$ .*

*Proof.* Factor  $c^d$  from the moment sum and apply [Theorem 21.7](#). Exactness on polynomials follows by linearity.  $\square$

## 21.4 Formal coefficient filters

For a formal power series  $f(X) = \sum_{m \geq 0} a_m X^m$ , define

$$(\mathcal{G}_{n,q} f)(X) := \sum_{k=0}^n \lambda_{n,k}^{(q)} f(q^k X). \quad (21.10)$$

The sum is finite, so this is well defined over any coefficient field in which the weights exist.

**Theorem 21.9** (Diagonal action and exact tail). *For every formal power series  $f(X) = \sum_{m \geq 0} a_m X^m$ ,*

$$[X^m] \mathcal{G}_{n,q} f = \left( \sum_{k=0}^n \lambda_{n,k}^{(q)} q^{km} \right) a_m, \quad (21.11)$$

$$\mathcal{G}_{n,q} f(X) = a_0 + (-1)^n q^{\binom{n+1}{2}} \sum_{r \geq 0} \begin{bmatrix} n+r \\ n \end{bmatrix}_q a_{n+1+r} X^{n+1+r}. \quad (21.12)$$

*Thus the filter deletes the coefficients of degrees  $1, \dots, n$  and multiplies every later coefficient by an explicit Gaussian polynomial.*

*Proof.* Substitution gives

$$\mathcal{G}_{n,q}f(X) = \sum_{k=0}^n \lambda_{n,k}^{(q)} \sum_{m \geq 0} a_m q^{km} X^m.$$

The outer sum is finite, so coefficient extraction may be interchanged with it, proving (21.11). Apply Theorem 21.7: the multiplier is 1 at  $m = 0$ , is 0 for  $1 \leq m \leq n$ , and at  $m = n + 1 + r$  is  $(-1)^n q^{\binom{n+1}{2}} \begin{bmatrix} n+r \\ n \end{bmatrix}_q$ . This gives (21.12).  $\square$

**Corollary 21.10** (Geometric Richardson extrapolation). *Let  $f$  be analytic at 0. The values*

$$f(z), f(qz), \dots, f(q^n z)$$

*produce the extrapolation formula*

$$\sum_{k=0}^n \lambda_{n,k}^{(q)} f(q^k z) = f(0) + O(z^{n+1}) \quad (z \rightarrow 0),$$

*and the complete asymptotic error is the convergent specialization of (21.12).*

*Proof.* Apply Theorem 21.9 to the Taylor series of  $f$ . The first nonconstant surviving term has degree  $n + 1$ .  $\square$

**Theorem 21.11** (A quantitative analytic bound). *Assume  $0 < q < 1$  and  $f(z) = \sum_{m \geq 0} a_m z^m$  with  $|a_m| \leq M R^{-m}$ . If  $|z| < R$ , then*

$$\left| \sum_{k=0}^n \lambda_{n,k}^{(q)} f(q^k z) - f(0) \right| \leq \frac{M q^{\binom{n+1}{2}} (|z|/R)^{n+1}}{(q; q)_n (1 - |z|/R)}. \quad (21.13)$$

*Proof.* For  $r \geq 0$ ,

$$\begin{bmatrix} n+r \\ n \end{bmatrix}_q = \frac{(q^{r+1}; q)_n}{(q; q)_n} \leq \frac{1}{(q; q)_n},$$

because every factor in  $(q^{r+1}; q)_n$  lies in  $(0, 1)$ . Take absolute values in (21.12), use the coefficient bound, and sum the resulting geometric series.  $\square$

*Remark 21.12.* The exact multiplier in (21.12), rather than merely the order estimate  $O(z^{n+1})$ , is the principal advantage of a geometric grid. The complete tail is controlled by one Gaussian row. This phenomenon is used repeatedly in geometric extrapolation, in formal power-series filters, and in the dyadic identities of the next chapter.

## Chapter 22

# Half-base arithmetic and Prouhet–Thue–Morse cancellation

The specialization  $q = 1/2$  turns factors  $1 - q^j$  into normalized Mersenne numbers. It is especially important in exact formulas for the Fabius and Rvachev functions. At this base, the finite  $q$ -binomial theorem produces a compact  $n + 1$ -point Prouhet functional, while the classical Thue–Morse product produces a  $2^n$ -point functional. Both annihilate all polynomials of degree below  $n$ , but their weights and node geometries are very different.

### 22.1 Mersenne normalization

Define

$$\mathfrak{m}_0 := 1, \quad \mathfrak{m}_n := \prod_{j=1}^n (2^j - 1) \quad (n \geq 1). \quad (22.1)$$

**Theorem 22.1** (Half-base factorial and Gaussian formulas). *For  $n \geq 0$  and  $0 \leq k \leq n$ ,*

$$(1/2; 1/2)_n = 2^{-\binom{n+1}{2}} \mathfrak{m}_n, \quad (22.2)$$

$$\left[ \begin{matrix} n \\ k \end{matrix} \right]_{1/2} = 2^{-k(n-k)} \frac{\mathfrak{m}_n}{\mathfrak{m}_k \mathfrak{m}_{n-k}}. \quad (22.3)$$

*Proof.* Each factor satisfies

$$1 - 2^{-j} = 2^{-j} (2^j - 1).$$

Multiplication for  $1 \leq j \leq n$  gives (22.2), because  $1 + 2 + \cdots + n = \binom{n+1}{2}$ . Substitute this expression into

$$\left[ \begin{matrix} n \\ k \end{matrix} \right]_{1/2} = \frac{(1/2; 1/2)_n}{(1/2; 1/2)_k (1/2; 1/2)_{n-k}}.$$

The power of 2 in the quotient is

$$-\binom{n+1}{2} + \binom{k+1}{2} + \binom{n-k+1}{2} = -k(n-k),$$

which proves (22.3). □



**Theorem 22.2** (Dyadic evaluation of the finite product). *For integers  $m, n \geq 0$ ,*

$$(2^m; 1/2)_n = \begin{cases} 0, & m < n, \\ (-1)^n \frac{\mathfrak{m}_m}{\mathfrak{m}_{m-n}}, & m \geq n. \end{cases} \quad (22.4)$$

*Equivalently,*

$$\sum_{k=0}^n (-1)^k 2^{-\binom{k}{2}} \begin{bmatrix} n \\ k \end{bmatrix}_{1/2} 2^{mk} = \begin{cases} 0, & m < n, \\ (-1)^n \frac{\mathfrak{m}_m}{\mathfrak{m}_{m-n}}, & m \geq n. \end{cases} \quad (22.5)$$

*Proof.* By definition,

$$(2^m; 1/2)_n = \prod_{j=0}^{n-1} (1 - 2^{m-j}).$$

If  $m < n$ , the factor with  $j = m$  is zero. If  $m \geq n$ , every factor is negative and

$$\prod_{j=0}^{n-1} (1 - 2^{m-j}) = (-1)^n \prod_{r=m-n+1}^m (2^r - 1) = (-1)^n \frac{\mathfrak{m}_m}{\mathfrak{m}_{m-n}}.$$

The finite  $q$ -binomial theorem at  $q = 1/2$  and  $z = 2^m$  turns the left side into the sum in (22.5).  $\square$

**Corollary 22.3** (Root locus). *The polynomial*

$$\sum_{k=0}^n (-1)^k 2^{-\binom{k}{2}} \begin{bmatrix} n \\ k \end{bmatrix}_{1/2} z^k$$

*has the simple roots  $1, 2, 4, \dots, 2^{n-1}$  and no others.*

*Proof.* It is  $(z; 1/2)_n = \prod_{j=0}^{n-1} (1 - z/2^j)$ , a degree- $n$  polynomial with exactly the displayed roots.  $\square$

## 22.2 A compressed geometric Prouhet functional

Define nodes and weights

$$\xi_k := -2^k, \quad w_{n,k} := (-1)^k 2^{-\binom{k}{2}} \begin{bmatrix} n \\ k \end{bmatrix}_{1/2} \quad (0 \leq k \leq n). \quad (22.6)$$

**Theorem 22.4** (Geometric Prouhet identity). *For every polynomial  $p$  of degree at most  $n$ ,*

$$\sum_{k=0}^n w_{n,k} p(\xi_k) = \mathfrak{m}_n[X^n] p(X). \quad (22.7)$$

*In particular, the functional on the left annihilates every polynomial of degree less than  $n$ .*

*Proof.* It suffices to take  $p(X) = X^d$  for  $0 \leq d \leq n$ . With  $q = 1/2$ ,

$$\begin{aligned} \sum_{k=0}^n w_{n,k} \xi_k^d &= (-1)^d \sum_{k=0}^n (-1)^k q^{\binom{k}{2}} \begin{bmatrix} n \\ k \end{bmatrix}_q q^{-kd} \\ &= (-1)^d (q^{-d}; q)_n \end{aligned}$$

by the finite  $q$ -binomial theorem. If  $d < n$ , the factor with index  $j = d$  in  $(q^{-d}; q)_n$  is zero. At  $d = n$ , base reversal gives

$$(q^{-n}; q)_n = (-1)^n q^{-\binom{n+1}{2}} (q; q)_n.$$

Therefore

$$(-1)^n (q^{-n}; q)_n = 2^{\binom{n+1}{2}} (1/2; 1/2)_n = \mathbf{m}_n$$

by (22.2). Linearity now proves (22.7).  $\square$

**Corollary 22.5** (Affine form). *For constants  $a, b$  and every polynomial  $p$  of degree at most  $n$ ,*

$$\sum_{k=0}^n w_{n,k} p(a + b\xi_k) = b^n \mathbf{m}_n[X^n] p(X).$$

*Proof.* Apply Theorem 22.4 to the polynomial  $X \mapsto p(a + bX)$ . Its leading coefficient is  $b^n[X^n]p(X)$ .  $\square$

*Remark 22.6.* The identity uses only  $n + 1$  geometrically spaced nodes. The price for this compression is that the weights are nonuniform rational numbers involving the half-base Gaussian row. The Thue–Morse construction below uses  $2^n$  equally weighted signs on an arithmetic grid instead.

## 22.3 The Thue–Morse product

For a nonnegative integer  $r$ , let  $s_2(r)$  be the sum of the binary digits of  $r$  and put

$$\tau_r := (-1)^{s_2(r)}.$$

This is the signed Thue–Morse sequence.

**Theorem 22.7** (Finite Thue–Morse product). *For every  $n \geq 0$ ,*

$$\sum_{r=0}^{2^n-1} \tau_r z^r = \prod_{j=0}^{n-1} (1 - z^{2^j}). \quad (22.8)$$

*Proof.* Every  $r$  with  $0 \leq r < 2^n$  has a unique binary expansion  $r = \sum_{j=0}^{n-1} \varepsilon_j 2^j$  with  $\varepsilon_j \in \{0, 1\}$ . Expanding the product on the right, choose either 1 or  $-z^{2^j}$  from the  $j$ th factor. The choice vector  $(\varepsilon_0, \dots, \varepsilon_{n-1})$  contributes

$$(-1)^{\varepsilon_0 + \dots + \varepsilon_{n-1}} z^{\varepsilon_0 + 2\varepsilon_1 + \dots + 2^{n-1}\varepsilon_{n-1}} = \tau_r z^r.$$

Summing over all binary choices gives the identity.  $\square$

**Theorem 22.8** (Prouhet cancellation and first surviving moment). *For  $n \geq 1$ ,*

$$\sum_{r=0}^{2^n-1} \tau_r r^d = \begin{cases} 0, & 0 \leq d < n, \\ (-1)^n 2^{\binom{n}{2}} n!, & d = n. \end{cases} \quad (22.9)$$

*More generally, for every polynomial  $p$  of degree at most  $n$ ,*

$$\sum_{r=0}^{2^n-1} \tau_r p(a + br) = (-1)^n 2^{\binom{n}{2}} n! b^n [X^n] p(X). \quad (22.10)$$

*Proof.* Let  $P_n(z) = \sum_{r < 2^n} \tau_r z^r$ . By [Theorem 22.7](#),  $P_n$  has a zero of order exactly  $n$  at  $z = 1$ , because every factor  $1 - z^{2^j}$  has a simple zero there. The Euler operator  $D = z \frac{d}{dz}$  satisfies

$$D^d P_n(1) = \sum_{r < 2^n} \tau_r r^d.$$

For  $d < n$ , this is zero because  $D^d$  is a linear combination of ordinary derivatives of orders at most  $d$ , all of which vanish at 1.

For  $d = n$ , the coefficient of the ordinary  $n$ th derivative in  $D^n$  is  $z^n$ , while all lower derivatives still vanish at 1. Hence

$$D^n P_n(1) = P_n^{(n)}(1).$$

The leading coefficient of the local expansion at 1 is the product of the first derivatives of the  $n$  factors:

$$P_n^{(n)}(1) = n! \prod_{j=0}^{n-1} \left. \frac{d}{dz} (1 - z^{2^j}) \right|_{z=1} = n! \prod_{j=0}^{n-1} (-2^j) = (-1)^n 2^{\binom{n}{2}} n!.$$

This proves [\(22.9\)](#). Apply it termwise to the polynomial  $X \mapsto p(a + bX)$ ; all terms below degree  $n$  vanish, and its leading coefficient is  $b^n [X^n]p(X)$ . This gives [\(22.10\)](#).  $\square$

**Corollary 22.9** (Equal power sums). *Partition  $\{0, 1, \dots, 2^n - 1\}$  according to the parity of  $s_2(r)$ . The two classes have equal sums of  $d$ th powers for every  $0 \leq d < n$ .*

*Proof.* Move the negative-sign terms in [\(22.9\)](#) to the other side.  $\square$

## 22.4 Two cancellation mechanisms

**Principle 22.10** (Geometric versus binary Prouhet filters). *There are two complementary exact functionals of order  $n$ :*

$$\begin{aligned} \sum_{k=0}^n (-1)^k 2^{-\binom{k}{2}} \left[ \begin{matrix} n \\ k \end{matrix} \right]_{1/2} p(-2^k) &= \mathfrak{m}_n[X^n]p, \\ \sum_{r=0}^{2^n-1} (-1)^{s_2(r)} p(r) &= (-1)^n 2^{\binom{n}{2}} n! [X^n]p, \end{aligned}$$

valid for  $\deg p \leq n$ . The first uses a Gaussian row on a geometric grid; the second uses uniform signs generated by binary digits on an arithmetic grid.

*Proof.* These are [Theorems 22.4](#) and [22.8](#).  $\square$

The common product mechanism is now visible. In the geometric case the annihilating polynomial is  $(q^{-d}; q)_n$  evaluated at integer  $d$ ; in the binary case it is the product  $\prod_{j < n} (1 - z^{2^j})$  and its order of vanishing at  $z = 1$ . Both encode a hierarchy of moment cancellations by forcing a product to acquire prescribed zeros.

## Chapter 23

# Geometric uniform sums, $q$ -cumulants, and the Fabius–Rvachev functions

A second bridge from  $q$ -products to the Fabius theory is probabilistic. Replace discrete binary digits by independent uniform coordinates and weight them geometrically. The resulting random series has a smooth compactly supported density, a Fourier transform given by an infinite product of sinc functions, and cumulants containing the factors  $1 - q^{2^m}$ . At  $q = 1/2$  its distribution function is the Fabius function and its centered density is the Rvachev up function [37, 39, 34, 38].

### 23.1 The geometric uniform random series

Let  $0 < q < 1$ , let  $U_0, U_1, \dots$  be independent random variables uniformly distributed on  $[0, 1]$ , and define

$$X_q := (1 - q) \sum_{j=0}^{\infty} q^j U_j. \quad (23.1)$$

**Theorem 23.1** (Convergence, support, symmetry, and self-similarity). *The series (23.1) converges absolutely for every outcome and satisfies  $0 \leq X_q \leq 1$ . Its topological support is  $[0, 1]$ , and*

$$\mathbb{E}X_q = \frac{1}{2}, \quad X_q \stackrel{d}{=} 1 - X_q. \quad (23.2)$$

If  $X'_q$  is an independent copy of  $X_q$ , then

$$X_q \stackrel{d}{=} (1 - q)U_0 + qX'_q. \quad (23.3)$$

Consequently its distribution function  $F_q(x) = \mathbb{P}(X_q \leq x)$  satisfies

$$F_q(x) = \int_0^1 F_q\left(\frac{x - (1 - q)u}{q}\right) du, \quad (23.4)$$

where  $F_q(y) = 0$  for  $y \leq 0$  and  $F_q(y) = 1$  for  $y \geq 1$ .

*Proof.* Because  $0 \leq U_j \leq 1$ ,

$$0 \leq (1 - q) \sum_{j=0}^{N-1} q^j U_j \leq (1 - q) \sum_{j=0}^{N-1} q^j \leq 1.$$

The tail is bounded by

$$(1 - q) \sum_{j=N}^{\infty} q^j = q^N,$$

so the series converges absolutely and takes values in  $[0, 1]$ .

Every  $x \in [0, 1]$  belongs to the support: if all of the first  $N$  variables lie sufficiently close to  $x$ , then their weighted partial sum lies close to  $x(1 - q^N)$ , and the remaining tail is at most  $q^N$ ; the corresponding finite cylinder has positive probability. No point outside  $[0, 1]$  can belong to the support.

Linearity of expectation and the geometric sum give

$$\mathbb{E}X_q = (1 - q) \sum_{j \geq 0} q^j \mathbb{E}U_j = \frac{1 - q}{2} \frac{1}{1 - q} = \frac{1}{2}.$$

Since  $1 - U_j$  are again independent uniform variables,

$$1 - X_q = (1 - q) \sum_{j \geq 0} q^j (1 - U_j)$$

has the same distribution as  $X_q$ . Splitting off the first term and reindexing the tail proves (23.3). Conditioning on  $U_0 = u$  then gives

$$\mathbb{P}((1 - q)u + qX'_q \leq x) = F_q\left(\frac{x - (1 - q)u}{q}\right),$$

and integration over  $u \in [0, 1]$  proves (23.4).  $\square$

## 23.2 Fourier product and cumulants

Use the entire function

$$\text{sinc } z := \begin{cases} \frac{\sin z}{z}, & z \neq 0, \\ 1, & z = 0. \end{cases}$$

**Theorem 23.2** (Characteristic function). *For real  $t$ ,*

$$\varphi_q(t) := \mathbb{E} e^{itX_q} = e^{it/2} \prod_{j=0}^{\infty} \text{sinc}\left(\frac{(1 - q)q^j t}{2}\right). \quad (23.5)$$

*The product converges uniformly on compact subsets of the complex  $t$ -plane and therefore defines an entire function.*

*Proof.* For a uniform variable  $U$  on  $[0, 1]$  and any complex  $s$ ,

$$\mathbb{E}e^{sU} = \int_0^1 e^{su} du = e^{s/2} \frac{\sinh(s/2)}{s/2}.$$

At  $s = ita$  this becomes  $e^{ita/2} \text{sinc}(at/2)$ . Independence gives the finite product for the partial sum. The phases combine because  $\sum_{j \geq 0} (1 - q)q^j = 1$ .

For compact  $t$ -sets, the arguments of the sinc factors are  $O(q^j)$ . Since  $\text{sinc } z = 1 + O(z^2)$  near 0, the series of deviations from 1 is bounded by a constant multiple of  $\sum q^{2j}$ , uniformly on the compact set. The standard uniform convergence criterion for infinite products proves locally uniform convergence to an entire function. Passing to the limit in the finite characteristic functions gives (23.5).  $\square$

Let  $Y_q = X_q - 1/2$ . We use Bernoulli numbers in the convention

$$\frac{z}{e^z - 1} = \sum_{m=0}^{\infty} B_m \frac{z^m}{m!}. \quad (23.6)$$

The cumulants  $\kappa_m(Y_q)$  are defined by

$$\log \mathbb{E} e^{tY_q} = \sum_{m \geq 1} \kappa_m(Y_q) \frac{t^m}{m!}$$

near  $t = 0$ .

**Theorem 23.3** ( $q$ -cumulants of the geometric uniform sum). *All odd cumulants of  $Y_q$  vanish, and for  $m \geq 1$ ,*

$$\kappa_{2m}(Y_q) = \frac{B_{2m}}{2m} \frac{(1-q)^{2m}}{1-q^{2m}}. \quad (23.7)$$

Equivalently,

$$\log \mathbb{E} e^{tY_q} = \sum_{m=1}^{\infty} \frac{B_{2m}}{2m(2m)!} \frac{(1-q)^{2m}}{1-q^{2m}} t^{2m} \quad (23.8)$$

for  $t$  in a neighborhood of 0.

*Proof.* For  $U$  uniform on  $[0, 1]$ ,

$$\mathbb{E} e^{t(U-1/2)} = \frac{\sinh(t/2)}{t/2}.$$

Differentiating the logarithm gives

$$\frac{d}{dt} \log \frac{\sinh(t/2)}{t/2} = \frac{1}{2} \coth(t/2) - \frac{1}{t}.$$

The Bernoulli generating function implies the classical Laurent expansion

$$\frac{1}{2} \coth(t/2) - \frac{1}{t} = \sum_{m=1}^{\infty} B_{2m} \frac{t^{2m-1}}{(2m)!}.$$

Integrating from 0 to  $t$  yields

$$\log \frac{\sinh(t/2)}{t/2} = \sum_{m=1}^{\infty} \frac{B_{2m}}{2m(2m)!} t^{2m}. \quad (23.9)$$

Now

$$Y_q = (1-q) \sum_{j \geq 0} q^j (U_j - 1/2).$$

Cumulant generating functions add under independence. Substitute  $t(1-q)q^j$  into (23.9), sum over  $j$ , and use  $\sum_{j \geq 0} q^{2mj} = 1/(1-q^{2m})$ . This proves (23.8); comparison of coefficients gives (23.7). Only even powers occur, so all odd cumulants vanish.  $\square$

**Definition 23.4** (Complete exponential Bell polynomials). The polynomials  $\mathcal{B}_n(x_1, \dots, x_n)$  are defined by

$$\exp \left( \sum_{r \geq 1} x_r \frac{t^r}{r!} \right) = \sum_{n \geq 0} \mathcal{B}_n(x_1, \dots, x_n) \frac{t^n}{n!}. \quad (23.10)$$

**Corollary 23.5** (All moments). *Let  $\kappa_r = \kappa_r(Y_q)$ , with the values in Theorem 23.3. Then*

$$\mathbb{E}(Y_q^n) = \mathcal{B}_n(\kappa_1, \dots, \kappa_n). \quad (23.11)$$

*The moments can also be generated recursively from  $\mu_0 = 1$  by*

$$\mu_n = \sum_{r=1}^n \binom{n-1}{r-1} \kappa_r \mu_{n-r}, \quad \mu_n := \mathbb{E}(Y_q^n). \quad (23.12)$$

*Finally,*

$$\mathbb{E}(X_q^n) = \sum_{j=0}^n \binom{n}{j} 2^{j-n} \mu_j. \quad (23.13)$$

*Proof.* The moment generating function is the exponential of the cumulant generating function. Comparing it with the defining identity (23.10) proves (23.11). If  $M(t) = \sum \mu_n t^n / n!$  and  $K(t) = \sum \kappa_r t^r / r!$ , then  $M = e^K$  and  $M' = K' M$ . Comparing the coefficient of  $t^{n-1}$  gives (23.12). The last formula is the binomial expansion of  $(Y_q + 1/2)^n$ .  $\square$

**Corollary 23.6** (The half-base cumulants). *At  $q = 1/2$ ,*

$$\kappa_{2m}(Y_{1/2}) = \frac{B_{2m}}{2m(2^{2m} - 1)}, \quad \kappa_{2m+1}(Y_{1/2}) = 0. \quad (23.14)$$

*Hence all moments of  $X_{1/2}$  are rational and are computable to arbitrary order by (23.12).*

*Proof.* In (23.7),

$$\frac{(1 - 1/2)^{2m}}{1 - 2^{-2m}} = \frac{1}{2^{2m} - 1}.$$

The Bernoulli numbers are rational, so the recurrence uses only rational operations.  $\square$

### 23.3 Superpolynomial Fourier decay and smoothness

**Theorem 23.7** (Log-Gaussian decay in frequency). *Put*

$$A = \frac{1-q}{2}, \quad b = \log(1/q) > 0.$$

*For sufficiently large  $|t|$ , with  $L = \log(A|t|)$ ,*

$$|\varphi_q(t)| \leq \exp\left(-\frac{L^2}{2b} + \frac{L}{2}\right). \quad (23.15)$$

*In particular, for every  $M \geq 0$ ,*

$$|t|^M \varphi_q(t) \in L^1(\mathbb{R}).$$

*Proof.* The phase  $e^{it/2}$  has absolute value 1, and  $|\operatorname{sinc} u| \leq \min(1, 1/|u|)$ . Choose

$$N = \lfloor L/b \rfloor + 1.$$

For  $0 \leq j < N$  we have  $A|t|q^j \geq 1$ . Hence

$$\begin{aligned} |\varphi_q(t)| &\leq \prod_{j=0}^{N-1} \frac{1}{A|t|q^j} \\ &= \exp\left(-NL + \frac{bN(N-1)}{2}\right). \end{aligned}$$

Because  $L \geq b(N-1)$ , the exponent is at most  $-bN(N-1)/2$ . For  $L$  sufficiently large,  $N \geq L/b$  and  $N-1 \geq L/b-1$ , so

$$-\frac{bN(N-1)}{2} \leq -\frac{L^2}{2b} + \frac{L}{2}.$$

This proves (23.15).

After the change of variables  $L = \log(At)$ , the right side is a Gaussian in  $L$  multiplied by  $e^{L/2}$ . It therefore dominates every negative power of  $t$ . More explicitly, for each  $M$  the exponent  $-L^2/(2b) + (M + 3/2)L$  tends to  $-\infty$  quadratically, which proves integrability of  $|t|^M \varphi_q(t)$ .  $\square$

**Theorem 23.8** (Smooth compactly supported density). *For every  $0 < q < 1$ , the distribution of  $X_q$  has a  $C^\infty$  density  $f_q$  supported on  $[0, 1]$ . For every  $m \geq 0$ ,*

$$f_q^{(m)}(x) = \frac{1}{2\pi} \int_{\mathbb{R}} (-it)^m e^{-itx} \varphi_q(t) dt. \quad (23.16)$$

*All derivatives of  $f_q$  vanish at the endpoints 0 and 1.*

*Proof.* By Theorem 23.7,  $t^m \varphi_q(t)$  is integrable for every  $m$ . Fourier inversion for a finite measure with integrable characteristic function gives the density

$$f_q(x) = \frac{1}{2\pi} \int_{\mathbb{R}} e^{-itx} \varphi_q(t) dt.$$

For completeness, one may obtain this directly by convolving the distribution with a centered Gaussian of variance  $\varepsilon$ : its density is the inverse transform of  $\varphi_q(t)e^{-\varepsilon t^2/2}$ . Dominated convergence as  $\varepsilon \downarrow 0$  yields the displayed  $f_q$  uniformly. Against every compactly supported continuous test function, the Gaussian convolutions converge both to integration against  $f_q$  and weakly to the original measure; hence that measure has density  $f_q$ .

Because  $t^m \varphi_q(t)$  is integrable, differentiation under the integral is justified by dominated convergence and gives (23.16). Thus  $f_q \in C^\infty(\mathbb{R})$ . The random variable is supported on  $[0, 1]$ , so the density is zero on the two open complementary half-lines. Smoothness then forces every derivative to have value 0 at both boundary points.  $\square$

## 23.4 Finite spline approximants

Put

$$X_{q,N} := (1-q) \sum_{j=0}^{N-1} q^j U_j, \quad a_j := (1-q)q^j. \quad (23.17)$$

For  $x \in \mathbb{R}$  and an integer  $r \geq 0$ , write  $x_+^r = (\max(x, 0))^r$ ; when  $r = 0$ , interpret  $x_+^0$  as the indicator of  $x > 0$  away from the knots.

The following nonidentically scaled uniform-sum formula is a box-spline form of the generalized Irwin–Hall distribution; compare Bradley and Gupta [35].



**Theorem 23.9** (Generalized uniform-sum spline and the weighted cube cut). *Let  $N \geq 1$ , let  $c_0, \dots, c_{N-1} > 0$ , let  $U_0, \dots, U_{N-1}$  be independent and uniform on  $[0, 1]$ , and put  $S_{\mathbf{c}} := \sum_{j=0}^{N-1} c_j U_j$ . Then the density of  $S_{\mathbf{c}}$  is*

$$f_{\mathbf{c}}(x) = \frac{1}{(N-1)! \prod_{j=0}^{N-1} c_j} \sum_{A \subseteq \{0, \dots, N-1\}} (-1)^{|A|} \left( x - \sum_{j \in A} c_j \right)_+^{N-1}, \quad (23.18)$$

*a compactly supported piecewise polynomial of degree at most  $N-1$ , and its distribution function is the volume of the corresponding section of the unit cube: for every  $t \in \mathbb{R}$ ,*

$$V_{\mathbf{c}}(t) := \text{vol} \left\{ u \in [0, 1]^N : \sum_{j=0}^{N-1} c_j u_j \leq t \right\} = \frac{1}{N! \prod_{j=0}^{N-1} c_j} \sum_{A \subseteq \{0, \dots, N-1\}} (-1)^{|A|} \left( t - \sum_{j \in A} c_j \right)_+^N. \quad (23.19)$$

*In particular, at the geometric weights  $c_j = a_j = (1-q)q^j$  of (23.17) the density of  $X_{q,N}$  is*

$$f_{q,N}(x) = \frac{1}{(N-1)! \prod_{j=0}^{N-1} a_j} \sum_{A \subseteq \{0, \dots, N-1\}} (-1)^{|A|} \left( x - \sum_{j \in A} a_j \right)_+^{N-1}, \quad (23.20)$$

*and its Fourier transform is the corresponding finite product in (23.5).*

*Proof.* The density of  $c_j U_j$  is  $c_j^{-1} \mathbf{1}_{[0, c_j]}$ , so  $f_{\mathbf{c}}$  is the convolution of  $N$  box densities and, for  $s > 0$ , its Laplace transform is

$$\prod_{j=0}^{N-1} \frac{1 - e^{-c_j s}}{c_j s},$$

which uses nothing about the numbers  $c_j$  beyond  $c_j > 0$ . On the other hand

$$\int_0^\infty e^{-sx} \frac{(x-c)_+^{N-1}}{(N-1)!} dx = \frac{e^{-cs}}{s^N},$$

so the Laplace transform of the right-hand side of (23.18) is

$$\frac{1}{s^N \prod_j c_j} \sum_A (-1)^{|A|} e^{-s \sum_{j \in A} c_j} = \prod_{j=0}^{N-1} \frac{1 - e^{-c_j s}}{c_j s},$$

because the subset sum factors as  $\prod_j (1 - e^{-c_j s})$ . Uniqueness of the Laplace transform proves (23.18), and piecewise polynomiality is immediate from the truncated-power representation.

Since  $(U_0, \dots, U_{N-1})$  is uniform on the cube,  $V_{\mathbf{c}}(t) = \mathbb{P}(S_{\mathbf{c}} \leq t) = \int_{-\infty}^t f_{\mathbf{c}}(x) dx$ . The sum in (23.18) is finite, so it may be integrated term by term, and

$$\int_{-\infty}^t (x-c)_+^{N-1} dx = \frac{(t-c)_+^N}{N} \quad (N \geq 1)$$

gives (23.19). Specializing  $c_j = a_j$  gives (23.20), and the Fourier product follows either by transforming the box convolutions or by the finite version of the proof of Theorem 23.2.  $\square$

**Remark 23.10** (The dyadic instance). For the integer weights  $c_j = 2^{n-j}$ ,  $1 \leq j \leq n$  – that is,  $2^n$  times the half-base weights  $a_{j-1} = 2^{-j}$  of (23.17) at  $q = 1/2$  – the map  $A \mapsto r = \sum_{j \in A} 2^{n-j}$  is binary expansion,

hence a bijection from the subsets of  $\{1, \dots, n\}$  onto  $\{0, 1, \dots, 2^n - 1\}$  with  $|A| = s_2(r)$ ; consequently  $(-1)^{|A|} = \tau_r$  and  $\prod_{j=1}^n c_j = 2^{\binom{n}{2}}$ , so (23.19) becomes

$$V_{\mathbf{c}}(t) = \frac{2^{-\binom{n}{2}}}{n!} \sum_{r=0}^{2^n-1} \tau_r (t-r)_+^n. \quad (23.21)$$

This is the cumulative counterpart of [Theorem 23.11](#), which records the same signs in the density. For  $0 \leq t \leq 1$  every term with  $r \geq 1$  vanishes and (23.21) reduces to  $t^n / (n! 2^{\binom{n}{2}})$ : that reduction is the simplex step of [Remark 24.11](#), and the terms it discards are exactly what makes (24.14) strict for  $q > 1/2$ .

**Theorem 23.11** (Dyadic Thue–Morse spline). *At  $q = 1/2$ ,*

$$f_{1/2,N}(x) = \frac{2^{\binom{N+1}{2}}}{(N-1)!} \sum_{r=0}^{2^N-1} (-1)^{s_2(r)} \left(x - \frac{r}{2^N}\right)_+^{N-1}. \quad (23.22)$$

*Thus, after translating by the mean  $(1 - 2^{-N})/2$ , the inverse Fourier transform of the  $N$ th partial sinc product is a Thue–Morse signed spline on the dyadic grid.*

*Proof.* Here  $a_j = 2^{-j-1}$ , so

$$\prod_{j=0}^{N-1} a_j = 2^{-\binom{N+1}{2}}.$$

Every subset  $A$  corresponds to a binary word  $(\varepsilon_0, \dots, \varepsilon_{N-1})$  and

$$\sum_{j \in A} 2^{-j-1} = \sum_{j=0}^{N-1} \varepsilon_j 2^{-j-1}.$$

Reversing the order of the bits identifies this number with  $r/2^N$  for a unique  $0 \leq r < 2^N$ , while  $|A|$  and  $s_2(r)$  have the same parity. Substitute these facts into (23.20).  $\square$

The constant  $\|f_q\|_\infty$  left implicit in (23.29) can be evaluated exactly. The mechanism is the fixed point (23.3) read as a convolution.

**Proposition 23.12** (Exact density on the central band, and the sharp sup bound). *Let  $0 < q < 1$  and let  $f_q$  be the density of [Theorem 23.8](#). Then:*

1. *for every  $x \in \mathbb{R}$ ,*

$$f_q(x) = \frac{1}{1-q} \left[ F_q\left(\frac{x}{q}\right) - F_q\left(\frac{x-1+q}{q}\right) \right]; \quad (23.23)$$

2.  $\|f_q\|_\infty \leq \frac{1}{1-q}$ ;

3. *if  $0 < q \leq 1/2$ , then*

$$f_q(x) = \frac{1}{1-q} F_q\left(\frac{x}{q}\right) \quad (0 \leq x \leq 1-q), \quad (23.24)$$

*so that  $f_q$  is identically  $1/(1-q)$  on the band  $[q, 1-q]$  and*

$$\|f_q\|_\infty = \frac{1}{1-q}; \quad (23.25)$$

4. if  $1/2 < q < 1$ , the inequality in (ii) is strict.

At  $q = 1/2$ , (23.23) is (23.34) and (23.24) is (23.35).

*Proof.* By (23.3) we may write  $X_q = (1 - q)U_0 + qX'_q$  with  $U_0$  uniform on  $[0, 1]$  independent of the copy  $X'_q$ . Conditionally on  $X'_q = y$  the variable  $X_q$  is uniform on  $[qy, qy + 1 - q]$ , whose density is the constant  $1/(1 - q)$ ; averaging over  $y$  gives

$$f_q(x) = \frac{1}{1 - q} \mathbb{P}(x - (1 - q) \leq qX'_q \leq x) \quad (23.26)$$

for almost every  $x$ , hence for every  $x$ : the left side is continuous by Theorem 23.8 and the right side is continuous because  $X'_q$  has no atoms. For the same reason the probability equals  $F_q(x/q) - F_q((x - 1 + q)/q)$ , which is (23.23), and a probability is at most 1, which is (ii).

Let  $q \leq 1/2$  and  $0 \leq x \leq 1 - q$ . Then  $x - (1 - q) \leq 0$ , so the event in (23.26) reduces to  $qX'_q \leq x$  and its probability is  $F_q(x/q)$ ; this is (23.24). If moreover  $x \geq q$  then  $x/q \geq 1$  and  $F_q(x/q) = 1$ , so  $f_q \equiv 1/(1 - q)$  on  $[q, 1 - q]$ , an interval that is nonempty precisely because  $q \leq 1/2$ ; with (ii) this gives (23.25).

Finally let  $q > 1/2$ . The window  $[x - (1 - q), x]$  has length  $1 - q < q$ , while  $qX'_q$  has support  $[0, q]$  by Theorem 23.1. An interval shorter than  $q$  cannot contain  $[0, q]$ , so its complement in  $[0, q]$  contains a nonempty relatively open set, of positive probability; hence the probability in (23.26) is  $< 1$  for every  $x$ . It is continuous and tends to 0 as  $|x| \rightarrow \infty$ , so its supremum is attained and is  $< 1$ , which is the strict inequality.  $\square$

**Corollary 23.13** (A coupling convergence bound, with its constant). *Almost surely,*

$$0 \leq X_q - X_{q,N} \leq q^N. \quad (23.27)$$

Consequently

$$F_q(x - q^N) \leq F_{q,N}(x) \leq F_q(x + q^N), \quad (23.28)$$

and in particular

$$\sup_x |F_{q,N}(x) - F_q(x)| \leq \|f_q\|_\infty q^N. \quad (23.29)$$

By Proposition 23.12,  $F_q$  is Lipschitz with constant  $1/(1 - q)$ , so

$$\sup_{x \in \mathbb{R}} |F_{q,N}(x) - F_q(x)| \leq \frac{q^N}{1 - q}, \quad (23.30)$$

and for  $0 < q \leq 1/2$  the constant  $1/(1 - q)$  is the exact Lipschitz constant of  $F_q$ . At  $q = 1/2$  this reads  $\|f\|_\infty = 2$  and

$$\sup_{x \in \mathbb{R}} |F_{1/2,N}(x) - F(x)| \leq 2^{1-N}. \quad (23.31)$$

*Proof.* The omitted tail is nonnegative and bounded by  $(1 - q) \sum_{j \geq N} q^j = q^N$ , which is (23.27). The inclusions

$$\{X_q \leq x - q^N\} \subseteq \{X_{q,N} \leq x\} \subseteq \{X_q \leq x + q^N\}$$

give (23.28). Since  $F_q$  has a bounded continuous density,  $|F_q(x + h) - F_q(x)| \leq \|f_q\|_\infty |h|$ ; applying this on both sides gives (23.29), and Proposition 23.12(ii) turns it into (23.30). Exactness of the Lipschitz constant for  $q \leq 1/2$  is (23.25), and  $q = 1/2$  gives (23.31).  $\square$

**Remark 23.14** (The two normalizations at  $q = 1/2$ ). The value  $\|f\|_\infty = 2$  is the Rvachev normalization seen through Proposition 24.1:  $f(x) = 2 \operatorname{up}(2x - 1) = 2F(1 - |2x - 1|)$  on  $[0, 1]$ , so  $\|\operatorname{up}\|_\infty = \operatorname{up}(0) = F(1) = 1$  and the probability normalization  $\int_{\mathbb{R}} \operatorname{up} = 1$  agrees with the classical normalization  $\operatorname{up}(0) = 1$ .

## 23.5 The Fabius and Rvachev specializations

Set  $q = 1/2$ , write  $X = X_{1/2}$ , and let

$$F(x) := \mathbb{P}(X \leq x),$$

extended by  $F(x) = 0$  for  $x \leq 0$  and  $F(x) = 1$  for  $x \geq 1$ .

**Theorem 23.15** (The probabilistic Fabius equation). *The function  $F$  is  $C^\infty$ , satisfies*

$$F(1 - x) = 1 - F(x), \quad (23.32)$$

$$F(x) = \int_{2x-1}^{2x} F(u) du, \quad (23.33)$$

$$F'(x) = 2F(2x) - 2F(2x - 1). \quad (23.34)$$

In particular, for  $0 \leq x \leq 1/2$ ,

$$F'(x) = 2F(2x). \quad (23.35)$$

Together with  $F(0) = 0$ ,  $F(1) = 1$ , and symmetry, this is the standard Fabius function.

*Proof.* Smoothness follows from Theorem 23.8. Since  $X$  has no atoms and  $X \stackrel{d}{=} 1 - X$ ,

$$F(1 - x) = \mathbb{P}(1 - X \leq 1 - x) = \mathbb{P}(X \geq x) = 1 - F(x),$$

which proves (23.32).

At  $q = 1/2$ , the fixed-point equation is  $X \stackrel{d}{=} (U + X')/2$ . Conditioning on  $U = u$  gives

$$F(x) = \int_0^1 F(2x - u) du = \int_{2x-1}^{2x} F(v) dv,$$

proving (23.33). Differentiate using the fundamental theorem of calculus to obtain (23.34). On  $0 \leq x \leq 1/2$  the term  $F(2x - 1)$  is zero, giving (23.35).  $\square$

Let  $f = F'$  and define the centered density

$$\text{up}(y) := \begin{cases} \frac{1}{2} f\left(\frac{y+1}{2}\right), & -1 \leq y \leq 1, \\ 0, & |y| > 1. \end{cases} \quad (23.36)$$

This is the Rvachev up function in the probability normalization  $\int_{\mathbb{R}} \text{up}(y) dy = 1$ .

**Theorem 23.16** (Infinite sinc product and the up equation). *With the Fourier convention  $\widehat{g}(t) = \int_{\mathbb{R}} e^{ity} g(y) dy$ ,*

$$\widehat{\text{up}}(t) = \prod_{k=1}^{\infty} \text{sinc}\left(\frac{t}{2^k}\right). \quad (23.37)$$

Moreover,

$$\text{up}'(y) = 2 \text{up}(2y + 1) - 2 \text{up}(2y - 1) \quad (y \in \mathbb{R}). \quad (23.38)$$

*Proof.* The random variable with density  $\text{up}$  is  $Y = 2X - 1$ . Writing  $V_j = 2U_j - 1$ , we have independent uniform variables on  $[-1, 1]$  and

$$Y = \sum_{j=0}^{\infty} 2^{-j-1} V_j.$$

The characteristic function of  $V_j$  is  $\text{sinc } t$ . Independence therefore gives

$$\mathbb{E}e^{itY} = \prod_{j=0}^{\infty} \text{sinc}(t/2^{j+1}),$$

which is (23.37).

From (23.36),  $f(x) = 2 \text{up}(2x - 1)$ . Differentiate (23.34) to obtain

$$f'(x) = 4f(2x) - 4f(2x - 1).$$

Set  $x = (y + 1)/2$ . Because  $\text{up}'(y) = f'((y + 1)/2)/4$ , substitution of  $f(t) = 2 \text{up}(2t - 1)$  gives

$$\text{up}'(y) = 2 \text{up}(2y + 1) - 2 \text{up}(2y - 1).$$

The identity remains valid outside the support because all boundary derivatives vanish. □

*Remark 23.17* (Where the  $q$ -objects enter). The parameter  $q$  controls four structures at once:

1. the geometric coordinate weights  $(1 - q)q^j$ ;
2. the factors  $1 - q^{2m}$  in the cumulants (23.7);
3. the geometric interpolation nodes and Gaussian residual moments of Chapter 21;
4. at  $q = 1/2$ , the Mersenne products, dyadic Thue–Morse splines, and the Fabius–Rvachev functional equations.

Thus the appearance of  $q$ -Pochhammer and Gaussian coefficients in exact Fabius formulas is not accidental: both arise from the same geometric scaling law.

## Chapter 24

# Dyadic Gaussian–Thue–Morse structure and the values of $F$

The three preceding chapters built the geometric filters, the half-base arithmetic and the geometric uniform law separately. This chapter runs them together in the one case where everything is exactly computable: the base  $q = 1/2$ , where the geometric uniform sum is the Fabius variable, the Prouhet filter is the Thue–Morse sign, and the resulting closed formula evaluates  $F$  at every inverse power of two.

**Proposition 24.1** (The up function is a Fabius tail). *For  $|y| \leq 1$ ,*

$$\text{up}(y) = F(1 - |y|), \quad (24.1)$$

*and in particular, for  $0 \leq t \leq 1$ ,*

$$\text{up}(t) = 1 - F(t) = \mathbb{P}(X \geq t). \quad (24.2)$$

*Proof.* Let  $-1 \leq y \leq 0$  and put  $x = (y + 1)/2 \in [0, 1/2]$ . By (23.35),  $f(x) = 2F(2x) = 2F(y + 1)$ , so (23.36) gives  $\text{up}(y) = \frac{1}{2}f(x) = F(1 + y) = F(1 - |y|)$ .

Differentiating the symmetry (23.32) gives  $f(1 - x) = f(x)$ , so  $\text{up}(-y) = \frac{1}{2}f((1 - y)/2) = \frac{1}{2}f((1 + y)/2) = \text{up}(y)$ ; thus  $\text{up}$  is even and (24.1) holds on all of  $[-1, 1]$ . Finally, for  $0 \leq t \leq 1$  the symmetry (23.32) turns  $F(1 - t)$  into  $1 - F(t)$ , and  $X$  has a density, so  $1 - F(t) = \mathbb{P}(X > t) = \mathbb{P}(X \geq t)$ .  $\square$

**Corollary 24.2** (Tail form of the moments). *For every  $n \geq 1$ ,*

$$\mathbb{E}X^n = n \int_0^1 t^{n-1} \text{up}(t) \, dt. \quad (24.3)$$

*Proof.* Since  $X \geq 0$ , Tonelli's theorem applied to the nonnegative integrand  $nt^{n-1} \mathbf{1}_{\{t < X\}}$  gives

$$\mathbb{E}X^n = \mathbb{E} \int_0^X nt^{n-1} \, dt = \int_0^1 nt^{n-1} \mathbb{P}(X \geq t) \, dt,$$

because  $0 \leq X \leq 1$ . Now apply (24.2).  $\square$

Write

Write

$$\Lambda(z) := \int_0^1 e^{zu} \, du = \frac{e^z - 1}{z}, \quad \Lambda(0) := 1, \quad (24.4)$$

an entire function with  $\Lambda(z) = \sum_{r \geq 0} z^r / (r+1)!$  and  $\Lambda(z) = 1 + \mathcal{O}(z)$  at the origin, whose zeros are exactly the points  $2\pi i k$  with  $k \in \mathbb{Z} \setminus \{0\}$ ; in particular  $\Lambda$  has no zero in the disc  $|z| < 2\pi$ . For  $q \in \mathbb{C}$  with  $|q| < 1$  set

$$\mathcal{A}_q(z) := \prod_{j=0}^{\infty} \Lambda((1-q)q^j z) = \sum_{n \geq 0} \alpha_n^{(q)} z^n. \quad (24.5)$$

**Theorem 24.3** (Moment product, dilation equation, coefficient recurrence). *Let  $q \in \mathbb{C}$  with  $|q| < 1$ . Then:*

1. *the product in (24.5) converges locally uniformly on  $\mathbb{C}$  and defines an entire function with  $\mathcal{A}_q(0) = 1$ ;*
2.  *$\mathcal{A}_q$  satisfies the dilation equation*

$$\mathcal{A}_q(z) = \Lambda((1-q)z) \mathcal{A}_q(qz), \quad (24.6)$$

*and is the unique formal power series with constant term 1 that does so; its coefficients satisfy  $\alpha_0^{(q)} = 1$  and*

$$(1-q^n) \alpha_n^{(q)} = \sum_{r=0}^{n-1} \frac{(1-q)^{n-r} q^r}{(n-r+1)!} \alpha_r^{(q)} \quad (n \geq 1); \quad (24.7)$$

3. *consequently each  $\alpha_n^{(q)}$  is a rational function of  $q$  with rational coefficients, its poles are roots of unity, and it is regular on the open unit disc;*
4. *for real  $0 < q < 1$  the function  $\mathcal{A}_q$  is the moment generating function of  $X_q$ :*

$$\mathcal{A}_q(z) = \mathbb{E} e^{zX_q} = \prod_{j=0}^{\infty} \Lambda((1-q)q^j z), \quad \alpha_n^{(q)} = \frac{\mathbb{E} X_q^n}{n!}, \quad (24.8)$$

*and  $\varphi_q(t) = \mathcal{A}_q(it)$  for every real  $t$ .*

*Proof.* (i) On a compact set  $K \subset \mathbb{C}$  the arguments  $(1-q)q^j z$  are  $\mathcal{O}(|q|^j)$  uniformly, and  $\Lambda(w) - 1 = \mathcal{O}(w)$  near the origin, so  $\sum_{j \geq 0} \sup_{z \in K} |\Lambda((1-q)q^j z) - 1|$  is dominated by a constant multiple of  $\sum_{j \geq 0} |q|^j < \infty$ . The standard criterion for infinite products gives locally uniform convergence to an entire function, and every factor equals 1 at  $z = 0$ .

(ii) Replacing  $z$  by  $qz$  shifts the index by one:

$$\Lambda((1-q)z) \mathcal{A}_q(qz) = \Lambda((1-q)z) \prod_{j \geq 0} \Lambda((1-q)q^{j+1}z) = \prod_{j \geq 0} \Lambda((1-q)q^j z) = \mathcal{A}_q(z),$$

which is (24.6). Comparing the coefficient of  $z^n$  in

$$\sum_{n \geq 0} \alpha_n^{(q)} z^n = \left( \sum_{s \geq 0} \frac{(1-q)^s z^s}{(s+1)!} \right) \left( \sum_{r \geq 0} \alpha_r^{(q)} q^r z^r \right)$$

gives  $\alpha_n^{(q)} = \sum_{r=0}^n (1-q)^{n-r} q^r \alpha_r^{(q)} / (n-r+1)!$ ; the term  $r = n$  on the right is  $q^n \alpha_n^{(q)}$ , and moving it to the left proves (24.7). Since  $1 - q^n \neq 0$  for  $n \geq 1$ , that recurrence determines every coefficient from its predecessors, which is the asserted formal uniqueness.

(iii) Induct on (24.7). One has  $\alpha_0^{(q)} = 1$ , and  $\alpha_n^{(q)}$  is obtained from  $\alpha_0^{(q)}, \dots, \alpha_{n-1}^{(q)}$  by rational operations with the single new denominator  $1 - q^n$ . Hence  $\alpha_n^{(q)} \in \mathbb{Q}(q)$  and its poles lie among the zeros of  $\prod_{j=1}^n (1 - q^j)$ , all of which are roots of unity, and none of those lies in  $|q| < 1$ .

(iv) Let  $0 < q < 1$ . By [Theorem 23.1](#) we have  $0 \leq X_q \leq 1$ , so  $\mathbb{E}X_q^n \leq 1$  and  $\sum_n |z|^n \mathbb{E}X_q^n / n! \leq e^{|z|}$ ; hence  $z \mapsto \mathbb{E}e^{zX_q}$  is entire and term-by-term integration is legitimate. For the truncation  $X_{q,N}$  of (23.17), independence gives

$$\mathbb{E}e^{zX_{q,N}} = \prod_{j=0}^{N-1} \int_0^1 e^{z(1-q)q^j u} du = \prod_{j=0}^{N-1} \Lambda((1-q)q^j z).$$

By (23.27),  $0 \leq X_q - X_{q,N} \leq q^N$  pointwise, so  $e^{zX_{q,N}} \rightarrow e^{zX_q}$  uniformly on the sample space while remaining bounded by  $e^{|z|}$ ; dominated convergence identifies the limit with  $\mathcal{A}_q(z)$  and proves (24.8). Substituting  $z = it$  reproduces  $\varphi_q$ , and comparison of Taylor coefficients gives  $\alpha_n^{(q)} = \mathbb{E}X_q^n / n!$ .  $\square$

*Remark 24.4* (Dilation, refinement, Mahler). Equation (24.6) carries three names in three literatures: it is a *dilation* or *refinement* equation in approximation theory and wavelet analysis, and a *Mahler functional equation* in transcendence theory and in the theory of automatic sequences. This book uses them interchangeably. Part (iii) is visible already in the first cases, which (24.7) produces as

$$\alpha_1^{(q)} = \frac{1}{2}, \quad \alpha_2^{(q)} = \frac{q+2}{12(q+1)}, \quad \alpha_3^{(q)} = \frac{1}{24(q+1)};$$

the half-base values are collected in [Example 24.6](#), and the systematic clearing of the denominators is [Definition 24.61](#).

**Corollary 24.5** (The half-base moment generating function). *Write  $X := X_{1/2}$ ,  $\mathcal{A} := \mathcal{A}_{1/2}$  and  $\alpha_n := \alpha_n^{(1/2)} = \mathbb{E}X^n / n!$ . Then  $\mathcal{A}$  is entire,  $\mathcal{A}(it) = \varphi_{1/2}(t)$ ,*

$$\mathcal{A}(z) = \prod_{j=1}^{\infty} \Lambda\left(\frac{z}{2^j}\right), \quad (24.9)$$

$$\mathcal{A}(2z) = \Lambda(z) \mathcal{A}(z), \quad (24.10)$$

and  $\alpha_0 = 1$  with

$$(2^n - 1) \alpha_n = \sum_{r=0}^{n-1} \frac{\alpha_r}{(n-r+1)!} \quad (n \geq 1). \quad (24.11)$$

*Proof.* Put  $q = 1/2$  in [Theorem 24.3](#). The product (24.8) becomes  $\prod_{j \geq 0} \Lambda(2^{-1}2^{-j}z)$ , which is (24.9); replacing  $z$  by  $2z$  in (24.6) gives (24.10). In (24.7) every summand acquires the factor  $(1/2)^{n-r}(1/2)^r = 2^{-n}$  and the left side is  $(1 - 2^{-n})\alpha_n$ ; multiplication by  $2^n$  gives (24.11). Finally  $\Lambda(is) = e^{is/2} \text{sinc}(s/2)$ , so (24.9) at  $z = it$  is (23.5) at  $q = 1/2$ ; this is the sinc product of [Theorem 23.16](#).  $\square$

**Example 24.6** (The first coefficients and the first dyadic values). Recurrence (24.11) gives

$$\alpha_0 = 1, \quad \alpha_1 = \frac{1}{2}, \quad \alpha_2 = \frac{5}{36}, \quad \alpha_3 = \frac{1}{36}, \quad \alpha_4 = \frac{143}{32400}, \quad \alpha_5 = \frac{19}{32400},$$

equivalently  $\mathbb{E}X^2 = 2! \alpha_2 = 5/18$  and  $\mathbb{E}X^4 = 4! \alpha_4 = 143/1350$ . These agree with the cumulant route of [Corollaries 23.5](#) and [23.6](#): for instance  $\kappa_2(Y_{1/2}) = B_2/(2 \cdot 3) = 1/36$ , so  $\mathbb{E}X^2 = \frac{1}{36} + \frac{1}{4} = \frac{5}{18}$ .

By [Theorem 24.9](#) below,  $F(2^{-n}) = 2^{-\binom{n}{2}} \alpha_n$ , so the same six rationals are the first inverse-dyadic values of the Fabius function:

$$F(1) = 1, \quad F\left(\frac{1}{2}\right) = \frac{1}{2}, \quad F\left(\frac{1}{4}\right) = \frac{5}{72}, \quad F\left(\frac{1}{8}\right) = \frac{1}{288},$$



$$F\left(\frac{1}{16}\right) = \frac{143}{2073600}, \quad F\left(\frac{1}{32}\right) = \frac{19}{33177600}.$$

The value  $F(1/2) = 1/2$  is forced by (23.32); the remaining entries are the first concrete instances of the rationality asserted, without values, by Corollary 23.6. The powers  $2^{-\binom{n}{2}}$  are what makes the values fall off so much faster than the coefficients.

*Remark 24.7* (Two routes to the same rationals). Unlike the moment–cumulant recursion (23.12), which is stated for the centred variable and needs Bernoulli input, the recursion (24.11) is a triangular rational recursion in the raw coefficients alone: one rational division per index, no Bernoulli numbers and no Bell polynomials. It is the cheaper and the numerically stable route to the values below.

**Lemma 24.8** (Derivative iterates of the Fabius function). *For every  $m \geq 0$ ,*

$$F^{(m)}(x) = 2^{\binom{m+1}{2}} F(2^m x) \quad (0 \leq x \leq 2^{-m}). \quad (24.12)$$

*In particular  $F^{(m)}(0) = 0$  for every  $m \geq 0$ .*

*Proof.* The case  $m = 0$  is trivial, and (23.35) is the case  $m = 1$ . Assume (24.12) for  $m$  and let  $0 \leq x \leq 2^{-m-1}$ , so that  $0 \leq 2^m x \leq 1/2$ . Differentiating and applying (23.35) at  $2^m x$  gives

$$F^{(m+1)}(x) = 2^{\binom{m+1}{2}} 2^m F'(2^m x) = 2^{\binom{m+1}{2} + m + 1} F(2^{m+1} x),$$

and  $\binom{m+1}{2} + m + 1 = \binom{m+2}{2}$ . Since  $F(0) = 0$ , evaluating (24.12) at  $x = 0$  gives  $F^{(m)}(0) = 0$ .  $\square$

**Theorem 24.9** (Half-moment evaluation). *For every  $n \geq 0$ ,*

$$F(2^{-n}) = 2^{-\binom{n}{2}} \alpha_n = \frac{\mathbb{E}X^n}{n! 2^{\binom{n}{2}}}. \quad (24.13)$$

*Proof.* For  $n = 0$  both sides are  $F(1) = 1$ ; let  $n \geq 1$ . By Lemma 24.8 we have  $F^{(j)}(0) = 0$  for  $0 \leq j < n$ , and  $F \in C^\infty$  by Theorem 23.15, so Taylor's formula with integral remainder at the point  $2^{-n}$  reduces to the remainder alone:

$$F(2^{-n}) = \frac{1}{(n-1)!} \int_0^{2^{-n}} (2^{-n} - t)^{n-1} F^{(n)}(t) dt = \frac{2^{\binom{n+1}{2}}}{(n-1)!} \int_0^{2^{-n}} (2^{-n} - t)^{n-1} F(2^n t) dt,$$

where (24.12) was used on the whole interval of integration. Substituting  $u = 2^n t$  produces the factor  $2^{-n(n-1)}$  from  $(2^{-n} - t)^{n-1}$  and the factor  $2^{-n}$  from  $dt$ , and

$$\binom{n+1}{2} - n(n-1) - n = \frac{n(n+1)}{2} - n^2 = -\binom{n}{2},$$

so that

$$F(2^{-n}) = \frac{2^{-\binom{n}{2}}}{(n-1)!} \int_0^1 (1-u)^{n-1} F(u) du.$$

By Proposition 24.1,  $F(u) = \text{up}(1-u)$  for  $0 \leq u \leq 1$ ; substituting  $t = 1-u$  and applying Corollary 24.2,

$$\frac{1}{(n-1)!} \int_0^1 t^{n-1} \text{up}(t) dt = \frac{\mathbb{E}X^n}{n(n-1)!} = \alpha_n. \quad \square$$

**Proposition 24.10** (Inverse-power values at a general base). *Let  $0 < q < 1$  and  $n \geq 0$ . Then*

$$F_q(q^n) \leq \frac{q^{\binom{n+1}{2}}}{(1-q)^n n!} \mathbb{E}(X_q^n), \quad (24.14)$$

with equality for every  $n$  if and only if  $q \leq 1/2$ ; for  $q > 1/2$  and  $n \geq 1$  the inequality is strict. Thus, for  $0 < q \leq 1/2$ ,

$$F_q(q^n) = \frac{q^{\binom{n+1}{2}}}{(1-q)^n n!} \mathbb{E}(X_q^n) = \frac{q^{\binom{n+1}{2}}}{(1-q)^n} \alpha_n^{(q)}, \quad (24.15)$$

with  $\alpha_n^{(q)}$  as in (24.5), and  $F_q(1 - q^n) = 1 - F_q(q^n)$ . At  $q = 1/2$  the prefactor is  $2^{-\binom{n}{2}}$  and (24.15) is (24.13).

*Proof.* For  $n = 0$  both sides are  $F_q(1) = 1$ ; let  $n \geq 1$ . Split the series (23.1) after  $n$  terms,

$$X_q = \sum_{j=0}^{n-1} a_j U_j + q^n X'_q, \quad a_j := (1-q)q^j, \quad X'_q := (1-q) \sum_{i \geq 0} q^i U_{n+i}, \quad (24.16)$$

with the  $a_j$  of (23.17) and  $X'_q$  a copy of  $X_q$  independent of  $U_0, \dots, U_{n-1}$ ; note  $\prod_{j < n} a_j = (1-q)^n q^{\binom{n}{2}}$ . Conditionally on  $X'_q = y$ , the event  $\{X_q \leq q^n\}$  reads  $\sum_{j < n} a_j U_j \leq t$  with  $t = q^n(1-y) \in [0, q^n]$ , so its conditional probability is the cube section

$$V(t) := \text{vol} \left\{ u \in [0, 1]^n : \sum_{j=0}^{n-1} a_j u_j \leq t \right\}.$$

Dropping the constraints  $u_j \leq 1$  enlarges the region to a simplex, so

$$V(t) \leq \frac{t^n}{n! \prod_{j < n} a_j} = \frac{t^n}{n! (1-q)^n q^{\binom{n}{2}}}, \quad (24.17)$$

with equality precisely when the simplex lies inside the cube, that is when  $t \leq \min_j a_j = (1-q)q^{n-1}$ : if  $t > (1-q)q^{n-1}$ , the point with  $u_{n-1} = t/a_{n-1} > 1$  and all other coordinates 0 lies in the simplex but not in the cube, and so does a neighbourhood of it, of positive volume. Averaging (24.17) over  $y$  with  $t = q^n(1-y)$  and using  $1 - X_q \stackrel{d}{=} X_q$  from (23.2),

$$F_q(q^n) \leq \frac{q^{n^2} \mathbb{E}[(1 - X_q)^n]}{n! (1-q)^n q^{\binom{n}{2}}} = \frac{q^{n^2 - \binom{n}{2}}}{n! (1-q)^n} \mathbb{E}(X_q^n),$$

and  $n^2 - \binom{n}{2} = \binom{n+1}{2}$ ; this is (24.14).

If  $q \leq 1/2$ , then every admissible threshold satisfies  $t \leq q^n = q \cdot q^{n-1} \leq (1-q)q^{n-1}$ , so (24.17) is an equality almost surely and (24.15) follows. If  $q > 1/2$  and  $n \geq 1$ , then  $(1-q)q^{n-1} < q^n$  and

$$\mathbb{P}(t > (1-q)q^{n-1}) = \mathbb{P}\left(1 - X'_q > \frac{1-q}{q}\right) > 0,$$

because  $(1-q)/q < 1$  and  $1 - X'_q$  has support  $[0, 1]$  by Theorem 23.1; on that event (24.17) is strict, and so therefore is the average. Finally  $X_q$  has no atoms and  $X_q \stackrel{d}{=} 1 - X_q$ , which gives  $F_q(1 - q^n) = 1 - F_q(q^n)$ .  $\square$

*Remark 24.11* (The half-base cube cut). At  $q = 1/2$ , [Proposition 24.10](#) is a second proof of [\(24.13\)](#) which uses neither [Proposition 24.1](#) nor the smoothness of  $F$ . In scaled coordinates it reads as follows. Write  $X = \sum_{j \geq 1} 2^{-j} V_j$  with  $V_1, V_2, \dots$  independent and uniform on  $[0, 1]$ , multiply by  $2^n$  and split the series after  $n$  terms:

$$2^n X = \sum_{j=1}^n c_j V_j + X', \quad c_j := 2^{n-j}, \quad X' := \sum_{i \geq 1} 2^{-i} V_{n+i}, \quad (24.18)$$

where  $X'$  is independent of  $V_1, \dots, V_n$  and distributed as  $X$ . Conditionally on  $X' = x$ , the event  $2^n X \leq 1$  is  $\sum_j c_j V_j \leq t$  with  $t := 1 - x \in [0, 1]$ . Each  $c_j$  is an integer with  $c_j \geq 1$ , so no upper constraint is active, the admissible region is the entire simplex, and its volume is  $t^n / (n! \prod_j c_j) = (1 - x)^n / (n! 2^{\binom{n}{2}})$  since  $\prod_{j=1}^n 2^{n-j} = 2^{\binom{n}{2}}$ . Averaging over  $X'$  and using  $X \stackrel{d}{=} 1 - X$  returns [\(24.13\)](#).

**Corollary 24.12** (Bell-polynomial form of the Fabius values). Put  $\kappa_1 := 1/2$  and  $\kappa_r := \kappa_r(Y_{1/2})$  for  $r \geq 2$ , the half-base cumulants [\(23.14\)](#). Then for every  $n \geq 0$

$$F(2^{-n}) = \frac{2^{-\binom{n}{2}}}{n!} \mathcal{B}_n(\kappa_1, \dots, \kappa_n) = 2^{-\binom{n}{2}} \sum_{\substack{\nu_1, \dots, \nu_n \geq 0 \\ \nu_1 + 2\nu_2 + \dots + n\nu_n = n}} \prod_{r=1}^n \frac{1}{\nu_r!} \left( \frac{\kappa_r}{r!} \right)^{\nu_r}. \quad (24.19)$$

Only  $\kappa_1$  and the even  $\kappa_r$  contribute, and every value  $F(2^{-n})$  is rational.

*Proof.* Cumulants of order  $r \geq 2$  are translation invariant, so  $X = Y_{1/2} + 1/2$  has  $\kappa_1(X) = \mathbb{E}X = 1/2$  by [\(23.2\)](#) and  $\kappa_r(X) = \kappa_r$  for  $r \geq 2$ . The moment generating function is the exponential of the cumulant generating function, so the defining identity [\(23.10\)](#) gives  $\mathbb{E}(X^n) = \mathcal{B}_n(\kappa_1, \dots, \kappa_n)$ , exactly as in the proof of [Corollary 23.5](#) but for the uncentred variable; now apply [\(24.13\)](#). The second expression is the explicit expansion of  $\mathcal{B}_n$  obtained by multiplying out  $\exp(\sum_r \kappa_r t^r / r!)$ , and the odd cumulants beyond the first vanish by [\(23.14\)](#).  $\square$

*Remark 24.13* (Sharpness of the hypothesis  $q \leq 1/2$ ). For  $q > 1/2$  the closed form [\(24.15\)](#) fails, and can fail grossly: at  $q = 4/5$  and  $n = 1$  its right side is  $q/(2(1 - q)) = 2$ , which is not even a probability. At  $q = 3/5$  the right sides for  $n = 1, 2, 3$  exceed the true values by  $1.3 \cdot 10^{-2}$ ,  $5.6 \cdot 10^{-4}$  and  $1.2 \cdot 10^{-5}$ ; in particular  $F_{3/5}(3/5) = 0.73738\dots$  against  $3/4$ . The reason is the one isolated in the proof: for  $q > 1/2$  the smallest coefficient  $(1 - q)q^{n-1}$  drops below the threshold  $q^n$  and the cube truncates the simplex, the deficit being exactly the inclusion–exclusion terms of [Theorem 23.9](#) that [\(24.18\)](#) annihilates.

## 24.1 Reciprocal nodes and the leading-coefficient filter

The filter  $\mathcal{G}_{n,q}$  of [\(21.10\)](#) reproduces the *constant* coefficient of a series and annihilates the degrees  $1, \dots, n$ . Its companion reproduces the coefficient of degree  $n$  and annihilates the degrees  $0, \dots, n - 1$ . That companion lives on the reciprocal nodes  $-q^{-k}$ , its weights are again a signed Gaussian row, and its residual tail is again controlled by a single Gaussian row. It is this second normalization, not the evaluation-at-zero one, that produces the exact dyadic formulas of [Chapter 23](#).

Throughout this section  $q$  lies in a field  $K$  and  $q^j \neq 1$  for  $1 \leq j \leq n$ ; from [Theorem 24.21](#) onwards  $K$  is also assumed to have characteristic zero. Put

$$x_k := -q^{-k}, \quad W_{n,k}(q) := (-1)^k q^{\binom{k}{2}} \begin{bmatrix} n \\ k \end{bmatrix}_q \quad (0 \leq k \leq n). \quad (24.20)$$

The nodes are distinct, because  $q^i = q^j$  with  $0 \leq i < j \leq n$  would force  $q^{j-i} = 1$ .

**Theorem 24.14** (Exact geometric moment transform). *For every integer  $d \geq 0$ ,*

$$\sum_{k=0}^n W_{n,k}(q) x_k^d = (-1)^d (q^{-d}; q)_n. \quad (24.21)$$

*In particular*

$$\sum_{k=0}^n W_{n,k}(q) x_k^d = 0 \quad (0 \leq d < n), \quad (24.22)$$

*and*

$$\sum_{k=0}^n W_{n,k}(q) x_k^n = q^{-\binom{n+1}{2}} (q; q)_n. \quad (24.23)$$

*Proof.* Since  $x_k^d = (-1)^d q^{-kd}$ , the left side of (24.21) equals

$$(-1)^d \sum_{k=0}^n (-1)^k q^{\binom{k}{2}} \begin{bmatrix} n \\ k \end{bmatrix}_q (q^{-d})^k,$$

which is  $(-1)^d (q^{-d}; q)_n$  by (3.1) at  $z = q^{-d}$ . If  $0 \leq d < n$ , the factor with  $j = d$  in

$$(q^{-d}; q)_n = \prod_{j=0}^{n-1} (1 - q^{j-d})$$

is zero, which gives (24.22). For  $d = n$ , extract  $-q^{j-n}$  from each factor:

$$(q^{-n}; q)_n = (-1)^n q^{-\sum_{j=0}^{n-1} (n-j)} \prod_{j=0}^{n-1} (1 - q^{n-j}) = (-1)^n q^{-\binom{n+1}{2}} (q; q)_n,$$

because  $\sum_{j=0}^{n-1} (n-j) = \binom{n+1}{2}$ . Multiplication by the outer sign  $(-1)^n$  proves (24.23).  $\square$

**Corollary 24.15** (Leading-coefficient functional). *If  $p \in K[x]$  has degree at most  $n$ , then*

$$[x^n]p(x) = \frac{q^{\binom{n+1}{2}}}{(q; q)_n} \sum_{k=0}^n W_{n,k}(q) p(-q^{-k}). \quad (24.24)$$

*Proof.* Write  $p(x) = \sum_{d=0}^n b_d x^d$  and apply the functional termwise. Every term with  $d < n$  vanishes by (24.22), and the term  $d = n$  contributes  $b_n$  by (24.23).  $\square$

**Remark 24.16** (Three faces of one weight row). The functional (24.24) is not a new interpolation device. It is the divided difference  $y[x_0, \dots, x_n]$  of Theorem 17.10 after the substitution  $q \mapsto q^{-1}$ ,  $x \mapsto -x$ ; in particular the closed evaluation of  $\prod_{j \neq k} (x_k - x_j)$  needed for a Lagrange derivation, and the uniqueness of a functional returning the leading coefficient, are already contained in that theorem and in Theorem 21.3. The Vandermonde determinant of the nodes differs only by  $x \mapsto -x$  and a column reversal from the alternant  $a_\delta(1, q^{-1}, \dots, q^{-n})$  evaluated inside the proof of Theorem 4.9. At  $q = 1/2$  the data (24.20) become the data (22.6):  $x_k = \xi_k$  and  $W_{n,k}(1/2) = w_{n,k}$ , so Theorem 22.4 is the case  $q = 1/2$  of Corollary 24.15.

**Theorem 24.17** (The reciprocal filter and its exact tail). *For a formal power series  $f(X) = \sum_{m \geq 0} a_m X^m$  over  $K$  put*

$$(\mathcal{D}_{n,q}f)(X) := \frac{q^{\binom{n+1}{2}}}{(q; q)_n} \sum_{k=0}^n W_{n,k}(q) f(x_k X). \quad (24.25)$$

Then

$$[X^m] \mathcal{D}_{n,q}f = \frac{q^{\binom{n+1}{2}}}{(q; q)_n} (-1)^m (q^{-m}; q)_n a_m, \quad (24.26)$$

$$(\mathcal{D}_{n,q}f)(X) = \sum_{r \geq 0} (-1)^r q^{-nr} \begin{bmatrix} n+r \\ n \end{bmatrix}_q a_{n+r} X^{n+r}. \quad (24.27)$$

Thus  $\mathcal{D}_{n,q}$  deletes the coefficients of degrees  $0, \dots, n-1$ , reproduces  $a_n$ , and multiplies every later coefficient by an explicit signed Gaussian polynomial. If  $\deg f \leq n$ , then  $(\mathcal{D}_{n,q}f)(1) = a_n$ , in agreement with [Corollary 24.15](#).

*Proof.* The outer sum in (24.25) is finite, so coefficient extraction commutes with it, and (24.26) is [Theorem 24.14](#). The multiplier vanishes for  $m < n$  by (24.22). Let  $m = n + r$  with  $r \geq 0$ . Extracting  $-q^{j-n-r}$  from each factor of  $(q^{-n-r}; q)_n$  gives

$$(q^{-n-r}; q)_n = (-1)^n q^{\binom{n}{2} - n(n+r)} (q^{r+1}; q)_n,$$

since  $\sum_{j=0}^{n-1} (j - n - r) = \binom{n}{2} - n(n+r)$  and  $\prod_{j=0}^{n-1} (1 - q^{n+r-j}) = (q^{r+1}; q)_n$ . Hence the sign becomes  $(-1)^{n+r} (-1)^n = (-1)^r$ , the total exponent of  $q$  becomes

$$\binom{n+1}{2} + \binom{n}{2} - n(n+r) = -nr,$$

and

$$\frac{(q^{r+1}; q)_n}{(q; q)_n} = \frac{(q; q)_{n+r}}{(q; q)_r (q; q)_n} = \begin{bmatrix} n+r \\ n \end{bmatrix}_q,$$

which is (24.27). □

**Remark 24.18** (The two exact tails). [Theorems 21.9](#) and [24.17](#) are companions, not restatements. Both multiply the surviving coefficients by one Gaussian row, but they extract different coefficients on different node sets:  $\mathcal{G}_{n,q}$  evaluates at 0 on the nodes  $q^k$  and returns  $a_0$  with tail multiplier  $(-1)^n q^{\binom{n+1}{2}} \begin{bmatrix} n+r \\ n \end{bmatrix}_q$ , whereas  $\mathcal{D}_{n,q}$  extracts the top coefficient on the nodes  $-q^{-k}$  and returns  $a_n$  with tail multiplier  $(-1)^r q^{-nr} \begin{bmatrix} n+r \\ n \end{bmatrix}_q$ . The  $r$ -dependent sign and the negative power of  $q$  have no counterpart in (21.12): reversing a polynomial exchanges the two functionals only in the finite-degree case, and a power series has no top coefficient to reverse. For  $0 < q < 1$  the estimate  $\begin{bmatrix} n+r \\ n \end{bmatrix}_q \leq 1/(q; q)_n$  used in [Theorem 21.11](#) applies verbatim to (24.27): if  $f$  is analytic on  $|z| \leq R$  with  $R > q^{-n}$  and  $M_R = \max_{|z|=R} |f|$ , then

$$|(\mathcal{D}_{n,q}f)(1) - a_n| \leq \frac{M_R}{(q; q)_n R^n} \cdot \frac{q^{-n}/R}{1 - q^{-n}/R}.$$

**Corollary 24.19** (Integrality and the exact 2-adic valuation). *For  $0 \leq k \leq n$ ,*

$$\begin{bmatrix} n \\ k \end{bmatrix}_{1/2} = 2^{-k(n-k)} \begin{bmatrix} n \\ k \end{bmatrix}_2. \quad (24.28)$$

The integer  $\begin{bmatrix} n \\ k \end{bmatrix}_2$  is odd; consequently  $2^{k(n-k)} \begin{bmatrix} n \\ k \end{bmatrix}_{1/2}$  is an odd integer and

$$\text{ord}_2 \begin{bmatrix} n \\ k \end{bmatrix}_{1/2} = -k(n-k). \quad (24.29)$$

*Proof.* Reciprocity, [Theorem 2.5\(ii\)](#), reads  $\begin{bmatrix} n \\ k \end{bmatrix}_{q^{-1}} = q^{-k(n-k)} \begin{bmatrix} n \\ k \end{bmatrix}_q$ ; taking  $q = 2$  gives (24.28). By [Corollary 2.4](#) the Gaussian coefficient lies in  $\mathbb{N}_0[q]$ , and by [Theorem 2.5\(iii\)](#) its constant term is 1; every other monomial  $q^j$  with  $j \geq 1$  is even at  $q = 2$ , so  $\begin{bmatrix} n \\ k \end{bmatrix}_2 \equiv 1 \pmod{2}$ . An odd integer has 2-adic valuation 0, so (24.29) follows from (24.28).  $\square$

*Remark 24.20* (Reading the valuation). Comparing (24.28) with (22.3) identifies the odd integer explicitly:

$$\begin{bmatrix} n \\ k \end{bmatrix}_2 = \frac{\mathfrak{m}_n}{\mathfrak{m}_k \mathfrak{m}_{n-k}},$$

and by [Theorem 6.1](#) it counts the  $k$ -dimensional subspaces of  $\mathbb{F}_2^n$ . Thus the half-base Gaussian row is a finite-geometric count divided by an explicit power of two, and the exponent of that power is  $k(n-k)$ , the dimension of the Grassmannian. In the same way (22.2) exhibits  $(1/2; 1/2)_n$  as a power-of-two normalization of a product of Mersenne numbers, with  $\text{ord}_2(1/2; 1/2)_n = -\binom{n+1}{2}$  because  $\mathfrak{m}_n$  is odd.

**Theorem 24.21** (Gaussian coefficient transfer). *Let  $K$  have characteristic zero, let  $q \in K^\times$  satisfy  $q^j \neq 1$  for  $1 \leq j \leq n$ , and let*

$$U(z) = \sum_{m \geq 0} u_m z^m, \quad u_0 = 1, \quad V(z) = \sum_{m \geq 0} v_m z^m$$

*be power series in  $K[[z]]$ . For each  $0 \leq k \leq n$  there is exactly one series  $\Psi_k(z) = \sum_{m \geq 0} c_{k,m} z^m \in K[[z]]$  satisfying the bridge relation*

$$\Psi_k(z) U(z) = e^{-z} V(-q^{-k} z), \quad (24.30)$$

*namely  $\Psi_k(z) = e^{-z} V(-q^{-k} z)/U(z)$ , the quotient being legitimate in  $K[[z]]$  because  $U$  has constant term 1. For that family, with the weights of (24.20),*

$$\sum_{k=0}^n W_{n,k}(q) c_{k,d} = 0 \quad (0 \leq d < n), \quad (24.31)$$

$$\sum_{k=0}^n W_{n,k}(q) c_{k,n} = q^{-\binom{n+1}{2}} (q; q)_n v_n. \quad (24.32)$$

*Of the two series, only the single coefficient  $v_n$  is ever seen;  $U$  enters through its constant term alone. In the self-referential case  $U(z) = A(-z)$ ,  $V = A$  with  $a_0 = 1$ , in which (24.30) reads  $\Psi_k(z) A(-z) = e^{-z} A(-q^{-k} z)$  and  $v_n = a_n$ , the theorem therefore extracts the  $n$ th coefficient of  $A$  from the family attached to  $A$  itself; that is the form used in [Theorem 24.28](#).*

*Proof.* Write  $S_d := \sum_{k=0}^n W_{n,k}(q) c_{k,d}$ . Comparing the coefficient of  $z^d$  on the two sides of (24.30) gives, for each fixed  $k$ ,

$$\sum_{i=0}^d u_{d-i} c_{k,i} = \sum_{j=0}^d \frac{(-1)^j}{j!} v_{d-j} (-q^{-k})^{d-j}. \quad (24.33)$$

Multiply (24.33) by  $W_{n,k}(q)$  and sum over  $k$ . Since  $x_k = -q^{-k}$ , the right-hand side becomes

$$\sum_{j=0}^d \frac{(-1)^j}{j!} v_{d-j} \sum_{k=0}^n W_{n,k}(q) x_k^{d-j}, \quad (24.34)$$

and the left-hand side becomes  $\sum_{i=0}^d u_{d-i} S_i$ .

Let  $0 \leq d < n$  and induct on  $d$ . Every exponent occurring in (24.34) satisfies  $0 \leq d - j \leq d < n$ , so the entire right-hand side vanishes by (24.22). On the left, the terms with  $i < d$  vanish by the induction hypothesis (for  $d = 0$  there are none), and the term  $i = d$  equals  $u_0 S_d = S_d$ . Hence  $S_d = 0$ , which is (24.31).

Now take  $d = n$ . In (24.34) every term with  $j \geq 1$  carries an exponent  $n - j < n$  and vanishes, while the term  $j = 0$  equals

$$v_n \sum_{k=0}^n W_{n,k}(q) x_k^n = q^{-\binom{n+1}{2}}(q; q)_n v_n$$

by (24.23). On the left, the terms with  $i < n$  vanish by (24.31) and the term  $i = n$  equals  $u_0 S_n = S_n$ . This proves (24.32).  $\square$

*Remark 24.22* (On the normalization  $u_0 = 1$ ). The proof uses  $u_0 = 1$  twice: once for the invertibility of  $U$ , and once at the last step of each induction stage. Only invertibility of  $u_0$  is actually needed: for arbitrary  $u_0 \neq 0$  the same argument gives  $u_0 S_d = 0$  for  $d < n$  and  $u_0 S_n = q^{-\binom{n+1}{2}}(q; q)_n v_n$ . This is not a restriction, because (24.30) is unchanged when the pair  $(U, V)$  is divided by  $u_0$ . In every application below the normalization is automatic,  $U$  being  $\mathcal{A}(-z)$  or  $A(-z)$  with  $A(0) = 1$ .

**Corollary 24.23** (Invariance under a common translation). *Under the hypotheses of Theorem 24.21, let  $\alpha \in K$  and put  $\tilde{\Psi}_k(z) = e^{\alpha z} \Psi_k(z)$ . Then*

$$\sum_{k=0}^n W_{n,k}(q) [z^n] \tilde{\Psi}_k(z) = \sum_{k=0}^n W_{n,k}(q) c_{k,n} = q^{-\binom{n+1}{2}}(q; q)_n v_n. \quad (24.35)$$

*Proof.* Convolution gives  $[z^n] \tilde{\Psi}_k = \sum_{j=0}^n \alpha^j c_{k,n-j} / j!$ . Weighting by  $W_{n,k}(q)$  and summing over  $k$ , the term with index  $j$  carries the factor  $S_{n-j}$ ; every  $j \geq 1$  has  $n - j < n$  and therefore vanishes by (24.31). Only  $j = 0$  survives, and (24.32) evaluates it.  $\square$

*Remark 24.24* (Two nested triangular systems). The proof runs two triangular cancellations in series. First the finite  $q$ -binomial theorem annihilates every node power of degree below  $n$ . Second, the coefficient comparison (24.33) propagates that annihilation successively through  $c_{k,0}, c_{k,1}, \dots, c_{k,n-1}$ . The first quantity surviving both is  $a_n$ . This double cancellation is what makes Theorem 24.28 possible, and Corollary 24.23 is the reason a free parameter may be inserted there.

## 24.2 The normalized Thue–Morse block

The Thue–Morse polynomial of Theorem 22.7 has a zero of order exactly  $k$  at  $z = 1$ . Dividing that zero out produces an entire function whose Taylor coefficients are precisely the Thue–Morse power sums, and which bridges  $\mathcal{A}$  to its own dilate: that bridge is the hypothesis (24.30) of Theorem 24.21 at  $q = 1/2$ , where  $q^{-k} = 2^k$ . Throughout,  $\mathcal{A} := \mathcal{A}_{1/2}$ ,  $\alpha_n := \alpha_n^{(1/2)}$  and  $\tau_r = (-1)^{s_2(r)}$ .

For  $k \geq 0$  put

$$\Psi_k(z) := e^{-z} \prod_{j=0}^{k-1} \Lambda(-2^j z) = \sum_{m \geq 0} c_{k,m} z^m, \quad (24.36)$$

and, for a translation parameter  $\theta \in \mathbb{C}$  and an integer  $d \geq 0$ , write

$$\tilde{T}_{k,d}(\theta) := \sum_{r=0}^{2^k-1} \tau_r (r + \theta)^d, \quad T_{k,d}(\theta) := \tilde{T}_{k,d}(\theta - 2^k) = \sum_{r=0}^{2^k-1} \tau_r (r - 2^k + \theta)^d. \quad (24.37)$$

**Lemma 24.25** (Thue–Morse blocks as dilation bridges). *For every  $k \geq 0$  the function  $\Psi_k$  is entire,*

$$\Psi_k(z) = e^{-2^k z} \prod_{j=0}^{k-1} \Lambda(2^j z), \quad (24.38)$$

$$\Psi_k(z) \mathcal{A}(-z) = e^{-z} \mathcal{A}(-2^k z), \quad (24.39)$$

and for every  $\theta \in \mathbb{C}$

$$e^{(1-\theta)z} \Psi_k(z) = \frac{1}{2^{\binom{k}{2}} z^k} \sum_{r=0}^{2^k-1} \tau_r e^{-(r+\theta)z}, \quad (24.40)$$

the singularity of the right-hand side at  $z = 0$  being removable. Consequently

$$[z^m] e^{(1-\theta)z} \Psi_k(z) = \frac{(-1)^{m+k}}{2^{\binom{k}{2}} (m+k)!} \tilde{T}_{k,m+k}(\theta) \quad (m \geq 0). \quad (24.41)$$

*Proof.*  $\Lambda$  is entire, hence so is  $\Psi_k$ . The reflection identity

$$\Lambda(-u) = \frac{e^{-u} - 1}{-u} = e^{-u} \frac{e^u - 1}{u} = e^{-u} \Lambda(u)$$

turns  $\prod_{j < k} \Lambda(-2^j z)$  into  $e^{-(2^k-1)z} \prod_{j < k} \Lambda(2^j z)$ , because  $\sum_{j < k} 2^j = 2^k - 1$ ; multiplying by  $e^{-z}$  gives (24.38).

Iterating (24.10) at the arguments  $-z, -2z, \dots, -2^{k-1}z$  gives

$$\mathcal{A}(-2^k z) = \mathcal{A}(-z) \prod_{j=0}^{k-1} \Lambda(-2^j z),$$

and multiplication by  $e^{-z}$  is (24.39).

Next,  $\Lambda(-2^j z) = (1 - e^{-2^j z})/(2^j z)$  and  $\sum_{j < k} j = \binom{k}{2}$ , so

$$e^{(1-\theta)z} \Psi_k(z) = \frac{e^{-\theta z}}{2^{\binom{k}{2}} z^k} \prod_{j=0}^{k-1} (1 - e^{-2^j z}).$$

Applying (22.8) with its variable set to  $e^{-z}$  turns the product into  $\sum_{r < 2^k} \tau_r e^{-rz}$ , and multiplication by  $e^{-\theta z}$  gives (24.40). The numerator vanishes to order exactly  $k$  at  $z = 0$  by (22.9), so the quotient is entire.

Finally, expand  $e^{-(r+\theta)z} = \sum_{p \geq 0} (-1)^p (r+\theta)^p z^p / p!$ . For  $p < k$  the polynomial  $X \mapsto (X+\theta)^p$  has degree  $p < k$ , so  $\tilde{T}_{k,p}(\theta) = 0$  by (22.10) with  $n = k, a = \theta, b = 1$ . Hence

$$\sum_{r < 2^k} \tau_r e^{-(r+\theta)z} = \sum_{p \geq k} \frac{(-1)^p z^p}{p!} \tilde{T}_{k,p}(\theta),$$

and dividing by  $2^{\binom{k}{2}} z^k$  and writing  $p = m + k$  gives (24.41).  $\square$

**Corollary 24.26** (Centred form and the coefficient dictionary). *For every  $k \geq 0$ ,*

$$\sum_{r=0}^{2^k-1} \tau_r e^{(r-2^k)z} = (-1)^k 2^{\binom{k}{2}} z^k \Psi_k(z), \quad (24.42)$$

and the Taylor coefficients of  $\Psi_k$  itself are the centred Thue–Morse power sums:

$$c_{k,m} = (-1)^k 2^{-\binom{k}{2}} \frac{T_{k,m+k}(0)}{(m+k)!} = \frac{(-1)^{m+k}}{2^{\binom{k}{2}} (m+k)!} \tilde{T}_{k,m+k}(1) \quad (m \geq 0). \quad (24.43)$$



*Proof.* Substituting  $z \mapsto e^z$  in (22.8) gives  $\sum_{r < 2^k} \tau_r e^{rz} = \prod_{j < k} (1 - e^{2^j z})$ , and  $1 - e^u = -u\Lambda(u)$ , so

$$\prod_{j=0}^{k-1} (1 - e^{2^j z}) = \prod_{j=0}^{k-1} (-2^j z \Lambda(2^j z)) = (-1)^k 2^{\binom{k}{2}} z^k \prod_{j=0}^{k-1} \Lambda(2^j z).$$

Multiplying by  $e^{-2^k z}$  and applying (24.38) proves (24.42). Its left-hand side is the exponential generating function  $\sum_{d \geq 0} T_{k,d}(0) z^d / d!$ , so comparing the coefficient of  $z^{m+k}$  gives the first equality in (24.43). For the second, note that  $2^k - 1 - r$  is the bitwise complement of  $r$  in  $k$  binary digits, whence  $s_2(2^k - 1 - r) = k - s_2(r)$  and  $\tau_{2^k-1-r} = (-1)^k \tau_r$ ; replacing  $r$  by  $2^k - 1 - r$  in  $T_{k,d}(0) = \sum_r \tau_r (r - 2^k)^d$  therefore gives  $T_{k,d}(0) = (-1)^{d+k} \tilde{T}_{k,d}(1)$ . Alternatively, the second equality is (24.41) at  $\theta = 1$ , where the factor  $e^{(1-\theta)z}$  is 1.  $\square$

**Remark 24.27** (Centred and raw coordinates). The two displays (24.40) and (24.42) are one identity in two coordinate systems. Indeed (24.38) says  $\Psi_k(z) = e^{-(2^k+1)z} \Psi_k(-z)$ , and substituting  $z \mapsto -z$  in (24.40) at  $\theta = -2^k$  returns (24.42). The centred sums  $T_{k,d}$  are the natural coefficients of  $\Psi_k$  itself, while the raw sums  $\tilde{T}_{k,d}(\theta)$  with a free  $\theta$  are what Corollary 24.23 feeds into Theorem 24.28; by (24.37) the dictionary between them is  $T_{k,d}(\theta) = (-1)^{d+k} \tilde{T}_{k,d}(1 - \theta)$ .

**Theorem 24.28** (Dyadic Gaussian–Thue–Morse coefficient formula). *For every  $n \geq 0$  and every  $\theta \in \mathbb{C}$ ,*

$$a_n = \frac{(-1)^n}{m_n} \sum_{k=0}^n \frac{[n]_{1/2}}{4^{\binom{k}{2}} (n+k)!} \sum_{r=0}^{2^k-1} (-1)^{s_2(r)} (r+\theta)^{n+k}. \quad (24.44)$$

*Every inner power sum depends on  $\theta$ ; the total does not.*

*Proof.* Fix  $\theta$  and set  $\tilde{\Psi}_k(z) = e^{(1-\theta)z} \Psi_k(z)$ . By (24.39) the family  $\Psi_0, \dots, \Psi_n$  satisfies the bridge relation (24.30) with  $q = 1/2$ , since then  $q^{-k} = 2^k$ ; and  $A$  has constant term 1 by Corollary 24.5. Hence Theorem 24.21 applies, and Corollary 24.23 with  $\alpha = 1 - \theta$  gives

$$\sum_{k=0}^n W_{n,k}(1/2) [z^n] \tilde{\Psi}_k(z) = 2^{\binom{n+1}{2}} \left(\frac{1}{2}; \frac{1}{2}\right)_n a_n = m_n a_n, \quad (24.45)$$

the last equality by (22.2). Comparing (24.20) with (22.6),

$$W_{n,k}(1/2) = (-1)^k 2^{-\binom{k}{2}} \left[ \begin{matrix} n \\ k \end{matrix} \right]_{1/2} = w_{n,k},$$

while (24.41) at  $m = n$  gives

$$[z^n] \tilde{\Psi}_k(z) = \frac{(-1)^{n+k}}{2^{\binom{k}{2}} (n+k)!} \sum_{r=0}^{2^k-1} \tau_r (r+\theta)^{n+k}.$$

In the product of the two, the signs combine to  $(-1)^k (-1)^{n+k} = (-1)^n$ , independent of  $k$ , and the powers of two combine to  $2^{-\binom{k}{2}} 2^{-\binom{k}{2}} = 4^{-\binom{k}{2}}$ . Substituting into (24.45) and dividing by  $m_n$  proves (24.44). The left side of (24.45) does not depend on  $\theta$ , by Corollary 24.23; this is the asserted invariance.  $\square$

**Example 24.29** (The first two cases). Take  $\theta = 1$ . For  $n = 1$  one has  $m_1 = 1$  and  $\begin{bmatrix} 1 \\ 0 \end{bmatrix}_{1/2} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}_{1/2} = 1$ ; the inner sums are  $1^1 = 1$  at  $k = 0$  and  $1^2 - 2^2 = -3$  at  $k = 1$ , so

$$a_1 = - \left( \frac{1}{1!} + \frac{-3}{2!} \right) = - \left( 1 - \frac{3}{2} \right) = \frac{1}{2}.$$

For  $n = 2$  one has  $m_2 = 3$  and the half-base Gaussian row  $1, \frac{3}{2}, 1$ ; the inner sums are  $1, 1^3 - 2^3 = -7$ , and  $1^4 - 2^4 - 3^4 + 4^4 = 160$ , so

$$a_2 = \frac{1}{3} \left( \frac{1}{2!} + \frac{3}{2} \cdot \frac{-7}{3!} + \frac{160}{4! \cdot 4!} \right) = \frac{1}{3} \left( \frac{1}{2} - \frac{7}{4} + \frac{5}{3} \right) = \frac{1}{3} \cdot \frac{5}{12} = \frac{5}{36},$$

in agreement with the value produced by (24.11).

**Theorem 24.30** (Inverse-dyadic values of the Fabius function). By Theorem 24.9,  $F(2^{-n}) = 2^{-\binom{n}{2}} a_n$  with  $a_n = \mathbb{E}(X^n)/n!$ . Consequently, for every  $n \geq 0$  and every  $\theta \in \mathbb{C}$ ,

$$F(2^{-n}) = \frac{(-1)^n}{2^{n^2} \left(\frac{1}{2}; \frac{1}{2}\right)_n} \sum_{k=0}^n \frac{\begin{bmatrix} n \\ k \end{bmatrix}_{1/2}}{4^{\binom{k}{2}} (n+k)!} \sum_{r=0}^{2^k-1} (-1)^{s_2(r)} (r + \theta)^{n+k}. \quad (24.46)$$

*Proof.* Only the closed form has to be proved.

For (24.46), multiply (24.44) by  $2^{-\binom{n}{2}}$  and use (22.2) together with  $\binom{n}{2} + \binom{n+1}{2} = n^2$ :

$$2^{\binom{n}{2}} m_n = 2^{\binom{n}{2} + \binom{n+1}{2}} \left(\frac{1}{2}; \frac{1}{2}\right)_n = 2^{n^2} \left(\frac{1}{2}; \frac{1}{2}\right)_n. \quad \square$$

The blocks of (24.46) may also be centred at their own right endpoints. This is the normalization in which the inner sums are the bare Taylor coefficients of the Thue–Morse blocks  $\Psi_k$ , and it costs one application of digit complementation.

The blocks of (24.46) may also be centred at their own right endpoints. This is the normalization in which the inner sums are the bare Taylor coefficients of the Thue–Morse blocks  $\Psi_k$ , it costs one application of digit complementation, and – unlike the raw normalization – it extends from the inverse powers  $2^{-n}$  to every dyadic point of  $[0, 1]$  at resolution  $2^{-n}$ , the numerator entering only through the length of the Thue–Morse block.

For integers  $m, k, d \geq 0$  and  $\theta \in \mathbb{C}$  define the *long* Thue–Morse block

$$T_{m;k,d}(\theta) := \sum_{r=0}^{m2^k-1} \tau_r (r - m2^k + \theta)^d, \quad \tau_r = (-1)^{s_2(r)}, \quad (24.47)$$

an empty sum being 0; thus  $T_{0;k,d} = 0$  and  $T_{1;k,d} = T_{k,d}$  is the centred power sum of (24.37).

**Corollary 24.31** (Centred form, at an arbitrary dyadic point). For  $n, m \geq 0$  and  $\theta \in \mathbb{C}$  put

$$\mathcal{N}_{m,n}(\theta) := \sum_{k=0}^n \frac{\begin{bmatrix} n \\ k \end{bmatrix}_{1/2}}{4^{\binom{k}{2}} (n+k)!} T_{m;k,n+k}(\theta), \quad \mathcal{N}_n(\theta) := \mathcal{N}_{1,n}(\theta). \quad (24.48)$$

Then  $\mathcal{N}_{m,n}(\theta)$  does not depend on  $\theta$ , and for every integer  $m$  with  $0 \leq m \leq 2^n$ ,

$$F\left(\frac{m}{2^n}\right) = \frac{\mathcal{N}_{m,n}(\theta)}{2^{\binom{n}{2}} m_n} = \frac{1}{2^{n^2} \left(\frac{1}{2}; \frac{1}{2}\right)_n} \sum_{k=0}^n \frac{\begin{bmatrix} n \\ k \end{bmatrix}_{1/2}}{4^{\binom{k}{2}} (n+k)!} \sum_{r=0}^{m2^k-1} \tau_r (r - m2^k + \theta)^{n+k}. \quad (24.49)$$

In particular  $\mathcal{N}_n(\theta) = \mathfrak{m}_n \alpha_n$  for every  $\theta$ , where  $\alpha_n = \mathbb{E}X^n/n!$  is the coefficient called  $a_n$  in [Theorem 24.28](#), and the case  $m = 1$  reads

$$F(2^{-n}) = \frac{\mathcal{N}_n(\theta)}{2^{\binom{n}{2}} \mathfrak{m}_n} = \frac{1}{2^{n^2} \left(\frac{1}{2}; \frac{1}{2}\right)_n} \sum_{k=0}^n \frac{[k]_{1/2}}{4^{\binom{k}{2}} (n+k)!} \sum_{r=0}^{2^k-1} \tau_r (r - 2^k + \theta)^{n+k}, \quad (24.50)$$

with no sign  $(-1)^n$  in front. The hypothesis  $m \leq 2^n$  cannot be dropped: at  $n = 2$ ,  $m = 5$  the right-hand side of (24.49) equals  $67/72$ , whereas  $F(5/4) = 1$ .

*Proof.* For  $n = 0$  the admissible numerators are  $m = 0$  and  $m = 1$ , and  $(\frac{1}{2}; \frac{1}{2})_0 = 1$ ; the single term  $k = 0$  is  $\sum_{r < m} \tau_r$ , which is 0 for  $m = 0$  and  $\tau_0 = 1$  for  $m = 1$ , in agreement with  $F(0) = 0$  and  $F(1) = 1$ . Assume  $n \geq 1$ .

*Step 1: the value is a moment functional.* Put

$$\mathcal{K}_m(z) := \sum_{h=0}^{m-1} \tau_h e^{(m-h-1)z}, \quad \mathcal{P}_m(z) := \mathcal{A}(z) \mathcal{K}_m(z) = \sum_{d \geq 0} p_{m,d} z^d, \quad (24.51)$$

so that  $\mathcal{P}_1 = \mathcal{A}$  and, by (24.5),

$$p_{m,n} = \frac{1}{n!} \sum_{h=0}^{m-1} \tau_h \mathbb{E}(X + m - h - 1)^n. \quad (24.52)$$

We claim that

$$F\left(\frac{m}{2^n}\right) = 2^{-\binom{n}{2}} p_{m,n} \quad (0 \leq m \leq 2^n). \quad (24.53)$$

Integrating (23.22) term by term, with  $\int_0^x (t - c)_+^{n-1} dt = (x - c)_+^n/n$ , gives the distribution function of the truncation  $X_{1/2,n}$  of (23.17):

$$F_{1/2,n}(x) = \frac{2^{\binom{n+1}{2}}}{n!} \sum_{r=0}^{2^n-1} \tau_r \left(x - \frac{r}{2^n}\right)_+^n.$$

By (24.18),  $X = X_{1/2,n} + 2^{-n}X'$  with  $X'$  an independent copy of  $X$ , so conditioning on  $X'$  and using  $x \mapsto 2^{-n}(m - x)$ ,

$$F\left(\frac{m}{2^n}\right) = \mathbb{E}\left[F_{1/2,n}(2^{-n}(m - X'))\right] = \frac{2^{\binom{n+1}{2}-n^2}}{n!} \sum_{r=0}^{2^n-1} \tau_r \mathbb{E}[(m - r - X')_+^n],$$

the factor  $2^{-n^2}$  coming from the  $n$ th power of  $2^{-n}$ . Since  $0 \leq X' \leq 1$  and  $m \leq 2^n$ , a term with  $r \geq m$  has  $m - r - X' \leq 0$  and drops out, while a term with  $r < m$  has  $m - r - X' \geq 0$  and keeps its positive part. With  $\binom{n+1}{2} - n^2 = -\binom{n}{2}$  and  $1 - X' \stackrel{d}{=} X$  from (23.2),

$$F\left(\frac{m}{2^n}\right) = \frac{2^{-\binom{n}{2}}}{n!} \sum_{r=0}^{m-1} \tau_r \mathbb{E}(m - r - X')^n = \frac{2^{-\binom{n}{2}}}{n!} \sum_{r=0}^{m-1} \tau_r \mathbb{E}(X + m - r - 1)^n,$$

which is (24.53) by (24.52). At  $m = 1$  this is [Theorem 24.9](#) again, since  $p_{1,n} = \alpha_n$ .

*Step 2: the long block is a dilated bridge.* Write  $r = h2^k + s$  with  $0 \leq h < m$  and  $0 \leq s < 2^k$ ; this is a bijection onto  $\{0, \dots, m2^k - 1\}$ , and the binary digits of  $h2^k$  and of  $s$  occupy disjoint positions, so  $\tau_{h2^k+s} = \tau_h \tau_s$ . Hence

$$\begin{aligned} \sum_{d \geq 0} T_{m;k,d}(0) \frac{z^d}{d!} &= e^{-m2^k z} \sum_{r=0}^{m2^k-1} \tau_r e^{rz} = e^{-m2^k z} \left( \sum_{h=0}^{m-1} \tau_h e^{h2^k z} \right) \left( \sum_{s=0}^{2^k-1} \tau_s e^{sz} \right) \\ &= \mathcal{K}_m(-2^k z) \left( e^{-2^k z} \sum_{s=0}^{2^k-1} \tau_s e^{sz} \right) = (-1)^k 2^{\binom{k}{2}} z^k \Psi_{m,k}(z), \end{aligned}$$

where  $\Psi_{m,k}(z) := \mathcal{K}_m(-2^k z) \Psi_k(z)$ ; the third equality uses  $\mathcal{K}_m(-2^k z) = e^{-(m-1)2^k z} \sum_{h < m} \tau_h e^{h2^k z}$  and the fourth is (24.42). Consequently

$$[z^n] \Psi_{m,k}(z) = \frac{(-1)^k}{2^{\binom{k}{2}} (n+k)!} T_{m;k,n+k}(0), \quad (24.54)$$

and multiplying (24.39) by  $\mathcal{K}_m(-2^k z)$  gives

$$\Psi_{m,k}(z) \mathcal{A}(-z) = e^{-z} \mathcal{K}_m(-2^k z) \mathcal{A}(-2^k z) = e^{-z} \mathcal{P}_m(-2^k z).$$

*Step 3: transfer.* Apply Theorem 24.21 at  $q = 1/2$ , where  $q^{-k} = 2^k$ , with  $U(z) = \mathcal{A}(-z)$ , whose constant term is 1 by Corollary 24.5, with  $V = \mathcal{P}_m$ , and with the family  $\Psi_{m,0}, \dots, \Psi_{m,n}$ , for which Step 2 is the bridge relation (24.30). By (24.32) and (22.2),

$$\sum_{k=0}^n W_{n,k}(1/2) [z^n] \Psi_{m,k} = 2^{\binom{n+1}{2}} \left( \frac{1}{2}; \frac{1}{2} \right)_n p_{m,n} = \mathfrak{m}_n p_{m,n},$$

and inserting (24.54) together with  $W_{n,k}(1/2) = (-1)^k 2^{-\binom{k}{2}} \begin{bmatrix} n \\ k \end{bmatrix}_{1/2}$  makes the two signs cancel and the two triangular powers of 2 combine into  $4^{-\binom{k}{2}}$ :

$$\mathcal{N}_{m,n}(0) = \mathfrak{m}_n p_{m,n}.$$

Dividing by  $2^{\binom{n}{2}} \mathfrak{m}_n$  and using (24.53) proves (24.49) at  $\theta = 0$ ; the second expression follows from  $2^{\binom{n}{2}} \mathfrak{m}_n = 2^{n^2} \left( \frac{1}{2}; \frac{1}{2} \right)_n$ , which is (22.2) with  $\binom{n}{2} + \binom{n+1}{2} = n^2$ .

*The free parameter.* Translating the inner argument multiplies the exponential generating function of Step 2 by  $e^{\theta z}$ , so  $[z^n] (e^{\theta z} \Psi_{m,k}) = (-1)^k 2^{-\binom{k}{2}} T_{m;k,n+k}(\theta) / (n+k)!$ , while Corollary 24.23 with  $\alpha = \theta$  leaves the weighted sum unchanged. Hence  $\mathcal{N}_{m,n}(\theta) = \mathcal{N}_{m,n}(0)$  for every  $\theta$ .

*Consistency with the raw form.* For  $m = 1$  the two normalizations are exchanged by digit complementation. Substituting  $r \mapsto 2^k - 1 - r$  in the inner sum of  $\mathcal{N}_n(1 - \theta)$  and using  $s_2(2^k - 1 - r) = k - s_2(r)$ , hence  $\tau_{2^k-1-r} = (-1)^k \tau_r$ , together with  $(2^k - 1 - r) - 2^k + (1 - \theta) = -(r + \theta)$ , gives

$$\sum_{r=0}^{2^k-1} \tau_r (r - 2^k + 1 - \theta)^{n+k} = (-1)^{n+2k} \sum_{r=0}^{2^k-1} \tau_r (r + \theta)^{n+k},$$

and  $(-1)^{n+2k} = (-1)^n$  does not depend on  $k$ , so it factors out of the sum over  $k$ :  $\mathcal{N}_n(1 - \theta)$  is  $(-1)^n$  times the sum in (24.44), which is  $\mathfrak{m}_n \alpha_n$ . This recovers the case  $m = 1$  a second time, and it is the  $k$ -independent sign referred to in Remark 24.33.  $\square$

*Remark 24.32* (Resolution and numerator are separated). In (24.49) the half-base Gaussian row  $\begin{bmatrix} n \\ k \end{bmatrix}_{1/2} / 4^{\binom{k}{2}}$  depends only on the resolution  $2^{-n}$ , while the numerator  $m$  enters only through the length  $m2^k$  of the Thue–Morse block. It is instructive to compare [Theorem 23.11](#), which writes the density of the truncation  $X_{1/2,N}$  as a Thue–Morse signed spline: there the Thue–Morse signs appear alone and the identity is exact only for the truncation, whereas here they appear weighted by a Gaussian row and the identity is exact for the limit function  $F$  itself. The formula returns 0 at  $m = 0$  and 1 at  $m = 2^n$ , and since  $F(m/2^n) = F(2m/2^{n+1})$  it is heavily overdetermined. The row  $n = 3$  is

$$0, \quad \frac{1}{288}, \quad \frac{5}{72}, \quad \frac{73}{288}, \quad \frac{1}{2}, \quad \frac{215}{288}, \quad \frac{67}{72}, \quad \frac{287}{288}, \quad 1,$$

symmetric about  $1/2$  as (23.32) requires.

*Remark 24.33* (The shift must be common, or geometric). The freedom in  $\theta$  in (24.44), (24.46) and (24.50) is [Corollary 24.23](#) at  $q = 1/2$ : one exponential  $e^{\alpha z}$  is pulled out of all  $n + 1$  blocks *simultaneously*, after which (24.31) annihilates every convolution term but one. It is not a symmetry of an individual block. To see what is and is not allowed, give each block its own shift and set

$$\mathcal{N}_n(\theta_0, \dots, \theta_n) := \sum_{k=0}^n \frac{\begin{bmatrix} n \\ k \end{bmatrix}_{1/2}}{4^{\binom{k}{2}} (n+k)!} \sum_{r=0}^{2^k-1} \tau_r(r - 2^k + \theta_k)^{n+k}. \quad (24.55)$$

Two one-parameter families of shifts leave this constant, and they are exactly the two displays above:

$$\mathcal{N}_n(\theta, \dots, \theta) = \mathfrak{m}_n \alpha_n, \quad \mathcal{N}_n(2^0 + \theta, \dots, 2^n + \theta) = (-1)^n \mathfrak{m}_n \alpha_n,$$

the first by [Corollary 24.31](#), the second by (24.44) after cancelling  $-2^k$  against  $2^k$ . The second family is genuinely block-dependent, and it survives only because digit complementation converts it into the first, at the cost of the  $k$ -independent sign  $(-1)^n$  computed in the proof of [Corollary 24.31](#).

These two families are not independent of each other, and they are not the only structured ones. Both are members of the two-parameter geometric family  $\theta_k = \eta - 2^k \vartheta$  of [Theorem 24.99](#), at  $\vartheta = 0$  and at  $\vartheta = -1$  respectively; and on the whole of that family the double cancellation of [Remark 24.24](#) survives and evaluates the sum in closed form,

$$\mathcal{N}_n(\eta - 2^0 \vartheta, \dots, \eta - 2^n \vartheta) = \mathfrak{m}_n \frac{A_n(\vartheta)}{n!},$$

a polynomial of degree  $n$  in  $\vartheta$ , free of  $\eta$ , whose value at  $\vartheta = 0$  is the invariant  $\mathfrak{m}_n \alpha_n$ . What fails away from  $\vartheta \in \{0, -1\}$  is therefore not the cancellation but the constancy: the transfer theorem still extracts exactly one coefficient, only of  $e^{\vartheta u} \mathcal{A}(u)$  rather than of  $\mathcal{A}(u)$ .

Outside that family, unequal shifts do destroy the extraction. For  $n = 3$  take  $(\theta_0, \theta_1, \theta_2, \theta_3) = (0, 1, 2, 3)$ , which is not of the form  $\eta - 2^k \vartheta$ : matching the first two entries forces  $\vartheta = \eta = -1$ , and that choice produces  $(0, 1, 3, 7)$  instead. One finds

$$\mathcal{N}_3(0, 1, 2, 3) = \frac{1}{32}, \quad \mathcal{N}_3(0, 1, 3, 7) = -\frac{7}{12}, \quad \mathfrak{m}_3 \alpha_3 = 21 \cdot \frac{1}{36} = \frac{7}{12},$$

the middle value being  $\mathfrak{m}_3 A_3(-1)/3!$ , as [Theorem 24.99](#) requires, and the first being nothing in particular.

*Remark 24.34* (Anatomy and conditioning of the dyadic formula). The ingredients of (24.46) separate cleanly, and each has a single identifiable origin. One factor  $2^{-\binom{k}{2}}$ , together with the row  $\begin{bmatrix} n \\ k \end{bmatrix}_{1/2}$ , is the geometric difference stencil  $w_{n,k}$  on the nodes  $-2^k$  of [Theorem 22.4](#); the second factor  $2^{-\binom{k}{2}}$  appears

when the order- $k$  Thue–Morse zero is removed in (24.40). Their product is the  $4^{-\binom{k}{2}}$  of the statement: the duplication is structural, not a misprint. The product  $\left(\frac{1}{2}; \frac{1}{2}\right)_n$  is the normalized geometric Vandermonde denominator of that stencil, equal by (22.2) to the odd Mersenne product  $\mathfrak{m}_n = \prod_{j=1}^n (2^j - 1)$  divided by  $2^{\binom{n+1}{2}}$ , and the remaining  $2^{-\binom{n}{2}}$  is the Jacobian  $\prod_{j=1}^n 2^{n-j}$  of the cube cut (24.18). The factorials  $(n+k)!$  are contributed by the exponential generating functions, and each inner sum is an integer, being a power sum of Thue–Morse signs whose first  $k$  moments vanish.

Arithmetically, therefore, four independent sources meet in one line: dyadic dilation powers, of which (24.28) isolates the part carried by the Gaussian row,  $\left[\begin{smallmatrix} n \\ k \end{smallmatrix}\right]_{1/2} = 2^{-k(n-k)} \left[\begin{smallmatrix} n \\ k \end{smallmatrix}\right]_2$  with an *odd* integer on the right (Corollary 24.19); the odd Mersenne product  $\mathfrak{m}_n$ ; ordinary factorials; and integer Thue–Morse power sums. The freedom in  $\theta$  is not an accident among those power sums: by Corollary 24.23 it says precisely that the Gaussian stencil has already annihilated everything below degree  $n$ , so only the top coefficient of the shifted test polynomial is ever seen.

The formula is an exact rational certificate, but it is ill conditioned in floating point, and the loss occurs *inside* the blocks rather than between them. The outer cancellation over  $k$  is mild: at  $\theta = 1$  the largest block of (24.46) exceeds the value of the whole sum by the factors

$$\begin{aligned} &3.00, 4.20, 4.69, 4.42, 4.09, \\ &3.84, 3.68, 3.59, 3.53, 3.50 \end{aligned}$$

for  $n = 1, \dots, 10$ , peaking at  $n = 3$  and decreasing thereafter, and other choices of  $\theta$  alter these ratios only by a bounded factor. The digits are lost within a single block, in the alternating Thue–Morse power sum: for  $k = n$  the largest summand exceeds  $\sum_{r < 2^k} (-1)^{s_2(r)} (r + \theta)^{n+k}$  by a factor 4.8 at  $n = 4$ ,  $5.2 \cdot 10^3$  at  $n = 8$ ,  $3.9 \cdot 10^6$  at  $n = 10$  and  $2.9 \cdot 10^{10}$  at  $n = 12$ , and this ratio grows without bound. For numerical work the triangular recursion (24.11) is preferable: it delivers  $a_n$ , hence  $F(2^{-n}) = 2^{-\binom{n}{2}} a_n$ , with one rational division per index and no cancellation at all.

**Proposition 24.35** (The block is a Taylor polynomial). *Put*

$$\mathcal{M}_n(\theta) := \frac{\mathcal{N}_n(\theta)}{2^{\binom{n}{2}} (1/2; 1/2)_n} = 2^n \alpha_n = 2^{\binom{n+1}{2}} F(2^{-n}), \quad (24.56)$$

and, for  $m \geq 0$ ,

$$\mathcal{T}_m(\theta, y) := 2^{-\binom{m+1}{2}} \sum_{n=0}^m \frac{(2^{m+1} y)^{m-n}}{(m-n)!} \mathcal{M}_n(\theta). \quad (24.57)$$

Then  $\mathcal{T}_m$  does not depend on  $\theta$ , contains no  $q$ -object, and is exactly the  $m$ th Taylor polynomial of  $F$  at the point  $2^{-m}$ :

$$\mathcal{T}_m(\theta, y) = \sum_{r=0}^m 2^{\binom{r+1}{2} - \binom{m-r}{2}} \alpha_{m-r} \frac{y^r}{r!} = \sum_{r=0}^m F^{(r)}(2^{-m}) \frac{y^r}{r!}. \quad (24.58)$$

*Proof.* Independence of  $\theta$  and the first equality in (24.56) are Corollary 24.23 combined with  $\mathcal{N}_n(0) = \mathfrak{m}_n \alpha_n$  and  $\mathfrak{m}_n = 2^{\binom{n+1}{2}} (1/2; 1/2)_n$ ; the second equality is Theorem 24.9. Substituting  $\mathcal{M}_n = 2^n \alpha_n$  into (24.57) and writing  $r = m - n$  gives the exponent

$$-\binom{m+1}{2} + (m+1)r + (m-r) = mr - \binom{m}{2} = \binom{r+1}{2} - \binom{m-r}{2},$$

which is the first equality in (24.58). For the second, Theorem 24.9 gives  $2^{-\binom{m-r}{2}} \alpha_{m-r} = F(2^{r-m})$ , while (24.12) at  $x = 2^{-m}$ , legitimate because  $r \leq m$ , gives  $F^{(r)}(2^{-m}) = 2^{\binom{r+1}{2}} F(2^{r-m})$ .  $\square$

*Remark 24.36.* [Proposition 24.35](#) explains [Corollary 24.23](#) concretely: a common translation perturbs each block by terms of lower degree, and those are exactly the terms annihilated by the order- $n$  geometric divided difference of [Theorem 22.4](#).

### 24.3 Composition and determinant forms

Expanding the product (24.8) instead of inverting it produces two further closed forms. The combinatorial input is an ordered geometric sum.

**Lemma 24.37** (Geometric simplex sum). *Let  $0 < |q| < 1$ , let  $r_1, \dots, r_\ell$  be positive integers, and put  $t_h := r_h + r_{h+1} + \dots + r_\ell$ . Then*

$$\sum_{1 \leq j_1 < j_2 < \dots < j_\ell} q^{j_1 r_1 + \dots + j_\ell r_\ell} = \prod_{h=1}^{\ell} \frac{q^{t_h}}{1 - q^{t_h}}. \quad (24.59)$$

*Proof.* Every strictly increasing  $\ell$ -tuple of positive integers has a unique gap representation  $j_i = i + g_1 + \dots + g_i$  with  $g_h \geq 0$ . Then

$$\sum_{i=1}^{\ell} j_i r_i = \sum_{i=1}^{\ell} i r_i + \sum_{h=1}^{\ell} g_h t_h,$$

because  $g_h$  occurs in  $j_i$  exactly for  $i \geq h$ . Summing the independent geometric series in  $g_1, \dots, g_\ell$  gives

$$q^{\sum_i i r_i} \prod_{h=1}^{\ell} \frac{1}{1 - q^{t_h}},$$

and  $\sum_i i r_i = \sum_h t_h$ , which converts the prefactor into  $\prod_h q^{t_h}$ .  $\square$

An ordered composition  $\rho = (r_1, \dots, r_\ell) \models n$  is a sequence of positive integers with  $r_1 + \dots + r_\ell = n$ ; write  $\ell(\rho) = \ell$  and  $s_j := r_1 + \dots + r_j$ .

**Theorem 24.38** (Composition representation). *For  $0 < q < 1$  and  $n \geq 1$ ,*

$$\alpha_n^{(q)} = \left( \frac{1-q}{q} \right)^n \sum_{\rho \models n} \prod_{j=1}^{\ell(\rho)} \frac{q^{s_j}}{(1 - q^{s_j})(r_j + 1)!}. \quad (24.60)$$

*In particular, at  $q = 1/2$ ,*

$$F(2^{-n}) = 2^{-\binom{n}{2}} \sum_{\rho \models n} \prod_{j=1}^{\ell(\rho)} \frac{1}{(2^{s_j} - 1)(r_j + 1)!}. \quad (24.61)$$

*Proof.* Expand each factor of (24.8) as  $\Lambda((1-q)q^j z) = \sum_{r \geq 0} (1-q)^r q^{j r} z^r / (r+1)!$  and collect the coefficient of  $z^n$ . A nonzero contribution chooses an index set  $0 \leq j_1 < \dots < j_\ell$  of factors and positive exponents  $r_1, \dots, r_\ell$  with  $\sum r_i = n$ , so that

$$\alpha_n^{(q)} = (1-q)^n \sum_{\substack{\ell \geq 1 \\ r_1 + \dots + r_\ell = n \\ r_i \geq 1}} \frac{1}{\prod_i (r_i + 1)!} \sum_{0 \leq j_1 < \dots < j_\ell} q^{\sum_i j_i r_i}.$$

Shifting  $j_i \mapsto j_i + 1$  multiplies the inner sum by  $q^{-n}$  and puts it in the form (24.59), so the inner sum equals  $q^{-n} \prod_h q^{t_h} / (1 - q^{t_h})$ . Reversing the composition  $(r_1, \dots, r_\ell) \mapsto (r_\ell, \dots, r_1)$  is a bijection of the compositions of  $n$  that turns the tail sums  $t_h$  into the prefix sums  $s_j$ . This proves (24.60). At  $q = 1/2$  the prefactor is 1 and  $q^{s_j} / (1 - q^{s_j}) = 1 / (2^{s_j} - 1)$ ; multiply by  $2^{-\binom{n}{2}}$  and apply [Theorem 24.9](#).  $\square$



**Theorem 24.39** (Lower-Hessenberg determinant). Put  $\beta_{n,k} := \frac{1}{(2^n - 1)(n - k + 1)!}$  for  $0 \leq k \leq n - 1$ , and let  $\mathcal{H}_n$  be the  $n \times n$  lower-Hessenberg matrix whose  $(i, k)$  entry is  $\beta_{i,k}$  for  $0 \leq k \leq i - 1$ , is  $-1$  for  $k = i$ , and is  $0$  for  $k > i$  (rows indexed by  $1 \leq i \leq n$ , columns by  $0 \leq k \leq n - 1$ ):

$$\mathcal{H}_n = \begin{pmatrix} \beta_{1,0} & -1 & 0 & \cdots & 0 \\ \beta_{2,0} & \beta_{2,1} & -1 & \ddots & \vdots \\ \beta_{3,0} & \beta_{3,1} & \beta_{3,2} & \ddots & 0 \\ \vdots & \vdots & \vdots & \ddots & -1 \\ \beta_{n,0} & \beta_{n,1} & \beta_{n,2} & \cdots & \beta_{n,n-1} \end{pmatrix}. \quad (24.62)$$

Then, for  $n \geq 1$ ,

$$\alpha_n = \det \mathcal{H}_n, \quad F(2^{-n}) = 2^{-\binom{n}{2}} \det \mathcal{H}_n. \quad (24.63)$$

*Proof.* Set  $D_0 := 1$  and  $D_n := \det \mathcal{H}_n$ . Note that  $\mathcal{H}_k$  is the leading  $k \times k$  principal submatrix of  $\mathcal{H}_n$  for  $k \leq n$ . Expand  $D_n$  along the last row, whose entries are  $\beta_{n,k}$  for  $0 \leq k \leq n - 1$ , and let  $M_{n,k}$  be the minor obtained by deleting row  $n$  and column  $k$ . In  $M_{n,k}$  the rows  $1, \dots, k$  have zero entries in every column of index greater than  $k$ , because the superdiagonal entry  $-1$  of row  $i \leq k$  sits in column  $i \leq k$ . Hence  $M_{n,k}$  is block lower triangular with diagonal blocks

$$(\text{rows } 1..k, \text{ columns } 0..k-1) = \mathcal{H}_k, \quad (\text{rows } k+1..n-1, \text{ columns } k+1..n-1),$$

and the second block is lower triangular with every diagonal entry equal to  $-1$ . Therefore  $\det M_{n,k} = (-1)^{n-1-k} D_k$ . The cofactor sign of the entry in row  $n$  and column  $k$ , that is, in matrix position  $(n, k+1)$ , is  $(-1)^{n+k+1}$ , so

$$D_n = \sum_{k=0}^{n-1} \beta_{n,k} (-1)^{n+k+1} (-1)^{n-1-k} D_k = \sum_{k=0}^{n-1} \beta_{n,k} D_k.$$

This is (24.11) written as  $\alpha_n = \sum_{k < n} \beta_{n,k} \alpha_k$ , with the same initial value. Induction gives  $D_n = \alpha_n$ , and Theorem 24.9 gives the second formula.  $\square$

*Remark 24.40.* Expanding  $\det \mathcal{H}_n$  over its nonzero Hessenberg paths reproduces (24.61): a path that descends from row  $n$  to row  $0$  using the steps  $r_1, \dots, r_\ell$  contributes exactly the corresponding composition term. The determinant and the composition sum are two descriptions of the same triangular resolvent, and the recursion (24.11) is its unrolled form.

## 24.4 A Fourier formula for the Fabius function

**Corollary 24.41** (Sinc-product representation of  $F$ ). For  $0 \leq x \leq 1$ ,

$$F(x) = \frac{1}{2} + \frac{1}{\pi} \int_0^\infty \frac{\sin((2x-1)t)}{t} \prod_{k=1}^\infty \operatorname{sinc}\left(\frac{t}{2^k}\right) dt, \quad (24.64)$$

the integral being absolutely convergent.

*Proof.* Let  $Y = 2X - 1$ , whose density is  $\widehat{\text{up}}$  and whose Fourier transform is  $\widehat{\text{up}}(t) = \prod_{k \geq 1} \operatorname{sinc}(t/2^k)$  by Theorem 23.16. Since  $\widehat{\text{up}}(t) = e^{-it} \varphi_{1/2}(2t)$ , Theorem 23.7 shows that  $\widehat{\text{up}}$  is integrable, and indeed decays faster than every negative power of  $t$ . Fourier inversion and evenness of  $\widehat{\text{up}}$  give

$$\text{up}(y) = \frac{1}{2\pi} \int_{\mathbb{R}} e^{-ity} \widehat{\text{up}}(t) dt = \frac{1}{\pi} \int_0^\infty \widehat{\text{up}}(t) \cos(ty) dt.$$



Because  $\text{up}$  is a probability density and is symmetric,  $F(x) = \mathbb{P}(Y \leq 2x - 1) = \frac{1}{2} + \int_0^{2x-1} \text{up}(y) \, dy$  for  $0 \leq x \leq 1$ . Integrating the last display in  $y$  over  $[0, 2x - 1]$  and interchanging the order of integration, which is permitted by absolute convergence, replaces  $\cos(ty)$  by  $\sin((2x - 1)t)/t$  and proves (24.64).  $\square$

*Remark 24.42.* The two exact descriptions of  $F$  are complementary. Formula (24.64) is global, analytic, and well suited to quadrature and to smoothness statements; formula (24.50) is finite, rational, and adapted to dyadic arguments. Their agreement encodes a highly nonlocal cancellation in the sinc integral. At  $x = 1/4$ , for instance, numerical evaluation of (24.64) reproduces  $5/72$  to more than twenty digits.

## 24.5 Digital cancellation in an arbitrary radix

The signs  $\tau_r = (-1)^{s_2(r)}$  are the case  $b = 2$ ,  $\omega = -1$  of a roots-of-unity construction available in every radix. Separating the radix from the special real signs that exist only at  $b = 2$  shows which half of Theorems 22.7 and 22.8 comes from binary digits and which half comes from having a digit expansion at all.

For  $b \geq 2$  let  $s_b(r)$  denote the sum of the base- $b$  digits of  $r$ , and for  $\omega \in \mathbb{C}$  put

$$\tau_r^{(b)} := \omega^{s_b(r)} \quad (r \geq 0), \quad (24.65)$$

so that  $\tau_r^{(2)} = \tau_r$  when  $\omega = -1$ .

**Theorem 24.43** (Radix- $b$  Prouhet filter). *Let  $b \geq 2$  and let  $\omega \in \mathbb{C}$  satisfy*

$$\omega^b = 1, \quad \omega \neq 1. \quad (24.66)$$

*Then for every  $k \geq 0$  the polynomial identity*

$$\sum_{r=0}^{b^k-1} \tau_r^{(b)} z^r = \prod_{j=0}^{k-1} \left( \sum_{a=0}^{b-1} \omega^a z^{ab^j} \right) \quad (24.67)$$

*holds, and for every  $\theta \in \mathbb{C}$ ,*

$$\sum_{r=0}^{b^k-1} \tau_r^{(b)} (r + \theta)^m = 0 \quad (0 \leq m < k), \quad (24.68)$$

$$\sum_{r=0}^{b^k-1} \tau_r^{(b)} (r + \theta)^k = \frac{k! \, b^{k(k+1)/2}}{(\omega - 1)^k}. \quad (24.69)$$

*Consequently, for every polynomial  $p$  of degree at most  $k$  and all  $\alpha, \beta \in \mathbb{C}$ ,*

$$\sum_{r=0}^{b^k-1} \tau_r^{(b)} p(\alpha + \beta r) = \frac{k! \, b^{k(k+1)/2}}{(\omega - 1)^k} \beta^k [X^k] p(X). \quad (24.70)$$

*Proof.* Every  $r$  with  $0 \leq r < b^k$  has a unique expansion  $r = a_0 + a_1 b + \cdots + a_{k-1} b^{k-1}$  with  $0 \leq a_j < b$ , and then  $s_b(r) = a_0 + \cdots + a_{k-1}$ . Hence

$$\tau_r^{(b)} z^r = \prod_{j=0}^{k-1} \omega^{a_j} z^{a_j b^j},$$

and summing independently over the  $k$  digits gives (24.67).

Substitute  $z \mapsto e^z$  in (24.67) and multiply by  $e^{\theta z}$ . The left side becomes the exponential generating function

$$\sum_{r=0}^{b^k-1} \tau_r^{(b)} e^{(r+\theta)z} = \sum_{m \geq 0} \left( \sum_{r=0}^{b^k-1} \tau_r^{(b)} (r+\theta)^m \right) \frac{z^m}{m!},$$

and the right side becomes  $e^{\theta z} \prod_{j=0}^{k-1} G_j(z)$ , where  $G_j(z) = \sum_{a=0}^{b-1} \omega^a e^{ab^j z}$ .

Because  $\omega^b = 1$  and  $\omega \neq 1$ , the finite geometric sum gives

$$G_j(0) = \sum_{a=0}^{b-1} \omega^a = \frac{\omega^b - 1}{\omega - 1} = 0.$$

Differentiating the identity  $\sum_{a=0}^{b-1} Z^a = (Z^b - 1)/(Z - 1)$  and multiplying by  $Z$  gives

$$\sum_{a=0}^{b-1} a Z^a = \frac{b Z^b (Z - 1) - Z(Z^b - 1)}{(Z - 1)^2},$$

so at  $Z = \omega$ , where  $\omega^b = 1$ ,

$$\sum_{a=0}^{b-1} a \omega^a = \frac{b}{\omega - 1} \neq 0. \quad (24.71)$$

Hence  $G'_j(0) = b^j \sum_{a < b} a \omega^a = b^{j+1}/(\omega - 1)$ , so each  $G_j$  has a zero of order exactly one at the origin and  $\prod_{j < k} G_j$  has a zero of order exactly  $k$ , with

$$[z^k] \prod_{j=0}^{k-1} G_j(z) = \prod_{j=0}^{k-1} \frac{b^{j+1}}{\omega - 1} = \frac{b^{k(k+1)/2}}{(\omega - 1)^k}.$$

Multiplying by  $e^{\theta z} = 1 + \mathcal{O}(z)$  changes neither the order of the zero nor the coefficient of  $z^k$ . Comparing the coefficients of  $z^m/m!$  for  $0 \leq m \leq k$  gives (24.68) and (24.69).

For (24.70), expand  $q(X) := p(\alpha + \beta X) = \sum_{m=0}^k \gamma_m X^m$ , a polynomial of degree at most  $k$  with  $\gamma_k = \beta^k [X^k]p$ . Applying (24.68) with  $\theta = 0$  to each  $m < k$  and (24.69) to  $m = k$  leaves only the term  $\gamma_k$ .  $\square$

**Corollary 24.44** (The binary case). *For  $b = 2$  and  $\omega = -1$ , identity (24.67) is (22.8), identity (24.70) is (22.10), and the top value (24.69) is*

$$\frac{k! 2^{k(k+1)/2}}{(-2)^k} = (-1)^k k! 2^{\binom{k}{2}},$$

in agreement with (22.9).

*Proof.* Substitute  $b = 2$  and  $\omega = -1$ , and use  $k(k+1)/2 - k = \binom{k}{2}$ .  $\square$

**Remark 24.45** (Rigidity of the binary radix). It is worth separating what in (24.46) needs the radix 2 from what does not. Less needs it than one might expect.

*The  $q$ -calculus is radix-free.* Chapter 21 is written at general  $q$ : for real  $b > 1$  the annihilator  $(z; 1/b)_n = \prod_{j=0}^{n-1} (1 - zb^{-j})$  has the roots  $1, b, \dots, b^{n-1}$  and supplies divided-difference weights on the nodes  $-1, -b, \dots, -b^n$ . This is Corollary 24.15 at  $q = 1/b$ , whose top moment (24.23) is

$$\mathfrak{m}_n^{(b)} := \prod_{j=1}^n (b^j - 1),$$

the Mersenne product  $m_n$  at  $b = 2$ . When  $b$  is a prime power the same  $b$  carries in addition a finite-field meaning through  $\begin{bmatrix} n \\ k \end{bmatrix}_b$  (Theorem 6.1), so that at such a radix three constructions share one parameter: digital cancellation (24.70), geometric interpolation on the nodes  $-b^k$ , and subspace enumeration. For  $b = 2$  this is the coincidence exploited in Principle 22.10.

The digital signs are radix-free. Theorem 24.43 produces, for every  $b \geq 2$  and every  $\omega$  with  $\omega^b = 1$ ,  $\omega \neq 1$ , a sign system  $\tau_r^{(b)} = \omega^{s_b(r)}$  whose block exponential sum has a zero of order exactly  $k$  at the origin — the property of (22.9) that removes the apparent pole in (24.40). Since (24.66) does not force  $\omega$  to be primitive, a composite  $b$  carries several inequivalent such systems;  $b = 4$ ,  $\omega = -1$  gives  $\tau_r^{(4)} = (-1)^{s_4(r)}$  with constant  $k! 4^{k(k+1)/2} / (-2)^k$ .

The coefficient formula is radix-free as well, in a shape that must be stated carefully. Put

$$Y_b := \sum_{j \geq 1} b^{-j} V_j = \frac{X_{1/b}}{b-1}, \quad a_n^{(b)} := \frac{\mathbb{E}(Y_b^n)}{n!},$$

with  $V_1, V_2, \dots$  independent and uniform on  $[0, 1]$ , so that  $Y_2 = X$  and  $a_n^{(2)} = a_n$ . Everything in Corollary 24.5 and lemma 24.25 runs verbatim with 2 replaced by  $b$ : the moment generating function satisfies  $A_b(bz) = \Phi(z)A_b(z)$ , and  $\Psi_k^{(b)}(z) = e^{-z} \prod_{j < k} \Phi(-b^j z)$  satisfies the bridge relation (24.30) at  $q = 1/b$ , since then  $q^{-k} = b^k$ . Expanding  $\prod_{j < k} (1 - e^{-b^j z})$  over subsets and applying Theorem 24.21 together with Corollary 24.23 gives, for every  $\theta \in \mathbb{C}$ ,

$$a_n^{(b)} = \frac{(-1)^n}{m_n^{(b)}} \sum_{k=0}^n \frac{\begin{bmatrix} n \\ k \end{bmatrix}_{1/b}}{b^{2\binom{k}{2}} (n+k)!} \sum_{r \in \mathcal{B}_k^{(b)}} (-1)^{\sigma(r)} (r+\theta)^{n+k}, \quad (24.72)$$

where  $\mathcal{B}_k^{(b)}$  is the set of  $r = \sum_{j < k} \varepsilon_j b^j$  with every  $\varepsilon_j \in \{0, 1\}$ , and  $\sigma(r) := \varepsilon_0 + \dots + \varepsilon_{k-1}$ . The inner sums of (24.72) are *not* the radix- $b$  Prouhet sums of Theorem 24.43: the refinement equation delivers the factor  $1 - y^{b^j}$ , whose digit alphabet is  $\{0, 1\}$ , whereas  $\tau^{(b)}$  arises from  $\sum_{a < b} \omega^a y^{ab^j}$ , whose digit alphabet is  $\{0, \dots, b-1\}$ . The two coincide exactly when  $\{0, 1\} = \{0, \dots, b-1\}$ , that is when  $b = 2$ ; there  $\mathcal{B}_k^{(2)} = \{0, 1, \dots, 2^k - 1\}$ ,  $(-1)^{\sigma(r)} = \tau_r$ ,  $b^{2\binom{k}{2}} = 4^{\binom{k}{2}}$ , and (24.72) is (24.44).

What is genuinely rigid is the passage from coefficients to values. The simplex cut in the proof of Theorem 24.30 uses only  $b^{n-j} \geq 1$  and a threshold at most 1, so it survives at every radix:

$$\mathbb{P}(Y_b \leq b^{-n}) = \frac{\mathbb{E}[(1 - Y_b)^n]}{n! b^{\binom{n}{2}}}.$$

Turning the right-hand side into the single moment  $\mathbb{E}(Y_b^n)$  uses the reflection  $1 - Y_b \stackrel{d}{=} Y_b$ . But  $Y_b$  has support  $[0, 1/(b-1)]$  while  $1 - Y_b$  has support  $[1 - 1/(b-1), 1]$ , and these agree only when  $1/(b-1) = 1$ . Only at  $b = 2$ , where the geometric mass  $\sum_{j \geq 1} b^{-j}$  is exactly 1 and  $Y_2 = X_{1/2}$  is symmetric about  $1/2$  by (23.2), does (24.72) evaluate a distribution function. For  $b \geq 3$  one still gets every value, since

$$F_{1/b}((b-1)b^{-n}) = \mathbb{P}(Y_b \leq b^{-n}) = \frac{1}{b^{\binom{n}{2}}} \sum_{m=0}^n (-1)^m \binom{n}{m} \frac{m!}{n!} a_m^{(b)} \cdot n! / n!,$$

a binomial combination of  $a_0^{(b)}, \dots, a_n^{(b)}$  rather than the single coefficient  $a_n^{(b)}$ . The  $q$ -binomial and the digital ingredients are therefore automatic at every radix; what singles out  $b = 2$  is the one normalization in which the dilation factor of (23.3), the digit alphabet of the refinement factor, and the right endpoint of the support all agree.

## 24.6 The geometric row as a summability kernel

The Lagrange row  $\lambda_{n,k}^{(q)}$  of (21.4) was used in Chapter 21 for exactness. It also has an exact and uniformly bounded  $\ell^1$  norm, which turns it into a regular summability kernel.

**Theorem 24.46** (Row mass, total variation, and the geometric Lebesgue constant). *Let  $0 < q < 1$ . The weights (21.4) alternate in sign,*

$$\operatorname{sgn} \lambda_{n,k}^{(q)} = (-1)^{n-k} \quad (0 \leq k \leq n), \quad (24.73)$$

and for every  $n \geq 0$

$$\sum_{k=0}^n \lambda_{n,k}^{(q)} = 1, \quad (24.74)$$

$$\sum_{k=0}^n |\lambda_{n,k}^{(q)}| = \frac{(-q; q)_n}{(q; q)_n} = \frac{(q^2; q^2)_n}{(q; q)_n^2}. \quad (24.75)$$

The right-hand side of (24.75) increases strictly with  $n$  to the finite limit

$$\mathcal{L}(q) := \frac{(-q; q)_\infty}{(q; q)_\infty} = \frac{(q^2; q^2)_\infty}{(q; q)_\infty^2}, \quad (24.76)$$

so that  $\sum_{k=0}^n |\lambda_{n,k}^{(q)}| < \mathcal{L}(q) < \infty$  for every  $n$ . At the half base,

$$\sum_{k=0}^n |\lambda_{n,k}^{(1/2)}| = 3 \prod_{r=2}^n \frac{1+2^{-r}}{1-2^{-r}} < \mathcal{L}(1/2) = 8.2559879357782500655 \dots < 9 \quad (n \geq 1). \quad (24.77)$$

*Proof.* By Theorem 21.1 and (21.4),

$$\sum_{k=0}^n \lambda_{n,k}^{(q)} z^k = \frac{1}{(q; q)_n} \prod_{j=1}^n (z - q^j). \quad (24.78)$$

Evaluating at  $z = 1$  gives  $(q; q)_n / (q; q)_n = 1$ , which is (24.74). For  $0 < q < 1$  the three factors  $(q; q)_n$ ,  $q^{\binom{n-k+1}{2}}$  and  $\begin{bmatrix} n \\ k \end{bmatrix}_q$  occurring in (21.4) are positive, which is (24.73); hence  $|\lambda_{n,k}^{(q)}| = (-1)^n (-1)^k \lambda_{n,k}^{(q)}$  and, by (24.78) at  $z = -1$ ,

$$\sum_{k=0}^n |\lambda_{n,k}^{(q)}| = \frac{(-1)^n}{(q; q)_n} \prod_{j=1}^n (-1 - q^j) = \frac{(-q; q)_n}{(q; q)_n}.$$

The second form in (24.75) follows from

$$(-q; q)_n (q; q)_n = \prod_{j=1}^n (1 + q^j) (1 - q^j) = \prod_{j=1}^n (1 - q^{2j}) = (q^2; q^2)_n.$$

Passing from  $n$  to  $n+1$  multiplies the quotient by  $(1 + q^{n+1}) / (1 - q^{n+1}) > 1$ , so it increases strictly; it converges because

$$\sum_{j \geq 1} \log \frac{1 + q^j}{1 - q^j} = \sum_{j \geq 1} \log \left( 1 + \frac{2q^j}{1 - q^j} \right) \leq \frac{2}{1 - q} \sum_{j \geq 1} q^j < \infty,$$

both products in (24.76) being convergent for  $0 < q < 1$ .

At  $q = 1/2$  the factor  $r = 1$  contributes  $(1 + 1/2)/(1 - 1/2) = 3$ , which is the displayed prefactor. For the bound, split the infinite product at  $r = 4$ : the factors  $r = 1, 2, 3$  give  $3 \cdot \frac{5}{3} \cdot \frac{9}{7} = \frac{45}{7}$ , while for  $r \geq 4$  one has  $x = 2^{-r} \leq 1/16$  and

$$\frac{1+x}{1-x} = 1 + \frac{2x}{1-x} \leq \exp\left(\frac{2x}{1-x}\right) \leq \exp\left(\frac{32x}{15}\right),$$

so, since  $\sum_{r \geq 4} 2^{-r} = 1/8$ , the tail is at most  $e^{4/15} < 1.3063$  and  $\mathcal{L}(1/2) \leq \frac{45}{7} e^{4/15} < 8.4 < 9$ . The decimal expansion in (24.77) is obtained by evaluating the product numerically.  $\square$

**Corollary 24.47** (Conditioning of geometric extrapolation). *Let  $0 < q < 1$ , let  $p$  be a polynomial of degree at most  $n$ , and let  $y_0, \dots, y_n$  be data with  $|y_k - p(q^k)| \leq \varepsilon$  for every  $k$ . Then*

$$\left| \sum_{k=0}^n \lambda_{n,k}^{(q)} y_k - p(0) \right| \leq \mathcal{L}(q) \varepsilon. \quad (24.79)$$

*The amplification of data error is therefore bounded independently of the number of nodes.*

*Proof.* By Theorem 21.3,  $\sum_k \lambda_{n,k}^{(q)} p(q^k) = p(0)$ ; subtract, take absolute values, and apply (24.75) together with (24.76).  $\square$

**Remark 24.48** (Geometric versus harmonic node sequences). The uniform bound is special to geometric nodes. For the harmonic sequence  $v_k = 1/(k+1)$ ,  $0 \leq k \leq n$ , the corresponding evaluation weights  $\ell_k(0) = \prod_{j \neq k} v_j / (v_j - v_k)$  have absolute sums

$$9, \quad 91.67, \quad 1.007 \cdot 10^3, \quad 1.151 \cdot 10^4, \quad 1.600 \cdot 10^6, \quad 3.495 \cdot 10^{10}$$

at  $n = 2, 4, 6, 8, 12, 20$  (computed in exact rational arithmetic), growing geometrically in  $n$ , whereas at  $q = 1/2$  the same quantity never reaches 9. Combined with Theorem 21.9, this says that the filters  $\mathcal{G}_{n,q}$  are exact and simultaneously uniformly bounded on the sampled values: raising the order of a geometric filter costs nothing in stability. This is the analytic half of Principle 24.52.

**Remark 24.49** (The half-base row in reversed coordinates). Put

$$W_{n,j} := \frac{(-1)^j 2^{-\binom{j+1}{2}}}{(1/2; 1/2)_n} \begin{bmatrix} n \\ j \end{bmatrix}_{1/2}, \quad 0 \leq j \leq n.$$

The symmetry  $\begin{bmatrix} n \\ j \end{bmatrix}_{1/2} = \begin{bmatrix} n \\ n-j \end{bmatrix}_{1/2}$  gives  $W_{n,j} = \lambda_{n,n-j}^{(1/2)}$ , and (24.78) becomes

$$\sum_{j=0}^n W_{n,j} z^j = \frac{(z/2; 1/2)_n}{(1/2; 1/2)_n}.$$

The two rows are reversals of one another, so Theorem 24.46 applies verbatim to either.

**Theorem 24.50** (Bounded-variation tail rows). *Let  $(x_m)_{m \geq 0}$  converge to  $L$  in a normed vector space. For each  $n$  let  $u_{n,0}, \dots, u_{n,n}$  be scalars and  $r(n,0), \dots, r(n,n)$  nonnegative integers with*

$$\sum_{j=0}^n u_{n,j} = 1, \quad \sup_n \sum_{j=0}^n |u_{n,j}| \leq M < \infty, \quad \min_{0 \leq j \leq n} r(n,j) \longrightarrow \infty.$$

*Then  $\sum_{j=0}^n u_{n,j} x_{r(n,j)} \rightarrow L$  as  $n \rightarrow \infty$ .*

*Proof.* Unit row mass gives

$$\left\| \sum_{j=0}^n u_{n,j} x_{r(n,j)} - L \right\| = \left\| \sum_{j=0}^n u_{n,j} (x_{r(n,j)} - L) \right\| \leq \sum_{j=0}^n |u_{n,j}| \|x_{r(n,j)} - L\|.$$

Given  $\varepsilon > 0$ , choose  $N$  with  $\|x_m - L\| < \varepsilon/M$  for all  $m \geq N$ . For all  $n$  large enough,  $r(n, j) \geq N$  for every  $j$ , so the right side is at most  $M \cdot \varepsilon/M = \varepsilon$ .  $\square$

**Corollary 24.51** (Gaussian averages of the spline approximants). *Let  $0 < q < 1$  and let  $F_{q,N}$  be the distribution function of  $X_{q,N}$ . For any integers  $r(n, j) \geq n$ ,*

$$\sup_{x \in \mathbb{R}} \left| \sum_{j=0}^n \lambda_{n,j}^{(q)} F_{q,r(n,j)}(x) - F_q(x) \right| \rightarrow 0 \quad (n \rightarrow \infty).$$

*Proof.* Apply [Theorem 24.50](#) in the Banach space of bounded functions on  $\mathbb{R}$  with the supremum norm, to the sequence  $x_m := F_{q,m}$ , which converges to  $F_q$  by [\(23.29\)](#). The hypotheses hold by [Theorem 24.46](#), with  $M = (-q; q)_\infty / (q; q)_\infty$ , and  $\min_j r(n, j) \geq n \rightarrow \infty$ .  $\square$

**Principle 24.52** (One polynomial, two roles). *The single finite product  $\prod_{j=1}^n (z - q^j)$ , normalized by  $(q; q)_n$ , controls both exactness and stability of the geometric row:*

1. *its roots  $q, q^2, \dots, q^n$  make the row an order- $n$  geometric finite-difference stencil, annihilating  $q^{kd}$  for  $1 \leq d \leq n$  ([Theorem 21.3](#)) and extracting one leading coefficient ([Theorem 22.4](#));*
2. *its endpoint values at  $z = 1$  and  $z = -1$  give unit row mass and the exact  $\ell^1$  norm  $(-q; q)_n / (q; q)_n$  ([Theorem 24.46](#)), which are precisely the two hypotheses of [Theorem 24.50](#).*

*Roots control vanishing moments; endpoint values control norms.*

*Proof.* Combine [Theorems 21.3, 22.4, 24.46](#) and [24.50](#).  $\square$

**Remark 24.53** (Where the analysis actually enters). The dyadic results of this chapter split along the boundary recommended in [Principle 20.18](#), and the split is sharper than it may look. Each of [Theorems 22.4, 24.21](#) and [24.30](#), [corollaries 24.23](#) and [24.31](#), and [proposition 24.35](#) is a finite identity between rational numbers, proved in polynomial rings and formal power series over  $\mathbb{Q}$ ; none of them uses a convergence hypothesis, and each survives mapping into an arbitrary commutative  $\mathbb{Q}$ -algebra, exactly as in [Remark 21.2](#). Analysis enters at precisely two points: the identification of  $\mathcal{A}$  with  $\mathbb{E}e^{zX}$  in [Theorem 24.3](#), and [Theorem 24.9](#), which converts the formal coefficient  $\alpha_n$  into the value  $F(2^{-n})$ . Every closed formula above is the algebraic core composed with that single analytic bridge. The formalization status of the individual statements is recorded in [Appendix D](#).

### 24.6.1 Negative parameters

The series [\(23.1\)](#) and the convergence estimate in the proof of [Theorem 23.1](#) use only  $|q| < 1$ , so  $X_q$  is a well-defined bounded random variable for every real  $q \in (-1, 1)$ . For negative  $q$  it is no longer supported in  $[0, 1]$ , but it is an affine image of a positive-parameter member of the same family.

**Proposition 24.54** (Negative parameters as affine positive laws). *Let  $0 < a < 1$  and set*

$$\lambda_a := \frac{1+a}{1-a}, \quad \beta_a := \frac{a}{1-a}, \quad \text{so that } \lambda_a = 1 + 2\beta_a. \quad (24.80)$$

Then

$$X_{-a} \stackrel{d}{=} \lambda_a X_a - \beta_a, \quad M_{-a}(t) = e^{-\beta_a t} M_a(\lambda_a t). \quad (24.81)$$

Consequently  $\mathbb{E}X_{-a} = \frac{1}{2}$  and  $X_{-a} \stackrel{d}{=} 1 - X_{-a}$ , and the topological support of  $X_{-a}$  is the interval

$$\left[ -\frac{a}{1-a}, \frac{1}{1-a} \right]. \quad (24.82)$$

*Proof.* Put  $V_j := U_j - \frac{1}{2}$ . The  $V_j$  are independent and symmetric, so the sequences  $(V_j)_{j \geq 0}$  and  $((-1)^j V_j)_{j \geq 0}$  have the same joint law. Since  $(1-q) \sum_{j \geq 0} q^j = 1$  for every  $|q| < 1$ ,

$$X_q - \frac{1}{2} = (1-q) \sum_{j \geq 0} q^j V_j.$$

Taking  $q = -a$  and using  $(1+a)(-a)^j = (1+a)a^j \cdot (-1)^j$ ,

$$\begin{aligned} X_{-a} - \frac{1}{2} &= (1+a) \sum_{j \geq 0} a^j (-1)^j V_j \stackrel{d}{=} (1+a) \sum_{j \geq 0} a^j V_j \\ &= \frac{1+a}{1-a} \left[ (1-a) \sum_{j \geq 0} a^j V_j \right] = \lambda_a \left( X_a - \frac{1}{2} \right), \end{aligned}$$

all three series converging absolutely. Hence  $X_{-a} \stackrel{d}{=} \lambda_a X_a - \frac{\lambda_a - 1}{2}$ , and  $\frac{\lambda_a - 1}{2} = \beta_a$  by (24.80). Taking moment generating functions gives the second identity in (24.81).

For the remaining assertions, recall from Theorem 23.1 that  $X_a$  has support  $[0, 1]$  and satisfies  $X_a \stackrel{d}{=} 1 - X_a$ . An increasing affine map carries the support onto its image, so the support of  $X_{-a}$  is  $[-\beta_a, \lambda_a - \beta_a]$ , and  $\lambda_a - \beta_a = \frac{(1+a)-a}{1-a} = \frac{1}{1-a}$ , which is (24.82). Finally  $\mathbb{E}X_{-a} = \frac{\lambda_a}{2} - \beta_a = \frac{(1+a)-2a}{2(1-a)} = \frac{1}{2}$ , and, using  $\lambda_a - 1 - \beta_a = \beta_a$ ,

$$1 - X_{-a} \stackrel{d}{=} 1 - \lambda_a X_a + \beta_a = \lambda_a(1 - X_a) - \beta_a \stackrel{d}{=} \lambda_a X_a - \beta_a \stackrel{d}{=} X_{-a}. \quad \square$$

For instance  $X_{-1/2} \stackrel{d}{=} 3X_{1/2} - 1$ , so the law of  $X_{-1/2}$  is the Fabius law stretched by 3 and supported on  $[-1, 2]$ ; likewise  $X_{-1/4} \stackrel{d}{=} \frac{5}{3}X_{1/4} - \frac{1}{3}$ .

## 24.6.2 Reciprocal germs and reciprocity

For  $|q| > 1$  the series (23.1) diverges and there is no random variable to expand. The functional equation (24.6) survives, however, and pins down a unique normalized germ.

**Theorem 24.55** (The Mahler equation with a normalization is rigid). *Let  $q \in \mathbb{C}$  with  $q \neq 0$  and  $|q| \neq 1$ , and put  $\rho := 2\pi \min(1, |q|)/|1-q|$ . Then there is exactly one function  $M$ , analytic on the disc  $|t| < \rho$ , with*

$$M(0) = 1, \quad M(t) = \mathcal{E}((1-q)t)M(qt) \text{ for } |t| < \rho, |qt| < \rho, \quad (24.83)$$

namely

$$M(t) = \begin{cases} \prod_{j=0}^{\infty} \mathcal{E}((1-q)q^j t), & |q| < 1, \\ \left[ \prod_{j=1}^{\infty} \mathcal{E}((1-q)q^{-j} t) \right]^{-1}, & |q| > 1. \end{cases} \quad (24.84)$$

Both products converge locally uniformly, and in the second case no factor vanishes on  $|t| < \rho$ .

*Proof.* Convergence of the first product is [Theorem 24.3](#); the second is identical after replacing  $q$  by  $q^{-1}$  in the estimate, since  $\sum_{j \geq 1} |q|^{-j} < \infty$ . When  $|q| > 1$  one has  $\rho = 2\pi/|1 - q|$ , so for  $|t| < \rho$  and  $j \geq 1$

$$|(1 - q)q^{-j}t| \leq |1 - q||q|^{-1}|t| < |1 - q||q|^{-1}\rho = \frac{2\pi}{|q|} < 2\pi.$$

Since every zero of  $\mathcal{E}$  has modulus at least  $2\pi$ , no factor vanishes on that disc and the reciprocal is analytic there.

That [\(24.84\)](#) solves [\(24.83\)](#) is the computation of [Theorem 24.3](#) in the first case. In the second case write  $P(t) = \prod_{j \geq 1} \mathcal{E}((1 - q)q^{-j}t)$  and split off  $j = 1$  after replacing  $t$  by  $qt$ :

$$P(qt) = \prod_{j \geq 1} \mathcal{E}((1 - q)q^{1-j}t) = \mathcal{E}((1 - q)t)P(t).$$

Taking reciprocals gives  $M(t) = \mathcal{E}((1 - q)t)M(qt)$  with  $M = P^{-1}$ .

For uniqueness, let  $M$  be any solution. If  $|q| < 1$ , then  $|q^k t| < \rho$  for all  $k \geq 0$ , so [\(24.83\)](#) may be iterated  $N$  times:

$$M(t) = \left[ \prod_{j=0}^{N-1} \mathcal{E}((1 - q)q^j t) \right] M(q^N t).$$

Let  $N \rightarrow \infty$ :  $q^N t \rightarrow 0$ , so  $M(q^N t) \rightarrow M(0) = 1$  by continuity, and the bracket converges to the first product in [\(24.84\)](#).

If  $|q| > 1$ , apply [\(24.83\)](#) at the point  $q^{-1}t$ , which again lies in the disc, to get  $M(q^{-1}t) = \mathcal{E}((1 - q)q^{-1}t)M(t)$ , that is,

$$M(t) = \frac{M(q^{-1}t)}{\mathcal{E}((1 - q)q^{-1}t)}.$$

Iterating  $N$  times,

$$M(t) = M(q^{-N}t) \left[ \prod_{j=1}^N \mathcal{E}((1 - q)q^{-j}t) \right]^{-1},$$

and letting  $N \rightarrow \infty$  gives the second line of [\(24.84\)](#). □

**Definition 24.56** (The reciprocal germ). For  $|q| > 1$ ,  $M_q$  denotes the unique germ of [Theorem 24.55](#), that is, the reciprocal product in [\(24.84\)](#); its Taylor coefficients are again written  $a_n(q) = d_n(q)/n!$ .

*Remark 24.57.* The symbol  $M_q$  is used in two senses. For real  $q$  with  $|q| < 1$  it is a genuine moment generating function and  $d_n(q) = \mathbb{E}X_q^n$ . For  $|q| > 1$  there is no underlying random variable, and  $d_n(q)$  are merely normalized Taylor coefficients. What is common to both regimes is [\(24.6\)](#), and therefore also [\(24.7\)](#): the derivation of the recurrence used the Mahler equation and nothing else, so a single rational function  $a_n$  of  $q$  governs both regimes.

**Theorem 24.58** (Reciprocity). For every  $q \in \mathbb{C}$  with  $q \neq 0$  and  $|q| \neq 1$ ,

$$M_q(t) M_{1/q}(-t) = 1 \tag{24.85}$$

as analytic germs at  $t = 0$ . Equivalently, at the level of coefficients,

$$\sum_{k=0}^n \binom{n}{k} (-1)^{n-k} d_k(q) d_{n-k}(1/q) = \delta_{n0} \quad (n \geq 0). \tag{24.86}$$



*Proof.* By symmetry it suffices to treat  $|q| < 1$ , so that  $|1/q| > 1$  and  $M_{1/q}$  is the germ of Definition 24.56. Writing  $Q = 1/q$  in (24.84) and evaluating at  $-t$ ,

$$M_{1/q}(-t)^{-1} = \prod_{j=1}^{\infty} \mathcal{E}((1-Q)Q^{-j}(-t)) = \prod_{j=1}^{\infty} \mathcal{E}(-(1-q^{-1})q^j t).$$

Now  $-(1-q^{-1})q^j = (q^{-1}-1)q^j = q^{j-1}-q^j = (1-q)q^{j-1}$ , so after the substitution  $i = j-1$  the right-hand side is  $\prod_{i \geq 0} \mathcal{E}((1-q)q^i t) = M_q(t)$ , which is (24.85).

For (24.86), multiply the two Taylor series

$$M_q(t) = \sum_{k \geq 0} d_k(q) \frac{t^k}{k!}, \quad M_{1/q}(-t) = \sum_{m \geq 0} (-1)^m d_m(1/q) \frac{t^m}{m!},$$

and read off the coefficient of  $t^n/n!$  in the product; by (24.85) it must equal  $\delta_{n0}$ .  $\square$

### 24.6.3 The spectral $q^2$ -Pochhammer factorization

Splitting  $\mathcal{E}$  into its centered part is the step that converts the geometric ratio  $q$  of the random series into the nome  $q^2$  of a  $q$ -Pochhammer product. From (23.5),

$$\mathcal{E}(z) = e^{z/2} \frac{e^{z/2} - e^{-z/2}}{z} = e^{z/2} \frac{\sinh(z/2)}{z/2}, \quad (24.87)$$

and Euler's product for the sine, in its hyperbolic form, is

$$\frac{\sinh(z/2)}{z/2} = \prod_{m=1}^{\infty} \left(1 + \frac{z^2}{4\pi^2 m^2}\right). \quad (24.88)$$

**Theorem 24.59** (Spectral Pochhammer factorization). *For  $|q| < 1$ ,*

$$M_q(t) = e^{t/2} \prod_{m=1}^{\infty} \left(-\frac{(1-q)^2 t^2}{4\pi^2 m^2}; q^2\right)_{\infty}, \quad (24.89)$$

*and for  $|q| > 1$ ,*

$$M_q(t) = e^{t/2} \prod_{m=1}^{\infty} \left(-\frac{(1-q^{-1})^2 t^2}{4\pi^2 m^2}; q^{-2}\right)_{\infty}^{-1}. \quad (24.90)$$

*In both cases the double product converges normally on compact subsets of the  $t$ -plane.*

*Proof.* Let  $|q| < 1$  and substitute (24.87) and (24.88) with  $z = (1-q)q^j t$  into (24.5). The exponential factors multiply to

$$\exp\left(\frac{t}{2}(1-q) \sum_{j \geq 0} q^j\right) = e^{t/2},$$

leaving the double product  $\prod_{j \geq 0} \prod_{m \geq 1} (1 + (1-q)^2 q^{2j} t^2 / (4\pi^2 m^2))$ . On  $|t| \leq T$  the sum of the absolute values of the perturbations is at most

$$\sum_{j \geq 0} \sum_{m \geq 1} \frac{|1-q|^2 |q|^{2j} T^2}{4\pi^2 m^2} = \frac{|1-q|^2 T^2}{24(1-|q|^2)} < \infty,$$

using  $\sum_m m^{-2} = \pi^2/6$ . The double product is therefore normally convergent and may be rearranged. Fixing  $m$  and multiplying over  $j$ ,

$$\prod_{j \geq 0} \left(1 + \frac{(1-q)^2 t^2}{4\pi^2 m^2} q^{2j}\right) = \left(-\frac{(1-q)^2 t^2}{4\pi^2 m^2}; q^2\right)_\infty$$

by the definition  $(x; p)_\infty = \prod_{j \geq 0} (1 - xp^j)$  with  $p = q^2$ . This proves (24.89).

Let  $|q| > 1$  and apply the same substitution to the product  $P(t) = \prod_{j \geq 1} \mathcal{E}((1-q)q^{-j}t)$  of (24.84), so that  $M_q = P^{-1}$ . Here the linear arguments sum to

$$\frac{t}{2}(1-q) \sum_{j \geq 1} q^{-j} = \frac{t}{2}(1-q) \frac{q^{-1}}{1-q^{-1}} = \frac{t}{2} \frac{1-q}{q-1} = -\frac{t}{2},$$

so  $P$  contributes  $e^{-t/2}$  and its reciprocal contributes  $e^{t/2}$ , as claimed. For the quadratic part, reindex by  $i = j - 1 \geq 0$  and use

$$(1-q)^2 q^{-2j} = (1-q)^2 q^{-2} q^{-2i} = (1-q^{-1})^2 (q^{-2})^i,$$

which is legitimate because  $(1-q)^2 q^{-2} = (q^{-1} - 1)^2 = (1 - q^{-1})^2$ . Fixing  $m$  and multiplying over  $i \geq 0$  gives  $(-(1-q^{-1})^2 t^2 / (4\pi^2 m^2); q^{-2})_\infty$ ; taking reciprocals proves (24.90). Normal convergence is the previous estimate with  $|q|^{-1} < 1$  in place of  $|q|$ .  $\square$

*Remark 24.60* (The nome, and where the  $q$ -objects finally appear). The geometric ratio  $q$  of the random series (23.1) becomes the nome  $q^2$  of the spectral product: at  $q = \pm \frac{1}{2}$  the nome is  $\frac{1}{4}$ , and at  $q = \pm \frac{1}{4}$  it is  $\frac{1}{16}$ . Reciprocal parameters exchange the zeros of (24.89) with the poles of (24.90), in accordance with Theorem 24.58. Theorem 24.59 is the point at which the two halves of this Part meet: the transform of the geometric uniform sum – and hence, at  $q = 1/2$ , of the Fabius law – is literally an infinite product of  $q$ -Pochhammer symbols in the nome  $q^2$ , with the  $m$ th factor supplied by the  $m$ th spectral line of Euler’s sine product.

Taking logarithms in (24.89) and expanding  $\log(1+x)$  gives, for small  $|t|$ ,

$$\log M_q(t) - \frac{t}{2} = \sum_{r \geq 1} \frac{(-1)^{r-1}}{r} \frac{\zeta(2r)}{1 - q^{2r}} \left( \frac{(1-q)^2 t^2}{4\pi^2} \right)^r,$$

which, by  $\zeta(2r) = (-1)^{r+1} B_{2r} (2\pi)^{2r} / (2(2r)!)$ , is exactly the cumulant expansion (23.8) of Theorem 23.3. The spectral factorization and the Bernoulli cumulants are two readings of the same product.

## 24.7 The moment polynomial and the endpoint values

The recurrence (24.7) carries apparent poles at all roots of unity. A single finite  $q$ -Pochhammer factor removes them.

**Definition 24.61** (The moment polynomial). For  $n \geq 0$  set

$$\mathcal{P}_n(q) := \frac{(q; q)_n}{(1-q)^n} a_n(q) = \frac{(q; q)_n}{(1-q)^n} \frac{d_n(q)}{n!}. \quad (24.91)$$

**Theorem 24.62** (Polynomiality and recurrence). *Each  $\mathcal{P}_n$  lies in  $\mathbb{Q}[q]$ . One has  $\mathcal{P}_0 = 1$  and, for  $n \geq 1$ ,*

$$\mathcal{P}_n(q) = \sum_{k=0}^{n-1} \frac{q^k}{(n-k+1)!} \frac{(q; q)_{n-1}}{(q; q)_k} \mathcal{P}_k(q), \quad (24.92)$$

and moreover

$$\deg \mathcal{P}_n \leq \binom{n}{2}, \quad \mathcal{P}_n(0) = \frac{1}{(n+1)!}. \quad (24.93)$$

The first cases are

$$\begin{aligned} \mathcal{P}_0 &= 1, & \mathcal{P}_1 &= \frac{1}{2}, & \mathcal{P}_2 &= \frac{q+2}{12}, & \mathcal{P}_3 &= \frac{q^2+q+1}{24}, \\ \mathcal{P}_4 &= \frac{(q^2+q+1)(6+3q+5q^2+2q^3-q^4)}{720}. \end{aligned}$$

*Proof.* Inverting (24.91) gives  $a_k(q) = (1-q)^k \mathcal{P}_k(q)/(q; q)_k$ . Substituting this into (24.7) and multiplying through by  $(q; q)_{n-1}/(1-q)^n$  turns the left-hand side into

$$\frac{(1-q^n)(q; q)_{n-1}}{(1-q)^n} a_n(q) = \frac{(q; q)_n}{(1-q)^n} a_n(q) = \mathcal{P}_n(q),$$

and the  $k$ th term on the right into  $\frac{q^k}{(n-k+1)!} \frac{(q; q)_{n-1}}{(q; q)_k} \mathcal{P}_k(q)$ . This is (24.92).

Each quotient  $(q; q)_{n-1}/(q; q)_k = \prod_{j=k+1}^{n-1} (1-q^j)$  occurring in (24.92) is a polynomial of degree  $\sum_{j=k+1}^{n-1} j = \binom{n}{2} - \binom{k+1}{2}$ . Assuming inductively that  $\mathcal{P}_k \in \mathbb{Q}[q]$  with  $\deg \mathcal{P}_k \leq \binom{k}{2}$  for all  $k < n$ , the  $k$ th summand is a polynomial of degree at most

$$k + \left( \binom{n}{2} - \binom{k+1}{2} \right) + \binom{k}{2} = \binom{n}{2},$$

because  $\binom{k+1}{2} - \binom{k}{2} = k$ . Hence  $\mathcal{P}_n \in \mathbb{Q}[q]$  and  $\deg \mathcal{P}_n \leq \binom{n}{2}$ . Setting  $q = 0$  in (24.92) kills every term with  $k \geq 1$  and leaves  $\mathcal{P}_n(0) = \mathcal{P}_0(0)/(n+1)! = 1/(n+1)!$ . The displayed values follow by evaluating (24.92) for  $n \leq 4$ .  $\square$

**Proposition 24.63** (Exact degree and the two leading coefficients). *Let  $B_m$  be the Bernoulli numbers in the convention (23.6), so that  $B_1 = -\frac{1}{2}$ . For every  $n \geq 0$ ,*

$$[q^{\binom{n}{2}}] \mathcal{P}_n(q) = \frac{(-1)^n B_n}{n!}, \quad (24.94)$$

and for every  $n \geq 2$ ,

$$[q^{\binom{n}{2}-1}] \mathcal{P}_n(q) = -\frac{(-1)^n B_n}{n!} + \frac{(-1)^{n-1} B_{n-1}}{2(n-1)!}. \quad (24.95)$$

Consequently

$$\deg \mathcal{P}_n = \begin{cases} \binom{n}{2}, & n \text{ even, or } n = 1, \\ \binom{n}{2} - 1, & n \geq 3 \text{ odd.} \end{cases} \quad (24.96)$$

*Proof.* Write  $R_n(q) := (q; q)_n/(1-q)^n = \prod_{j=2}^n (1+q+\cdots+q^{j-1})$ , a polynomial of degree  $\sum_{j=2}^n (j-1) = \binom{n}{2}$ . Factoring the top power out of each of its  $n-1$  factors,

$$R_n(q) = q^{\binom{n}{2}} \prod_{j=2}^n (1+q^{-1}+\cdots+q^{-(j-1)}) = q^{\binom{n}{2}} (1+(n-1)q^{-1}+O(q^{-2})) \quad (24.97)$$

as  $q \rightarrow \infty$ . Since  $\mathcal{P}_n = R_n a_n$  is a polynomial of degree at most  $\binom{n}{2}$ , the rational function  $a_n = \mathcal{P}_n/R_n$  is regular at  $q = \infty$ ; let

$$a_n(q) = L_n + M_n q^{-1} + O(q^{-2}) \quad (q \rightarrow \infty) \quad (24.98)$$

be its expansion there. Multiplying (24.97) by (24.98) and comparing coefficients gives

$$[q^{\binom{n}{2}}]\mathcal{P}_n = L_n, \quad [q^{\binom{n}{2}-1}]\mathcal{P}_n = M_n + (n-1)L_n \quad (n \geq 2). \quad (24.99)$$

It remains to compute  $L_n$  and  $M_n$ . Multiply (24.7) by  $q^{-n}$ :

$$(q^{-n} - 1)a_n(q) = \sum_{k=0}^{n-1} \frac{(-1)^{n-k}(1-q^{-1})^{n-k}}{(n-k+1)!} a_k(q), \quad (24.100)$$

using  $q^{-n}q^k(1-q)^{n-k} = (-1)^{n-k}(1-q^{-1})^{n-k}$ . Letting  $q \rightarrow \infty$  in (24.100) gives

$$-L_n = \sum_{k < n} (-1)^{n-k} \frac{L_k}{(n-k+1)!},$$

that is,

$$\sum_{k=0}^n \frac{(-1)^{n-k}}{(n-k+1)!} L_k = 0 \quad (n \geq 1), \quad L_0 = 1. \quad (24.101)$$

Set  $L(s) = \sum_{n \geq 0} L_n s^n$  and  $\Psi(s) = \sum_{r \geq 0} (-s)^r / (r+1)! = (1 - e^{-s})/s$ . Relation (24.101) says precisely that  $L(s)\Psi(s) = 1$ , whence

$$L(s) = \frac{s}{1 - e^{-s}} = \frac{-s}{e^{-s} - 1} = \sum_{n \geq 0} B_n \frac{(-s)^n}{n!}$$

by (23.6). Thus  $L_n = (-1)^n B_n / n!$ , which with (24.99) is (24.94).

For  $M_n$ , note first that  $a_1 \equiv \frac{1}{2}$ , so  $M_0 = M_1 = 0$ . For  $n \geq 2$  the term  $q^{-n}a_n(q)$  on the left of (24.100) is  $O(q^{-2})$ , so comparing coefficients of  $q^{-1}$  there gives

$$-M_n = \sum_{k=0}^{n-1} \frac{(-1)^{n-k}(M_k - (n-k)L_k)}{(n-k+1)!},$$

that is,  $[s^n](M\Psi) = [s^n](L\Theta)$  for  $n \geq 2$ , where  $M(s) = \sum M_n s^n$  and  $\Theta(s) = \sum_{r \geq 0} (-1)^r r s^r / (r+1)!$ . Since  $\sum_{r \geq 0} (r+1)(-s)^r / (r+1)! = e^{-s}$ , we have  $\Theta = e^{-s} - \Psi$ . The two remaining coefficients are checked directly:  $[s^0](M\Psi) = M_0 = 0 = [s^0](L\Theta)$ , while  $[s^1](M\Psi) = M_1 = 0$  and  $[s^1](L\Theta) = L_0\Theta_1 = -\frac{1}{2}$ . Hence  $M\Psi = L\Theta + \frac{s}{2}$ , and since  $1/\Psi = L$ ,

$$M = L^2\Theta + \frac{sL}{2} = L^2e^{-s} - L + \frac{sL}{2}.$$

Differentiating  $L = s/(1 - e^{-s})$  gives  $sL' = L - L^2e^{-s}$ , so  $L^2e^{-s} = L - sL'$  and therefore  $M = -sL' + \frac{sL}{2}$ . Reading off coefficients,  $M_n = -nL_n + \frac{1}{2}L_{n-1}$ , and substituting into (24.99),

$$[q^{\binom{n}{2}-1}]\mathcal{P}_n = -nL_n + \frac{1}{2}L_{n-1} + (n-1)L_n = -L_n + \frac{1}{2}L_{n-1},$$

which is (24.95).

Finally,  $B_n \neq 0$  for  $n$  even and  $B_n = 0$  for odd  $n \geq 3$ , while  $L_1 = \frac{1}{2} \neq 0$ . So (24.94) is nonzero exactly for  $n$  even and for  $n = 1$ ; and when  $n \geq 3$  is odd, (24.95) reduces to  $\frac{1}{2}L_{n-1} = B_{n-1}/(2(n-1)!) \neq 0$ , since  $n-1$  is even and at least 2. This is (24.96).  $\square$

**Example 24.64.** For  $n = 3, \dots, 8$  the degrees of  $\mathcal{P}_n$  are

$$2, 6, 9, 15, 20, 28, \quad \text{against} \quad \binom{n}{2} = 3, 6, 10, 15, 21, 28,$$

so the bound (24.93) is missed by exactly one at each odd  $n$ . The leading coefficients of  $\mathcal{P}_4, \mathcal{P}_6, \mathcal{P}_8$  are  $-\frac{1}{720}, \frac{1}{30240}, -\frac{1}{1209600}$ , in agreement with  $B_4/4!, B_6/6!, B_8/8!$ .

**Corollary 24.65** (Pochhammer-cleared endpoint form and the half-base dictionary). *For  $0 < q \leq \frac{1}{2}$  and  $n \geq 0$ ,*

$$F_q(q^n) = \frac{q^{\binom{n+1}{2}}}{(q; q)_n} \mathcal{P}_n(q). \quad (24.102)$$

At  $q = 1/2$  this is a dictionary between the moment polynomial and the values of the Fabius function:

$$\mathcal{P}_n(1/2) = \mathfrak{m}_n F(2^{-n}), \quad (24.103)$$

where  $\mathfrak{m}_n = \prod_{j=1}^n (2^j - 1)$  is the Mersenne product (22.1).

*Proof.* By (24.91),  $\alpha_n^{(q)}/(1-q)^n = \mathcal{P}_n(q)/(q; q)_n$ , so (24.102) is (24.15) rewritten. At  $q = \frac{1}{2}$ , (22.2) gives  $(1/2; 1/2)_n = 2^{-\binom{n+1}{2}} \mathfrak{m}_n$ , and the two powers of 2 in (24.102) cancel, leaving  $F(2^{-n}) = \mathcal{P}_n(1/2)/\mathfrak{m}_n$ .  $\square$

The first values, with  $F(2^{-n}) = 2^{-\binom{n}{2}} \alpha_n$  from Theorem 24.9 and  $\alpha_n$  as in Example 24.6, are

| $n$                  | 1             | 2              | 3               | 4                     | 5                     |
|----------------------|---------------|----------------|-----------------|-----------------------|-----------------------|
| $\mathcal{P}_n(1/2)$ | $\frac{1}{2}$ | $\frac{5}{24}$ | $\frac{7}{96}$  | $\frac{1001}{46080}$  | $\frac{4123}{737280}$ |
| $\mathfrak{m}_n$     | 1             | 3              | 21              | 315                   | 9765                  |
| $F(2^{-n})$          | $\frac{1}{2}$ | $\frac{5}{72}$ | $\frac{1}{288}$ | $\frac{143}{2073600}$ | $\frac{19}{33177600}$ |

**Remark 24.66** (The Pochhammer denominator is not imported notation). Corollary 24.65 closes the circle of this part. The finite  $q$ -Pochhammer symbol  $(q; q)_n$  in (24.102) was not chosen for cosmetic reasons: by Definition 24.61 and Theorem 24.62 it is exactly the pole-clearing factor of the recurrence (24.7), whose denominators  $1 - q^n$  are forced by the self-similarity (23.3). At  $q = 1/2$  it specializes, through Theorem 22.1, to the Mersenne product  $\mathfrak{m}_n$  of Chapter 22 — the same constant that Theorem 24.21 produces on the right of (24.32), and that normalizes the dyadic numerator  $\mathcal{N}_n(\theta) = \mathfrak{m}_n \alpha_n$  of Corollary 24.31. The three occurrences are one occurrence: the geometric Vandermonde denominator of the order- $n$  stencil on the nodes  $-2^k$ , the pole-clearing factor of the moment recurrence, and the Mersenne product are the same object read three ways.

## 24.8 The universal division-free definition

The factorial quotient is a formula, not a definition: it presupposes that every factor  $1 - q^j$  with  $j \leq n$  is invertible. Either recurrence of Theorem 2.3 is free of that hypothesis, and free of subtraction as well. It therefore defines the Gaussian coefficient over an arbitrary commutative semiring, and it is this universal object, rather than the quotient, that makes the root-of-unity specializations of Chapter 16 and the layered architecture of Principle 20.18 legitimate.

**Definition 24.67** (Universal Gaussian polynomials). Let  $u$  be an indeterminate. Define  $G_{n,k}(u)$  for  $n \in \mathbb{N}_0$  and  $0 \leq k \leq n$  by  $G_{0,0}(u) = 1$  and

$$G_{n+1,k}(u) = G_{n,k}(u) + u^{n+1-k} G_{n,k-1}(u) \quad (0 \leq k \leq n+1), \quad (24.104)$$

with the convention that  $G_{n,k}(u) = 0$  whenever  $k < 0$  or  $k > n$ . In the displayed range the exponent  $n+1-k$  is nonnegative, so no negative power occurs. For a commutative semiring  $R$  and  $a \in R$ , write  $G_{n,k}(a) \in R$  for the element produced by the same recursion with  $u$  replaced by  $a$ .

**Theorem 24.68** (Existence, uniqueness, and functoriality). (i) *There is exactly one family satisfying (24.104) with  $G_{0,0} = 1$ , and every  $G_{n,k}(u)$  lies in  $\mathbb{N}_0[u]$ .*

(ii)  *$G_{n,k}(u) = \begin{bmatrix} n \\ k \end{bmatrix}_q$  under  $q = u$ ; that is, the factorial quotient is the value of the universal polynomial.*

(iii) *If  $\phi: R \rightarrow S$  is a homomorphism of commutative semirings, then*

$$\phi(G_{n,k}(a)) = G_{n,k}(\phi(a)) \quad (a \in R, 0 \leq k \leq n).$$

*Hence the Gaussian coefficient is definable, and specializable, with no division, no subtraction, and no invertibility hypothesis on the factors  $1 - a^j$ .*

*Proof.* (i) Row  $n+1$  is determined by row  $n$  through (24.104), and row 0 is prescribed; induction on  $n$  gives existence and uniqueness. The recursion uses only 0, 1, addition, multiplication, and nonnegative powers of  $u$ , so every  $G_{n,k}$  lies in  $\mathbb{N}_0[u]$ .

(ii) Replacing  $n$  by  $n+1$  in the second recurrence (2.2) of Theorem 2.3 gives

$$\begin{bmatrix} n+1 \\ k \end{bmatrix}_q = \begin{bmatrix} n \\ k \end{bmatrix}_q + q^{n+1-k} \begin{bmatrix} n \\ k-1 \end{bmatrix}_q,$$

which is (24.104) with  $u = q$ , and the boundary value  $\begin{bmatrix} 0 \\ 0 \end{bmatrix}_q = 1$  agrees. The uniqueness argument of (i) is valid over any commutative ring, in particular over the field  $\mathbb{Q}(u)$ ; hence the two families coincide there, and therefore in  $\mathbb{N}_0[u]$  by Corollary 2.4.

(iii) Apply  $\phi$  to (24.104). A semiring homomorphism preserves 0, 1, sums, products, and hence powers, so the family  $(\phi(G_{n,k}(a)))$  satisfies the same boundary condition and the same recursion in  $S$  with parameter  $\phi(a)$ . Uniqueness in  $S$ , again by (i), gives the assertion.  $\square$

**Corollary 24.69** (Denominator-free factorial identities). *Let  $R$  be a commutative ring and let  $a \in R$ . For  $0 \leq k \leq n$ ,*

$$(a; a)_k G_{n,k}(a) = (a^{n-k+1}; a)_k, \quad (24.105)$$

$$(a; a)_n = (a; a)_k (a; a)_{n-k} G_{n,k}(a). \quad (24.106)$$

*No factor  $1 - a^j$  is assumed invertible; both identities hold even when every product displayed in them vanishes. Thus Proposition 2.2 is a consequence of a polynomial identity rather than a prerequisite for the definition.*

*Proof.* By Theorem 24.68(ii),  $G_{n,k}(u) = (u; u)_n / ((u; u)_k (u; u)_{n-k})$  in  $\mathbb{Q}(u)$ . Both sides of (24.106) with  $a = u$  are polynomials, so the equality holds in  $\mathbb{Z}[u]$ . By Proposition 1.2 with  $m = n - k$ ,

$$(u; u)_n = (u; u)_{n-k} (u^{n-k+1}; u)_k;$$

since  $\mathbb{Z}[u]$  is an integral domain and  $(u; u)_{n-k} \neq 0$ , cancelling it in (24.106) gives (24.105) in  $\mathbb{Z}[u]$ . Finally apply the unique ring homomorphism  $\mathbb{Z}[u] \rightarrow R$  with  $u \mapsto a$ .  $\square$

**Example 24.70** (A primitive root of unity). Let  $a = e^{2\pi i/3}$ , so  $(a; a)_3 = (1 - a)(1 - a^2)(1 - a^3) = 0$  while  $(a; a)_2 = 3$ . The quotient  $\begin{bmatrix} 6 \\ 3 \end{bmatrix}_q$  at  $q = a$  displays  $0/0$ , but the polynomial value is unambiguous:  $G_{6,3}(a) = 2$ , in agreement with the  $q$ -Lucas congruence of [Chapter 16](#). Identity (24.106) reads  $0 = 0 \cdot 0 \cdot 2$ , and it is true. For a nondegenerate instance take  $n = 5$ ,  $k = 2$ : here  $G_{5,2}(a) = 1$  and (24.105) reads  $(a; a)_2 \cdot 1 = (a^4; a)_2$ , both sides equal to 3.

*Remark 24.71.* The transport statement [Theorem 24.68\(iii\)](#) is what licenses steps 1–2 and 6 of [Principle 20.18](#): an identity established without division descends to finite characteristic, to quotient rings, to truncated power-series rings, and to roots of unity with no cancellation theorem interposed. The same observation applies verbatim to [Theorem 3.1](#), whose proof is an induction from [Theorem 2.3](#) and therefore valid over every commutative ring, and to the reversed row of [Chapter 21](#); compare [Remark 21.2](#).

## 24.9 The Möbius function of the subspace lattice

Counting is only the first half of the finite-field picture. The subspaces of  $\mathbb{F}_Q^n$  form a lattice, and its Möbius function is again a Gaussian datum: a sign and a quadratic power of  $Q$ . This is the structural reason for the shape of the inverse kernel in [Theorem 3.12](#).

**Theorem 24.72** (Möbius function of the subspace lattice). *Let  $Q$  be a prime power and let  $V$  be a  $d$ -dimensional vector space over  $\mathbb{F}_Q$ . In the lattice of subspaces of  $V$ , ordered by inclusion,*

$$\mu(0, V) = (-1)^d Q^{\binom{d}{2}}. \quad (24.107)$$

*More generally, if  $U \subseteq W$  are subspaces and  $r = \dim(W/U)$ , then*

$$\mu(U, W) = (-1)^r Q^{\binom{r}{2}}.$$

*Proof.* The correspondence  $X \mapsto X/U$  is an isomorphism from the interval  $[U, W]$  onto the lattice of subspaces of  $W/U$ , and the Möbius function of a poset depends only on the isomorphism type of the interval. It therefore suffices to prove (24.107).

Induct on  $d$ . For  $d = 0$  the value is  $\mu(0, 0) = 1$ , as asserted. Let  $d \geq 1$  and assume the formula for all smaller dimensions. The defining relation for the Möbius function of the interval  $[0, V]$  is

$$\sum_{X \subseteq V} \mu(0, X) = 0,$$

the sum running over all subspaces of  $V$ . Grouping the subspaces by dimension and using [Theorem 6.1](#) together with the induction hypothesis,

$$\mu(0, V) = - \sum_{j=0}^{d-1} \begin{bmatrix} d \\ j \end{bmatrix}_Q (-1)^j Q^{\binom{j}{2}}.$$

By [Corollary 3.2](#) at base  $Q$ , which is (3.1) at  $z = 1$ ,

$$\sum_{j=0}^d \begin{bmatrix} d \\ j \end{bmatrix}_Q (-1)^j Q^{\binom{j}{2}} = (1; Q)_d = 0 \quad (d \geq 1),$$

since the factor with index 0 in  $(1; Q)_d$  vanishes. Hence the truncated sum through  $j = d - 1$  equals the negative of the omitted term  $j = d$ , and therefore  $\mu(0, V) = (-1)^d Q^{\binom{d}{2}}$ .  $\square$

*Remark 24.73* (Where this kernel is met again). The inverse kernel  $(-1)^{n-k} q^{\binom{n-k}{2}} \begin{bmatrix} n \\ k \end{bmatrix}_q$  of [Theorem 3.12](#) is exactly  $\mu(U, W)$  of [Theorem 24.72](#) with  $r = n - k$ , multiplied by the number  $\begin{bmatrix} n \\ k \end{bmatrix}_Q$  of intermediate subspaces: for  $q = Q$  a prime power,  $q$ -binomial inversion is Möbius inversion over the subspace lattice for functions depending only on dimension, and the polynomial identity then extends it to an indeterminate  $q$ . The reversed Gaussian row  $N_{n,k}(q)$  of (21.1) carries the additional monomial  $q^{n-k}$ , because  $\binom{n-k}{2} + (n-k) = \binom{n-k+1}{2}$  and its nodes are  $q^1, \dots, q^n$  rather than  $q^0, \dots, q^{n-1}$ . This accounts for the sign and the quadratic exponent that recur throughout [Chapter 21](#).

**Proposition 24.74** (Recurrence for the translated Thue–Morse power sums). *For all  $k, d \geq 0$ ,*

$$M_{0,d}(c) = c^d, \quad (24.108)$$

and

$$M_{k+1,d}(c) = - \sum_{j=1}^d \binom{d}{j} 2^{kj} M_{k,d-j}(c). \quad (24.109)$$

Consequently all the quantities  $M_{k,d}(c)$  with  $k \leq n$  and  $d \leq 2n$  needed in (24.46) can be tabulated with  $O(n^3)$  arithmetic operations, rather than the  $2^k$  terms of the defining sum (24.37).

*Proof.* The base case is the single term  $r = 0$ . For the recurrence, split  $\{0, \dots, 2^{k+1} - 1\}$  into  $\{0, \dots, 2^k - 1\}$  and its translate by  $2^k$ . Adding  $2^k$  to an integer  $r < 2^k$  turns off no binary digit and turns on exactly one, so  $\tau_{r+2^k} = -\tau_r$ . Hence

$$M_{k+1,d}(c) = \sum_{r=0}^{2^k-1} \tau_r \left( (r+c)^d - (r+c+2^k)^d \right) = - \sum_{j=1}^d \binom{d}{j} 2^{kj} \sum_{r=0}^{2^k-1} \tau_r (r+c)^{d-j},$$

by the binomial theorem, the term  $j = 0$  cancelling. This is (24.109). Filling the table in increasing  $k$  and  $d$  costs  $O(d)$  operations per entry, hence  $O(n \cdot n \cdot n)$  in total for the stated ranges.  $\square$

## 24.10 A ring-level residual principle

[Theorem 21.6](#) is stated for the Lagrange weights of  $N$  distinct nodes in a field, and the free target  $x$  carried by both sides of (21.7) is what makes it a statement about interpolation. The mechanism it uses is narrower than its hypotheses. The field, the distinctness of the nodes, the Lagrange normalization of the weights, and the invertibility of the differences  $v_i - v_j$  can all be discarded if one asks only for the moments of a finite weight system: the single hypothesis that the first  $p$  moments vanish already determines every later moment. In that form the statement holds over an arbitrary commutative ring, and both exact tails of this part become instances of it.

**Theorem 24.75** (All-degree residual from the first surviving moment). *Let  $R$  be a commutative ring and let  $x_0, \dots, x_p \in R$  and  $w_0, \dots, w_p \in R$  be arbitrary elements. Put*

$$M_m := \sum_{i=0}^p w_i x_i^m \quad (m \geq 0), \quad (24.110)$$

with the convention  $x^0 = 1$  for every  $x \in R$ . If

$$M_0 = M_1 = \dots = M_{p-1} = 0, \quad M_p = K, \quad (24.111)$$



then

$$M_{p+r} = K h_r(x_0, \dots, x_p) \quad (r \geq 0), \quad (24.112)$$

where  $h_r$  is the complete homogeneous symmetric polynomial in the  $p+1$  displayed variables, evaluated in  $R$ . No node is assumed distinct from another, no weight and no difference  $x_i - x_j$  is assumed invertible, and  $R$  may have zero divisors. For  $p=0$  the hypothesis (24.111) is empty,  $K = w_0$ , and (24.112) is the definition of  $M_r$ .

*Proof.* Work in  $R[[t]]$  and put

$$M(t) = \sum_{m \geq 0} M_m t^m, \quad E(t) = \prod_{i=0}^p (1 - x_i t).$$

Each factor  $1 - x_i t$  is invertible in  $R[[t]]$ , with inverse  $\sum_{m \geq 0} x_i^m t^m$ . Interchanging the finite sum over  $i$  with the sum over  $m$  in (24.110) gives  $M(t) = \sum_{i=0}^p w_i (1 - x_i t)^{-1}$ , and therefore

$$E(t)M(t) = \sum_{i=0}^p w_i \prod_{j \neq i} (1 - x_j t). \quad (24.113)$$

The right-hand side of (24.113) is a polynomial in  $t$  of degree at most  $p$ , each of its  $p+1$  summands being a product of  $p$  linear factors.

Compute the same coefficients from the left. By (4.3) with  $z = -t$  one has  $E(t) = \sum_{l=0}^{p+1} (-1)^l e_l t^l$ , where  $e_l = e_l(x_0, \dots, x_p)$ , so

$$[t^m](E(t)M(t)) = \sum_{l=0}^m (-1)^l e_l M_{m-l}$$

involves only  $M_0, \dots, M_m$ . For  $m < p$  every moment occurring here has index at most  $m < p$  and vanishes by (24.111); for  $m = p$  the only surviving term is  $e_0 M_p = K$ . Comparing with the degree bound just obtained,

$$E(t)M(t) = K t^p.$$

Finally  $E(t)$  has constant term 1, and by (4.4) its inverse in  $R[[t]]$  is  $\sum_{r \geq 0} h_r(x_0, \dots, x_p) t^r$ . Multiplying the last display by that inverse gives

$$M(t) = K t^p \sum_{r \geq 0} h_r(x_0, \dots, x_p) t^r,$$

and comparison of the coefficients of  $t^{p+r}$  is (24.112). □

*Remark 24.76* (The exact scope of the hypotheses). Three comments delimit the statement.

- (i) *The hypotheses are satisfiable with non-invertible data.* In  $R = \mathbb{Z}/12\mathbb{Z}$  take nodes  $(x_0, x_1, x_2) = (0, 1, 2)$  and weights  $(w_0, w_1, w_2) = (1, -2, 1)$ . Then  $M_0 = M_1 = 0$  and  $K = M_2 = 2$ , a zero divisor; no weight and no node difference is a unit. Here  $h_r(0, 1, 2) = h_r(1, 2) = 2^{r+1} - 1$  and  $M_{2+r} = 2^{r+2} - 2 = 2h_r(0, 1, 2)$ , an identity of integers and hence one in every quotient of  $\mathbb{Z}$ .
- (ii) *Coincident nodes make the conclusion empty rather than false.* Group the nodes by their distinct values  $v_1, \dots, v_s$  and let  $W_j$  be the sum of the weights carried by the value  $v_j$ , so that  $M_m = \sum_j W_j v_j^m$ . If two nodes coincide then  $s \leq p$ , and the first  $s$  hypotheses say that the  $s \times s$  Vandermonde matrix of  $v_1, \dots, v_s$  annihilates  $(W_1, \dots, W_s)$ . Whenever its determinant  $\prod_{i < j} (v_j - v_i)$  is not a zero divisor – in particular over a field – the adjugate identity forces  $W_1 = \dots = W_s = 0$ , hence  $K = 0$ , and both sides of (24.112) vanish. Distinctness is thus not needed as a hypothesis, and adding it would exclude nothing of substance.

- (iii) *Neither residual theorem contains the other.* [Theorem 21.6](#) is not a corollary of [Theorem 24.75](#): its two sides carry a free target  $x$ , and its right-hand side involves the  $(N + 1)$ -variable symmetric function  $h_r(x, v_1, \dots, v_N)$ , of which [\(24.112\)](#) sees only the specialization  $x = 0$ . Conversely [\(24.112\)](#) is not a corollary of [\(21.7\)](#), which presupposes a field, distinct nodes, and one particular weight vector. The present development uses [\(21.7\)](#) only at  $x = 0$ , in the proof of [Theorem 21.7](#), and [Remark 24.80](#) (i) recovers exactly that use from [\(24.112\)](#).

**Corollary 24.77** (Geometric Gaussian residual for arbitrary weights). *Let  $R$  be a commutative ring, let  $c, q \in R$ , and let  $w_0, \dots, w_p \in R$  satisfy [\(24.111\)](#) on the nodes  $x_i = cq^i$ . Then*

$$\sum_{i=0}^p w_i (cq^i)^{p+r} = K c^r \begin{bmatrix} p+r \\ p \end{bmatrix}_q \quad (r \geq 0), \quad (24.114)$$

the Gaussian coefficient being read in  $R$  as the value at  $q$  of the universal polynomial of [Definition 24.67](#), so that no factor  $1 - q^j$  is assumed invertible.

*Proof.* By [Theorem 24.75](#) the left-hand side of [\(24.114\)](#) equals  $K h_r(c, cq, \dots, cq^p)$ . Since  $h_r$  is homogeneous of degree  $r$ , scaling every variable by  $c$  multiplies it by  $c^r$ , so  $h_r(c, cq, \dots, cq^p) = c^r h_r(1, q, \dots, q^p)$ .

It remains to identify  $h_r(1, q, \dots, q^p)$ . In the notation of [Chapter 4](#) this is  $S_r^{p+1}(\mathbf{w})$  for  $w_i = q^{i-1}$ , which by [Theorem 4.6](#) (ii) with  $n = p + 1$  and  $k = r$  equals  $\begin{bmatrix} p+r \\ r \end{bmatrix}_q$ , hence  $\begin{bmatrix} p+r \\ p \end{bmatrix}_q$  by the symmetry [Theorem 2.5](#) (i). Both sides of

$$h_r(1, q, \dots, q^p) = \begin{bmatrix} p+r \\ p \end{bmatrix}_q \quad (24.115)$$

lie in  $\mathbb{Z}[q]$  – the left one by inspection of [\(4.4\)](#), the right one by [Theorem 24.68](#) (i),(ii) – and they agree in  $\mathbb{Q}(q)$ , so they agree in  $\mathbb{Z}[q]$ . Applying the evaluation homomorphism  $\mathbb{Z}[q] \rightarrow R$  with  $q \mapsto q$  and using the functoriality [Theorem 24.68](#) (iii) transports [\(24.115\)](#) to  $R$ .  $\square$

**Remark 24.78.** [Corollary 21.8](#) is the case of [\(24.114\)](#) in which the weights are the Lagrange row  $\lambda_{n,k}^{(q)}$ , after the deflation described in [Remark 24.80](#) (i); what is new here is that no property of the weight row is used beyond [\(24.111\)](#), and that  $c, q$  and the  $w_i$  need not lie in a field.

**Corollary 24.79** (Formal-series filter for arbitrary weights). *Let  $R$  be a commutative ring and let  $x_0, \dots, x_p$  and  $w_0, \dots, w_p$  in  $R$  satisfy [\(24.111\)](#). For every  $f(z) = \sum_{m \geq 0} a_m z^m \in R[[z]]$ ,*

$$\sum_{i=0}^p w_i f(x_i z) = K \sum_{r \geq 0} a_{p+r} h_r(x_0, \dots, x_p) z^{p+r}. \quad (24.116)$$

On a geometric node set  $x_i = cq^i$  the multiplier of  $a_{p+r}$  is  $K c^r \begin{bmatrix} p+r \\ p \end{bmatrix}_q$ .

*Proof.* The left-hand side is a sum of *finitely many* elements of  $R[[z]]$ , so coefficient extraction commutes with it:

$$[z^m] \sum_{i=0}^p w_i f(x_i z) = a_m \sum_{i=0}^p w_i x_i^m = a_m M_m.$$

By [\(24.111\)](#) this vanishes for  $m < p$ , and for  $m = p + r$  it is  $K a_{p+r} h_r(x_0, \dots, x_p)$  by [Theorem 24.75](#). The last assertion is [Corollary 24.77](#).  $\square$

**Remark 24.80** (One principle, two exact tails). The two exact tails of this part are instances of [\(24.116\)](#). This does not merge them – what keeps them apart is recorded in [Remark 24.18](#) – but it does replace two separate residual computations by one.

- (i) *Evaluation at zero, after deflation.* The Lagrange row  $\lambda_{n,k}^{(q)}$  has  $M_0 = 1$ , so (24.111) fails for it. Adjoin one further node  $x_{n+1} = 0$  with weight  $w_{n+1} = -1$ . The augmented system has  $n + 2$  nodes, so  $p = n + 1$ ; its moments are  $M_0 = 1 - 1 = 0$  and  $M_m = 0$  for  $1 \leq m \leq n$  by (21.5), since  $0^m = 0$  there. Its value of  $K$  can be read off (24.113) without any prior knowledge of the row: among the  $n + 2$  summands only the one indexed by the adjoined node has degree  $n + 1$ , namely  $-\prod_{k=0}^n (1 - q^k t)$ , whose coefficient of  $t^{n+1}$  is  $-(-1)^{n+1} q^{\binom{n+1}{2}}$ , so

$$K = (-1)^n q^{\binom{n+1}{2}}.$$

A variable equal to 0 contributes the factor 1 to (4.4), so

$$h_r(1, q, \dots, q^n, 0) = h_r(1, q, \dots, q^n),$$

and (24.114) returns (21.8) while (24.116) returns (21.12). Running the same argument on the nodes  $cq^k$  together with 0 returns Corollary 21.8. This is a proof of Theorem 21.7 that does not pass through Theorem 21.6 and uses no symmetric-function orthogonality.

- (ii) *Extraction of the top coefficient.* The data (24.20) satisfy (24.111) with  $p = n$  and  $K = q^{-\binom{n+1}{2}}(q; q)_n$ , by (24.22) and (24.23); no deflation is needed, because that row already annihilates the constants. Its nodes are  $-q^{-k} = c\tilde{q}^k$  with  $c = -1$  and  $\tilde{q} = q^{-1}$ , so (24.115) gives

$$h_r(-1, -q^{-1}, \dots, -q^{-n}) = (-1)^r \begin{bmatrix} n+r \\ n \end{bmatrix}_{q^{-1}} = (-1)^r q^{-nr} \begin{bmatrix} n+r \\ n \end{bmatrix}_q,$$

the last step being the reciprocity Theorem 2.5 (ii) with  $k(n-k) = nr$ . Multiplying (24.116) by  $q^{\binom{n+1}{2}}/(q; q)_n$  therefore reproduces (24.27) exactly, giving a second proof of Theorem 24.17 in which no shifted factorial  $(q^{-n-r}; q)_n$  is ever rewritten.

The two instances differ in precisely one structural respect: the first has a nonzero moment of order 0 and must be deflated by one extra node, the second does not. Everything else — the vanishing block, the single surviving constant  $K$ , and the Gaussian row that multiplies the tail — is supplied by Theorem 24.75.

## 24.11 Cells, a local reduction, and the digit expansion

Formula (24.49) evaluates  $F$  exactly, but only at dyadic rationals; formula (24.64) evaluates it at every point, but only by quadrature. This section supplies the missing third description: an exact, digit-driven expansion valid at every point of  $[0, 1)$ , whose partial sums are finite rational combinations of the moments of  $X$  and whose remainder is itself a value of  $F$ , at an argument smaller by a factor  $2^{-N}$ . The mechanism is the moment functional of Corollary 24.31 carried through with a fractional part attached.

**Definition 24.81** (Shifted moment polynomials). For  $n \geq 0$  and  $\theta \in \mathbb{C}$  put

$$\mathfrak{a}_n(\theta) := \mathbb{E}(X + \theta)^n = \sum_{j=0}^n \binom{n}{j} \mathbb{E}X^{n-j} \theta^j, \quad (24.117)$$

a monic polynomial of degree  $n$  with

$$\mathfrak{a}_n(0) = \mathbb{E}X^n = n! \alpha_n$$

and exponential generating function

$$\sum_{n \geq 0} \mathfrak{a}_n(\theta) z^n / n! = e^{\theta z} \mathcal{A}(z).$$

Thus  $(a_n)$  is the Appell sequence attached to  $\mathcal{A}$ : differentiating the generating function gives  $a'_n = n a_{n-1}$ , multiplying it by  $e^{cz}$  gives the binomial translation law, of which (23.13) is the instance  $c = 1/2$  for the centred variable, and  $X \stackrel{d}{=} 1 - X$  from (23.2) gives the reflection  $a_n(-1 - \theta) = (-1)^n a_n(\theta)$ , the moment counterpart of the  $k$ -independent digit-complementation sign  $(-1)^n$  computed at the end of the proof of Corollary 24.31. Only (24.117) itself is used below; in this notation the numerator coefficient (24.52) reads  $p_{m,n} = \frac{1}{n!} \sum_{h < m} \tau_h a_n(m - h - 1)$ .

**Lemma 24.82** (Spline expectation at an arbitrary argument). *Write  $y_+ := \max(y, 0)$ . For every  $n \geq 1$  and every real  $t$ ,*

$$F\left(\frac{t}{2^n}\right) = \frac{2^{-\binom{n}{2}}}{n!} \sum_{r=0}^{2^n-1} \tau_r \mathbb{E}(t - r - X)_+^n. \quad (24.118)$$

*Proof.* Use the scaled splitting (24.18):  $2^n X = \sum_{j=1}^n c_j V_j + X'$  with  $c_j = 2^{n-j}$ , with  $V_1, \dots, V_n$  independent and uniform on  $[0, 1]$ , and with  $X'$  independent of them and distributed as  $X$ . Since  $(V_1, \dots, V_n)$  is uniform on the unit cube, conditioning on  $X'$  gives

$$\mathbb{P}(2^n X \leq t \mid X') = \mathbb{P}\left(\sum_{j=1}^n c_j V_j \leq t - X' \mid X'\right) = V_c(t - X'),$$

where  $V_c$  is the weighted cube cut (23.19). For the integer weights  $c_j = 2^{n-j}$  that cut is (23.21), an identity valid at every real argument. Substituting it, averaging over  $X'$ , and exchanging the expectation with the finite sum – legitimate because the sum has  $2^n$  terms, each bounded on the bounded range of  $X'$  – gives (24.118), with  $X$  written for  $X'$ .  $\square$

**Remark 24.83** (What the unrestricted argument buys). At an integer  $t = m$  with  $0 \leq m \leq 2^n$  every term with  $r \geq m$  of (24.118) vanishes and every term with  $r < m$  loses its positive part; that is exactly the computation performed in Step 1 of the proof of Corollary 24.31, and it produces (24.53). The content of the lemma is that no restriction on  $t$  is needed. At a non-integer  $t$  one term keeps its positive part, and Theorem 24.84 identifies it.

**Theorem 24.84** (Exact formula on every dyadic cell). *Let  $n \geq 1$ , let  $m$  be an integer with  $0 \leq m \leq 2^n$ , and let  $\theta$  be real with  $0 \leq \theta < 1$  and  $m + \theta \leq 2^n$ . Then*

$$F\left(\frac{m + \theta}{2^n}\right) = \tau_m F\left(\frac{\theta}{2^n}\right) + \frac{2^{-\binom{n}{2}}}{n!} \sum_{h=0}^{m-1} \tau_h a_n(m - h - 1 + \theta), \quad (24.119)$$

*the sum being empty when  $m = 0$ . At  $\theta = 0$  the first term vanishes and (24.119) is (24.53).*

*Proof.* Apply Lemma 24.82 at  $t = m + \theta$  and classify the  $2^n$  terms, using  $0 \leq X \leq 1$  and  $0 \leq \theta < 1$ .

If  $r \geq m + 1$ , then  $t - r - X \leq m + \theta - (m + 1) = \theta - 1 < 0$  and the term vanishes.

If  $r \leq m - 1$ , then  $t - r - X \geq m + \theta - (m - 1) - 1 = \theta \geq 0$ , so the positive part is inactive and

$$\mathbb{E}(m + \theta - r - X)_+^n = \mathbb{E}((1 - X) + m - r - 1 + \theta)^n = a_n(m - r - 1 + \theta),$$

because  $1 - X \stackrel{d}{=} X$  by (23.2). These terms contribute the sum in (24.119).

There remains the term  $r = m$ , which occurs in the range  $0 \leq r \leq 2^n - 1$  precisely when  $m \leq 2^n - 1$ . In that case its contribution is  $\tau_m 2^{-\binom{n}{2}} \mathbb{E}(\theta - X)_+^n / n!$ . Apply Lemma 24.82 once more, now at  $t = \theta$ : since  $0 \leq \theta < 1$ , every index  $r \geq 1$  gives  $\theta - r - X \leq \theta - 1 < 0$  and drops out, so only  $r = 0$  survives and

$$F\left(\frac{\theta}{2^n}\right) = \frac{2^{-\binom{n}{2}}}{n!} \mathbb{E}(\theta - X)_+^n. \quad (24.120)$$

The  $r = m$  term is therefore  $\tau_m F(\theta/2^n)$ , which is the first summand of (24.119).

If instead  $m = 2^n$ , then  $\theta = 0$  by the hypothesis  $m + \theta \leq 2^n$ ; the index  $r = m$  is absent from the range of summation, and the first summand of (24.119) is  $\tau_{2^n} F(0) = 0$ , so the formula again holds. This proves (24.119) in all cases. Finally, at  $\theta = 0$  one has  $F(0) = 0$  and  $\mathfrak{a}_n(m - h - 1)$  in place of  $\mathfrak{a}_n(m - h - 1 + \theta)$ , so (24.119) becomes  $F(m/2^n) = 2^{-\binom{n}{2}} p_{m,n}$  by (24.52), which is (24.53).  $\square$

*Remark 24.85* (Both hypotheses on  $\theta$  are needed, and how sharp they are). The restriction  $m + \theta \leq 2^n$  cannot be dropped: at  $n = 1, m = 2, \theta = 1/4$  the right-hand side of (24.119) is  $-F(1/8) + \mathfrak{a}_1(5/4) - \mathfrak{a}_1(1/4) = 1 - \frac{1}{288} = \frac{287}{288}$ , whereas  $F(9/8) = 1$ . The restriction  $\theta < 1$  can be relaxed to  $\theta \leq 1$  but no further. At  $\theta = 1$  with  $m + 1 \leq 2^n$  the identity persists, because  $\mathfrak{a}_n(0) = n! \alpha_n$  and (24.13) turn the extra term  $\tau_m 2^{-\binom{n}{2}} \mathfrak{a}_n(0)/n!$  of (24.119) at the numerator  $m + 1$  into  $\tau_m F(2^{-n})$ . At  $\theta = 3/2$  it fails: for  $n = 2, m = 1$  the right-hand side is  $-F(3/8) + \frac{1}{4} \mathfrak{a}_2(3/2) = -\frac{73}{288} + \frac{145}{144} = \frac{217}{288}$ , whereas  $F(5/8) = \frac{215}{288}$ .

Let  $x \in [0, 1)$  and let  $x = 0.b_1 b_2 b_3 \dots$  in base two be its canonical binary expansion, the one that is not eventually constant equal to 1. Put

$$m_j := \lfloor 2^j x \rfloor, \quad \theta_j := 2^j x - m_j \in [0, 1) \quad (j \geq 0), \quad (24.121)$$

so that  $m_0 = 0$  and  $\theta_0 = x$ . The binary digits of  $m_j$  are  $b_1 \dots b_j$ ; hence

$$m_j = 2m_{j-1} + b_j, \quad \tau_{m_j} = (-1)^{b_j} \tau_{m_{j-1}}, \quad \theta_{j-1} = \frac{b_j + \theta_j}{2} \quad (j \geq 1), \quad (24.122)$$

the middle relation because appending a digit 0 leaves  $s_2$  unchanged and appending a digit 1 raises it by one.

**Example 24.86** (A non-dyadic argument). Let  $x = 1/3$ , whose binary expansion is  $0.\overline{01}$ . Then  $b_j = 1$  exactly for even  $j$ , and  $\theta_j = 1/3$  for even  $j$ ,  $\theta_j = 2/3$  for odd  $j$ , while  $m_1 = 0, m_3 = 2, m_5 = 10$ , with  $\tau_{m_1} = 1, \tau_{m_3} = -1, \tau_{m_5} = 1$ . Hence (24.153) reads

$$F\left(\frac{1}{3}\right) = \frac{\mathfrak{a}_2(1/3)}{4} - \frac{\mathfrak{a}_4(1/3)}{2^6 \cdot 4!} + \frac{\mathfrak{a}_6(1/3)}{2^{15} \cdot 6!} - \dots = \frac{13}{72} - \frac{2429}{6220800} + \frac{198007}{8427641241600} - \dots,$$

whose partial sums are

$$S_2 = \frac{13}{72} = 0.180555\dots, \quad S_4 = 0.1801650913\dots, \quad S_6 = 0.18016511480153\dots,$$

against  $F(1/3) = 0.1801651148014819\dots$  obtained from (24.64). Three terms already agree with  $F(1/3)$  to within  $5.2 \cdot 10^{-14}$ . The guaranteed bound of (24.154) at  $N = 6$  is  $2^{-15}/6! = 4.2 \cdot 10^{-8}$ ; it is far from sharp here because the residual is  $F(1/192)$  and  $1/192$  lies well inside the interval  $[0, 2^{-6}]$  on which  $F$  is extremely flat.

*Remark 24.87* (Three exact descriptions of  $F$ ). The remark following Corollary 24.41 contrasts two exact descriptions of  $F$ : the global analytic one (24.64) and the finite rational one (24.49), which applies only at dyadic arguments. Formula (24.153) is a third, and it is the only one that is simultaneously exact, elementary, and valid at every argument. Its cost structure is also different from both. The  $j$ th term needs the moments  $\mathbb{E}X^0, \dots, \mathbb{E}X^j$ , which (24.11) produces through  $\mathbb{E}X^n = n! \alpha_n$  at one rational division each; the accuracy is governed by the superfactorial weight  $2^{-\binom{N}{2}}/N!$  of (24.154), which is  $7.8 \cdot 10^{-21}$  at  $N = 10$  and  $2.6 \cdot 10^{-76}$  at  $N = 20$ , uniformly in  $x$ ; and only the indices  $j$  with  $b_j = 1$  contribute a term, so the number of nonzero terms is the binary weight of the truncated argument and not its length. By contrast (24.49) needs a Thue–Morse block of length  $m2^k$ , and (24.64) needs quadrature of an oscillatory integral.

The truncation  $X_{q,N}$  of (23.17) discards a nonnegative tail, so Corollary 23.13 is a one-sided coupling and its rate  $q^N$  is the length of the discarded interval. Adding back the *mean* of that tail centres the coupling and removes the first-order term; the resulting rate is second order in  $q^N$ , and it can be computed exactly.

**Proposition 24.88** (Midpoint-centred truncations, and an exact second-order rate). *Let  $0 < q < 1$  and  $N \geq 1$ , and put*

$$X_{q,N}^{\text{mid}} := X_{q,N} + \frac{1}{2}q^N, \quad F_{q,N}^{\text{mid}}(x) := \mathbb{P}(X_{q,N}^{\text{mid}} \leq x) = F_{q,N}(x - \frac{1}{2}q^N). \quad (24.123)$$

At the half base write  $S_p := F_{1/2,p}^{\text{mid}}$ . Then:

1. for every  $p \geq 1$  and every real  $x$ ,

$$S_p(x) = \frac{2^{-\binom{p}{2}}}{p!} \sum_{r=0}^{2^p-1} \tau_r \left(2^p x - \frac{1}{2} - r\right)_+^p, \quad \tau_r = (-1)^{s_2(r)}; \quad (24.124)$$

2. for every  $N \geq 2$ ,

$$\sup_{x \in \mathbb{R}} |F_{q,N}^{\text{mid}}(x) - F_q(x)| \leq \frac{q^{2N-1}}{24(1-q^2)}; \quad (24.125)$$

3. if moreover  $0 < q \leq 1/2$ , then (24.125) is an equality, the extreme value being attained at  $x^* := q(1-q)$ :

$$F_{q,N}^{\text{mid}}(x^*) - F_q(x^*) = -\frac{q^{2N-1}}{24(1-q^2)}. \quad (24.126)$$

At  $q = 1/2$  this reads  $\sup_{x \in \mathbb{R}} |S_p(x) - F(x)| = \frac{1}{9} 4^{-p}$  for  $p \geq 2$ , attained at  $x = 1/4$ ;

4. the hypothesis  $N \geq 2$  cannot be dropped: at  $q = 1/2$  and  $N = 1$  the variable  $X_{1/2,1}^{\text{mid}} = \frac{1}{2}U_0 + \frac{1}{4}$  is uniform on  $[1/4, 3/4]$ , so  $S_1(1/4) = 0$  while  $F(1/4) = 5/72$  by Remark 24.32, whereas the right-hand side of (24.125) would be  $1/36 < 5/72$ .

*Proof.* (i). By Remark 23.10 with  $n = p$ , the weights  $c_j = 2^{p-j}$  are  $2^p$  times the half-base weights  $a_{j-1} = 2^{-j}$ , so  $V_{\mathbf{c}}(t) = \mathbb{P}(2^p X_{1/2,p} \leq t) = F_{1/2,p}(2^{-p}t)$  and (23.21) reads

$$F_{1/2,p}(2^{-p}t) = \frac{2^{-\binom{p}{2}}}{p!} \sum_{r=0}^{2^p-1} \tau_r (t - r)_+^p.$$

By (24.123),  $S_p(x) = F_{1/2,p}(x - 2^{-p-1})$ , which is the above at  $t = 2^p x - \frac{1}{2}$ . This is (24.124).

*An exact error integral.* Iterating (23.3)  $N$  times gives  $X_q \stackrel{d}{=} X_{q,N} + q^N X'_q$  with  $X'_q$  an independent copy of  $X_q$ ; at  $q = 1/2$  this is (24.18) after scaling. Write  $Y := X_{q,N}$ , let  $g_{q,N}$  be its density (Theorem 23.9), and put

$$H_q(u) := \mathbf{1}_{[1/2, \infty)}(u) - F_q(u),$$

which vanishes for  $u < 0$  and for  $u > 1$ . Conditioning on  $Y$ ,

$$F_q(x) = \mathbb{E}\left[F_q(q^{-N}(x - Y))\right], \quad F_{q,N}^{\text{mid}}(x) = \mathbb{P}(Y \leq x - \frac{1}{2}q^N) = \mathbb{E}\left[\mathbf{1}_{[1/2, \infty)}(q^{-N}(x - Y))\right],$$

so, substituting  $u = q^{-N}(x - y)$ ,

$$F_{q,N}^{\text{mid}}(x) - F_q(x) = \mathbb{E}\left[H_q(q^{-N}(x - Y))\right] = q^N \int_0^1 g_{q,N}(x - q^N u) H_q(u) du. \quad (24.127)$$

The two moments of  $H_q$ . Since  $\mathbb{E}X_q = \int_0^1 (1 - F_q)$  and  $\mathbb{E}X_q = 1/2$  by (23.2),

$$\int_0^1 H_q(u) \, du = \frac{1}{2} - \int_0^1 F_q(u) \, du = \frac{1}{2} - (1 - \mathbb{E}X_q) = 0. \quad (24.128)$$

Integrating by parts,  $\int_0^1 u F_q(u) \, du = \frac{1}{2} - \frac{1}{2} \mathbb{E}X_q^2$ , whence

$$\int_0^1 (u - \frac{1}{2}) H_q(u) \, du = \int_{1/2}^1 (u - \frac{1}{2}) \, du - \int_0^1 (u - \frac{1}{2}) F_q(u) \, du = \frac{1}{2} \mathbb{E}X_q^2 - \frac{1}{8} = \frac{1}{2} \text{Var}(X_q).$$

By (23.7) at  $m = 1$ , with  $B_2 = 1/6$ ,  $\text{Var}(X_q) = \kappa_2(Y_q) = (1 - q)/(12(1 + q))$ , so

$$\int_0^1 (u - \frac{1}{2}) H_q(u) \, du = \frac{1 - q}{24(1 + q)}. \quad (24.129)$$

Finally  $H_q \leq 0$  on  $[0, 1/2)$  and  $H_q \geq 0$  on  $[1/2, 1]$ , so  $(u - \frac{1}{2}) H_q(u) = |u - \frac{1}{2}| |H_q(u)|$  pointwise and (24.129) also computes  $\int_0^1 |u - \frac{1}{2}| |H_q(u)| \, du$ .

*The density and its Lipschitz constant.* For  $M \geq 1$  write  $X_{q,M} = (1 - q)U_0 + qX_{q,M-1}$  with the two summands independent and  $X_{q,0} := 0$ . Conditioning on the second summand, which contributes the uniform density  $1/(1 - q)$  on a window of length  $1 - q$ , gives

$$g_{q,M}(x) = \frac{1}{1 - q} \left[ F_{q,M-1}\left(\frac{x}{q}\right) - F_{q,M-1}\left(\frac{x - 1 + q}{q}\right) \right], \quad (24.130)$$

the difference of two values of a distribution function; hence  $0 \leq g_{q,M} \leq 1/(1 - q)$ , the finite- $M$  form of Proposition 23.12(ii). If  $M \geq 2$  then  $F_{q,M-1}$  has the bounded density  $g_{q,M-1}$  and is therefore Lipschitz with constant  $1/(1 - q)$ , so (24.130) shows that  $g_{q,M}$  is Lipschitz with constant

$$\text{Lip}(g_{q,M}) \leq \frac{1}{q(1 - q)^2} \quad (M \geq 2). \quad (24.131)$$

(ii). Let  $N \geq 2$ . By (24.128) we may subtract the constant  $g_{q,N}(x - \frac{1}{2}q^N)$  inside (24.127), so that

$$\begin{aligned} |F_{q,N}^{\text{mid}}(x) - F_q(x)| &\leq q^N \int_0^1 |g_{q,N}(x - q^N u) - g_{q,N}(x - \frac{1}{2}q^N)| \\ &\quad \cdot |H_q(u)| \, du \\ &\leq \frac{q^{2N}}{q(1 - q)^2} \int_0^1 |u - \frac{1}{2}| |H_q(u)| \, du. \end{aligned}$$

by (24.131). Now insert (24.129):

$$\frac{q^{2N-1}}{(1 - q)^2} \cdot \frac{1 - q}{24(1 + q)} = \frac{q^{2N-1}}{24(1 - q^2)},$$

which is (24.125).

(iii). Let  $0 < q \leq 1/2$  and  $N \geq 2$ . We claim that  $g_{q,N}$  is affine with slope exactly  $1/(q(1 - q)^2)$  on  $I := [x^* - q^N, x^*]$ , where  $x^* = q(1 - q)$ .

First, for  $M \geq 1$ ,

$$g_{q,M}(y) = \frac{1}{1 - q} \quad \text{for (almost) every } y \in [1 - q - q^M, 1 - q]. \quad (24.132)$$



For  $M = 1$  the variable  $X_{q,1} = (1 - q)U_0$  is uniform on  $[0, 1 - q]$  and  $1 - q - q \geq 0$  because  $q \leq 1/2$ , so the interval lies in the support and the density is  $1/(1 - q)$  almost everywhere on it. For  $M \geq 2$  use (24.130): such a  $y$  has  $y - 1 + q \leq 0$ , and  $X_{q,M-1}$  has a density, so  $F_{q,M-1}((y - 1 + q)/q) = 0$ ; moreover  $y/q \geq (1 - q)/q - q^{M-1} \geq 1 - q^{M-1}$  because  $(1 - q)/q \geq 1$ , while  $X_{q,M-1} \leq 1 - q^{M-1}$ , so  $F_{q,M-1}(y/q) = 1$ . This proves (24.132), and hence  $F_{q,M}$  is affine with slope  $1/(1 - q)$  on  $[1 - q - q^M, 1 - q]$ .

Now let  $x \in I$ . Then  $x \leq q(1 - q) < 1 - q$ , so  $x - 1 + q < 0$  and the second term of (24.130) vanishes, giving  $g_{q,N}(x) = F_{q,N-1}(x/q)/(1 - q)$ . As  $x$  runs through  $I$ , the point  $x/q$  runs through  $[1 - q - q^{N-1}, 1 - q]$ , on which  $F_{q,N-1}$  is affine with slope  $1/(1 - q)$  by (24.132) with  $M = N - 1 \geq 1$ . Composing the two affine maps proves the claim.

Write therefore  $g_{q,N}(x^* - q^N u) = g_{q,N}(x^*) - q^N u / (q(1 - q)^2)$  for  $0 \leq u \leq 1$  and substitute into (24.127). The constant term is annihilated by (24.128), which also gives  $\int_0^1 u H_q = \int_0^1 (u - \frac{1}{2}) H_q$ , so by (24.129)

$$F_{q,N}^{\text{mid}}(x^*) - F_q(x^*) = -\frac{q^{2N}}{q(1 - q)^2} \int_0^1 (u - \frac{1}{2}) H_q(u) du = -\frac{q^{2N-1}}{24(1 - q^2)},$$

which is (24.126); with (ii) it makes (24.125) an equality. At  $q = 1/2$  one has  $x^* = 1/4$  and  $q^{2N-1}/(24(1 - q^2)) = 2^{1-2N}/18 = \frac{1}{9}4^{-N}$ .

(iv). The computation is the one displayed in the statement.  $\square$

*Remark 24.89* (What the midpoint buys). Against (23.30) the gain is a full extra factor  $q^{N-1}(1 - q)/(24(1 + q))$ , and at the half base the gain is  $9 \cdot 2^N$ : the plain truncation obeys  $\sup_x |F_{1/2,N} - F| = 2^{-N}$ , while  $\sup_x |S_N - F| = \frac{1}{9}4^{-N}$ . The mechanism is visible in (24.127): the shift  $\frac{1}{2}q^N$  is exactly the one that makes the kernel  $H_q$  have vanishing mean, after which only its first moment  $\frac{1}{2} \text{Var}(X_q)$  survives. The same identity shows why no further shift helps: the next obstruction is the second moment of  $H_q$ , which cannot be removed by a translation.

## 24.12 An off-grid continuation of the dyadic formula

**Corollary 24.31** evaluates  $F$  at the dyadic points  $m/2^n$  with  $0 \leq m \leq 2^n$ , and **Remark 24.32** shows that the hypothesis  $m \leq 2^n$  cannot simply be dropped: at  $n = 2$ ,  $m = 5$  the right-hand side of (24.49) returns  $67/72$  instead of  $F(5/4) = 1$ . Both restrictions disappear at once if the block length  $m2^k$  is replaced by a cutoff that is *clamped* at the top. The resulting function is defined for every real argument, agrees with  $F$  on the grid, and is an explicit Gaussian average of the midpoint splines of **Proposition 24.88**.

**Definition 24.90** (The clamped midpoint cutoff). For an integer  $p \geq 1$  and  $x \in \mathbb{R}$  put

$$N_p(x) := \min\left(2^p, \max\left(0, \left\lfloor 2^p x + \frac{1}{2} \right\rfloor\right)\right) \in \{0, 1, \dots, 2^p\}. \quad (24.133)$$

**Lemma 24.91** (Cutoff form of the midpoint spline). *Let  $p \geq 1$ . For every real  $x$ ,*

$$\frac{1}{p!} \sum_{r=0}^{N_p(x)-1} \tau_r\left(r - 2^p x + \frac{1}{2}\right)^p = (-1)^p 2^{\binom{p}{2}} S_p(x), \quad (24.134)$$

*an empty sum being 0. Moreover, if  $0 \leq n \leq p$  and  $m$  is an integer with  $0 \leq m \leq 2^n$ , then*

$$N_p\left(\frac{m}{2^n}\right) = m 2^{p-n}. \quad (24.135)$$



*Proof.* Put  $t := 2^p x - \frac{1}{2}$ . In (24.124) the term of index  $r$  is nonzero exactly when  $t - r > 0$ ; since  $r$  is an integer this says  $r < \lceil t \rceil$ , so the terms that survive are those with  $0 \leq r < \min(2^p, \max(0, \lceil t \rceil))$ .

We check that  $\lceil t \rceil$  may be replaced by  $\lfloor t + 1 \rfloor = \lfloor 2^p x + \frac{1}{2} \rfloor$ . The two integers agree unless  $t \in \mathbb{Z}$ , in which case  $\lfloor t + 1 \rfloor = \lceil t \rceil + 1$ ; the single extra index is then  $r = t$ , whose contribution to (24.134) is  $\tau_t(t - t)^p = 0$  because  $p \geq 1$ . Neither side of (24.134) changes, so the summation range in (24.134) is exactly the active range of (24.124).

Consequently, using  $(t - r)^p = (-1)^p(r - t)^p$  and  $r - t = r - 2^p x + \frac{1}{2}$ ,

$$S_p(x) = \frac{2^{-\binom{p}{2}}}{p!} \sum_{r=0}^{2^p-1} \tau_r(t - r)_+^p = \frac{(-1)^p 2^{-\binom{p}{2}}}{p!} \sum_{r=0}^{N_p(x)-1} \tau_r(r - 2^p x + \frac{1}{2})^p,$$

which is (24.134) after multiplying by  $(-1)^p 2^{\binom{p}{2}}$ .

For (24.135),  $2^p \cdot m/2^n = m2^{p-n}$  is a nonnegative integer, so  $\lfloor m2^{p-n} + \frac{1}{2} \rfloor = m2^{p-n}$ ; and  $0 \leq m2^{p-n} \leq 2^n 2^{p-n} = 2^p$ , so neither clamp is active.  $\square$

*Remark 24.92* (The clamp repairs the hypothesis  $m \leq 2^n$ ). The upper clamp in (24.133) is not cosmetic. Without it the right-hand side of (24.134) would continue to accumulate terms for  $x > 1$  and would describe a signed global extension rather than a bounded distribution function; the failure recorded in Remark 24.32 at  $n = 2$ ,  $m = 5$  is precisely the case  $N_{n+k}(x)$  replaced by an unclamped  $m2^k > 2^{n+k}$ . The lower clamp plays the symmetric role for  $x < 0$ .

**Theorem 24.93** (Off-grid continuation of the dyadic formula). *For  $n \geq 1$  define, for every  $x \in \mathbb{R}$ ,*

$$D_n(x) := \frac{1}{2^{n^2} (1/2; 1/2)_n} \sum_{k=0}^n \frac{\begin{bmatrix} n \\ k \end{bmatrix}_{1/2}}{4^{\binom{k}{2}} (n+k)!} \sum_{r=0}^{N_{n+k}(x)-1} \tau_r(r - 2^{n+k}x + \frac{1}{2})^{n+k}, \quad (24.136)$$

with  $N_p$  as in (24.133), and put

$$\omega_{n,j} := \frac{(-1)^j 2^{-\binom{j+1}{2}}}{(1/2; 1/2)_n} \begin{bmatrix} n \\ j \end{bmatrix}_{1/2} \quad (0 \leq j \leq n). \quad (24.137)$$

Then:

1. for every  $x \in \mathbb{R}$ ,

$$D_n(x) = \sum_{j=0}^n \omega_{n,j} S_{2n-j}(x); \quad (24.138)$$

2.  $\omega_{n,j} = W_{n,j} = \lambda_{n,n-j}^{(1/2)}$  in the notation of Remark 24.49; in particular  $\sum_{j=0}^n \omega_{n,j} = 1$ ,  $\text{sgn } \omega_{n,j} = (-1)^j$ ,  $\sum_{j=0}^n \omega_{n,j} z^j = (z/2; 1/2)_n / (1/2; 1/2)_n$  and  $\sum_{j=0}^n |\omega_{n,j}| = (-1/2; 1/2)_n / (1/2; 1/2)_n$ ;

3.  $D_n$  interpolates  $F$  on the grid of resolution  $2^{-n}$ :

$$D_n\left(\frac{m}{2^n}\right) = F\left(\frac{m}{2^n}\right) \quad (m \in \mathbb{Z}, 0 \leq m \leq 2^n); \quad (24.139)$$

4.  $D_n$  is a continuous piecewise polynomial of degree at most  $2n$ , its knots are the odd multiples of  $2^{-2n-1}$ , it satisfies  $D_n(1 - x) = 1 - D_n(x)$ , and

$$D_n(x) = 0 \quad (x \leq 2^{-2n-1}), \quad D_n(x) = 1 \quad (x \geq 1 - 2^{-2n-1}). \quad (24.140)$$

*Proof.* (i). Apply [Lemma 24.91](#) with  $p = n + k$ : the inner sum of (24.136), divided by  $(n + k)!$ , equals  $(-1)^{n+k} 2^{\binom{n+k}{2}} S_{n+k}(x)$ . Hence

$$D_n(x) = \frac{1}{(1/2; 1/2)_n} \sum_{k=0}^n (-1)^{n+k} 2^{\binom{n+k}{2} - n^2 - 2\binom{k}{2}} \begin{bmatrix} n \\ k \end{bmatrix}_{1/2} S_{n+k}(x),$$

since  $4^{\binom{k}{2}} = 2^{2\binom{k}{2}}$ . Substitute  $j = n - k$ , so that  $n + k = 2n - j$ . The sign is  $(-1)^{2n-j} = (-1)^j$ ; the Gaussian coefficient is unchanged because  $\begin{bmatrix} n \\ n-j \end{bmatrix}_{1/2} = \begin{bmatrix} n \\ j \end{bmatrix}_{1/2}$ , by the symmetry visible in (22.3); and the exponent of 2 becomes

$$\binom{2n-j}{2} - 2\binom{n-j}{2} - n^2 = \frac{1}{2}(2n-j)(2n-j-1) - (n-j)(n-j-1) - n^2 = -\frac{1}{2}j(j+1) = -\binom{j+1}{2},$$

an identity in  $n$  and  $j$  verified by expanding both sides. This is exactly (24.137), and (24.138) follows.

(ii). The displayed  $\omega_{n,j}$  is the row  $W_{n,j}$  of [Remark 24.49](#), where it is identified with  $\lambda_{n,n-j}^{(1/2)}$  and its generating polynomial is computed. Unit mass is (24.74), the sign pattern is (24.73) read in reversed coordinates, and the  $\ell^1$  norm is (24.75), all at  $q = 1/2$ .

(iii). Let  $0 \leq m \leq 2^n$ . By (24.135) with  $p = n + k \geq n$  we have  $N_{n+k}(m/2^n) = m2^k$ , while  $2^{n+k} \cdot m/2^n = m2^k$ ; hence (24.136) becomes

$$\frac{1}{2^{n^2} (1/2; 1/2)_n} \sum_{k=0}^n \frac{\begin{bmatrix} n \\ k \end{bmatrix}_{1/2}}{4^{\binom{k}{2}} (n+k)!} \sum_{r=0}^{m2^k-1} \tau_r \left(r - m2^k + \frac{1}{2}\right)^{n+k},$$

which is the right-hand side of (24.49) at  $\theta = \frac{1}{2}$ . By [Corollary 24.31](#) its value is  $F(m/2^n)$ .

(iv). Each  $S_p$  is, by (24.124), a continuous piecewise polynomial of degree  $p$  whose knots are the numbers  $(2r+1)/2^{p+1}$ ; the degrees occurring in (24.138) are  $n, \dots, 2n$  and the finest knot set is that of  $S_{2n}$ . The variable  $X_{1/2,p}^{\text{mid}}$  of (24.123) is symmetric about  $\frac{1}{2}$ , because  $X_{1/2,p}$  is symmetric about its mean  $(1 - 2^{-p})/2$  and  $(1 - 2^{-p})/2 + 2^{-p-1} = \frac{1}{2}$ ; as it has no atoms,  $S_p(1-x) = 1 - S_p(x)$ , and the unit row mass of (ii) transfers this to  $D_n$ . Finally  $X_{1/2,p}^{\text{mid}}$  takes values in  $[2^{-p-1}, 1 - 2^{-p-1}]$ , so  $S_p = 0$  on  $(-\infty, 2^{-p-1}]$  and  $S_p = 1$  on  $[1 - 2^{-p-1}, \infty)$ ; both hold simultaneously for all  $p \in \{n, \dots, 2n\}$  exactly on the two rays of (24.140), the binding index being  $p = 2n$ .  $\square$

*Remark 24.94* (Which part of the dyadic formula is being continued). The separation of [Remark 24.32](#) survives the continuation: in (24.136) the half-base Gaussian row  $\begin{bmatrix} n \\ k \end{bmatrix}_{1/2} / 4^{\binom{k}{2}}$  still depends only on the resolution  $2^{-n}$ , and the argument  $x$  enters only through the block length  $N_{n+k}(x)$  and the common shift. What changes is that the block length is now a step function of a real variable instead of an integer parameter, and (24.138) identifies the resulting object as a *fixed* Gaussian average of box splines — one Lagrange row of [Chapter 21](#) applied to the family  $S_{2n}, S_{2n-1}, \dots, S_n$ . The choice  $\theta = \frac{1}{2}$  of the free shift in (24.49) is what forces the midpoint centring: no other value of  $\theta$  produces a combination of distribution functions.

**Theorem 24.95** (Uniform half-base Toeplitz limit, with the sharp rate). *For every  $n \geq 2$ ,*

$$\sup_{x \in \mathbb{R}} |D_n(x) - F(x)| \leq \frac{1}{9 \cdot 16^n} \frac{(-2; 1/2)_n}{(1/2; 1/2)_n} < \frac{2\mathcal{L}(1/2)}{3} 16^{-n} < 6 \cdot 16^{-n}, \quad (24.141)$$

with  $\mathcal{L}(1/2) = 8.2559879357782500655 \dots$  as in (24.77). In particular  $D_n \rightarrow F$  uniformly on the whole real line, and the convergence is geometric of ratio  $16^{-1}$ .

*Proof.* By [Theorem 24.93\(i\)](#) and the unit row mass of [Theorem 24.93\(ii\)](#),

$$D_n(x) - F(x) = \sum_{j=0}^n \omega_{n,j} (S_{2n-j}(x) - F(x)) \quad (x \in \mathbb{R}).$$

For  $0 \leq j \leq n$  one has  $2n - j \geq n \geq 2$ , so [Proposition 24.88\(iii\)](#) at  $q = 1/2$  applies to every summand and gives  $\sup_x |S_{2n-j} - F| = \frac{1}{9} 4^{-(2n-j)}$ . Hence

$$\sup_{x \in \mathbb{R}} |D_n(x) - F(x)| \leq \frac{1}{9} \sum_{j=0}^n |\omega_{n,j}| 4^{j-2n} = \frac{16^{-n}}{9} \sum_{j=0}^n |\omega_{n,j}| 4^j.$$

By [Theorem 24.93\(ii\)](#) the signs alternate,  $|\omega_{n,j}| = (-1)^j \omega_{n,j}$ , so the last sum is the value at  $z = -4$  of the row polynomial of [Remark 24.49](#):

$$\sum_{j=0}^n |\omega_{n,j}| 4^j = \sum_{j=0}^n \omega_{n,j} (-4)^j = \frac{(-2; 1/2)_n}{(1/2; 1/2)_n}. \quad (24.142)$$

This proves the first inequality in [\(24.141\)](#).

For the second, split off the first two factors of the numerator: for  $n \geq 2$ ,

$$(-2; 1/2)_n = \prod_{i=0}^{n-1} (1 + 2^{1-i}) = 3 \cdot 2 \cdot \prod_{h=1}^{n-2} (1 + 2^{-h}) = 6 (-1/2; 1/2)_{n-2}.$$

Both infinite products in [\(24.76\)](#) converge at  $q = 1/2$ , by [Theorem 24.46](#); since  $(-1/2; 1/2)_m$  increases in  $m$  to  $(-1/2; 1/2)_\infty$  and  $(1/2; 1/2)_m$  decreases in  $m$  to  $(1/2; 1/2)_\infty > 0$ , we get

$$\frac{(-2; 1/2)_n}{(1/2; 1/2)_n} = 6 \frac{(-1/2; 1/2)_{n-2}}{(1/2; 1/2)_n} < 6 \frac{(-1/2; 1/2)_\infty}{(1/2; 1/2)_\infty} = 6 \mathcal{L}(1/2).$$

Dividing by 9 gives  $\frac{2}{3} \mathcal{L}(1/2) = 5.5039919571 \dots < 6$ . □

*Remark 24.96* (The first row, and what the abstract theorem gives). The hypothesis  $n \geq 2$  in [Theorem 24.95](#) is necessary. For  $n = 1$  the decomposition [\(24.138\)](#) reads  $D_1 = 2S_2 - S_1$ , and  $S_1$  is outside the range of [Proposition 24.88\(iii\)](#); indeed  $S_1(1/4) = 0$  and  $S_2(1/4) = 1/16$ , so  $D_1(1/4) = 1/8$ , whereas  $F(1/4) = 5/72$  by [Remark 24.32](#). Thus

$$D_1(1/4) - F(1/4) = \frac{1}{8} - \frac{5}{72} = \frac{1}{18},$$

against the value  $6/144 = 1/24$  that the right-hand side of [\(24.141\)](#) would take at  $n = 1$ . Replacing the term  $j = 1$  by the exact bound  $\sup_x |S_1 - F| = 5/72$  restores a valid estimate  $|D_1 - F| \leq 1/12$ .

It is worth separating what is and is not an instance of the abstract machinery. That  $\sum_j \omega_{n,j} F_{1/2, r(n,j)} \rightarrow F$  uniformly, for any exponents  $r(n, j) \geq n$ , is [Corollary 24.51](#) at  $q = 1/2$ , itself an application of [Theorem 24.50](#) with the two hypotheses supplied by [Theorem 24.46](#); that argument gives no rate beyond the rate of its input, and applied to [\(23.31\)](#) it yields only  $O(2^{-n})$ . What [Theorem 24.95](#) adds is, first, that the average being estimated is the concrete off-grid formula [\(24.136\)](#) — an object that exists only because of the clamped cutoff — and second, that feeding the sharp midpoint rate [\(24.126\)](#) into the same sum improves the exponent from  $2^{-n}$  to  $16^{-n}$ . The improvement is entirely due to the input: the row is unchanged, and [\(24.142\)](#) is the same endpoint evaluation as [\(24.75\)](#), taken at  $z = -4$  instead of  $z = -1$ .

Both shift patterns evaluated so far – the common shift of (24.50) and the shift  $2^k + \theta$  of (24.44) – belong to a single two-parameter family, on which the weighted sum is not constant at all, but is a polynomial: the Appell polynomial attached to the Fabius law. The mechanism is the decoupling  $U \neq V$  permitted by Theorem 24.21.

**Definition 24.97** (Fabius Appell polynomials). For  $n \geq 0$  and  $\theta \in \mathbb{C}$  put

$$A_n(\theta) := \mathbb{E}(X + \theta)^n = \sum_{j=0}^n \binom{n}{j} \mathbb{E}X^{n-j} \theta^j, \quad (24.143)$$

a monic polynomial of degree  $n$  in  $\theta$  with  $A_n(0) = \mathbb{E}X^n = n! \alpha_n$ . Since  $0 \leq X \leq 1$ , the exchange of expectation and summation in

$$\sum_{n \geq 0} A_n(\theta) \frac{z^n}{n!} = \mathbb{E} e^{z(X+\theta)} = e^{\theta z} \mathcal{A}(z) \quad (24.144)$$

is legitimate for every  $z \in \mathbb{C}$ , by (24.8) at  $q = 1/2$ . The roman  $A_n$  is not the calligraphic  $\mathcal{A}$  of Corollary 24.5: the latter is one entire function of  $z$ , the former the coefficient sequence of its exponential shift, and  $A_n(0)/n! = \alpha_n$  recovers it.

**Proposition 24.98** (Appell and reflection laws). For  $n \geq 1$  and  $\theta, c \in \mathbb{C}$ ,

$$A'_n(\theta) = n A_{n-1}(\theta), \quad A_n(\theta + c) = \sum_{j=0}^n \binom{n}{j} A_{n-j}(\theta) c^j, \quad (24.145)$$

$$A_n(-1 - \theta) = (-1)^n A_n(\theta). \quad (24.146)$$

*Proof.* Differentiating (24.144) in  $\theta$  multiplies it by  $z$ , which shifts the coefficient index by one; multiplying it by  $e^{cz}$  and expanding the product of the two exponential series gives the translation law. For the reflection, (23.2) at  $q = 1/2$  gives  $1 - X \stackrel{d}{=} X$ , whence

$$A_n(-1 - \theta) = \mathbb{E}(X - 1 - \theta)^n = (-1)^n \mathbb{E}(1 - X + \theta)^n = (-1)^n \mathbb{E}(X + \theta)^n. \quad \square$$

**Theorem 24.99** (Geometric block shifts and the Appell polynomials). For every  $n \geq 0$  and all  $\theta, \eta \in \mathbb{C}$ ,

$$\sum_{k=0}^n \frac{[n]_{1/2}}{4^{\binom{k}{2}} (n+k)!} \sum_{r=0}^{2^k-1} \tau_r (r - 2^k + \eta - 2^k \theta)^{n+k} = \mathfrak{m}_n \frac{A_n(\theta)}{n!}; \quad (24.147)$$

the left-hand side depends on  $\theta$ , but not on  $\eta$ . Equivalently, in the centred normalization  $\eta = 0$ ,

$$\boxed{\frac{A_n(\theta)}{n!} = \frac{2^{-(n+1)}}{(\frac{1}{2}; \frac{1}{2})_n} \sum_{k=0}^n \frac{[n]_{1/2}}{4^{\binom{k}{2}} (n+k)!} \sum_{r=0}^{2^k-1} \tau_r (r - 2^k(1 + \theta))^{n+k}}, \quad (24.148)$$

and, in the raw normalization, for all  $\theta, \eta \in \mathbb{C}$ ,

$$\boxed{\frac{A_n(\theta)}{n!} = \frac{(-1)^n 2^{-(n+1)}}{(\frac{1}{2}; \frac{1}{2})_n} \sum_{k=0}^n \frac{[n]_{1/2}}{4^{\binom{k}{2}} (n+k)!} \sum_{r=0}^{2^k-1} \tau_r (r + \eta + 2^k \theta)^{n+k}}. \quad (24.149)$$

The specialization  $\theta = 0$  of (24.147) is  $\mathcal{N}_n(\eta) = \mathfrak{m}_n \alpha_n$ , that is (24.50); the specialization  $\theta = -1$  is (24.44), with  $\eta$  in the role of the dummy parameter there.

*Proof.* Apply [Theorem 24.21](#) at  $q = 1/2$ , where  $q^{-k} = 2^k$ , with

$$U(z) = \mathcal{A}(-z), \quad V(u) = e^{\theta u} \mathcal{A}(u).$$

The normalization  $u_0 = 1$  holds because  $\mathcal{A}(0) = 1$  ([Corollary 24.5](#)), and  $v_n = A_n(\theta)/n!$  by (24.144). The unique family satisfying the bridge relation (24.30) is

$$\Psi_k^\theta(z) := \frac{e^{-z} V(-2^k z)}{\mathcal{A}(-z)} = e^{-2^k \theta z} \frac{e^{-z} \mathcal{A}(-2^k z)}{\mathcal{A}(-z)} = e^{-2^k \theta z} \Psi_k(z),$$

the last step by the dilation bridge (24.39).

Multiply the centred Thue–Morse identity (24.42) by  $e^{(\eta-2^k\theta)z}$ :

$$\sum_{r=0}^{2^k-1} \tau_r e^{(r-2^k+\eta-2^k\theta)z} = (-1)^k 2^{\binom{k}{2}} z^k e^{\eta z} \Psi_k^\theta(z).$$

The left-hand side is the exponential generating function  $\sum_{d \geq 0} T_{k,d}(\eta - 2^k\theta) z^d / d!$  of the centred power sums (24.37), so comparing the coefficient of  $z^{n+k}$  gives

$$[z^n](e^{\eta z} \Psi_k^\theta(z)) = \frac{(-1)^k}{2^{\binom{k}{2}} (n+k)!} T_{k,n+k}(\eta - 2^k\theta).$$

By [Corollary 24.23](#) with  $\alpha = \eta$ , together with (24.32) and the Mersenne evaluation (22.2),

$$\sum_{k=0}^n W_{n,k}(1/2) [z^n](e^{\eta z} \Psi_k^\theta) = 2^{\binom{n+1}{2}} \left(\frac{1}{2}; \frac{1}{2}\right)_n \frac{A_n(\theta)}{n!} = m_n \frac{A_n(\theta)}{n!},$$

and the same corollary states that this value does not depend on  $\eta$ . Inserting  $W_{n,k}(1/2) = (-1)^k 2^{-\binom{k}{2}} [n]_{1/2}$  makes the two signs cancel and combines the two triangular powers of two into  $4^{-\binom{k}{2}}$ ; that is (24.147).

Putting  $\eta = 0$ , writing  $r - 2^k - 2^k\theta = r - 2^k(1 + \theta)$  and dividing by  $m_n = 2^{\binom{n+1}{2}} \left(\frac{1}{2}; \frac{1}{2}\right)_n$  gives (24.148). For (24.149), replace  $\theta$  by  $-1 - \theta$  in (24.147): the inner argument becomes  $r - 2^k + \eta + 2^k(1 + \theta) = r + \eta + 2^k\theta$ , while the right-hand side becomes  $m_n A_n(-1 - \theta)/n! = (-1)^n m_n A_n(\theta)/n!$  by (24.146).

Finally,  $\theta = 0$  in (24.147) is (24.50), because  $A_n(0) = n! \alpha_n$  and the inner argument is then  $r - 2^k + \eta$ ; and  $\theta = -1$  turns the inner argument into  $r + \eta$  and the right-hand side into  $(-1)^n m_n \alpha_n$ , which is (24.44).  $\square$

*Remark 24.100* (Why  $\theta$  is a variable and  $\eta$  a dummy). [Corollary 24.23](#) says that a shift common to all  $n + 1$  blocks is invisible; [Theorem 24.99](#) says that a shift scaled by  $2^k$  in the  $k$ th block is visible, and is seen exactly once, in the top coefficient of  $e^{\theta u} \mathcal{A}(u)$ . The distinction is precisely the decoupling  $U \neq V$  of [Theorem 24.21](#): the self-referential instance  $V = \mathcal{A}$  attached to  $U(z) = \mathcal{A}(-z)$  can only ever return  $\alpha_n$ , whereas the factor  $e^{\theta u}$  inserted into  $V$  alone survives the double cancellation of [Remark 24.24](#). The two displays (24.148) and (24.149) are one identity in the two coordinate systems of [Remark 24.27](#), exchanged by the digit complementation carried out in the last paragraph of the proof of [Corollary 24.31](#).

**Example 24.101** (The case  $n = 2$ ). Here  $m_2 = 3$  and the half-base Gaussian row is  $1, \frac{3}{2}, 1$ . At  $\eta = 0$  the three inner sums of (24.148) are

$$(1 + \theta)^2, \quad -8(1 + \theta)^3 + (1 + 2\theta)^3, \quad c^4 - (1 - c)^4 - (2 - c)^4 + (3 - c)^4 \quad \text{with } c = 4(1 + \theta),$$

that is  $\theta^2 + 2\theta + 1$ ,  $-12\theta^2 - 18\theta - 7$  and  $384\theta^2 + 480\theta + 160$ . Weighted by  $1/2!$ ,  $\frac{3}{2}/3!$  and  $1/(4 \cdot 4!)$  they sum to

$$\frac{3}{2}\theta^2 + \frac{3}{2}\theta + \frac{5}{12} = 3 \left( \frac{\theta^2}{2} + \frac{\theta}{2} + \frac{5}{36} \right) = m_2 \frac{A_2(\theta)}{2!},$$

as (24.147) asserts. At  $\theta = 1$  the inner sums are the integers 4,  $-37$ ,  $1024$ , the terms are  $2$ ,  $-\frac{37}{4}$ ,  $\frac{32}{3}$ , the total is  $41/12$ , and  $A_2(1)/2! = 41/36 = \frac{1}{2} + \frac{1}{2} + \frac{5}{36}$ . At  $\theta = 0$  they are  $1$ ,  $-7$ ,  $160$  with total  $5/12 = 3\alpha_2$ , which is Example 24.29 once more.

### 24.13 A digit-sparse expansion at an arbitrary argument

The dyadic formula (24.49) is exact but confined to the grid; the sinc integral (24.64) reaches every  $x$  but is an integral of an infinite product. Between them lies a third representation: an exact series whose  $j$ th term is a finite rational expression attached to the  $j$ th binary digit of  $x$ , which terminates on the grid and converges superexponentially off it.

**Lemma 24.102** (Spline representation at a real argument). *For every  $n \geq 1$  and every real  $t$  with  $0 \leq t \leq 2^n$ ,*

$$F\left(\frac{t}{2^n}\right) = \frac{2^{-\binom{n}{2}}}{n!} \sum_{r=0}^{2^n-1} \tau_r \mathbb{E}(t - r - X)_+^n, \quad (24.150)$$

where  $y_+ = \max(y, 0)$ .

*Proof.* By (24.18) we may write  $X = X_{1/2,n} + 2^{-n}X'$  with  $X'$  an independent copy of  $X$ , so conditioning on  $X'$  gives  $F(y) = \mathbb{E}[F_{1/2,n}(y - 2^{-n}X')]$  for every real  $y$ , where  $F_{1/2,n}$  is the distribution function of the truncation (23.17) at  $q = 1/2$ . Integrating (23.22) term by term, with  $\int_0^y (u - c)_+^{n-1} du = (y - c)_+^n/n$ ,

$$F_{1/2,n}(y) = \frac{2^{-\binom{n+1}{2}}}{n!} \sum_{r=0}^{2^n-1} \tau_r \left(y - \frac{r}{2^n}\right)_+^n.$$

Put  $y = t/2^n$  and extract the factor  $2^{-n}$  from each positive part,  $(2^{-n}(t - X' - r))_+^n = 2^{-n^2}(t - X' - r)_+^n$ . With  $\binom{n+1}{2} - n^2 = -\binom{n}{2}$  and  $X' \stackrel{d}{=} X$  this is (24.150).  $\square$

**Corollary 24.103** (Local reduction identity). *For every  $n \geq 1$  and every  $\theta \in [0, 1)$ ,*

$$F\left(\frac{1+\theta}{2^n}\right) + F\left(\frac{\theta}{2^n}\right) = \frac{2^{-\binom{n}{2}}}{n!} A_n(\theta). \quad (24.151)$$

*Proof.* Both  $t = \theta$  and  $t = 1 + \theta$  lie in  $[0, 2] \subseteq [0, 2^n]$ , so Lemma 24.102 applies to each. Since  $0 \leq X \leq 1$ , a term with  $r \geq 1$  in the sum for  $t = \theta$  has  $\theta - r - X \leq \theta - 1 < 0$  and drops out, leaving

$$F\left(\frac{\theta}{2^n}\right) = \frac{2^{-\binom{n}{2}}}{n!} \mathbb{E}(\theta - X)_+^n.$$

In the sum for  $t = 1 + \theta$  a term with  $r \geq 2$  has  $1 + \theta - r - X \leq \theta - 1 < 0$  and drops out; the term  $r = 0$  has  $1 + \theta - X \geq \theta \geq 0$ , so its positive part is the plain power, and  $1 - X \stackrel{d}{=} X$  by (23.2) turns  $\mathbb{E}(1 + \theta - X)^n$  into  $\mathbb{E}(X + \theta)^n = A_n(\theta)$ ; the term  $r = 1$  carries  $\tau_1 = -1$  and contributes  $-\mathbb{E}(\theta - X)_+^n$ . Hence

$$F\left(\frac{1+\theta}{2^n}\right) = \frac{2^{-\binom{n}{2}}}{n!} \left( A_n(\theta) - \mathbb{E}(\theta - X)_+^n \right),$$

and adding the two displays proves (24.151).  $\square$

Let  $x \in [0, 1)$  and let  $x = \sum_{j \geq 1} b_j 2^{-j}$ ,  $b_j \in \{0, 1\}$ , be its canonical binary expansion, the one that is not eventually constant equal to 1. Put

$$m_j := \lfloor 2^j x \rfloor, \quad \theta_j := 2^j x - m_j \in [0, 1), \quad m_0 = 0, \quad \theta_0 = x, \quad (24.152)$$

the binary *prefix* and *tail* of  $x$  at depth  $j$ ; then  $m_j = 2m_{j-1} + b_j$  and  $\theta_{j-1} = (b_j + \theta_j)/2$ , and the floor in (24.152) selects the canonical expansion automatically.

**Theorem 24.104** (Digit-sparse Appell expansion). *For every  $x \in [0, 1)$ , with the notation (24.152),*

$$F(x) = \sum_{j=1}^{\infty} b_j \tau_{m_{j-1}} \frac{2^{-\binom{j}{2}}}{j!} A_j(\theta_j). \quad (24.153)$$

Writing  $S_N(x)$  for the sum of the first  $N$  terms,

$$F(x) - S_N(x) = \tau_{m_N} F\left(\frac{\theta_N}{2^N}\right), \quad |F(x) - S_N(x)| \leq \frac{2^{-\binom{N}{2}}}{N!}. \quad (24.154)$$

The series therefore converges absolutely and uniformly on  $[0, 1)$ , and at a dyadic rational  $m/2^n$ , taken in its canonical terminating expansion, it terminates after  $n$  terms.

*Proof.* Put  $R_N := \tau_{m_N} F(\theta_N/2^N)$ , so that  $R_0 = \tau_0 F(x) = F(x)$ . Fix  $j \geq 1$ . If  $b_j = 0$  then  $m_j = 2m_{j-1}$ , and doubling shifts every binary digit without creating or destroying one, so  $\tau_{m_j} = \tau_{m_{j-1}}$ ; moreover  $\theta_{j-1} = \theta_j/2$ , hence  $\theta_{j-1}/2^{j-1} = \theta_j/2^j$  and  $R_{j-1} = R_j$ , which is also the  $j$ th term of (24.153), namely 0.

If  $b_j = 1$  then  $m_j = 2m_{j-1} + 1$ , so  $s_2(m_j) = s_2(m_{j-1}) + 1$  and  $\tau_{m_j} = -\tau_{m_{j-1}}$ , while  $\theta_{j-1}/2^{j-1} = (1 + \theta_j)/2^j$ . Therefore

$$R_{j-1} - R_j = \tau_{m_{j-1}} \left[ F\left(\frac{1 + \theta_j}{2^j}\right) + F\left(\frac{\theta_j}{2^j}\right) \right] = \tau_{m_{j-1}} \frac{2^{-\binom{j}{2}}}{j!} A_j(\theta_j)$$

by Corollary 24.103, which is again the  $j$ th term. Summing over  $j = 1, \dots, N$  telescopes to  $R_0 - R_N = S_N(x)$ , which is the first half of (24.154).

For the bound,  $0 \leq \theta_N < 1$  and  $F$  is nondecreasing, so  $F(\theta_N/2^N) \leq F(2^{-N}) = 2^{-\binom{N}{2}} \alpha_N$  by Theorem 24.9, and  $\alpha_N = \mathbb{E}X^N/N! \leq 1/N!$  because  $0 \leq X \leq 1$ . This bound is independent of  $x$  and tends to 0, so  $S_N \rightarrow F$  uniformly on  $[0, 1)$  and (24.153) holds. The convergence is absolute because  $0 \leq X + \theta_j < 2$  gives  $|A_j(\theta_j)| \leq 2^j$ , so the terms are dominated by the summable sequence  $2^{j-\binom{j}{2}}/j!$ .

Finally, if  $x = m/2^n$  then  $2^j x \in \mathbb{Z}$  for every  $j \geq n$ , so  $b_j = 0$  and  $\theta_j = 0$  there; hence  $R_N = \tau_{m_N} F(0) = 0$  for  $N \geq n$  and  $F(x) = S_N(x) = S_n(x)$ .  $\square$

**Corollary 24.105** (Digit-sparse half-base  $q$ -series). *For every  $x \in [0, 1)$ , with the notation (24.152),*

$$F(x) = \sum_{j=1}^{\infty} \frac{b_j \tau_{m_{j-1}}}{2^{j^2} \left(\frac{1}{2}; \frac{1}{2}\right)_j} \sum_{k=0}^j \frac{\begin{bmatrix} j \\ k \end{bmatrix}_{1/2}}{4^{\binom{k}{2}} (j+k)!} \sum_{r=0}^{2^k-1} \tau_r (r - 2^k(1 + \theta_j))^{j+k}, \quad (24.155)$$

with the convergence, residual and termination properties of Theorem 24.104.

*Proof.* Insert (24.148) with  $n = j$  and  $\theta = \theta_j$  into (24.153). The two triangular powers of two combine as  $2^{-\binom{j}{2}} 2^{-(j+1)} = 2^{-j^2}$ . This is a termwise replacement by an exact finite identity, so every analytic assertion is unchanged.  $\square$



*Remark 24.106* (Three ways to evaluate  $F$ ). The monograph now evaluates  $F$  by three mechanisms. [Corollary 24.31](#) is finite, rational and exact, but only on the dyadic grid; [Corollary 24.41](#) is global and analytic, an integral of an infinite sinc product; [Corollary 24.105](#) is global and *arithmetic*, each term being a finite half-base Gaussian sum indexed by one binary digit of  $x$ . The three agree where they overlap: at  $x = 2^{-n}$  only the digit  $b_n$  is nonzero, with  $m_{n-1} = 0$  and  $\theta_n = 0$ , so (24.153) collapses to  $F(2^{-n}) = 2^{-\binom{n}{2}} A_n(0)/n! = 2^{-\binom{n}{2}} \alpha_n$ , which is [Theorem 24.9](#).

At  $x = \frac{1}{3} = 0.\overline{01}_2$  the odd digits vanish and  $\theta_j = \frac{2}{3}$  or  $\frac{1}{3}$  according as  $j$  is odd or even, so the first six terms contain only three nonzero summands:

$$S_2 = \frac{13}{72} = 0.18055\dots, \quad S_4 = \frac{1120771}{6220800} = 0.18016509\dots, \quad S_6 = 0.1801651148015\dots,$$

against  $F(\frac{1}{3}) = 0.180165114801481906955\dots$ : thirteen correct digits from three terms, where (24.154) guarantees only  $2^{-15}/6! < 4.3 \cdot 10^{-8}$ . Unlike (24.46), whose conditioning is analysed in [Remark 24.34](#), the outer series here has no cancellation at all for  $x < \frac{1}{2}$  beyond the Thue–Morse signs  $\tau_{m_{j-1}}$ ; the ill-conditioning of the inner blocks is inherited unchanged, so for numerical work one evaluates  $A_j(\theta_j)$  from (24.143) and the recurrence (24.11) rather than from (24.155).



# Appendix A

## Formula atlas

This appendix is a navigation table, not a substitute for hypotheses and proofs. Each formula points back to the theorem where its exact domain and derivation are given.

| Name                      | Formula                                                                                                                                                                 | Location                         |
|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|
| Concatenation             | $(a; q)_{m+n} = (a; q)_m (aq^m; q)_n$                                                                                                                                   | <a href="#">Proposition 1.2</a>  |
| Dissection                | $(a; q)_{rn} = \prod_{j=0}^{r-1} (aq^j; q^r)_n$                                                                                                                         | <a href="#">Proposition 1.3</a>  |
| Dissection with remainder | $(a; q)_{rn+u} = \prod_{s < u} (aq^s; q^r)_{n+1} \prod_{s \geq u} (aq^s; q^r)_n$                                                                                        | <a href="#">Corollary 1.4</a>    |
| Base reversal             | $(a; q)_n = (-a)^n q^{\binom{n}{2}} (a^{-1} q^{1-n}; q)_n$                                                                                                              | <a href="#">Proposition 1.6</a>  |
| Negative index            | $(a; q)_{-n} = 1 / (aq^{-n}; q)_n$                                                                                                                                      | <a href="#">Proposition 1.8</a>  |
| Infinite logarithm        | $\log(a; q)_\infty = - \sum_{r \geq 1} a^r / (r(1 - q^r))$                                                                                                              | <a href="#">Theorem 1.16</a>     |
| Gaussian product          | $\begin{bmatrix} n \\ k \end{bmatrix}_q = (q; q)_n / ((q; q)_k (q; q)_{n-k})$                                                                                           | <a href="#">Proposition 2.2</a>  |
| Pascal I                  | $\begin{bmatrix} n \\ k \end{bmatrix}_q = q^k \begin{bmatrix} n-1 \\ k \end{bmatrix}_q + \begin{bmatrix} n-1 \\ k-1 \end{bmatrix}_q$                                    | <a href="#">Theorem 2.3</a>      |
| Pascal II                 | $\begin{bmatrix} n \\ k \end{bmatrix}_q = \begin{bmatrix} n-1 \\ k \end{bmatrix}_q + q^{n-k} \begin{bmatrix} n-1 \\ k-1 \end{bmatrix}_q$                                | <a href="#">Theorem 2.3</a>      |
| Reciprocity               | $\begin{bmatrix} n \\ k \end{bmatrix}_q = q^{k(n-k)} \begin{bmatrix} n \\ k \end{bmatrix}_{q^{-1}}$                                                                     | <a href="#">Theorem 2.5</a>      |
| Finite theorem            | $(z; q)_n = \sum_k (-1)^k q^{\binom{k}{2}} \begin{bmatrix} n \\ k \end{bmatrix}_q z^k$                                                                                  | <a href="#">Theorem 3.1</a>      |
| Reciprocal theorem        | $(z; q)_{n+1}^{-1} = \sum_{k \geq 0} \begin{bmatrix} n+k \\ k \end{bmatrix}_q z^k$                                                                                      | <a href="#">Theorem 3.6</a>      |
| $q$ -Vandermonde          | $\begin{bmatrix} m+n \\ r \end{bmatrix}_q = \sum_k q^{\binom{m-k}{2} + \binom{r-k}{2}} \begin{bmatrix} m \\ k \end{bmatrix}_q \begin{bmatrix} n \\ r-k \end{bmatrix}_q$ | <a href="#">Theorem 3.8</a>      |
| $q$ -binomial inversion   | $b_n = \sum_k \begin{bmatrix} n \\ k \end{bmatrix}_q a_k \iff a_n = \sum_k (-1)^{n-k} q^{\binom{n-k}{2}} \begin{bmatrix} n \\ k \end{bmatrix}_q b_k$                    | <a href="#">Theorem 3.12</a>     |
| Infinite theorem          | $\sum_{n \geq 0} (a; q)_n z^n / (q; q)_n = (az; q)_\infty / (z; q)_\infty$                                                                                              | <a href="#">Theorem 8.1</a>      |
| Euler I                   | $(z; q)_\infty = \sum_{n \geq 0} (-1)^n q^{\binom{n}{2}} z^n / (q; q)_n$                                                                                                | <a href="#">Corollary 8.2</a>    |
| Euler II                  | $1 / (z; q)_\infty = \sum_{n \geq 0} z^n / (q; q)_n,  z  < 1$                                                                                                           | <a href="#">Corollary 8.2</a>    |
| $q$ -Gauss                | ${}_2\phi_1(a, b; c; q, c/(ab)) = (c/a, c/b; q)_\infty / (c, c/(ab); q)_\infty$                                                                                         | <a href="#">Theorem 9.6</a>      |
| Heine                     | ${}_2\phi_1(a, b; c; q, z) = \frac{(b, az; q)_\infty}{(c, z; q)_\infty} {}_2\phi_1(c/b, z; az; q, b)$                                                                   | <a href="#">Theorem 9.5</a>      |
| $q$ -Pfaff–Saalschütz     | Terminating balanced ${}_3\phi_2$ product evaluation                                                                                                                    | <a href="#">Theorem 9.9</a>      |
| $q$ -Chu–Vandermonde      | Two terminating ${}_2\phi_1$ evaluations                                                                                                                                | <a href="#">Corollary 9.10</a>   |
| Ramanujan ${}_1\psi_1$    | ${}_1\psi_1(a; b; q, z) = (q, b/a, az, q/(az); q)_\infty / (b, q/a, z, b/(az); q)_\infty$                                                                               | <a href="#">Theorem 10.2</a>     |
| Finite triple product     | $(z; q)_N (q/z; q)_N = \sum_{k=-N}^N (-1)^k q^{\binom{k}{2}} \begin{bmatrix} 2N \\ N+k \end{bmatrix}_q z^k$                                                             | <a href="#">Theorem 13.3</a>     |
| Jacobi triple product     | $\sum_{n \in \mathbb{Z}} (-1)^n q^{\binom{n}{2}} z^n = (q, z, q/z; q)_\infty$                                                                                           | <a href="#">Theorem 13.1</a>     |
| Pentagonal theorem        | $(q; q)_\infty = \sum_{n \in \mathbb{Z}} (-1)^n q^{n(3n-1)/2}$                                                                                                          | <a href="#">Theorem 13.7</a>     |
| $q$ -derivative           | $D_q f(x) = (f(x) - f(qx)) / ((1 - q)x)$                                                                                                                                | <a href="#">Proposition 11.1</a> |
| Jackson integral          | $\int_0^a f(x) d_q x = (1 - q)a \sum_{n \geq 0} q^n f(aq^n)$                                                                                                            | <a href="#">Theorem 11.11</a>    |
| $q$ -gamma                | $\Gamma_q(x) = (1 - q)^{1-x} (q; q)_\infty / (q^x; q)_\infty$                                                                                                           | <a href="#">Theorem 12.2</a>     |
| Cyclotomic factorization  | $\begin{bmatrix} n \\ k \end{bmatrix}_q = \prod_d \Phi_d(q)^{\lfloor n/d \rfloor - \lfloor k/d \rfloor - \lfloor (n-k)/d \rfloor}$                                      | <a href="#">Theorem 16.1</a>     |
| $q$ -Lucas                | $\begin{bmatrix} ad+b \\ rd+s \end{bmatrix}_q \equiv \begin{bmatrix} a \\ r \end{bmatrix}_q \begin{bmatrix} b \\ s \end{bmatrix}_q \pmod{\Phi_d(q)}$                    | <a href="#">Theorem 16.7</a>     |
| $q$ -Babbage              | $\begin{bmatrix} an \\ bn \end{bmatrix}_q \equiv \begin{bmatrix} a \\ b \end{bmatrix}_{q^{n^2}} \pmod{\Phi_n(q)^2}$                                                     | <a href="#">Theorem 16.10</a>    |

| Name                   | Formula                                                                                                              | Location                         |
|------------------------|----------------------------------------------------------------------------------------------------------------------|----------------------------------|
| Negative upper index   | $\begin{bmatrix} -m \\ k \end{bmatrix}_q = (-1)^k q^{-mk - \binom{k}{2}} \begin{bmatrix} m+k-1 \\ k \end{bmatrix}_q$ | <a href="#">Proposition 17.2</a> |
| Geometric Newton basis | $P(x) = \sum_k c_k (-1)^k q^{\binom{k}{2}} (x; q^{-1})_k$                                                            | <a href="#">Theorem 17.10</a>    |
| Central reduction      | $\begin{bmatrix} 2k \\ k \end{bmatrix}_{q^2} / (-q; q)_{2k} = (q; q^2)_k / (q^2; q^2)_k$                             | <a href="#">Lemma 18.1</a>       |
| Bernoulli expansion    | All-orders expansion of $\log(\begin{bmatrix} n \\ k \end{bmatrix}_{e^{-t}} / \binom{n}{k})$                         | <a href="#">Theorem 19.2</a>     |

## Appendix B

# Limit and specialization dictionary

### B.1 Elementary limits

For fixed nonnegative integers, the following transitions are direct consequences of the product formulas:

$$\begin{aligned} \lim_{q \rightarrow 1} [n]_q &= n, & \lim_{q \rightarrow 1} (1-q)^{-n} (q^a; q)_n &= (a)_n, \\ \lim_{q \rightarrow 1} \begin{bmatrix} n \\ k \end{bmatrix}_q &= \binom{n}{k}, & \lim_{n \rightarrow \infty} \begin{bmatrix} n \\ k \end{bmatrix}_q &= \frac{1}{(q; q)_k} \quad (|q| < 1), \\ \lim_{q \rightarrow 1^-} \Gamma_q(x) &= \Gamma(x), & \lim_{q \rightarrow 0} \begin{bmatrix} n \\ k \end{bmatrix}_q &= 1 \quad (0 \leq k \leq n). \end{aligned}$$

Each has been proved in the main text; the relevant references are [Propositions 1.19](#) and [19.11](#), [corollary 2.6](#), and [Theorems 12.5](#) and [19.6](#).

Transitions in the size parameter, rather than in  $q$ , are already represented above by  $\lim_{n \rightarrow \infty} \begin{bmatrix} n \\ k \end{bmatrix}_q = 1/(q; q)_k$ . Their central counterpart is

$$\lim_{N \rightarrow \infty} \begin{bmatrix} 2N \\ N+k \end{bmatrix}_q = \frac{1}{(q; q)_\infty} \quad (k \in \mathbb{Z} \text{ fixed, } 0 < |q| < 1),$$

which is [Theorem 19.7](#) when  $k = 0$  and follows for every fixed  $k$  by the same argument; it is what carries the finite triple product [\(13.3\)](#) to Jacobi's [\(13.2\)](#). See [Theorem 13.3](#) and [corollary 13.5](#), and compare [Proposition 14.5](#), which records the same stabilization coefficientwise.

### B.2 Classical shadows of the main summations

Under the standard scaling  $a = q^A$ ,  $b = q^B$ , and  $q \rightarrow 1^-$ , terminating basic hypergeometric sums tend to ordinary hypergeometric sums. The mechanism is

$$\frac{(q^A; q)_n}{(1-q)^n} \longrightarrow (A)_n, \quad \frac{(q; q)_n}{(1-q)^n} \longrightarrow n!.$$

Thus:

1. the finite  $q$ -binomial theorem tends to Newton's binomial theorem;
2.  $q$ -Chu–Vandermonde tends to Chu–Vandermonde;

3.  $q$ -Pfaff–Saalschütz tends to Pfaff–Saalschütz;
4. the Jackson  $q$ -beta integral tends to Euler’s beta integral;
5. [Theorem 18.2](#) tends to [Corollary 18.3](#);
6. [Theorem 18.4](#) tends, at the level of its Lambert tail and prefactor, to [Theorem 18.7](#).

### B.3 Roots of unity

At a root of unity, factorial quotients should be replaced by polynomial formulas before evaluation. The basic dictionary is:

$$\begin{aligned} \Phi_d(q) \mid \begin{bmatrix} n \\ k \end{bmatrix}_q &\iff k \bmod d > n \bmod d, \\ \begin{bmatrix} ad + b \\ rd + s \end{bmatrix}_q &\equiv \begin{pmatrix} a \\ r \end{pmatrix} \begin{bmatrix} b \\ s \end{bmatrix}_q \pmod{\Phi_d(q)}, \\ \begin{bmatrix} n \\ k \end{bmatrix}_q \Big|_{q=-1} &\text{ is given by [Corollary 16.8](#).} \end{aligned}$$

These statements turn apparently singular substitutions into finite enumerations and are the algebraic basis of the subset cyclic-sieving theorem.

## Appendix C

# Proof-dependency guide

The monograph can be read linearly, but several shorter routes are useful.

### Route to Gaussian combinatorics

finite products  $\longrightarrow$   $q$ -Pascal  $\longrightarrow$  finite  $q$ -binomial theorem  $\longrightarrow$  partitions, paths, subspaces.

Read [Chapters 1 to 3, 5 and 6](#).

### Route to product–sum identities

infinite  $q$ -binomial theorem  $\longrightarrow$  Heine transformation  $\longrightarrow$   $q$ -Gauss,  
negative indices and bilateral convergence  $\longrightarrow$   ${}_1\psi_1$ ,  
Euler expansions  $\longrightarrow$  Jacobi triple product  $\longrightarrow$  Bailey pairs.

Read [Chapters 8 to 10, 13 and 15](#).

### Route to arithmetic

$1 - q^n = \prod_{d|n} \Phi_d(q) \longrightarrow$  carry criterion  $\longrightarrow$   $q$ -Lucas  $\longrightarrow$  cyclic sieving and  $q$ -Babbage.

Read [Chapter 16](#).

### Route to analytic deformation

$\log(a; q)_\infty \longrightarrow$  Lambert series and dilogarithms  $\longrightarrow$   $q$ -gamma  
 $\longrightarrow$  central sums and Bernoulli asymptotics.

Read [Chapters 1, 12, 18 and 19](#).

### Route to geometric filters and the Fabius bridge

reversed Gaussian rows  $\longrightarrow$  geometric Lagrange moments  $\longrightarrow$  exact coefficient filters  
 $\longrightarrow$  dyadic Prouhet cancellation  $\longrightarrow$  Fabius–Rvachev sinc products.

Read [Chapters 21 to 23](#).

## Appendix D

# Formalization status

Every mathematical result in this book is intended, eventually, to have a machine-checked proof in the Lean development that accompanies the Fabius project [38]. This appendix records how far that has got, result by result, so that what remains can be planned rather than rediscovered.

The classification is deliberately severe, and it is the same rule the project uses to decide whether a claim may leave its quarantine boundary. A result counts as formalized only when a current Lean declaration proves the *exact* statement: same hypotheses, same domain, same normalization, same conclusion. A related definition, a theorem with stronger hypotheses or a weaker conclusion, an executable computation, a numerical match, or an apparently immediate derivation is *not* an exact counterpart. In particular a Lean lemma that appears as a step inside the proof below does not formalize the result the proof concludes.

**exact** a named Lean declaration proves the same statement; the declaration and its module are given.

**partial** Lean has the result only in a special case, at a fixed parameter, on a restricted domain, or with a weaker conclusion. The named declaration is the nearest one, and the shortfall is real.

**none** nothing in the development proves it, even partially.

**n/a** the environment states a convention or an interface rather than a mathematical claim.

Of the 183 labelled results, 31 are **exact**, 25 are **partial**, 124 are **none**, and 3 are n/a. The formalized part is concentrated exactly where the Fabius development needed it: the finite Gaussian-coefficient core, the geometric Lagrange and filter calculus, half-base and Thue–Morse cancellation, and the geometric uniform law. The complementary modules `CompleteHomogeneousGenerating` and `SymmetricFunctionGenerating` now formalize both finite algebraic halves of the weighted generating-product theorem as formal-power-series identities; its analytic evaluation and convergence clause is not formalized. The finite-family theorem in `CompleteHomogeneousAsymptotics` separately transfers coordinatewise Big-O bounds through a fixed complete homogeneous degree, without asserting formal-series convergence. Most of the classical analytic layer — nonterminating basic hypergeometric summation, Jacobi triple product, Bailey inversion and the Bailey lemma, and the Rogers–Ramanujan identities — remains untouched; the algebraic  $q$ -difference annihilation input to that chapter is now exact. The former claim that every infinite  $q$ -Pochhammer symbol in the development was a finite `Finset.range` product is now false: `LimitConditionNumber.qPochhammerInf` is the genuine real  $\prod'_{j \geq 0} (1 - aq^j)$ ; for  $0 \leq q < 1$  the factors are multipliable for every real  $a$ , the finite symbols converge to it, and the positive-factor specializations used for  $(q; q)_\infty$  and  $(-q; q)_\infty$  are strictly positive. This real contractive-base product layer does not formalize the book’s general complex analytic or basic-hypergeometric infinite-product theory.

The new weighted symmetric-function tranche and the triangular-transform refactor in this edition were statically checked against the complete public declaration lists:

`QBinomialCauchy.lean`: one definition and four theorems;  
`SymmetricFunctionOrthogonality.lean`: one definition and six theorems;  
`CompleteHomogeneousGenerating.lean`: one definition and six theorems;  
`CompleteHomogeneousAsymptotics.lean`: one theorem;  
`FiniteTriangularTransform.lean`: one definition and one theorem;  
`SymmetricFunctionTransform.lean`: four definitions and five theorems;  
`SymmetricFunctionGenerating.lean`: two definitions and six theorems;  
`QDifferenceAnnihilation.lean`: four theorems;  
`QBinomialInversionSpecializations.lean`: two definitions and four theorems;  
`GeometricCompleteHomogeneous.lean`: six theorems;  
`GeometricLagrangeCompleteHomogeneous.lean`: five theorems;  
`GeometricLagrangeQMoments.lean`: one definition and 37 theorems;  
`FinitePolynomialFilterExactness.lean`: five theorems;  
`QuarterCatalanGerm.lean`: two definitions and nine theorems;  
`QuarterCatalanRichardson.lean`: three definitions and 15 theorems.

The added finite geometric  $q$ -layer has the following exhaustive public surface.

- The six theorems of `GeometricCompleteHomogeneous.lean` are `Fabius.completeHomogeneousEval_geometric`, `Fabius.completeHomogeneousEval_scaled_geometric`, `Fabius.completeHomogeneousEvalOn_range_pow_eq_gaussianBinomial`, `Fabius.completeHomogeneousEvalOn_range_pow_eq_gaussianBinomial_degree`, `Fabius.gaussianBinomial_add_symm`, and `Fabius.gaussianBinomial_symm_via_completeHomogeneous`. They prove both principal-specialization orientations, common scaling, and both generic Gaussian symmetry laws over every commutative semiring, without division, distinctness, or cancellation.
- The five theorems of `GeometricLagrangeCompleteHomogeneous.lean` are `Fabius.completeHomogeneousEvalOn_geometric_range`, `Fabius.sum_geometricLagrangeWeight_mul_pow_succ_add_eq_gaussianBinomial`, `Fabius.geometricLagrangeQMoment_eq_residual_gaussianBinomial`, `Fabius.completeHomogeneousEvalOn_geometric_range_eq_qBinomial`, and `Fabius.geometricLagrangeQMoment_eq_residual_qBinomial_via_completeHomogeneous`. The first is semiring-level; the field statement uses finite-node injectivity; the rational quotient bridges retain their explicit nonzero-Pochhammer or  $0 < q < 1$  assumptions.
- `GeometricLagrangeQMoments.lean` has the one definition `Fabius.geometricLagrangeQMoment` and exactly 37 theorems: `Fabius.geometricLagrangeQMoment_eq_weightPolynomial_eval`, `Fabius.geometricLagrangeQMoment_eq_forwardRichardson_eval`, `Fabius.geometricRootPolynomial_inv_eval_pow_mul_signedPowers`, `Fabius.geometricRootPolynomial_inv_eval_pow_mul_triangular`, `Fabius.geometricRootPolynomial_inv_eval_one_mul_triangular`, `Fabius.geometricLagrangeQMoment_eq_qPochhammer`, `Fabius.geometricLagrangeQMoment_zero`, `Fabius.geometricLagrangeQMoment_eq_zero`, `Fabius.geometricRootPolynomial_inv_eval_pow_eq_qPochhammer_of_le`, `Fabius.geometricLagrangeQMoment_eq_residual_qPochhammer`, `Fabius.qPochhammer_self_add`, `Fabius.qPochhammer_self_pos_of_pos_of_lt_one`, `Fabius.qBinomial_pos_of_pos_of_lt_one`, `Fabius.gaussianBinomial_eq_qBinomial_of_pos_of_lt_one`, `Fabius.qPochhammer_pow_pos_of_pos_of_lt_one`, `Fabius.qPochhammer_tail_div_self_eq_qBinomial`, `Fabius.geometricLagrangeQMoment_eq_residual_qBinomial`, `Fabius.geometricLagrangeQMoment_firstUncanceled`, `Fabius.negOnePow_mul_geometricLagrangeQMoment_eq_positiveResidual`, `Fabius.negOnePow_mul_geometricLagrangeQMoment_pos`, `Fabius.qPochhammer_self_succ`, `Fabius.qBinomial_succ_succ_of_pos_of_lt_one`, `Fabius.qBinomial_succ_succ_of_pos_of_lt_one`, `Fabius.qBinomial_theorem_of_pos_of_lt_one`,

`Fabius.sum_qBinomial_triangular_succ_eq_neg_qPochhammer`, `Fabius.abs_geometricLagrangeWeight_eq_qBinomial`, `Fabius.abs_geometricLagrangeWeight_eq_sign_mul`, `Fabius.abs_geometricLagrangeWeight_complement_eq_qBinomial`, `Fabius.sum_abs_geometricLagrangeWeight_eq_qPochhammer_ratio`, `Fabius.neg_qPochhammer_div_self_eq_prod`, `Fabius.sum_abs_geometricLagrangeWeight_eq_prod`, `Fabius.quarterGeometricLagrangeQMoment_eq_qPochhammer`, `Fabius.quarterGeometricLagrangeQMoment_eq_zero`, `Fabius.quarterGeometricLagrangeQMoment_eq_residual_qPochhammer`, `Fabius.quarterGeometricLagrangeQMoment_eq_residual_qBinomial`, `Fabius.quarterGeometricLagrangeQMoment_firstUncancelled`, and `Fabius.sum_abs_quarterGeometricLagrangeWeight_eq_qPochhammer_ratio`. These are finite rational identities. Quotient statements retain their nonzero denominators, and positivity, sign, and absolute-value formulas retain  $0 < q < 1$ ; the module makes no analytic convergence or remainder claim.

- The five theorems of `FinitePolynomialFilterExactness.lean` are `Fabius.polynomialFilter_response_eq`, `Fabius.polynomialFilter_exact`, `Fabius.normalizedGeometricRootPolynomial_filter_exact`, `Fabius.forwardGeometricRichardsonPolynomial_filter_exact`, and `Fabius.forwardGeometricRichardsonPolynomial_filter_firstUncancelled`. The first two are arbitrary commutative-semiring response and exactness laws; the field-valued geometric forms retain their nonzero-base and normalization- denominator hypotheses and expose the first surviving mode.
- `QuarterCatalanGerm.lean` has the two definitions `Fabius.quarterCatalanCoefficient` and `Fabius.quarterCatalanGermSeries`, and the nine theorems `Fabius.quarterCatalanCoefficient_zero`, `Fabius.quarterCatalanCoefficient_succ_eq_report`, `Fabius.quarterCatalanGermSeries_coeff`, `Fabius.quarterCatalanGermSeries_coeff_succ`, `Fabius.quarterCatalanGermSeries_constantCoeff`, `Fabius.quarterCatalanGermSeries_equation`, `Fabius.powerSeries_quadratic_injectiveOn_zeroConstant`, `Fabius.eq_quarterCatalanGermSeries_of_equation`, and `Fabius.existsUnique_quarterCatalanGermSeries`. This is the unique zero-constant solution of  $D + 4D^2 = (4/9)X$  in  $\mathbb{Q}[[X]]$ , with its complete coefficient formula, but no convergence or real-inverse identification.
- `QuarterCatalanRichardson.lean` has the three definitions `Fabius.finiteRescaleFilter`, `Fabius.geometricRichardsonPowerSeriesFilter`, and `Fabius.quarterCatalanRichardsonFilter`, and the 15 theorems `Fabius.finiteRescaleFilter_coeff`, `Fabius.geometricRichardsonPowerSeriesFilter_coeff`, `Fabius.geometricRichardsonPowerSeriesFilter_coeff_zero`, `Fabius.geometricRichardsonPowerSeriesFilter_coeff_eq_zero`, `Fabius.geometricRichardsonPowerSeriesFilter_coeff_eq_qPochhammer`, `Fabius.geometricRichardsonPowerSeriesFilter_coeff_eq_qBinomial`, `Fabius.geometricRichardsonPowerSeriesFilter_firstUncancelled_coeff_of_nonzero`, `Fabius.geometricRichardsonPowerSeriesFilter_firstUncancelled_coeff`, `Fabius.quarterCatalanRichardsonFilter_coeff`, `Fabius.quarterCatalanRichardsonFilter_coeff_zero`, `Fabius.quarterCatalanRichardsonFilter_coeff_eq_zero`, `Fabius.quarterCatalanRichardsonFilter_coeff_eq_zero_of_le`, `Fabius.quarterCatalanRichardsonFilter_coeff_eq_qBinomial`, `Fabius.quarterCatalanRichardsonFilter_coeff_succ_eq_qBinomial`, and `Fabius.quarterCatalanRichardsonFilter_firstUncancelled_coeff`. These are coefficientwise formal-power-series identities only: no convergence, real error sign, remainder estimate, or analytic acceleration is asserted.

Finally, `Fabius.sum_lagrangeEvalWeight_mul_pow_card_add_zero` in `LagrangeResidualMo`



ments. Lean gives every higher evaluation-at-zero moment of a nonempty distinct field-valued node family as the negative signed nodal product times its complete homogeneous function. Its nonempty premise is exactly what removes the exceptional empty-row  $0^0$  term.

The theorem statements were checked at arbitrary parameters and degrees, including the empty and degree-zero boundaries. Weighted inversion is now **exact**. Weighted generating products and reciprocal finite products are each **partial**: their formal power-series identities are proved, but their analytic evaluation and convergence clauses are not. The nearby  $q$ -Pfaff–Saalschütz summation remains **none** and is not inferred from a related finite identity.

## D.1 By chapter

| Chapter                                                                  | exact | partial | none | n/a |
|--------------------------------------------------------------------------|-------|---------|------|-----|
| The algebra of $q$ -shifted factorials                                   | 1     | 2       | 12   | 0   |
| $q$ -integers and Gaussian coefficients                                  | 1     | 2       | 5    | 0   |
| Finite $q$ -binomial theorems and inversion                              | 9     | 1       | 0    | 0   |
| Weighted binomial coefficients and symmetric functions                   | 3     | 2       | 3    | 0   |
| Partitions, words, paths, and inversion statistics                       | 1     | 1       | 7    | 0   |
| Finite fields, Grassmannians, and rank enumeration                       | 0     | 1       | 5    | 0   |
| Symmetry and unimodality                                                 | 0     | 0       | 3    | 0   |
| The infinite $q$ -binomial theorem                                       | 0     | 0       | 5    | 0   |
| Basic hypergeometric series                                              | 1     | 0       | 5    | 0   |
| Bilateral series and Ramanujan's ${}_1\psi_1$ sum                        | 0     | 0       | 2    | 0   |
| $q$ -difference calculus and the quantum binomial                        | 0     | 0       | 10   | 0   |
| The $q$ -gamma and $q$ -beta functions                                   | 0     | 0       | 8    | 0   |
| Jacobi's triple product                                                  | 0     | 0       | 6    | 0   |
| Partitions and product–sum identities                                    | 0     | 0       | 4    | 0   |
| Bailey pairs and the Rogers–Ramanujan identities                         | 1     | 0       | 6    | 0   |
| Cyclotomic factorization and congruences                                 | 0     | 0       | 9    | 0   |
| Negative upper indices and geometric Newton interpolation                | 0     | 2       | 7    | 0   |
| Central $q$ -binomial sums and Lambert series                            | 0     | 0       | 7    | 0   |
| Asymptotic regimes and Bernoulli expansions                              | 0     | 0       | 8    | 0   |
| Computation, certification, and formal proof architecture                | 0     | 2       | 8    | 3   |
| Geometric interpolation and exact $q$ -filters                           | 6     | 2       | 1    | 0   |
| Half-base arithmetic and Prouhet–Thue–Morse cancellation                 | 6     | 3       | 0    | 0   |
| Geometric uniform sums, $q$ -cumulants, and the Fabius–Rvachev functions | 2     | 7       | 3    | 0   |
| <b>Total</b>                                                             | 31    | 25      | 124  | 3   |

## D.2 By result

| Result                                                  | Status       | Lean counterpart                                                         |
|---------------------------------------------------------|--------------|--------------------------------------------------------------------------|
| <i>The algebra of <math>q</math>-shifted factorials</i> |              |                                                                          |
| <a href="#">Proposition 1.2</a>                         | <b>exact</b> | <code>finiteQPochhammerIn_add</code><br><code>FiniteQBinomialCore</code> |

| Result                                          | Status  | Lean counterpart                                                                                                                                                                |
|-------------------------------------------------|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Proposition 1.3                                 | none    | —                                                                                                                                                                               |
| Corollary 1.4                                   | none    | —                                                                                                                                                                               |
| Proposition 1.6                                 | none    | —                                                                                                                                                                               |
| Proposition 1.8                                 | none    | —                                                                                                                                                                               |
| Corollary 1.9                                   | partial | qPochhammer_two_pow_eq_mersenne_div<br>HalfQBinomial                                                                                                                            |
| Theorem 1.10                                    | none    | —                                                                                                                                                                               |
| Proposition 1.11                                | none    | —                                                                                                                                                                               |
| Theorem 1.12                                    | none    | —                                                                                                                                                                               |
| Proposition 1.14                                | none    | —                                                                                                                                                                               |
| Proposition 1.15                                | none    | —                                                                                                                                                                               |
| Theorem 1.16                                    | partial | tsum_log_one_sub_geom<br>EulerLogTransform                                                                                                                                      |
| Corollary 1.17                                  | none    | —                                                                                                                                                                               |
| Theorem 1.18                                    | none    | —                                                                                                                                                                               |
| Proposition 1.19                                | none    | —                                                                                                                                                                               |
| <i>q-integers and Gaussian coefficients</i>     |         |                                                                                                                                                                                 |
| Proposition 2.2                                 | partial | qPochhammer_tail_div_self_eq_qBinomial<br>GeometricLagrangeQMoments                                                                                                             |
| Theorem 2.3                                     | exact   | gaussianBinomial_succ_succ_alt<br>FiniteQBinomialCore                                                                                                                           |
| Corollary 2.4                                   | none    | —                                                                                                                                                                               |
| Theorem 2.5                                     | partial | gaussianBinomial_symm<br>FiniteQBinomialCore (HalfQBinomial)                                                                                                                    |
| Corollary 2.6                                   | none    | —                                                                                                                                                                               |
| Proposition 2.8                                 | none    | —                                                                                                                                                                               |
| Proposition 2.9                                 | none    | —                                                                                                                                                                               |
| Theorem 2.10                                    | none    | —                                                                                                                                                                               |
| <i>Finite q-binomial theorems and inversion</i> |         |                                                                                                                                                                                 |
| Theorem 3.1                                     | exact   | finite_qBinomial_theorem<br>FiniteQBinomialCore                                                                                                                                 |
| Corollary 3.2                                   | exact   | sum_gaussianBinomial_alternating_mul_pow,<br>sum_gaussianBinomial_alternating<br>QBinomialInversion; arbitrary commutative ring, with the<br>row-zero boundary included         |
| Corollary 3.4                                   | exact   | sum_pow_sum_powersetCard_eq_<br>gaussianBinomial, sum_pow_sum_powersetCard_<br>Icc_eq_gaussianBinomial<br>BitPositionQBinomial; total zero-based and literal one-based<br>forms |

| Result                                                        | Status  | Lean counterpart                                                                                                                                                                                                      |
|---------------------------------------------------------------|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Theorem 3.6                                                   | partial | prod_one_sub_qPow_X_mul_gaussianBinomialGeneratingSeries<br>SymmetricFunctionGenerating; exact formal identity in $R[[X]]$ over every commutative ring, but no analytic evaluation or convergence proof for $ z  < 1$ |
| Theorem 3.8                                                   | exact   | gaussianBinomial_add_vandermonde, gaussianBinomial_add_vandermonde',<br>QBinomialVandermonde                                                                                                                          |
| Corollary 3.9                                                 | exact   | gaussianBinomial_add_central<br>QBinomialVandermonde                                                                                                                                                                  |
| Corollary 3.10                                                | exact   | gaussianBinomial_two_mul_int_shifted_central_fsum<br>QBinomialVandermonde, using gaussianBinomialInt from FiniteQBinomialCore; total in $k \in \mathbb{Z}$ , with support contained in $0 \leq \ell \leq N$           |
| Theorem 3.12                                                  | exact   | gaussianBinomial_inversion<br>QBinomialTransform; module-valued, hence in particular ring-valued sequences                                                                                                            |
| Corollary 3.13                                                | exact   | gaussianBinomialKernel_left_orthogonality, gaussianBinomialKernel_right_orthogonality<br>QBinomialInversion; both total lcc convolution orders                                                                        |
| Theorem 3.15                                                  | exact   | finite_qCauchy_identity<br>QBinomialCauchy; reflected identity and q-Bernstein partition-of-unity strengthenings also exposed                                                                                         |
| <i>Weighted binomial coefficients and symmetric functions</i> |         |                                                                                                                                                                                                                       |
| Theorem 4.2                                                   | exact   | elementarySymmetricEval_option_succ, completeHomogeneousEval_option_succ<br>SymmetricFunctionOrthogonality, CompleteHomogeneous; both adjoining-variable recurrences, all degrees                                     |

| Result                                                    | Status  | Lean counterpart                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-----------------------------------------------------------|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Theorem 4.3                                               | partial | <code>elementarySymmetricGeneratingSeries_eq_prod</code> , <code>elementarySymmetricGeneratingSeries_neg_mul_</code><br><code>completeHomogeneousGeneratingSeries</code> , <code>completeHomogeneousGeneratingSeries_eq_invOfUnit_</code><br><code>elementarySymmetricGeneratingSeries_neg_prod_one_sub_mul_</code><br><code>completeHomogeneousGeneratingSeriesOn</code> , <code>completeHomogeneousGeneratingSeriesOn_eq_invOfUnit_prod</code><br><code>SymmetricFunctionGenerating</code> and <code>CompleteHomogeneousGenerating</code> ; complementary finite-family formal surfaces prove the elementary product, elementary–complete reciprocity, and the canonical complete-homogeneous inverse, but not analytic evaluation or convergence under $ wiz  < 1$ |
| Corollary 4.4                                             | exact   | <code>sum_elementarySymmetricEval_mul_</code><br><code>completeHomogeneousEval</code><br><code>SymmetricFunctionOrthogonality</code> ; arbitrary finite family, including empty family and degree zero                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Theorem 4.6                                               | partial | <code>completeHomogeneousEval_geometric</code><br><code>GeometricCompleteHomogeneous</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Lemma 4.8                                                 | none    | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Theorem 4.9                                               | none    | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Corollary 4.10                                            | none    | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Theorem 4.12                                              | exact   | <code>weightedSymmetricFunction_inversion</code><br><code>SymmetricFunctionTransform</code> ; whole-function iff over a commutative ring, strengthened to sequences in any additive commutative monoid carrying the module structure                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <i>Partitions, words, paths, and inversion statistics</i> |         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Theorem 5.1                                               | none    | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Theorem 5.3                                               | none    | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Corollary 5.4                                             | partial | <code>qBinomial_symm</code><br><code>HalfQBinomial</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Corollary 5.5                                             | none    | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Theorem 5.6                                               | exact   | <code>gaussianBinomial_add_central_min</code><br><code>QBinomialVandermonde</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Theorem 5.7                                               | none    | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Proposition 5.9                                           | none    | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Proposition 5.10                                          | none    | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Corollary 5.11                                            | none    | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <i>Finite fields, Grassmannians, and rank enumeration</i> |         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

| Result                                                              | Status  | Lean counterpart                                                                                            |
|---------------------------------------------------------------------|---------|-------------------------------------------------------------------------------------------------------------|
| Theorem 6.1                                                         | none    | —                                                                                                           |
| Theorem 6.2                                                         | partial | gaussianBinomial_succ_succ<br>FiniteQBinomialCore                                                           |
| Theorem 6.3                                                         | none    | —                                                                                                           |
| Theorem 6.4                                                         | none    | —                                                                                                           |
| Theorem 6.5                                                         | none    | —                                                                                                           |
| Proposition 6.7                                                     | none    | —                                                                                                           |
| <i>Symmetry and unimodality</i>                                     |         |                                                                                                             |
| Lemma 7.1                                                           | none    | —                                                                                                           |
| Proposition 7.2                                                     | none    | —                                                                                                           |
| Theorem 7.3                                                         | none    | —                                                                                                           |
| <i>The infinite <math>q</math>-binomial theorem</i>                 |         |                                                                                                             |
| Theorem 8.1                                                         | none    | —                                                                                                           |
| Corollary 8.2                                                       | none    | —                                                                                                           |
| Corollary 8.3                                                       | none    | —                                                                                                           |
| Proposition 8.4                                                     | none    | —                                                                                                           |
| Proposition 8.5                                                     | none    | —                                                                                                           |
| <i>Basic hypergeometric series</i>                                  |         |                                                                                                             |
| Proposition 9.2                                                     | none    | —                                                                                                           |
| Theorem 9.5                                                         | none    | —                                                                                                           |
| Theorem 9.6                                                         | none    | —                                                                                                           |
| Lemma 9.7                                                           | exact   | finite_qCauchy_second_identity<br>QBinomialCauchy; arbitrary commutative ring and all<br>parameters/degrees |
| Theorem 9.9                                                         | none    | —                                                                                                           |
| Corollary 9.10                                                      | none    | —                                                                                                           |
| <i>Bilateral series and Ramanujan's <math>{}_1\psi_1</math> sum</i> |         |                                                                                                             |
| Proposition 10.1                                                    | none    | —                                                                                                           |
| Theorem 10.2                                                        | none    | —                                                                                                           |
| <i><math>q</math>-difference calculus and the quantum binomial</i>  |         |                                                                                                             |
| Proposition 11.1                                                    | none    | —                                                                                                           |
| Theorem 11.2                                                        | none    | —                                                                                                           |
| Lemma 11.3                                                          | none    | —                                                                                                           |
| Theorem 11.4                                                        | none    | —                                                                                                           |
| Theorem 11.6                                                        | none    | —                                                                                                           |
| Corollary 11.7                                                      | none    | —                                                                                                           |
| Corollary 11.9                                                      | none    | —                                                                                                           |
| Proposition 11.10                                                   | none    | —                                                                                                           |

| Result                                                            | Status | Lean counterpart                                                                                                                                                                                              |
|-------------------------------------------------------------------|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Theorem 11.11                                                     | none   | —                                                                                                                                                                                                             |
| Corollary 11.12                                                   | none   | —                                                                                                                                                                                                             |
| <i>The <math>q</math>-gamma and <math>q</math>-beta functions</i> |        |                                                                                                                                                                                                               |
| Theorem 12.2                                                      | none   | —                                                                                                                                                                                                             |
| Proposition 12.3                                                  | none   | —                                                                                                                                                                                                             |
| Theorem 12.4                                                      | none   | —                                                                                                                                                                                                             |
| Theorem 12.5                                                      | none   | —                                                                                                                                                                                                             |
| Theorem 12.7                                                      | none   | —                                                                                                                                                                                                             |
| Corollary 12.8                                                    | none   | —                                                                                                                                                                                                             |
| Theorem 12.9                                                      | none   | —                                                                                                                                                                                                             |
| Proposition 12.10                                                 | none   | —                                                                                                                                                                                                             |
| <i>Jacobi's triple product</i>                                    |        |                                                                                                                                                                                                               |
| Theorem 13.1                                                      | none   | —                                                                                                                                                                                                             |
| Corollary 13.2                                                    | none   | —                                                                                                                                                                                                             |
| Theorem 13.7                                                      | none   | —                                                                                                                                                                                                             |
| Corollary 13.8                                                    | none   | —                                                                                                                                                                                                             |
| Theorem 13.9                                                      | none   | —                                                                                                                                                                                                             |
| Corollary 13.10                                                   | none   | —                                                                                                                                                                                                             |
| <i>Partitions and product–sum identities</i>                      |        |                                                                                                                                                                                                               |
| Theorem 14.1                                                      | none   | —                                                                                                                                                                                                             |
| Proposition 14.3                                                  | none   | —                                                                                                                                                                                                             |
| Corollary 14.4                                                    | none   | —                                                                                                                                                                                                             |
| Proposition 14.5                                                  | none   | —                                                                                                                                                                                                             |
| <i>Bailey pairs and the Rogers–Ramanujan identities</i>           |        |                                                                                                                                                                                                               |
| Lemma 15.2                                                        | exact  | qDifference_sum_eval <sub>2</sub> _eq_zero_of_degree_lt<br>qDifferenceAnnihilation; semiring coefficients, arbitrary<br>scalar extension, commutative-ring target; no division or<br>distinct-node hypothesis |
| Theorem 15.4                                                      | none   | —                                                                                                                                                                                                             |
| Corollary 15.5                                                    | none   | —                                                                                                                                                                                                             |
| Lemma 15.6                                                        | none   | —                                                                                                                                                                                                             |
| Theorem 15.7                                                      | none   | —                                                                                                                                                                                                             |
| Corollary 15.8                                                    | none   | —                                                                                                                                                                                                             |
| Theorem 15.9                                                      | none   | —                                                                                                                                                                                                             |
| <i>Cyclotomic factorization and congruences</i>                   |        |                                                                                                                                                                                                               |
| Theorem 16.1                                                      | none   | —                                                                                                                                                                                                             |
| Corollary 16.2                                                    | none   | —                                                                                                                                                                                                             |
| Corollary 16.4                                                    | none   | —                                                                                                                                                                                                             |

| Result                                                           | Status  | Lean counterpart                                  |
|------------------------------------------------------------------|---------|---------------------------------------------------|
| Lemma 16.6                                                       | none    | —                                                 |
| Theorem 16.7                                                     | none    | —                                                 |
| Corollary 16.8                                                   | none    | —                                                 |
| Theorem 16.9                                                     | none    | —                                                 |
| Theorem 16.10                                                    | none    | —                                                 |
| Corollary 16.12                                                  | none    | —                                                 |
| <i>Negative upper indices and geometric Newton interpolation</i> |         |                                                   |
| Proposition 17.2                                                 | none    | —                                                 |
| Theorem 17.3                                                     | partial | gaussianBinomial_succ_succ<br>FiniteQBinomialCore |
| Corollary 17.4                                                   | none    | —                                                 |
| Theorem 17.5                                                     | none    | —                                                 |
| Theorem 17.7                                                     | none    | —                                                 |
| Corollary 17.8                                                   | none    | —                                                 |
| Theorem 17.9                                                     | partial | finite_qBinomial_theorem<br>FiniteQBinomialCore   |
| Theorem 17.10                                                    | none    | —                                                 |
| Corollary 17.11                                                  | none    | —                                                 |
| <i>Central q-binomial sums and Lambert series</i>                |         |                                                   |
| Lemma 18.1                                                       | none    | —                                                 |
| Theorem 18.2                                                     | none    | —                                                 |
| Corollary 18.3                                                   | none    | —                                                 |
| Theorem 18.4                                                     | none    | —                                                 |
| Theorem 18.6                                                     | none    | —                                                 |
| Theorem 18.7                                                     | none    | —                                                 |
| Proposition 18.8                                                 | none    | —                                                 |
| <i>Asymptotic regimes and Bernoulli expansions</i>               |         |                                                   |
| Lemma 19.1                                                       | none    | —                                                 |
| Theorem 19.2                                                     | none    | —                                                 |
| Corollary 19.4                                                   | none    | —                                                 |
| Theorem 19.6                                                     | none    | —                                                 |
| Theorem 19.7                                                     | none    | —                                                 |
| Proposition 19.9                                                 | none    | —                                                 |
| Corollary 19.10                                                  | none    | —                                                 |
| Proposition 19.11                                                | none    | —                                                 |
| <i>Computation, certification, and formal proof architecture</i> |         |                                                   |
| Algorithm 20.1                                                   | n/a     |                                                   |
| Theorem 20.2                                                     | partial | gaussianBinomial<br>FiniteQBinomialCore           |

| Result                                                          | Status  | Lean counterpart                                                                            |
|-----------------------------------------------------------------|---------|---------------------------------------------------------------------------------------------|
| Algorithm 20.4                                                  | n/a     |                                                                                             |
| Theorem 20.5                                                    | none    | —                                                                                           |
| Theorem 20.7                                                    | none    | —                                                                                           |
| Corollary 20.8                                                  | none    | —                                                                                           |
| Lemma 20.10                                                     | none    | —                                                                                           |
| Corollary 20.11                                                 | none    | —                                                                                           |
| Lemma 20.12                                                     | partial | finiteQPochhammerIn_self_mul_<br>gaussianBinomial<br>FiniteQBinomialCore                    |
| Theorem 20.13                                                   | none    | —                                                                                           |
| Corollary 20.15                                                 | none    | —                                                                                           |
| Theorem 20.17                                                   | none    | —                                                                                           |
| Principle 20.18                                                 | n/a     |                                                                                             |
| <i>Geometric interpolation and exact <math>q</math>-filters</i> |         |                                                                                             |
| Theorem 21.1                                                    | exact   | reversed_finite_qBinomial_theorem<br>GeometricQBinomialLagrange                             |
| Theorem 21.3                                                    | exact   | geometricLagrangeWeight_eq_<br>gaussianBinomial_div<br>GeometricQBinomialLagrange           |
| Corollary 21.4                                                  | exact   | geometricLagrangeWeight_eq_product<br>GeometricLagrangeWeights                              |
| Theorem 21.6                                                    | exact   | sum_lagrangeEvalWeight_mul_pow_card_add<br>LagrangeResidualMoments                          |
| Theorem 21.7                                                    | exact   | sum_geometricLagrangeWeight_mul_pow_eq_<br>gaussianBinomial<br>GeometricQBinomialLagrange   |
| Corollary 21.8                                                  | partial | sum_geometricLagrangeWeight_mul_scaled_<br>geometric_pow_of_pos<br>GeometricResidualMoments |
| Theorem 21.9                                                    | exact   | coeff_geometricSeriesFilter<br>GeometricPowerSeriesFilter                                   |
| Corollary 21.10                                                 | partial | geometricLagrangeAnalyticSeriesFilter_eq_<br>constant_add_gaussi<br>AnalyticSeriesFilter    |
| Theorem 21.11                                                   | none    | —                                                                                           |
| <i>Half-base arithmetic and Prouhet–Thue–Morse cancellation</i> |         |                                                                                             |
| Theorem 22.1                                                    | exact   | halfQPochhammer_eq_mersenne_div<br>HalfQBinomial                                            |
| Theorem 22.2                                                    | exact   | qPochhammer_two_pow_eq_ite<br>HalfQBinomial                                                 |
| Corollary 22.3                                                  | partial | halfQBinomial_sum_eq_zero_iff<br>HalfQBinomial                                              |



| Result                                                                       | Status  | Lean counterpart                                                               |
|------------------------------------------------------------------------------|---------|--------------------------------------------------------------------------------|
| Theorem 22.4                                                                 | exact   | halfQBinomial_negativeDyadic_polynomial_sum_eq_mersenne<br>HalfQBinomial       |
| Corollary 22.5                                                               | partial | halfQBinomial_negativeDyadic_polynomial_sum_eq_mersenne<br>HalfQBinomial       |
| Theorem 22.7                                                                 | exact   | prod_one_sub_pow_two_pow<br>ThueMorseProductIdentity                           |
| Theorem 22.8                                                                 | exact   | sum_thueMorseSign_mul_affine_pow_eq_zero<br>ThueMorseSparseProuhet             |
| Corollary 22.9                                                               | partial | sum_thueMorseSign_mul_affine_pow_eq_zero<br>ThueMorseSparseProuhet             |
| Principle 22.10                                                              | exact   | halfQBinomial_negativeDyadic_polynomial_sum_eq_mersenne<br>HalfQBinomial       |
| <i>Geometric uniform sums, q-cumulants, and the Fabius–Rvachev functions</i> |         |                                                                                |
| Theorem 23.1                                                                 | partial | geometricUniformSeries_mem_Icc<br>GeometricUniformLaw                          |
| Theorem 23.2                                                                 | partial | charFun_geometricUniformDistribution_prefix_sinc<br>GeometricSincFactorization |
| Theorem 23.3                                                                 | none    | —                                                                              |
| Corollary 23.5                                                               | partial | completeBellPolynomial<br>MomentCumulantAlgebra                                |
| Corollary 23.6                                                               | none    | —                                                                              |
| Theorem 23.7                                                                 | partial | norm_rvachevFourierProduct_le_decayGauge_abs<br>GlobalDecayEnvelope            |
| Theorem 23.8                                                                 | partial | geometricUniformDistribution_eq_withDensity<br>GeometricUniformCDF             |
| Theorem 23.9                                                                 | none    | —                                                                              |
| Theorem 23.11                                                                | partial | fabiusUniformSpline_eq_tsum_positivePart<br>FabiusUniformSpline                |
| Corollary 23.13                                                              | partial | abs_uniformPartialSum_sub_weightedCoordinateSum_le<br>FabiusUniformSpline      |
| Theorem 23.15                                                                | exact   | weightedSumCDF_eq_fabiusReal<br>ProbabilityRepresentation                      |
| Theorem 23.16                                                                | exact   | rvachevFourier_eq_product<br>FourierProduct                                    |

# Bibliography

- [1] G. E. Andrews, *The Theory of Partitions*, Addison–Wesley, 1976; Cambridge University Press reprint, 1998.
- [2] G. E. Andrews, *q-Series: Their Development and Application in Analysis, Number Theory, Combinatorics, Physics, and Computer Algebra*, CBMS Regional Conference Series in Mathematics 66, American Mathematical Society, 1986.
- [3] G. E. Andrews,  $q$ -analogs of the binomial coefficient congruences of Babbage, Wolstenholme and Glaisher, *Discrete Mathematics* **204** (1999), 15–25.
- [4] G. E. Andrews, R. Askey, and R. Roy, *Special Functions*, Encyclopedia of Mathematics and its Applications 71, Cambridge University Press, 1999.
- [5] W. N. Bailey, *Generalized Hypergeometric Series*, Cambridge Tracts in Mathematics and Mathematical Physics 32, Cambridge University Press, 1935.
- [6] D. M. Bressoud, *Proofs and Confirmations: The Story of the Alternating Sign Matrix Conjecture*, MAA Spectrum, Mathematical Association of America, 1999.
- [7] S. Chern, K. Dilcher, and L. Jiu, New central  $q$ -binomial identities, preprint, arXiv:2608.05498, 2026.
- [8] NIST Digital Library of Mathematical Functions, Chapter 17, “ $q$ -Hypergeometric and Related Functions,” F. W. J. Olver et al., editors, <https://dlmf.nist.gov/17>, accessed August 2026.
- [9] N. J. Fine, *Basic Hypergeometric Series and Applications*, Mathematical Surveys and Monographs 27, American Mathematical Society, 1988.
- [10] G. Gasper, Lecture notes for an introductory minicourse on  $q$ -series, preprint, arXiv:math/9509223, 1995.
- [11] G. Gasper and M. Rahman, *Basic Hypergeometric Series*, second edition, Encyclopedia of Mathematics and its Applications 96, Cambridge University Press, 2004.
- [12] J. Goldman and G.-C. Rota, On the foundations of combinatorial theory IV: finite vector spaces and Eulerian generating functions, *Studies in Applied Mathematics* **49** (1970), 239–258.
- [13] E. Heine, *Handbuch der Kugelfunctionen*, second edition, G. Reimer, Berlin, 1878.
- [14] M. E. H. Ismail, *Classical and Quantum Orthogonal Polynomials in One Variable*, Encyclopedia of Mathematics and its Applications 98, Cambridge University Press, 2005.
- [15] F. H. Jackson, On  $q$ -functions and a certain difference operator, *Transactions of the Royal Society of Edinburgh* **46** (1908), 253–281.

- [16] V. Kac and P. Cheung, *Quantum Calculus*, Universitext, Springer, 2002.
- [17] R. Koekoek, P. A. Lesky, and R. F. Swarttouw, *Hypergeometric Orthogonal Polynomials and Their  $q$ -Analogues*, Springer Monographs in Mathematics, Springer, 2010.
- [18] J. Konvalina, Generalized binomial coefficients and the subset-subspace problem, *Advances in Applied Mathematics* **21** (1998), 228–240.
- [19] B. A. Kupershmidt,  $q$ -Newton binomial: from Euler to Gauss, *Journal of Nonlinear Mathematical Physics* **7** (2000), 244–262.
- [20] K. Lau, S. Lee, and K. Ono, Formalized  $q$ -series: the Rogers–Ramanujan identities and beyond, preprint, arXiv:2607.01544, 2026.
- [21] I. G. Macdonald, *Symmetric Functions and Hall Polynomials*, second edition, Oxford Mathematical Monographs, Oxford University Press, 1995.
- [22] K. M. O’Hara, Unimodality of Gaussian coefficients: a constructive proof, *Journal of Combinatorial Theory, Series A* **53** (1990), 29–52.
- [23] M. Petkovšek, H. S. Wilf, and D. Zeilberger,  $A = B$ , A K Peters, Wellesley, MA, 1996.
- [24] V. Reiner, D. Stanton, and D. White, The cyclic sieving phenomenon, *Journal of Combinatorial Theory, Series A* **108** (2004), 17–50.
- [25] L. J. Rogers, Second memoir on the expansion of certain infinite products, *Proceedings of the London Mathematical Society* **25** (1894), 318–343.
- [26] G.-C. Rota, On the foundations of combinatorial theory I: theory of Möbius functions, *Zeitschrift für Wahrscheinlichkeitstheorie und Verwandte Gebiete* **2** (1964), 340–368.
- [27] B. E. Sagan, Congruence properties of  $q$ -analogs, *Advances in Mathematics* **95** (1992), 127–143.
- [28] L. J. Slater, *Generalized Hypergeometric Functions*, Cambridge University Press, 1966.
- [29] R. P. Stanley, *Enumerative Combinatorics*, Volume 1, second edition, Cambridge Studies in Advanced Mathematics 49, Cambridge University Press, 2012.
- [30] R. P. Stanley, *Enumerative Combinatorics*, Volume 2, Cambridge Studies in Advanced Mathematics 62, Cambridge University Press, 1999.
- [31] A. Straub, A  $q$ -analog of Ljunggren’s binomial congruence, *DMTCS Proceedings, FPSAC 2011*, 897–902; arXiv:1103.3258.
- [32] D. Zeilberger, Kathy O’Hara’s constructive proof of the unimodality of the Gaussian polynomials, *American Mathematical Monthly* **96** (1989), 590–602.
- [33] W. Zudilin, Congruences for  $q$ -binomial coefficients, *Annals of Combinatorics* **23** (2019), 1123–1135.
- [34] J. Arias de Reyna, An infinitely differentiable function with compact support: definition and properties, *Revista de la Real Academia de Ciencias Exactas, Físicas y Naturales de Madrid* **76** (1982), 21–38; English translation, arXiv:1702.05442.
- [35] D. M. Bradley and R. C. Gupta, On the distribution of the sum of  $n$  non-identically distributed uniform random variables, preprint, arXiv:math/0411298, 2004.

- [36] M. Eberl, Combinatorial  $q$ -analogues, *Archive of Formal Proofs*, 2024, current document revision February 2026, [https://isa-afp.org/entries/Combinatorial\\_Q\\_Analogues.html](https://isa-afp.org/entries/Combinatorial_Q_Analogues.html).
- [37] J. Fabius, A probabilistic example of a nowhere analytic  $C^\infty$ -function, *Zeitschrift für Wahrscheinlichkeitstheorie und Verwandte Gebiete* **5** (1966), 173–174.
- [38] V. Reshetnikov, *ProveIt: Fabius Function development and documentation*, GitHub repository, 2026, <https://github.com/VladimirReshetnikov/ProveIt>.
- [39] V. A. Rvachev, Compactly supported solutions of functional-differential equations and their applications, *Russian Mathematical Surveys* **45** (1990), no. 1, 87–120.
- [40] H. Zhong and L. Zhao,  $q$ -binomial identities finder, *Advances in Applied Mathematics* **172** (2026), article 102965; arXiv:2408.00186.

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